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Northern Nut Growers Association Incorporated 39th Annual Report

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[Illustration]

NORTHERN

NUT GROWERS

ASSOCIATION

INCORPORATED

39th Annual Report

[Illustration]

CONVENTION AT NORRIS, TENN.

SEPTEMBER 13-15

1948

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OFFICERS OF THE ASSOCIATION 1949

President--H. F. Stoke, 1436 Watts Avenue, Roanoke, Virginia

Vice-President--Dr. L. H. MacDaniels, Dept. of Floriculture and Ornamental Horticulture, Cornell University, Ithaca, New York

Treasurer--Sterling A. Smith, 630 West South Street, Vermilion, Ohio

Secretary--J. C. McDaniel, Tennessee Dept. of Agriculture, State Office Bldg., Nashville 3, Tennessee

Directors include above officers _plus_: John Davidson, 234 E. Second Street, Xenia, Ohio; and Clarence A. Reed, 7309 Piney Branch Road, N. W., Washington 12, D.C.

Dean--Dr. W. C. Deming, 31 S. Highland, West Hartford 7, Connecticut

Nominating Committee--Dr. H. L. Crane, Harry R. Weber, Dr. William L. Rohrbacher, J. Ford Wilkinson, George L. Slate

EXECUTIVE APPOINTMENTS

Press and Publications--Editorial Section: Dr. Lewis E. Theiss, Dr. W. C. Deming, Dr. L. H. MacDaniels, C. A. Reed, Dr. A. S. Colby, George L. Slate, Dr. J., Russell Smith Publicity Section: Dr. J. Russell Smith, C. A. Reed, Dr. A. S. Colby, Carrol D. Bush, A. A. Bungart, J. C. McDaniel Printing Section: John Davidson, Harry R. Weber, J. C. McDaniel

Program--H. L. Crane, R. P. Allaman, George L. Slate, C. A. Reed, J. C. McDaniel, Raymond E. Silvis

Place of Meeting--Dr. A. S. Colby, J. F. Wilkinson, D. C. Snyder, Carl F. Walker, H. H. Corsan

Varieties and Contests--Spencer B. Chase, G. J. Korn, J. F. Wilkinson, Gilbert Becker, A. G. Hirschi, L. Walter Sherman, C. A. Reed, Dr. L. H. MacDaniels, Dr. J. Russell Smith Standards and Judging section of this committee: Dr. L. H. MacDaniels, Spencer B. Chase, C. A. Reed, Dr. J. Russell Smith

Survey and Research--R. E. Silvis, plus the state and foreign vice-presidents

Membership--Mrs. Harry Weber, Mrs. Blaine McCollum, Mrs. Stephen Bernath

Exhibits--R. P. Allaman, Carl Weschcke, Fayette Etter, A. G. Hirschi, G. J. Korn, J. F. Wilkinson, G. L. Smith, Seward Berhow, Royal Oakes, H. H. Corsan, G. H. Corsan

Necrology--Mrs. Herbert Negus, Mrs. Wm. Rohrbacher, Miss Jeannette F. Johns, Barbara Sly

Audit--Dr. Wm. Rohrbacher, E. P. Gerber, Raymond E. Silvis

Finance--Harry Weber, D. C. Snyder, Carl Weschcke, Sterling Smith

Legal Advisers--Sargent Wellman, Harry Weber

Official Journal--American Fruit Grower, 1370 Ontario St., Cleveland 13, Ohio

State and Foreign Vice-Presidents

Alabama LOVIC ORR

Alberta, Canada A. L. YOUNG

Arkansas A. C. HALE

British Columbia, Canada J. U. GELLATLY

California DR. THOMAS R. HAIG

Connecticut GEORGE D. PRATT, JR.

Delaware LEWIS WILKINS

Denmark COUNT F. M. KNUTH

District of Columbia GEORGE U. GRAFF

Ecuador, South America F. A. COLWELL

Florida C. A. AVANT

Georgia WM. J. WILSON

Idaho J. E. MCGORAN

Illinois ROYAL OAKES

Indiana FORD WALLICK

Iowa IRA M. KYHL

Kansas DR. CLYDE GRAY

Kentucky DR. C. A. MOSS

Manitoba, Canada A. H. YOUNG

Maryland BLAINE MCCOLLUM

Massachusetts I. W. SHORT

Mexico FEDERICO COMPEAN

Michigan GILBERT BECKER

Minnesota R. E. HODGSON

Mississippi JAMES R. MEYER

Missouri RALPH RICHTERKESSING

Nebraska GEORGE BRAND

New Hampshire MATTHEW LAHTI

New Jersey MRS. ALAN R. BUCKWALTER

New Mexico REV. TITUS GEHRING

New York GEORGE SALZER

North Carolina DR. R. T. DUNSTAN

North Dakota HOMER L. BRADLEY

Ohio A. A. BUNGART

Oklahoma A. G. HIRSCHI

Ontario, Canada G. H. CORSAN

Oregon HARRY L. PEARCY

Pennsylvania R. P. ALLAMAN

Prince Edward Island, Canada ROBERT SNAZELLE

Rhode Island PHILIP ALLEN

South Carolina JOHN T. BREGGER

South Dakota HERMAN RICHTER

Tennessee THOMAS G. ZARGER

Texas KAUFMAN FLORIDA

Utah HARLAN D. PETTERSON

Vermont A. W. ALDRICH

Virginia H. R. GIBBS

Washington CARROLL D. BUSH

West Virginia WILBERT M. FRYE

Wisconsin NORMAN KOELSCH

CONSTITUTION

of the

NORTHERN NUT GROWERS ASSOCIATION, INCORPORATED

(As read at the annual meeting, Guelph, Ontario, September 5, 1947, and adopted September 13, 1948, at Norris, Tennessee)

NAME

ARTICLE I. This Society shall be known as the Northern Nut Growers Association, Incorporated. It is strictly a non-profit organization.

PURPOSES

ARTICLE II. The purposes of this Association shall be to promote interest in the nut bearing plants; scientific research in their breeding and culture; standardization of varietal names the dissemination of information concerning the above and such other purposes as may advance the culture of nut bearing plants, particularly in the North Temperate Zone.

MEMBERS

ARTICLE III. Membership in this Association shall be open to all persons interested in supporting the purposes of the Association. Classes of members are as follows: Annual members, Contributing members, Life members, Honorary members, and Perpetual members. Applications for membership in the Association shall be presented to the secretary or the treasurer in writing, accompanied by the required dues.

OFFICERS

ARTICLE IV. The elected officers of this Association shall consist of a President, Vice-president, a Secretary and a Treasurer or a combined Secretary-treasurer as the Association may designate.

BOARD OF DIRECTORS

ARTICLE V. The Board of Directors shall consist of six members of the Association who shall be the officers of the Association and the two preceding elected presidents. If the offices of Secretary and Treasurer are combined, the three past presidents shall serve on the Board of Directors.

There shall be a State Vice-president for each state, dependency, or country represented in the membership of the Association, who shall be appointed by the President.

AMENDMENTS TO THE CONSTITUTION

ARTICLE VI. This constitution may be amended by a two-thirds vote of the members present at any annual meeting, notice of such amendment having been read at the previous annual meeting, or copy of the proposed amendments having been mailed by the Secretary or by any member to each member thirty days before the date of the annual meeting.

BY-LAWS

(Revised and adopted at Norris, Tennessee, September 13, 1948)

SECTION I.--MEMBERSHIP

Classes of membership are defined as follows:

ARTICLE 1. _Annual members._ Persons who are interested in the purposes of the Association who pay annual dues of Three Dollars ($3.00).

ARTICLE 2. _Contributing members._ Persons who are interested in the purposes of the Association who pay annual dues of Ten Dollars ($10.00) or more.

ARTICLE 3. _Life members._ Persons who are interested in the purposes of the Association who contribute Seventy Five Dollars ($75.00) to its support and who shall, after such contribution, pay no annual dues.

ARTICLE 4. _Honorary members._ Those whom the Association has elected as honorary members in recognition of their achievements in the special fields of the Association and who shall pay no dues.

ARTICLE 5. _Perpetual members._ "Perpetual" membership is eligible to any one who leaves at least five hundred dollars to the Association and such membership on payment of said sum to the Association shall entitle the name of the deceased to be forever enrolled in the list of members as "Perpetual" with the words "In Memoriam" added thereto. Funds received therefor shall be invested by the Treasurer in interest bearing securities legal for trust funds in the District of Columbia. Only the interest shall be expended by the Association. When such funds are in the treasury the Treasurer shall be bonded. Provided: that in the event the Association becomes defunct or dissolves, then, in that event, the Treasurer shall turn over any funds held in his hands for this purpose for such uses, individuals or companies that the donor may designate at the time he makes the bequest of the donation.

SECTION II.--DUTIES OF OFFICERS

ARTICLE 1. The President shall preside at all meetings of the Association and Board of Directors, and may call meetings of the Board of Directors when he believes it to be to the best interests of the Association. He shall appoint the State Vice-presidents; the standing committees, except the Nominating Committee, and such special committees as the Association may authorize.

ARTICLE 2. Vice-president. In the absence of the President, the Vice-president shall perform the duties of the President.

ARTICLE 3. Secretary. The Secretary shall be the active executive officer of the Association. He shall conduct the correspondence relating to the Association's interests, assist in obtaining memberships and otherwise actively forward the interests of the Association, and report to the Annual Meeting and from time to time to meetings of the Board of Directors as they may request.

ARTICLE 4. Treasurer. The Treasurer shall receive and record memberships, receive and account for all moneys of the Association and shall pay all bills approved by the President or the Secretary. He shall give such security as the Board of Directors may require or may legally be required, shall invest life memberships or other funds as the Board of Directors may direct, subject to legal restrictions and in accordance with the law, and shall submit a verified account of receipts and disbursements to the Annual meeting and such current accounts as the Board of Directors may from time to time require. Before the final business session of the Annual Meeting of the Association, the accounts of the Treasurer shall be submitted for examination to the Auditing Committee appointed by the President at the opening session of the Annual Meeting.

ARTICLE 5. The Board of Directors shall manage the affairs of the Association between meetings. Four members, including at least two elected officers, shall be considered a quorum.

SECTION III.--ELECTIONS

ARTICLE 1. The Officers shall be elected at the Annual Meeting and hold office for one year beginning immediately following the close of the Annual Meeting.

ARTICLE 2. The Nominating Committee shall present a slate of officers on the first day of the Annual Meeting and the election shall take place at the closing session. Nominations for any office may be presented from the floor at the time the slate is presented or immediately preceding the election.

ARTICLE 3. For the purpose of nominating officers for the year 1949 and thereafter, a committee of five members shall be elected annually at the preceding Annual Meeting.

ARTICLE 4. A quorum at a regularly called Annual Meeting shall be fifteen (15) members and must include at least two of the elected officers.

ARTICLE 5. All classes of members whose dues are paid shall be eligible to vote and hold office.

SECTION IV.--FINANCIAL MATTERS

ARTICLE 1. The fiscal year of the Association shall extend from October 1st through the following September 30th. All annual memberships shall begin October 1st.

ARTICLE 2. The names of all members whose dues have not been paid by January 1st shall be dropped from the rolls of the Society. Notices of non-payment of dues will be mailed to delinquent members on or about December 1st.

ARTICLE 3. The Annual Report shall be sent to only those members who have paid their dues for the current year. Members whose dues have not been paid by January 1st shall be considered delinquent. They will not be entitled to receive the publication or other benefits of the Association until dues are paid.

SECTION V.--MEETINGS

ARTICLE 1. The place and time of the Annual Meeting shall be selected by the membership in session or, in the event of no selection being made at this time, the Board of Directors shall choose the place and time for the holding of the annual convention. Such other meetings as may seem desirable may be called by the President and Board of Directors.

SECTION VI.--PUBLICATIONS

ARTICLE 1. The Association shall publish a report each fiscal year and such other publications as may be authorized by the Association.

ARTICLE 2. The publishing of the report shall be the responsibility of the Committee on Publications.

SECTION VII.--AWARDS

ARTICLE 1. The Association may provide suitable awards for outstanding contributions to the cultivation of nut bearing plants and suitable recognition for meritorious exhibits as may be appropriate.

SECTION VIII.--STANDING COMMITTEES

As soon as practicable after the Annual Meeting of the Association, the President shall appoint the following standing committees:

1. Membership 2. Auditing 3. Publications 4. Survey 5. Program 6. Research 7. Exhibit 8. Varieties and Contests

SECTION IX.--REGIONAL GROUPS AND AFFILIATED SOCIETIES

ARTICLE 1. The Association shall encourage the formation of regional groups of its members, who may elect their own officers and organize their own local field days and other programs. They may publish their proceedings and selected papers in the yearbooks of the parent society subject to review of the Association's Committee on Publications.

ARTICLE 2. Any independent regional association of nut growers may affiliate with the Northern Nut Growers Association provided one-fourth of its members are also members of the Northern Nut Growers Association. Such affiliated societies shall pay an annual affiliation fee of $3.00 to the Northern Nut Growers Association. Papers presented at the meetings of the regional society may be published in the proceedings of the parent society subject to review of the Association's Committee on Publications.

SECTION X.--AMENDMENTS TO BY-LAWS

ARTICLE 1. These by-laws may be amended at any Annual Meeting by a two-thirds vote of the members present provided such amendments shall have been submitted to the membership in writing at least thirty days prior to that meeting.

PROCEEDINGS of the Thirty-ninth Annual Convention of the Northern Nut Growers Association, Inc.

Meeting at NORRIS, TENNESSEE SEPTEMBER 13-15, 1948

The meeting was called to order by President John Davidson at 8:45 o'clock, a. m.

Address of Welcome

GEORGE F. GANT, General Manager, Tennessee Valley Authority, Knoxville, Tennessee

Mr. President, ladies and gentlemen: It is a distinct pleasure to welcome you to Norris and to the Tennessee Valley. You have had very fine weather here, and we hope that you will enjoy the climate and the scenery and the fishing and the pleasures of this part of the country during your short stay.

The Northern Nut Growers Association is a much older organization than I had thought, and it is much older than the Tennessee Valley Authority, but a review of some of the things, you have done and some of the interests you have expressed from time to time indicate that we have many interests in common, your organization and the TVA.

You are concerned with experimentation of new and better ways of growing tree crops. You are concerned with the environment in which tree crops must find a place in our economy and in our culture, because, as I understand it, your interest goes beyond mere economics to the full use of trees.

Now, the Tennessee Valley Authority is likewise concerned with experimentation. As a matter of fact, it is an experiment, a new and different way of achieving a better use of natural resources.

There is nothing new in what the TVA does. There are no activities conducted by TVA that have not been or are not being conducted by other agencies all over the country and which have been conducted by Federal agencies for many, many years. The TVA has no new regulatory or coercive functions. As a matter of fact, the TVA has no coercive functions. It has no new or unique or different governmental functions.

There is only one thing that is different about TVA, and that is the way in which it approaches the job of resource use on an overall basis.

Now, I might illustrate that by referring to the construction of dams and reservoirs. In the Tennessee Valley the TVA builds dams and reservoirs to prevent floods, to produce a navigable channel, to produce power, and in its reservoirs it also has the responsibility of achieving the best uses of reservoirs and reservoir lands in the interests of fish and wild life, in the interests of recreation, and in the interests of malaria control.

Now, the unique fact here is not that these things are going on or being done, at least in part, through a Federal agency, but that one Federal agency is responsible for achieving a balance between all of these activities and with the administrative responsibility for doing that. In other efforts the situation is different, with as many as eight agencies having something to do with the development of some one of these activities in a way which might or might not be integrated.

Now, the second illustration, I think, is that unity can be accomplished only if all of the agencies which are concerned with the use of resources have an environment in which they can work effectively. The Federal Government is not and should not in the Tennessee Valley be developing all of these resources itself. It feels that the unified development of the resources depends upon the participation of the people of the Tennessee Valley and their institutions, the local and the state agencies. There can't be unity any more if local agencies are conducting one program and a Federal agency conducting another program, than there can be if several Federal agencies are conducting several programs.

Consequently, the Tennessee Valley Authority, except for the operation of these huge new facilities which have been added to the resources of the Tennessee Valley, conducts its activities in collaboration with local and state agencies. That not only avoids the expense of duplication, but it achieves the collaboration, the participation, the active interest of the people in getting a full job done.

That is true in the field of forestry. Forestry has a particular role in the Tennessee Valley. First of all, the TVA is concerned with the effective use and control of water, not only in the river channel itself, but on the land. Forestry, together with engineering and agriculture, must come together, not only come together within the administrative framework of TVA, but within the framework of what our colleges and state departments are doing and with what the land owners are doing in these watersheds.

Further than that, the TVA is fully aware that watershed protection cannot be achieved except within the economy of the region. That means that the best use of forest lands from the economic point of view, from the productive point of view, as well as from the conservation point of view, must be taken into account.

For these reasons the TVA is concerned not only with multiple-purpose dams, but with multiple-purpose land use. These activities are not conducted directly by TVA, but in cooperation with the land grant colleges and with the appropriate state departments.

I think and I hope that as you review the several activities which are going on in the Tennessee Valley area that you will keep these characteristics of TVA in mind. We are very happy to have you here. I hope that many of you will be able to extend your visit or to come back and see us another time.

* * * * *

President John Davidson: I am personally very glad to have heard this talk. I know quite a bit more about the fundamental principles of the work underlying TVA than I did before.

Dr. MacDaniels, will you say a word on behalf of the Association?

+Response+

Dr. L. H. MacDaniels: Mr. President and members of the Northern Nut Growers Association, I am sure that I voice the sentiment of all of the Association to you, Mr. Gant, and all of the Tennessee Valley Authority our very great appreciation of your allowing us to come and meet with you and use the very fine facilities which are available here in Upper Norris Park.

As far as I am concerned, and probably I am in the same situation as most of you in the North; we have heard a lot about the Tennessee Valley Authority, but mostly it is bandied around in the newspapers and usually connected with some sort of a political argument of one kind or another. And I think that to come here and to see the place and to live in the cabins and drive through the forests, to swim in the lake, as some of us did yesterday afternoon, went far away around the bend, and went in swimming--I think you might improve the mud bottom in some places, which is not too good, but it reminds us of our youth, at least--and to fish in the lakes, although not too successfully. After we have done that we certainly know much more about what sort of a development the Tennessee Valley Authority is.

Another thing, as a member of the Northern Nut Growers Association and as you are members, I think we all appreciate what the Tennessee Valley Authority has done for the Northern Nut Growers Association. The Tennessee Valley Authority has been the first, you might say, large agency which has taken northern nut growing seriously and has used the knowledge which has been developed by this Association in an extensive way in the planting and developing of new varieties, developing of new techniques in the use of the plants, the nut trees and the persimmons, and what not, with which the Northern Nut Growers Association has been concerned.

As we drive up the valley here and we see these thousands of walnut seedlings which are still to be used and see the plantings which you will see more intimately later, we can realize just how extensively the Tennessee Valley Authority has been concerned with the development of our forest resources and particularly these plants which are of economic value, inasmuch as they are nut trees, and their relationship to wildlife and a project of this kind in which forest resources and tree resources are to be made use of.

I have noticed that you did mention fishing as one of the things that has been developed by the Tennessee Valley Authority. I also am reminded of the fact that

some of us, including our president, tried to go out and exercise some of these fish, without much success, and I have been trying to think of the reason. I know, as far as we are concerned, we used all the plugs and spinners and floating baits and sinking baits, and I went completely through my tackle box and pulled out the one that we call the "Christmas tree," a big bunch of spoons with a place to put a minnow on the end, and we dragged that around, almost swamped the motor, but did get around; didn't catch anything.

It reminds me of an incident there at Cornell. We have a director, who was head of the Pomology Department at that time. He had a dog that wasn't disciplined very well, he wouldn't come when he was called, and so on. The foreman out at the orchard had a dog that was very well disciplined. He'd say, "Go get my hat," and he'd get the hat, and "Go quickly," and he'd go quickly. And this head of the department asked the foreman, "Well, how was it that you trained this dog, and how do you train a dog, anyway?"

"Well," he said, "first of all, you have got to know more than the dog." Perhaps that's the case with some of us and the fish. Anyway, we didn't catch any fish.

I don't care to say any more, except, Mr. Gant, to express our appreciation to you for the excellent facilities which you have furnished.

President Davidson: Thank you, Dr. MacDaniels.

I believe the next order is the little talk by myself.

President's Address

JOHN DAVIDSON, Xenia, Ohio

When I was notified that this Association, in session at Guelph, had named me as its president, I was surprised and deeply honored. I suppose there is not a single member of this body who does not have the feeling that the Northern Nut Growers Association is "different," unique, and, very special: Here are all kinds: scientists and rule-of-thumb planters, experienced professionals and inexperienced amateurs, conservationists and hobbyists, all bent on one objective--to enlighten Americans and themselves on the values and opportunities that lie in the study and practice of planting forest trees which bear crops--specifically, nuts.

But the interest of most of our members is rather broader than our name would indicate. Forest _crops_, not merely nuts, are the logical outgrowth in interest that such an organization as ours stimulates. Dr. Zimmerman's work with papaws is a case in point. Mr. Wilkinson's work with the Lamb curly walnut is another. The persimmon, the papaw, the mulberry, the haws, the juneberries--you are likely to find them all, sooner or later, among the nut trees of our members. You will hear presently about a wood from one of our nut trees that is so valuable, _and so possible to grow_, that we may presently be planting for extraordinarily beautiful and valuable _timber_.

Patience is what it takes, and faith. Trees are an example to us. If we could only look at the procession of the centuries with the eyes of the sequoias, we should see

creation moving on marvelously with magnificent fruitfulness, and we should take courage.

Has the process of evolution been more successful with plants than with the human race? Should benevolent creation _fail at its highest point_? Certainly it should not. Nevertheless it certainly will fail there so long as so large a body of the race is undernourished, ill-born, hopelessly submerged--dragging downward rather than lifting upward.

Who knows the total answer? Education, of course, is a part of it--in industry, in eugenics, in moral responsibility. But you can't preach education effectively to a starving or half-starved man or child. The multiplication of population, the better distribution of goods throughout the world (which means in the end the avoidance of extremes of over and under-production)--these are the world's next greatest problems. I personally have the feeling that we are on the verge of an almost unthinkable increase in food productiveness through chemurgy's better and more complete use of plant life. We shall yet learn to gauge population to food supplies and food to population. Both are essential.

We need more plant breeders and more organic chemists at work on food supply all over the world. We need more people of good will and long vision, fewer political and social parasites; more producers.

Singularly, at the very moment of writing these words, a letter from a well known plant breeder is dropped upon my desk. In it he turns down the idea of an hypothetical executive position which most people would regard as promotion. The importance and interest of his work is so great _in its own right_ that he would not think of changing.

That is what I mean. We need more of his kind in the world. It is hoped that in this Association such men may find the kindredship and comradeship they so richly earn.

This was the spirit with which our Association was organized by Dr. Robert Morris, Dr. Deming, and a few far-sighted men in the early days of this century and carried on by them, by Mr. Reed, Dr. Zimmerman, Professor Neilson and their kind since. We salute them all. Their works follow and honor them by their multiplied fruits.

I shall not take the time in this full program to review the events of the past year. Some of our speakers will do this far better than I. But I wish to greet our visitors and the new members who may not have been with us before. We hope you will feel very much at home in our family of kindred minds.

Also, these remarks would not be complete without recognition of the efforts of those who unselfishly and unstintingly have given of their time and strength to this important work: our Secretary, Joe McDaniel! You all know him by his exceptional service to us all. (Let's rise and give him a hand.) And while we are on our feet--one of the best treasurers any organization ever had, efficient, kindly, but a veritable watch-dog of the Treasury, Mr. Snyder! Also a hand to the members of our important committees, Mr. Chase, Dr. MacDaniels, Mr. Slate, Mr. Stoke--I can't name or praise them all as they deserve. The NNGA could not possibly be what it is without them.

And now let us get on to the business before us.

Secretary's Report

J. C. McDANIEL, Nashville, Tennessee

The membership of the Association seems to be increasing fairly steadily. When I checked the mailing list early last October, it had 667 names, as compared with 691 listed in the 37th Annual Report. When I left Nashville last week, the number had increased to 742, according to my stenographer's latest count. There have been some discontinued memberships, as will happen almost every year in any organization, but the new members have more than compensated for them, in numbers.

We did not add up a total on all the mail sent out in response to inquiries, but it has been voluminous. Close to 800 requests for our nut nursery list have been received solely as a result of Mr. Stoke's _Southern Agriculturist_ chestnut article in last February's issue, and they are still trickling in. Some new memberships have resulted from these contacts, but more have come as a result of our column in the _American Fruit Grower_, and a Chinese chestnut article in _The Flower Grower_ early last spring, which gave our Association a boost.

Some members have said they did not find their _American Fruit Grower_ subscriptions of much value to them, particularly since the inauguration of _The Nutshell_, our news bulletin which has been issued four times since the last annual meeting. I will take some of the blame for this, since as editor of _The Nutshell_, I am somewhat in the position of competing with myself as columnist for the _Fruit Grower_. Space is limited in the latter publication, too, and sometimes publication of the "Nut Growers News" column is deferred a month or two, and again, I have been known to miss a deadline. Most of the columns, as in the previous years, are digests of material previously given in our Annual Reports. This practice seems to be justified as a matter of keeping nut news before the orcharding public and as a means of attracting some new memberships for the Association. I do not know of a better conditioned list of prospects than the more than 150,000 _American Fruit Grower_ subscribers all over the continent, who are at least interested in some kind of fruiting trees or plants. In that many, by the law of averages, are many with some interest in nuts. Several hundred will write to the secretary or other N.N.G.A. members who are mentioned during the year, and at least a few score normally will join us.

This does not minimize the desirability of having other publicity outlets. More of you who have a knack at writing should try your own contributions to national, regional or even community-wide publications. Even short letters to the editor, in such cases, may be read by "kindred spirits," and you will be read by men and women whose interest in nut trees (even though it may have been a dormant interest) will be stimulated to the extent of becoming N.N.G.A. members. Then it is up to our officers, the program committee members, and our contributors to keep them interested enough to renew their memberships another year!

Your comments on _The Nutshell_ have been quite flattering to its editor. You _all_ can help make it a better publication by contributing short original observations or clippings of good items on hardy nut trees from other sources.

There is a continuing shortage apparent in the supply of good named varieties of hardy nut trees in nearly all areas. This seems particularly the case with Chinese chestnuts. Few propagators at present have them in even enough quantity to catalogue, and the demand which has been built up by the good publicity on chestnuts exhausts most nurseries' supplies each spring before all orders can be filled. Our nursery list in the Winter issue of _The Nutshell_ has gone to some 2,000 people and has helped the nurserymen to sell out their trees quickly. We hope this will lead to a sound expansion in the commercial propagation of _good_ nut trees.

I should again call attention to our affiliation with the American Horticultural Society. This enables our members in good standing to receive their good quarterly publication, _The National Horticultural Magazine_, for only $3.50 a year. You may obtain your affiliate membership through our Treasurer, or directly from the American Horticultural Society, Room 821, Washington Loan and Trust Building, Washington 4, D. C.

* * * * *

President Davidson: You have heard the Secretary's report. Has anyone any revisions or modifications of this report to suggest?

Dr. MacDaniels: I move acceptance with thanks.

(The motion was seconded, a vote taken, and the motion carried unanimously.)

President Davidson: If the Secretary will also read the Treasurer's report, we will proceed with it.

Mr. McDaniel: Mr. Snyder wrote recently, regretting that he would miss this meeting (for reasons of health). He says he can not accept the position of Treasurer another year.

Treasurer's Report for Year September 1, 1947 to September 1, 1948

D. C. SNYDER, Center Point, Iowa

INCOME

Dues $1,396.00 Reports sold 153.75 Bond Dividends 25.00 Advertising 5.00 Miss Jones' Postage Acc't. 36.85 C. A. Reed Typesetting 32.50 Miscellaneous 7.60 _____ $1,656.70

DISBURSEMENTS

Fruit Grower Subscriptions 100.80 Reports, Stationery etc. 1,105.06 Secretary's expense 100.30 Treasurer's expense 58.17 Reporting Guelph Meeting 25.00 Miscellaneous 15.60 Bank service charges and checks returned N.G. 12.90 _____ 1,417.83 _____

Balance gained during year 238.87 On hand September 1, 1947 1,790.44 Paid out for Bonds 1,100.00 _____ 680.44 _____ Cash total on hand, September 1, 1948

(subject to minor bank service charges and checks which may be returned) $ 919.31
Bonds in box at Peoples Bank & Trust Company $2,500.00

* * * * *

President Davidson: You have heard the Treasurer's Report. Any remarks? It is a very good report. It shows that the organization is creeping up financially and in very good condition due to the continuous care that the Secretary and the Treasurer both have used in keeping up with our membership, keeping dues paid up, and so on. I will entertain a motion to accept our Treasurer's Report.

Dr. MacDaniels: I so move.

(The motion was seconded.)

Mr. O'Rourke: It should be accepted for audit.

Dr. MacDaniels: I will accept the amendment.

President Davidson: It is moved now, then, that the report be accepted for audit. Are there any remarks?

(A vote was taken on the motion, and it was carried unanimously.)

President Davidson: The next order of business is the regular business meeting of the Association. I think perhaps the first thing we should do might be to proceed with the election of a Nominating Committee and the Auditing Committee. I believe both, if I am not misinformed, are elective and not appointive. The chair will entertain nominations for the Nominating Committee.

+Nominating Committee Elected+

(The following were nominated for the Nominating Committee: Dr. H. L. Crane, Harry R. Weber, Dr. Wm. L. Rohrbacher, J. F. Wilkinson, George L. Slate. Upon motion that the Secretary cast a unanimous ballot for those nominated, vote was taken and motion carried unanimously.)

President Davidson: Am I correct in saying that the Auditing Committee is elective, rather than appointive by the Executive Committee?

Mr. Silvis: I understood it was three members and just appointed.

Mr. McDaniel: Yes, under Article I of the by-laws, it is appointed.

President Davidson: In that case we will do nothing about that now.

I think perhaps we might proceed with a few resolutions or motions before going to the further order of business. The chair will entertain a motion that the Association give its thanks to Mrs. Baker and her committee of the ladies for their entertainment of last evening and for future entertainment.

Mr. Weber: I so move, Mr. President.

(The motion was seconded, a vote called for, and the motion carried unanimously.)

President Davidson: Also the chair will entertain a motion that the Secretary be instructed to send Dr. Deming our usual affectionate greetings and assure him that his beloved association is still carrying on in the spirit of the founders.

Mr. McDaniel: By the way, I have a letter from Dr. Deming. Should I read that?

President Davidson: That would be fine if you would, yes.

+A Letter from Dr. Deming+

(Secretary's note: We substitute a more recent letter, dated May 9, 1949).

"... You are giving me much consolation for all _my_ broken promises to get out the annual report at an early date. I suggest that you have a lawyer draw up a contract for the printer to get out the report at a given date or forfeit so much per day for all delay. If you don't do that the printer will put you off for something that will give him a little more profit. I don't know that we ever got out a report in plenty of time for the members to get their orders in early or get other benefits from the report if it arrived before planting time.

"I note in the announcement of our Connecticut state medical society that it scheduled a recess of 15 minutes or so at intervals for members to 'view the exhibits.' It looks to me like a good idea....

"Congratulations on the fast work of Joe, Jr. The idea is to get plenty of limbs before letting him bear. Have you tried the sweet buckeye on him? [See page 181.]

"We have Spring here, too, as well as you in Nashville, and it is good.

"I get awfully tired after very little exertion. I'll be 87 on September 1. Too old to undertake any obligations.

"Best luck.

"Yours,

s/W. C. DEMING"

President Davidson: That is expressed beautifully, as usual. May I have that motion?

Dr. Crane: It has been moved and seconded that the Secretary be instructed to send Dr. Deming our affectionate greetings and assure him that his beloved association is still carrying on in the spirit of the founders.

(A vote on the motion was taken, and it was carried unanimously.)

President Davidson: Another, that the Association accept with deep regrets the resignation of D. C. Snyder, and that the Secretary be instructed to send him our affectionate greetings and thanks for his long, efficient and outstanding services as Treasurer of this body. Are you in favor of such a motion?

Mr. Weber: Take out the accepting the resignation part, and the rest will be O.K.

President Davidson: That is right. As amended then, with the omission of that "accepting the resignation."

(A vote was taken on the motion, and it was carried unanimously.)

+Clarence A. Reed Elected Honorary Member+

President Davidson: One more. The chair will entertain a motion that the Secretary be instructed to send C. A. Reed our greetings and as a small measure of the esteem we have for him and in recognition of his long and extraordinary services to this Association, we elect him a life member there-of.

Dr. MacDaniels: I think it should be an "honorary member" rather than a "life member." A life member contributes $75.

President Davidson: I believe that is correct, an honorary member. With that amendment, then.

Dr. MacDaniels: I would so move, Mr. President.

Dr. Crane: Second the motion.

(A vote was taken on the motion, and it was carried unanimously.)

Dr. Crane: Mr. President, I would like at this time, if I may, to say a few remarks in regard to Mr. Reed. I saw him last Friday afternoon, and he asked me to convey to the Association his very deep regrets that he was unable to attend. He had planned to attend, but his doctor said absolutely no. So he has learned from experience that he has got to pay more attention to his doctor's orders than he has in the past.

He wanted me to tell the members of the Association that although he wasn't here in body he was in spirit and in mind.

President Davidson: That's fine. I think perhaps we should proceed first with the reports of committees.

The Finance Committee. Mr. Weshcke is not here. Mr. Weber is next in order on that committee. I presume there would be nothing special to report at this time.

Mr. Weber: Nothing.

President Davidson: Press and Publication. Mr. Stoke is chairman of that committee. Mr. Stoke is not present at this time. Dr. MacDaniels, would you have anything to say in the matter of Press and Publications Committee? Have you any recommendations or reports to make?

Dr. MacDaniels: Mr. Chairman, I hadn't planned to make any report. As a matter of fact, I had very little to do with the work of the Publications Committee this year. I have been rather happy that it has been handled otherwise, and I think our thanks are due to our Secretary, who has carried the brunt, in fact, almost the entire burden of the publication of the proceedings. Also of _The Nutshell_. That occurred through a series of circumstances which I don't wish to outline here. I think probably the chief determining factor was that the contract for printing was awarded to a firm in Nashville, which almost automatically made it at least convenient and expedient to have the matter handled in Nashville. I believe you will concur in that general opinion.

Mr. MacDaniel: Yes.

Dr. MacDaniels: So that our Secretary has had an unusually heavy burden which we should not expect him to carry again.

President Davidson: Thank you, Dr. MacDaniels.

The chair will entertain a motion to accept Dr. MacDaniels' report on behalf of the Press and Publication Committee.

(It was so moved and seconded, a vote taken and motion carried unanimously.)

President Davidson: On Varieties and Contests. Mr. Zarger is not going to be with us, I am afraid, and if there is any other member of that committee present who has something to say on the matter of variety and contests, we would be very glad to hear from him. I don't hear anything, so we will proceed to the next one.

The report of the Survey Committee. Mr. Silvis is chairman of that committee, and I will say on his behalf that he was raring to go and would have gone if it had been the feeling on the part of some of the other members that a survey was timely at that time. It happened that that was not the feeling, it was not a good year to make a survey, and on that account I wrote to Mr. Silvis that possibly it would be well to put off any important survey for the year 1947.

Do you have anything to say, Mr. Silvis, in addition to this?

Mr. Silvis: Well, on the cuff, no, and off the cuff I would like to make this remark, that I just had one question I was going to require every member to answer to me for, and that was what kind of a nut tree should I plant, and thereby try to establish a zone between frost-free dates for various locations or states or territories. It didn't develop.

I received as late as last week John Bregger's note explaining why it was his reply came late. But I do want to make this remark, and for our able Secretary's first issue of _The Nutshell_ I know this to be a fact, that with it, it's the nuts, and without it, it's hell.

President Davidson: What shall we do with Mr. Silvis's report? We have some action to take presently on the matter of survey in addition to this report. Could I have a motion to accept the report of the Survey Committee?

Dr. Crane: So move.

Mr. Weber: Second.

(A vote was taken on the motion, and it was carried unanimously.)

President Davidson: Mr. Chase disappeared again. He is chairman of the Program Committee. We all have evidence of what he has been doing. Perhaps his program is sufficient to report.

Mrs. S. H. Graham is chairman of the Membership Committee. I think Mrs. Graham is not here, so perhaps we can pass on.

Report of the Necrology Committee fortunately is blank.

Mr. McDaniel: There is one that I know of. Mr. Schuster of Oregon passed away last winter.

President Davidson: I think that points out a little weakness in our organization. The death of Mr. Schuster should have been reported and some notice of it taken, perhaps.

Mr. Stoke, you are here as chairman of the Exhibits Committee. Would you like to say something?

Mr. Stoke: I don't know that I have anything to say. The exhibits speak for themselves back there. I wish to thank those who made contributions to that exhibit, and some still came in this morning that you haven't seen. I think it's been fine cooperation.

I feel an apology is due for not getting out more publicity on behalf of the committee. I had hoped that another copy of _The Nutshell_ would be out before this meeting so I could make another call for exhibits, but it wasn't, and I didn't get my material in to our Secretary in time for the earlier one.

Mr. McDaniel: I believe we did have a notice in the summer issue.

Mr. Stoke: Yes, there was a notice. At any rate, we have had exhibits here all the way from Georgia to New York. I am not sure whether they have any from Canada or not. I think it makes a very nice display, and I certainly appreciate your cooperation.

Dr. MacDaniels: In connection with these exhibits, we were driving along talking to Mr. Slate about the desirability of the Northern Nut Growers Association sending an exhibit to the Harvest Show of the Massachusetts Horticultural Society. That was done about ten years ago, and the Society gave us a silver medal at that time. I know from talking with Mr. Nehrling that they would be pleased to have such an exhibit put on, and I think that if we could take much of the material from our exhibits here and send it there that that would make an acceptable exhibit, and we almost assuredly would get not only considerable publicity out of that, because it would be an exhibit of the Northern Nut Growers Association, but we might also get either a cash award or a medal. I think if we work behind the scenes, if we preferred the cash we could get that, which would be of some value to the Association.

Now, I speak of this merely to bring it to your attention and to point out that any of the personally furnished exhibits that you wish to turn over for that purpose, you may arrange with Mr. Stoke for that.

(Further discussion on the details of sending in the above-mentioned exhibits.)

Dr. MacDaniels: I would move this Association favored sending an exhibit to the Massachusetts Horticultural Society Harvest show, provided material is available.

Dr. Crane: Second the motion.

Dr. Silvis: May I make this remark and also be in the form of a motion, that those exhibitors report immediately at the adjournment of this session to Mr. Stoke and make known to him whether yes or no, whether their exhibits can be sent up.

President Davidson: Do you make that motion in the form of an amendment?

Dr. MacDaniels: I will include that in the motion.

Dr. Crane: I accept it.

(A vote was taken on the motion as amended, and it was carried unanimously.)

President Davidson: Place of Meeting Committee. I judge that that committee is not ready to report, is it, Mr. Slate, for this following meeting?

Mr. Slate: The chairman didn't realize until just before we were ready to leave that he was a member of that committee. I have given the matter some thought on the way down, and in the previous years I have usually gone fishing for invitations some time before the meeting. I did drop a line overboard a few days ago, but I didn't catch anything more than I caught in this big lake up here.

Now, from previous experience I don't believe we can consider going to the Middle West. Mr. Snyder, Mr. Becker in Michigan, and Dr. Colby at Illinois, have not thought that they had enough material to make it worth while to go out there. That throws it back to the East, and we have been to some of the better places in recent years; Ronoake, Virginia, Hershey, Pennsylvania, and Boston.

I think there are two places that we should consider. I think we should consider Beltsville and the New York City region. We all know that there is plenty of material at Beltsville. We have not been there for some time. And in the New York City region we have the plantings of Gilbert Smith, who is probably 85 or 90 miles above New York. He is not far from Poughkeepsie where I am sure there are ample facilities for handling the crowd. Then there may be possibly some of Dr. Graves' plantings that would be worth seeing on a field trip.

Now, of course, the committee will be very glad to receive invitations from anyone here and consider them, and we will make the final report at the final business session at the time of the banquet, I believe. But between now and then I want you to consider the matter rather seriously and let me know what you are thinking about.

President Davidson: I think it would be desirable, if it were possible, for Mr. Slate to wire the proper authorities at Beltsville or Poughkeepsie.

Mr. Weber: Mr. President, one of our members is Mr. Bernath, who has been quite faithful in attending nearly all our meetings, and he has, I imagine, much of interest to show to the members, and he is located near Poughkeepsie. I am just throwing that out for the members to think over as to what they would think about Poughkeepsie as a possible meeting place.

President Davidson: That's worth listening to.

Would it be advisable, do you think, for Mr. Slate at the expense of the Association to wire to Poughkeepsie or to Beltsville to see whether an invitation is available or not?

Mr. Slate: Those places are well represented now.

Mr. Weber: I imagine Mr. Bernath can speak for himself.

Mr. Bernath: I don't know, I think if we could delay it another year, Mr. Smith is going to retire from the State School, and he will have plenty of time. I am very busy, and he will have loads of time on his hands, and then he can give it his attention. I think that would be all right next year.

Mr. Slate: That's up to the Association to decide.

Mr. Bernath: We would like to have you come at that time.

Mr. Slate: Beltsville is very well represented in Dr. Crane.

Mr. Weber: Mr, Chairman, in view of what Mr. Bernath says, I'd accept Mr. Bernath's suggestion and have Poughkeepsie on the list for the year following.

Mr. Bernath: That's right.

President Davidson: Dr. Crane may have something.

Dr. Crane: Mr. President and members of the Association, we'd like to have the Association meeting at Beltsville again. However, we have had four years of May freezes in Beltsville Station, and I am going to tell you all is not in any too good condition. A lot of it has been pulled, and we have had to replant an awful lot of the stuff that is now just planted this year. We lost a lot of the plantings that were made last year because of injury. As you folks probably know that have been there before, we labored under very great difficulties on soil conditions in that we have mostly sands and gravel.

So we are kind of in a mess there right now. We'd be glad to have the Association meet at Beltsville, and we have right good facilities there for meetings, but as far as any plantings in the area, a lot of the work we are doing, we are kind of going through a period of change right now and getting re-established, and I want you to know the situation.

President Davidson: Well, we have been forewarned. It's a case, I judge, of not being unwilling to see us, but you are not so anxious, for us to see you, is that it?

Dr. Crane: I wouldn't want you to come there under false hopes that you would see a lot.

Mr. Gravatt: I would like to say we have done quite a lot of work in breeding chestnuts and also work with forest types, crossing American chestnuts and Chinese. But I agree quite with Dr. Crane, that we haven't so much to show you there. Of course, it's a dog-gone good thing to get familiar with these diseases and see what you are up against, because all through the history of nut culture, and so forth, one of the basic defects has been the failure to appreciate the importance of insect and disease factors. And we are very much in need of more basic research along those lines, but I agree with Dr. Crane that at present we have a limited amount to show you there.

Of course, there is the Plant Industry Station there with a lot of experimental work, greenhouse work and all sorts of basic research work, fertilization work, and so forth,

going on there. A lot of people like to come to Washington. Our plantings are pretty much the same condition as Dr. Crane's and not a display proposition such as you have here at TVA.

President Davidson: Suppose we regard this report, then, as temporary and hear more from you later.

I think that concludes the reports.

The Board of Directors, unless there is some other order of business to be taken up, have some recommendations to make to the Association. One is the recommendation that the Association place the annual membership fee at $3, the supporting membership fee at $10 and the life membership fee at $75. They didn't wish to take the responsibility of doing anything more than referring that matter to this Association.

Dr. MacDaniels: That could be handled in the by-laws under the constitution.

President Davidson: We still also have another rather important matter that's been referred to the Association, and that is the matter of a sufficient amount of remuneration to permit our Secretary to hire a stenographer to do the extra amount of work that is gradually accumulating in that office. The resolution that is referred to you calls for a payment of 50 cents per member to the Secretary for this purpose.... We have no right to be set up so that the work of the Association would encroach upon a person's job as it is set up at the present time. That recommendation was that it was contingent, of course, upon raising the dues to $3.00 and take 50 cents of that to offset the stenographic help and try to re-organize our affairs between the Secretary and Treasurer so that as much as possible of the routine mailing, and routine stenographic work would be carried in this way.

(Discussion on the above recommendation.)

Mr. Weber: I move that the additional remuneration be granted, 50 cents per member, to the Secretary.

Mr. Smith: I will second the motion.

Mr. Fisher: I'd like to make an amendment to that, that the dues be raised to $3.00 in order to make this possible.

Mr. Weber: I will accept the amendment.

Mr Smith: And I will second the motion contingent, of course, to the raising of the dues.

(Vote taken, motion carried unanimously.)

President Davidson: We will appoint a Resolutions Committee.

+Resolutions Committee+

Sterling Smith, H. L. Crane, Raymond E. Silvis, H. F. Stoke.

President Davidson: I think so far as I know that's everything except the report of the Committee on the Constitution. Unless I hear otherwise we will proceed with that

report.

(Discussion on Constitution.)

(Constitution and by-laws approved as set out in another part of this report, the Constitution having first been read at 1947 meeting)

President Davidson: As I understand it, then, this constitution, unless we make some other provision, is in effect as of now.

Mr. Weber: Now with these by-laws in effect there will have to be a fresh nominating committee elected for the next year.

Mr. Smith: Mr. President, I make a motion, if it's in order, that the Nominating Committee as elected previously for this meeting also continue and serve for next year.

Mr. Clarke: Second the motion.

(Vote taken, motion carried unanimously.)

President Davidson: There is one other matter that was brought up at the directors' meeting, and inasmuch as the directors did not have a quorum, it should be voted through here, I think, and that is that a motion is in order to pay Mrs. Gibbs $25 for her services as stenographer at our meeting. That was done, I believe, at Guelph, and it involves a lot of important work.

Mr. Korn: I second the motion.

(Vote taken, motion carried unanimously.)

President Davidson: Shall we adjourn, with a continuance of the business meeting at the banquet?

(Recess taken until 1:00 o'clock p. m.)

+Monday Afternoon Session+

President Davidson: Shall we come to order?

We now come to the interesting part of our program, and we will listen first to Mr. Quick of West Virginia, who will take the place of Mr. Sayers, the State Forester at Charleston, West Virginia. Mr. Quick.

The Development and Propagation of Blight Resistant Chestnut in West Virginia

RALPH H. QUICK, Conservation Commission, Charleston, West Virginia

Mr. Quick: Ladies and gentlemen of the Association, your guests and friends: In substituting for the State Forester of West Virginia I realize that I am undertaking a big job. A few of you know Mr. Wilson Sayers, who is the State Forester, and those of you who do may assure the rest of the group what a big job I am undertaking, because I feel that I am in pretty good-sized shoes.

The subject that has been assigned is The Development and Propagation of Blight Resistant Chestnut in West Virginia. Now, being a forester, I am perhaps interested

in blight resistant chestnut from a little different standpoint than the majority of this group. As representing the Conservation Commission of that state I might say that we are interested primarily from the game-food viewpoint. Now, that's a little bit different, I expect, than most of you have been thinking about, or some of you, at least. But that is the standpoint from which we are interested.

So I would like to go along with you this afternoon and discuss some of the things that we have done and some of the things that we are learning--there are a few yet--that lead us along that line to believe that we can do something with blight-resistant chestnuts in West Virginia as a game food. We are just at the beginning, so to speak--that is, the Conservation Commission of that state is just at the beginning of our study. We have been fooling with it a little off and on since back in the middle '30's, but interest has lagged and then has picked up again two or three times.

I am sure that as far as the production of good strains of blight-resistant chestnut, better strains of Chinese, and so on, that there are people in West Virginia who are more capable of telling you what has been done from a private viewpoint than anyone with the Conservation Commission, but we are interested in learning about it and producing it in large numbers for a game food, and, of course, if we are interested in distributing from our nursery over the state for that purpose, we are interested in producing better strains of blight-resisting chestnut as we go.

Along back in the 1920's a few plantations, or a few trees were planted in the state by what was then the old Fish and Game Commission, and the records have been lost, as has been true in many other states. But then, apparently, the beginning was made. In going over some of those early plantings I will only have time to hit the high spots and the ones in which we are particularly interested in our line, but the first ones were back there somewhere in the '20's.

One of the best plantations, the one that we are particularly interested in at the present time, is in Jackson County, West Virginia, and it is of the University of Nanking strain, and there were 34 trees planted there back in 1926, and we are told that they were planted from 2-0[1] stock, from nuts that came from China in 1924. Twenty-six of those trees survived, and we think they are pretty good nuts. You may be interested to know that that plantation now averages 22 feet in height and has an average diameter at breast height of 8 inches. The spacing in that plantation was 26 by 26 feet.

Now, we can't take credit, nor do we want to take credit, for that plantation. The state agency had nothing to do with it. It was put in there through the cooperation of the gentlemen from Beltsville, but we are very much interested in that plantation; so interested that we have gone to the owner, along with the permission of the fellows from Beltsville, and sewed the thing up for a five year period, during which time we hope to get the seed and to improve our own strains and establish blocks of our own on state-owned land under different conditions and on different sites where we expect in the future to be able to secure seed for our use and production at the nursery.

In the first few years that this plantation that we are speaking of in Jackson County produced, not many people paid much attention to it or attached much significance to it. The man who had charge of it gave the nuts away for experimental purposes or for any reason that anybody happened to ask for them, and shipped a lot of them free. But along in the early 1940's he began to find out what he had, and he started selling seed and made a pretty good thing out of it.

Last year was the first year that we had gotten seed from that plantation. We got 75 pounds of good nuts taken in the fall of 1947.

We have another orchard, another plantation that led us to become interested, I guess, in producing blight-resistant chestnut as a game food and along forestry lines, and that is the orchard that we have on nursery property. It was one of the early ones, and I expect one of the earliest in the state, but it was planted along back in 1936, fifty-one trees.

When we started in this we didn't know anything about it at all, so we have built up our small knowledge in the last few years. But it didn't take us long to realize that our orchard on our nursery property was of badly crossed material, and it had some very undesirable trees. If we succeeded in doing anything with them as a game food we would have to eliminate, and only last year did we get around to the place where we could secure authority to eliminate the undesirable species. We have about half of the stand left now, but we are pretty sure that the trees that we do have are of good strain.

It might be interesting for you to note--maybe some of you can top it--we were interested when this orchard was planted, in what would happen if the trees were planted and allowed to grow as a forest stand. So they were planted in six-by-six spacing. Of course, we got a lot of self-pruning and a lot of competition, as we would in forests by the trees growing up and competing with each other and reaching for height and light. Some of them died and some were so badly suppressed that they failed to make any growth at all. But there is one tree that we still have in that orchard that we are proud of, not from the standpoint of nut production, nor does it produce a very good nut as far as the human taste is concerned. But it has made a single stick that far surpasses any other tree we have in the orchard. It looks like a forest tree. In 1945--it might be hard for you to believe--it grew nine feet. That isn't an exaggeration. It was measured. We thought that was a lot better than fair growth. Of course, it hasn't made any growth like that since, and I don't think it ever did before, but it just had the push to go and went nine feet in one growing season.

Leaving that orchard for a few minutes, there were 38 plantings of from 10 to 50 trees each made by the Soil Conservation Service and the Division of Forest Pathology of the Bureau of Plant Industry in the spring of 1939. These were examined by Dr. Diller of that Bureau in the spring of 1940 and in 1947. He has told us that he graded those plantings as he found them, 10 being good, and he said the next 15 were only fair and he put 13 down as total failures.

Of those 13 that failed--from the forestry standpoint now, remember--he said that 7 of the failures were due to poor site selection, three were suppressed by surrounding

hardwoods and other competing growth, and three had been destroyed by cattle.

[Footnote 1: Meaning, two years old, not transplanted in the nursery.--Ed.]

+A Commercial Chestnut Nurseryman+

I don't know whether any of you know of--I expect you do--the Gold Chestnut Nursery in West Virginia near Cowen, and it is owned and operated by Mr. Arthur A. Gold. He has been interested in blight-resistant chestnuts from a commercial standpoint, selling from his nursery for a good many years. He has worked with us some in the Conservation Commission and has given us the benefit of his experience. And if any of you have the opportunity I think you would be interested in seeing Mr. Gold's nursery. He was an old-time nurseryman that handled most of the conifers found in a commercial nursery, but in the last few years he has gone into chestnut production almost entirely, and if you have an opportunity, I am sure Mr. Gold would welcome you to his nursery in Webster County.

The Game Division of the Conservation Commission of West Virginia established three or four small plantings on the state forests in 1938 and 1939, but they had low survival. Dr. Diller in going around with some of us and checking on those has found that we were back there where all of us were trying to find something and trying to learn something and that we made many mistakes and that we picked poor stock, for one thing, and poor sites for another thing, but the great disadvantage and the biggest limiting factor was our poor selection of sites there in the beginning.

In handling chestnuts that you people handle maybe in small or large quantities where all of your time can be devoted to that particular thing, you probably have a lot of things that you do that we don't have time to do because at the nursery in West Virginia we are interested primarily in producing conifers and other forest trees for the reforestation of abandoned land. So in handling this Chinese chestnut as a game food we are working on a sideline. We have to pick it up as fast as we can do the job and do as much as we can and learn about as much as we can. And, of course, we learn slower than people who have the time to spend and perhaps the money to spend at it. But we are limited in those two respects.

But seed collections are made, and we find it necessary in collecting from two of the orchards that we are now using for seed to collect twice a day in the season that the nuts are ripening, because both of those orchards which we prize are close to forest lands and squirrel country, and they really give us a race for it. The fact of the matter is the orchard at the nursery has attracted the squirrels on that particular side of the mountain. I have hunted on opening day and killed my limit of squirrels without going outside of the residence and been back at work time at eight o'clock. It really attracts them on that side of the hill. We are going to compete with the squirrels, but as you will see, we have just about given up that orchard as a seed source.

We find it necessary to treat the seed, of course, before we plant it. Many of you people, of course, go into the spraying end of it before the nut ever develops. We haven't the time or the money right now to go into it that way, so we try to take care of the nut after we collect it and bring it in.

I expect it is not necessary for me to go into any of the details on any of the methods that may be used to get rid of the weevil, because you are all familiar with that. Maybe it suffices to say that we at the nursery now are using the hot water treatment. The little weevil is found in there and not always apparent. In fact, most of the time it isn't apparent that the nut is infested, but they are, and if we take measures to kill the weevil we haven't any germination of the weevil. We used gas once, but we are limited in that at present. It is a lot more expensive.

We have, in the first few years that we tried to produce chestnuts at the nursery, stratified them. We got along pretty well with that in damp sand, we got along fairly well in sawdust, and we got along especially well with damp sphagnum moss. But in the end we determined that we are getting better results if we plant the nuts as they are collected. In other words, the seed was taken from the orchard, treated to kill the weevil and put in the ground in the fall.

Now, you can't get away with that everywhere. Our orchard is far enough away from the nursery that we don't have any rodent damage. We have had some trouble from skunks, and they finally find out that the nuts are in there in a row where we have planted them, and they go right down and get them. But we have no trouble from mice or rats. We are far away from woodland and buildings.

We find that some people have trouble with wind or water erosion. We don't have that. So we can get by and do a better job and produce better trees by sowing nuts in the fall, and we sow them in the fall, just as if we were sowing black walnuts for production and distribution over the state.

By the next fall when we are ready to distribute those seedlings as 1-0 stock we find that we have produced seedlings of about 14 inches in height as 1-0 stock. From what I have seen that isn't a bad size to produce as 1-0 stock, though it is better in some places. We find, too, in the spring before germination, that in our particular section of the state along the Ohio River valley we sometimes get a dry spring and find it necessary to irrigate that land where we planted the chestnuts, just as the seed beds where we planted pine, in order to keep the ground moist and keep it in a condition where seeds will germinate freely.

We weed our chestnuts just as we do every row planted in the nursery, cultivate with the tractor about three times in a season, which is all the time we have to give to it, and hand weed it once. Perhaps it ought to have a little more than that. Some seasons I am sure it should, but that's about the time we are allowed or the time that we can allot to that.

I hope, Mr. Davidson, you will check me here on this time. I don't want to get too far out and upset the schedule.

President Davidson: All right, if necessary.

Mr. Quick: In distributing, the seedlings or blight-resistant chestnut seed in West Virginia we began back in 1943 putting them out in quantity. We had to limit them, the only thing in the nursery we had to limit the amount as to seed. That was because everybody in the state became very much interested, and the Conservation

Commission makes those available to any land owner in the state free of charge if he will plant them as a game food but not under other circumstances. He can't use them for ornamentals, and he can't use them for shade purposes in his yard. But he can receive a limited number if he is willing to use them for game. So in scattering them over the state, so many people wanted so many of them that if we didn't watch we'd have all of our chestnuts planted in three or four, or half a dozen spots in the state, and we are interested in learning as much as we can by having them put out at different elevations, different sites and under different conditions, so we had to limit it to ten to an individual in 1943. We have gradually upped that as our production has gone up, from 15 to 20, then 40, and this year we are offering 50 to any land owner in the State of West Virginia.

Now you can see why we are interested in trying to improve the nut. If we are going to distribute them all over the state, let's distribute a good nut, a nut that is not only a heavy bearer for the game, but a nut, too, that is fit for human consumption.

In our site recommendations we have been trying to follow pretty well the ideas of the boys from Beltsville, and we found out that what they have been telling us is just about right. In other words, we are setting our chestnuts in the cove types, moist with gentle slope, preferably on the north, and we are getting better growth there. It doesn't mean as far as we are concerned that it doesn't grow well on drier land and on rich hill-tops but the growth is so much greater when it's put in good ground and under those conditions. In other words, it needs a tulip poplar site; where tulip poplar is growing or has recently grown might be one way to select a site for our chestnuts.

In these five year now that we have been distributing these chestnuts we have distributed something like 200,000. Now, we know that all of those seedlings haven't been good strains, but they have been the best we could do at that time as we were going along. We hope to learn from you people, and we hope you can give us help in improving our strains so that we can distribute better chestnuts over the state.

We haven't had a good system of checking up, until the present time, on plantings that have been made in the past, but we are initiating a system just now wherein all plantations that have been made from forest stock will have regular examination all over the state of West Virginia, and we are including chestnuts in that. We have made some checks in the state on certain selected sites and have found out, strange enough, that these little plantations that are spotted around on the farms, if they were put in correctly and handled properly according to our instructions, have given us a survival of about 80 to 85 per cent, which is, as you will remember, about the percentage in the Nanking strain planting in Jackson County, 26 out of the 34 original trees. That seems strange, but it has proved true all over the state in the few checks that we have made. But we are going into it and checking these plantations and by so doing I believe we can eliminate a good many of our own troubles, along with your help.

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President Davidson: Thank you, Mr. Quick for a very interesting paper.

Is Professor Moore, present? Our next talk will be on The Present Status of the Chestnut in Virginia, by Professor R. C. Moore of the Virginia Polytechnic Institute of Blacksburg, Virginia. Professor Moore.

The Present Status of the Chestnut in Virginia

R.C. MOORE, Department of Horticulture Virginia Agricultural Experiment Station

Briefly reviewing the past, Virginia has been in the same position as many other states in regard to the large number of native American chestnuts that once grew wild before the blight epidemic occurred. Most of the chestnuts were found on loose, open type soils rather than on heavy limestone soil. In mountainous parts of the state, considerable income was obtained from the sale of wild chestnuts. Men, women, and children gathered these nuts and traded them at the stores for merchandise. One small country store, in Floyd County, southwest Virginia, assembled and shipped between sixty and eighty thousand pounds annually. A small town, Stuart, in Patrick County, shipped three carloads daily during the peak of the season. These nuts found their way to city markets, where chestnut roasters were as commonly seen as popcorn poppers. Since many of these native chestnut trees grew in forests or on wasteland, there was little expense involved except in the time required to gather them. The demand was good but frequently the sale price was rather low, especially during years when the crop was heavy.

After blight destroyed the wild trees, a considerable amount of timber was cut from the dead trees. At present this wood has largely decayed beyond usefulness except for firewood, although in some areas it is being gathered for pulpwood. Sprouts have arisen from the bases of the trunks and have borne nuts, but blight sooner or later destroys those sprouts.

Chinkapins are found in many counties of Virginia, especially on shale or sandy loam soils. Blight affects chinkapins to a considerable extent; but because of their bushy type of growth, new shoots arise to replace blighted shoots, thus perpetuating the plants so that they have not died out. Chinkapins are gathered by children for eating and for sale along the roadside, but at present they have little total economic value.

+The Asiatic Chestnuts+

Since the native American chestnuts passed out of existence, there has been a gradually accumulating interest in the Asiatic species, especially Chinese chestnuts, which appear superior, in blight resistance and nut quality to the Japanese species. The growing of these Chinese chestnuts is such a new enterprise that its problems are not fully solved nor its opportunities fully explored.

The earlier plantings of seedling Chinese chestnut trees were made by cooperating growers and nurserymen. They were interested in a forest type chestnut that might replace the dead native trees. A few of these plantings were made under semi-forest conditions, on cut-over timber land or on dry ridges. The first lesson that was learned was that the Chinese chestnut is an orchard type tree requiring rather fertile soil and ample moisture. It would not compete favorably with most native forest trees, but

rather was a slow growing, shallow rooted type of tree. Under these unfavorable growing conditions the trees tended to be small and to sprout from the bases of the trunks. The weakest seedlings died.

In other cases the trees were planted in yards, back lots, along the sides of ravines, or in other locations where the soil was fertile and moist. Under these favorable conditions most seedlings have grown and produced crops of nuts, especially when the trees were pruned and competing weeds and brush were mowed. Very few of these first seedlings of the Chinese chestnuts showed much promise although a few of them were fairly satisfactory.

Several old Japanese chestnut trees have been observed. One of these is estimated to be 50 years of age with a trunk diameter of 18 inches and a height of about 50 feet. It is growing in a very fertile spot and heavy crops in the past have broken its limbs. Chinkapins growing nearby appeared to have supplied pollen. Recently the nearest chinkapins were cleared away and hence at present the nuts fail to fill well. Another large tree in eastern Virginia produces many burs but the nuts fail to develop, indicating self-sterility. The nuts of both trees are rather coarse and of poor quality.

More recent plantings have been rather widely scattered over the state, although the total number of trees is not large and no one person has planted many trees. One large general nursery, serving this area, reported sales last spring of 196 Chinese seedling trees to thirty-five different customers. The largest single sale was for fifty trees. Several customers purchased only one tree each.

+Problems Encountered+

In visiting and corresponding with individuals who are growing Chinese chestnuts, I have made a few observations, as to problems that have arisen.

+1. Site and Soil.+ The most successful trees from the standpoint of growth and production were those growing on fertile, well drained soil in which moisture was plentiful. The Chinese chestnut tree appears to be shallow rooted and to require good growing conditions. Dry ridges were unfavorable for growth, and in bottom land the trees frequently were subjected to late spring freezing of tender shoots.

2. Blight injury to the trees and weevil damage to the nuts seemed to be the most serious enemies of chestnuts. Seedlings varied considerably in their resistance to blight. Some of them showed no indications of blight; others were damaged but outgrew the injury; and a few trees were weakened and died.

Weevils appeared to be quite prevalent. One grower reported almost 100% wormy nuts. It is my understanding that a spray program has been developed for control of the weevil. Mr. H. F. Stoke of Roanoke believes that the Illinois No. 31-4 chestnut (a hybrid) is resistant to weevil, probably because of its thick burs and closely set spines.

+3. Cultural Care.+ Chinese chestnuts benefited from pruning; it being especially important to cut away the sprouts at the bases of the trunks. Mowing weeds and brush around the trees seemed helpful. Applications of nitrate of soda stimulated more rapid growth of young trees, and in limited amounts benefited the older trees. It

appears, however, that there may be a danger of overstimulation which increases the hazard of limb breakage by snow and ice, especially in the case of younger trees. The largest crops of nuts, however, were frequently produced on trees of only moderate vigor.

+4. Freezing damage to the bark of the trunks and large limbs.+ This occurred in the VPI Horticultural Department planting in 1945, when a temperature of about 17 deg.F. occurred after the trees had started growth in the spring. This injury appeared as a darkening of the outer bark and cambium. Trees that were severely damaged became weakened and tended to sprout vigorously from the bases of their trunks. Other trees overcame a slight injury with little apparent ill-effect.

+5. Seedlings or Varieties.+ The question is whether to grow seedlings or grafted varieties. Seedlings are more easily propagated, the nursery plants less expensive, and the trees longer lived on the average; but seedling trees and nuts are quite variable. Named varieties are difficult to propagate, the nursery plants expensive, and stock-scion incompatability may occur; but the trees and nuts are uniform. Seedlings serve a useful purpose in developing new varieties; but with more planting of superior varieties and a fuller understanding of propagation methods, and of cultural care, chestnut growing on a commercial scale may be more likely to become a reality.

+Future Prospects+

For the present, at least, it appears that growing Chinese chestnuts may be limited to small specialty plantings rather than any large commercial enterprise. The trees seem well adapted to yard and back lot planting as ornamentals and to furnish the family with nuts. Also hobbyists and specialists find them to be interesting plants with which to work.

The industry is new and involves uncertainties and risks, which a commercial grower should not be expected to assume. Further study is needed to clear up the uncertainties, especially as to production costs, markets, and profits to be expected. As additional trees come into bearing over a wider area, a better understanding may be had of the economic value of these chestnuts in the various sections of the state. There is a market for high quality chestnuts, but it remains to be seen whether there will be sufficient profit with the risks involved to attract commercial growers.

+Summary+

In conclusion, the following points are to be stressed in regard to growing chestnuts in Virginia:

(1) Chinese chestnuts are adapted for home planting or for planting by hobbyists and specialists; but their commercial prospects as yet are uncertain.

(2) The trees require fertile soil with ample moisture but should not be planted in frost pockets.

(3) Weevils and blight have been the most serious pests.

(4) Seedlings serve a useful purpose in developing new varieties; but greater progress should be expected from growing superior named varieties.

(5) Additional study is needed to determine the profit prospects, to evaluate varieties, and to work out details of cultural practices, harvesting, and storage of nuts on a variety basis.

Although the chestnut blight has destroyed the native _Castanea dentata_ trees, it is hoped that breeding programs may produce a blight resistant, hardy tree, of a size that will lend itself to orchard planting and cultural practices, and which will be regularly productive of high quality nuts.

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President Davidson: Thank you, Mr. Moore.

The next thing on the program is the talk by Mr. G. S. Jones of Phenix City, Alabama, on Growing Chestnuts in Lee County, Alabama.

Growing Chinese Chestnuts in Lee County, Alabama

G. S. JONES, R.F.D. 1, Phenix City, Alabama

Ever since childhood, chestnuts have held a fascination for me. How well I remember the delightful Sunday afternoon trips we used to make in the fall up on Earkett's Hill to gather a few small nuts from some native trees which often had been burned by woods fires. I occasionally revisit this area to see these trees, which are in better condition now than then. Native chestnuts were never, to my knowledge, very abundant in our area and are now indeed scarce, but I still hear of a few living trees, some of which grow as far south as North Florida.

I first became interested in Chinese chestnuts from an article I read in the early '30's in a Department of Agriculture yearbook which I think had been written by Mr. Gravatt. This article told about these trees being introduced into this country because of their high resistance to blight. Until this time I had heard little about chestnut blight. In order to find out more about these trees I wrote Mr. Gravatt, who in reply said seedling trees were available for distribution on an experimental basis. I applied for some of these, more, I must admit, to get them to grow on our place just to have some chestnuts than with any thought of disease resistance. When these trees came in the spring of 1934 I even had some trouble in getting permission to set them in an open field near the house, for chestnuts were considered as a tree of minor importance, to be grown in some out of the way place.

These trees were set in sandy loam soil with a porous yellow subsoil in a field of medium elevation which has excellent air drainage so I have had little damage from cold injury. The soil is of fair fertility for the Upper Costal Plain area. Of the trees sent me, fourteen of the ML selection, originating, I am informed by Mr. Gravatt, from seed obtained in Anhwei Province of China, and 10 MO selection originating in Chekiang Province were set in my orchard. Only two of these failed to survive, leaving a total of twenty-two. These were cultivated with the field crops, mostly cotton and corn, and I must admit didn't have much individual attention for several

years. I even left the side branches to minimize injury from the mule and plow used in cultivation. Some leaves and trash were put around them at times and they received some benefit from the fertilizer of the row crops. I mention this to show that my chestnuts grew quite well though only moderately fertilized, but receiving good cultivation while young. I might mention that I set two trees in stiff Piedmont clay soil a few miles above here, to try them under woodland conditions. These have never done well, although one had burs but I found no nuts. Other trees which I observe have not been given cultivation grow very slowly, although I have not seen any tried on what I would consider _good_ woodland areas.

My trees, spaced about 40 x 40 ft., have grown quite rapidly so that now some of the limbs are almost touching. Tree ML No. 2, which is about average size, measured last fall in diameter 12-1/2 inches, in height 24 feet, with a limb spread of 30 feet. By 1943 the trees were getting so large that cultivation was discontinued. An attempt is made to keep all litter possible in the orchard, which, with the shade of the trees, has caused much of the soil to become loose and mellow. Since our sandy soil is very low in calcium I applied limestone one time at the rate of about 1500 lbs. per acre. This I hoped would improve the texture of the soil and make better conditions for growing bur clover between the trees. Basic slag which contains about 10% phosphate was applied at the rate of about 600 lbs. per acre in the early '40's. For the last four or five years I applied about 200 lbs. of guano (4-10-7 usually) and 200 lbs of basic slag annually. Since 1944 I have been adding about 50 lbs. of minor mineral elements to the above mixture. Whether it is a coincidence or not I cannot say, but the next year after applying these elements my yields increased from 430 lbs. the previous year to 961 lbs. and have remained high ever since. Minor mineral elements show beneficial results on our garden crops, and I am inclined to believe they are needed, since our soil is so sandy and porous, and especially the soil that has been cultivated so long. Since my trees have produced so well with this moderate fertilization, I have made no check against higher rates of application. In fact I am against the use of large amounts of mineral guanos since I know certain tender shrubs and plants are injured by their use and some soil bacteria and animal life are also harmfully affected, according to reports I have read.

Three of my trees bore a few nuts at four years. No record of yields was kept until the seventh year or 1942, in which I gathered about 328 lbs. of nuts. After that my records show for 1943, 554 lbs., 1944--430 lbs; 1945--961 lbs; 1946--1722 lbs; 1947--1554 lbs. No individual tree records were kept except in a few cases. I kept a rough record by looking at the burs at the end of the season, and classed trees as excellent, good, or poor producers, along with other characteristics of the trees. However, I know several of my trees produced over 100 lbs. each in 1946 and one tree, ML No. 2, of which I kept a record by weight, in 1947 produced a little over 150 lbs. of nuts.

[A note from Mr. Jones early in 1949 reports a crop of 1,836 pounds of chestnuts harvested from his 21 trees in 1948, the largest yield to date. His ML No. 2 tree produced 165 pounds.]

Nuts on a few of my trees begin ripening the latter part of August, but September is the heavy month, with some extending to the middle of October. Their early ripening period while the weather is usually hot and dry, I think tends to cause damage to nuts from the effects of the hot sun and rapid drying. Damage to the nuts and consequent spoilage can be kept at a minimum if they are gathered promptly, which should be daily.

+Preparing Chestnuts for Market+

Here is how I generally handle my crop. As soon as the nuts are gathered I put them in a container with water and remove the nuts that float. This eliminates practically all spoiled nuts and those beginning to spoil. Those that sink are then placed in coarse mesh burlap bags (about 25 lbs. to the bag) which are tied near the top. These bags are laid on a slatted platform under a shade tree and pressed out flat, so nuts will not be thicker than 2 or 3 inches. These bags are thoroughly wet with water once or twice daily, depending on the weather, until I can carry them to cold storage and store at 30 deg.F., or they are marketed fresh, advising buyer of the perishable nature of these nuts. Last year my nuts kept excellently in cold storage, and after remaining there about six weeks had dried sufficiently to keep much better after taking out than when they were fresh.

Nuts for planting purposes can be kept in excellent condition for several weeks by spreading them thinly between layers of damp sphagnum moss and storing in a cool place. This cannot be allowed to get very wet or sprouting will begin. While holding the nuts out of cold storage I attempt to keep sufficient moisture available so the nuts are not allowed to dry much, and yet have plenty of ventilation to keep them from heating or souring. Until I began using this method, a large percentage of my nuts began spoiling soon after gathering, which caused me much discouragement, as I did not want to offer such a product for sale. Since then my losses still run around 12%, but this could be reduced still further by more prompt gathering and by the elimination of several trees which retain nuts in the burs to a large extent.

I have been able to dispose of my nuts quite easily in near-by Columbus, Ga. and for the last few years have had quite a demand for nuts to use in planting.

My orchard as a whole has been very healthy, showing no blight signs that I can detect, although there is little chance of exposure to blight in my section. One tree is slowly dying, which may be due to cold injury, as it comes into leaf early and also ripens very early. So far I have noticed no damage from chestnut weevils. As my trees are seedlings, there is quite a bit of variation in size of nuts and production of individual trees.

+Undesirable Traits in Seedling Trees+

I might mention some undesirable traits which I notice in my trees. First, I would place retention of nuts in the burs as the worst trouble. This is quite bad in five or six of my trees. Next, nuts too dry and loose in the hull at time of falling, which is present in four or five trees, some of which retain nuts in the burs and some which do not. The dry textured nuts seem to spoil more easily than plump well filled ones. Some trees produce too small nuts but the trees which produce extra large nuts do not

usually yield nearly so heavily as those producing small to medium size nuts. I consider too early ripening as undesirable, for those that ripen later are usually better keepers, but this does not always hold true as some of the later ripening ones are also poor keepers.

This year my trees have an excellent crop of burs and show promise of a good average yield on each tree. Considering all things, I am highly pleased with my Chinese chestnuts and believe they have a good future in our section if no greater troubles arise than I now know of although there is much room for improvement.

+Other Tree Crops+

Although Chinese chestnuts are my largest producing tree crop, I am working with a number of other trees and shrubs for both nut and fruit production, as well as other purposes. I have several Thomas black walnuts which I set about 1938. Three of these have grown quite rapidly and are beginning to produce nice crops of nuts, although the kernels have a tendency to be spongy at times.

Of course, I have a small orchard of budded pecans, which do so well in our section. These trees, which are young, are just coming into production. Some other nut trees which I am trying in field plantings include native chestnuts, chinkapins, hazel nuts, native black walnuts, and scaly bark hickory (_Carya ovata_). Since most of these are young and grow so slowly, I cannot say much about their production yet. I have also planted quite a large number of white oaks from a high production tree in hopes of producing acorns for hogs and wild life, also some cork oaks on an experimental basis.

Among non-nut producers I am trying honeylocust, persimmons, and mulberries. I also grow catalpa and black locust for fence posts. This makes no mention of the great variety of native timber trees such as pines, tulip poplar, and others which I try to protect from fires so as to get as great a variety of trees as possible to use for various purposes. I also encourage the growth of ornamental trees and shrubs such as dogwood, redbud, and holly to add beauty to the landscape in season.

Dr. J. Russell Smith's book, "Tree Crops" has been a great inspiration to me along these lines, and I am attempting to study and use as many trees, shrubs, and plants here on my place as possible because I believe we can live easier and better and make better use of the land both for ourselves and nature when we learn how to use our various native plants to the best advantage along with many of the exotic ones.

I might end by saying that I would much rather work in the shade of trees than in the open sun and benefit by their long life and varied uses than to depend so heavily on short lived crops which often require such intensive care.

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President Davidson: Thank you, Mr. Jones. A very interesting paper with details that are worth listening to.

Professor J. C. Moore of the Department of Horticulture, Alabama Polytechnic Institute, will give us a talk on Processed Chestnuts on the Market throughout the

Year.

Processed Chestnuts on the Market throughout the Year

J. C. MOORE, Alabama Polytechnic Institute, Auburn, Ala.

Professor Moore: Mr. President, members of the Association: I have a few packages here that I just wanted to pass around after we get through with a short discourse on processed chestnuts. It might be somewhat of an inspiration to look while I talk a few minutes about it.

These nuts, of course, have been put up from the 1947 crop, but I have nuts put up in 1945 that are still in fair shape. The quality on the 1945 product is not too good. The quality on the 1947 product is excellent when the nut is hot. For instance, a toasted chestnut, I think, has a quality that no other nut has. When the nut sits in a bag sealed for several weeks and gets cold it still is good, but it doesn't have quite the crispness that it has when it is really fresh and hot.

We were very much disappointed with Chinese chestnuts when they first began to bear at Auburn. We got some plants from Mr. Gravatt and the Bureau of Plant Industry in Beltsville in 1938. They were planted; some of them started bearing in 1941. The nuts were large in size; the trees seemed to be perfectly healthy. The early bearing habit gave us a great deal of encouragement. Then we sampled these nuts, and the quality was not good. While the nuts were green and in storage the nuts decomposed in just a few days' time.

The first nuts that we harvested in 1941 were picked, placed in paper bags, set in the office, and we forgot about them, because they were not good when we put them in the bags, and we just put them back for our record purposes. A few days afterwards they were moldy and ruined. In 1942 we had a little better crop, but again the nuts rotted. In 1943 we had a still larger crop, and the nuts rotted again. We did not know how to take care of those nuts at the time.

In 1944 Mr. L. S. Holden was with the Soil Conservation Service. He was transferred to Auburn at the time I was transferred down into Haiti to do some work on rubber production, and he took my place at Auburn on the hillculture project. In the fall of 1944 Mr. Holden had an idea that he could can those chestnuts and preserve them. So he took the nuts, cracked the hull off of the nut, ground it with a little food chopper, and placed the nuts in cans, pints and quarts, put them in a pressure cooker at 15 pounds pressure and cooked them for 15 minutes.

During the fall of 1944, or after the crop was produced, Mr. Holden left Auburn, and he told me when he left that he had sent some of the samples to different parts of the United States and had gotten favorable replies from the samples that he had sent out. That gave me a renewed courage, and along with that in 1945 we sold quite a few raw nuts on the market at Auburn. Those nuts sold just like hot cakes for 40 cents a pound. There were quite a few comments came back to us about those nuts. They were the most beautiful nuts the people had ever seen, and several different ones made comments that the nuts toasted had excellent quality and the nuts boiled had excellent quality, and raw nuts after they were cured had an excellent quality.

Those few different peoples comment on the material and Mr. Holden's work that he had done on canning gave me an idea that maybe he had something, and I have worked since that time trying to perfect a product that would be edible from the hand from a cellophane-bag standpoint. At the present time we have a plan worked out whereby we can produce large quantities of Chinese chestnuts in Alabama.

The thing that is going to confront us in the near future is the marketing possibility. We have to handle Chinese chestnuts rapidly if we put them on the market raw. This processed method that we have has been worked out to perfection, we think, for cold storage purposes.

Now, you can put Chinese chestnuts raw in cellophane bags and seal them with a hot iron. These bags are not sealed. It is a non-sealable cellophane. I didn't get hold of the type of cellophane that you can seal. They are unsealed. They have been in this package about a week, and the nuts are in good shape. On cold storage I have held those nuts for 40 days. Last year was the first time that I tried them in sealed cellophane, but sealed in cellophane bags in cold storage last year they remained perfectly good for 40 days. At that time the cold storage plant went bad, and, of course, the nuts molded.

We think that on the cold storage proposition, and if you have followed food processing and cold storage possibilities on strawberry shortcake, strawberry pies, apple pies and other types of cold storage products, I think when you go to the locker and pick out a little bag of lima beans in a cold storage locker or any other kind of cold packed foods, if you see a pack that looks attractive, chestnuts, after you get accustomed to their flavor especially, it will be a difficult thing for you to fail to pick up a bag of chestnuts and walk out with them among your other grocery purchases. That type of marketing has possibilities throughout the year.

With that possibility from last year this crop came in. We had an excellent crop. I contacted Mr. Harris, who is one of the professors working with food processing at Auburn, and we went over the work quite carefully together, what I had done and the possibilities for the work in the future, and with some suggestions from him and with his help we think we have just about fixed a product that will be a permanent thing on the grocery shelves throughout the year.

Up to the present time all of the nuts that were canned in cans with the shells on developed throughout the year somewhat of a soured condition. When you opened the can and smelled, the odor was foul. When you cracked the shell and tasted the nut, the flesh had just the least bit of a foul odor. Mr. Harris suggested that probably that was a flat sour. We weren't sure that it was flat sour, but we haven't had the bacteria check to find out whether it was caused by one of the thermophilic bacteria or not, but we are pretty confident that it was a flat sour that caused the foul odor. With careful heating and careful drying we have developed some products here that I think have a possibility, and these products will maintain their quality throughout the year.

+Nuts Cured Before Canning+

I have canned chestnuts that have been canned for three years, and the quality is just as good as it was a month after they were canned. The product, however, when it is canned green does not have the quality that it does when it is canned after curing. The way we handle these, to begin with, is to take the nuts from the field, put them on a woven wire and elevate the wire so that air can go under and over, cure at room temperature for about three days. If you cure longer than three days you will lose quite a few of your nuts. That is a rapid cure. We have not tried curing under cooler conditions to see if we can eliminate part of the damage that is caused by deterioration, but curing the nuts rapidly you get a deterioration on quite a few of the nuts after the third or fourth day. If you take the raw nuts three days cured rapidly where the air can circulate over and under, the quality is excellent raw, and I have those nuts cured for three days in cellophane bags on cold storage that can be sold throughout the year. Those nuts must be heated enough to stop the deterioration, whatever it is. It may be a physiological condition, I am not sure, it may be a vitamin reaction, I am not sure, but when the nut dries too fast it turns white on the inside, gets hard, loses its flavor, and it is no good.

This nut (indicating) canned in cans, I will give you the treatment for it. I told you we cured them on those drying racks for three days. Then we put them in a pressure cooker and run the temperature up to about 10 pounds pressure for 30 minutes, take them out of the pressure cooker and hull them, and at that stage they hull quite easily. The hull itself will turn loose from the nut quite easily if you heat it a little while before you try to hull. A machine which can thresh the hulls off very easily will be simple to develop. After the shell is taken off, then they are put in an oven (a drying oven that has an automatic control at 270 degrees), for about 10 minutes in order to evaporate the excess moisture that you get in the steaming process. Then they are put in the cans hot, set back into the oven and heated for just a few moments to get your temperature up again and you put lids on at a boiling temperature. You get quite a vacuum created by sealing them hot. We have had as high as fourteen and a half pounds of vacuum on those cans the third day after they were canned, and if you can get a vacuum like that by sealing the nuts hot, you can preserve their quality for a long period.

I don't care if you open any bag that's here and taste these products. You will find that the ones with the shells off are much better than the ones with the shells on. I believe you will find that. However, the quality of the nut with the shell on is excellent.

[Illustration: Mr. Hardy and some chestnuts prepared for storage (Courtesy Southern Agriculturist)]

Chestnut Growing in the Southeast

Max B. Hardy,[2] Leeland Farms, Leesburg, Ga.

+Introduction+

Just about forty years ago the first blight resistant chestnuts were introduced into the Southeast. This event was to have more far-reaching effects than could be foreseen at that time, as is illustrated by the present extensive interest in the growing

of these chestnuts as an orchard crop.

Chestnut blight, a fungus disease of the native American chestnut (_Castanea dentata_ (Marsh) Borkh), first appeared on Long Island in 1904 and destroyed this magnificent nut and timber tree. A Phytophthora root disease added its toll so that a bearing tree of this species is a rarity in the East at the present time. The U. S. Department of Agriculture began making introductions of two species of chestnut from the Orient in 1906, both of which were resistant to the blight which was then destroying the native American chestnut. Of the two species, the Japanese chestnut (_C. crenata_ Sieb. and Zuce.) and the Chinese chestnut (_C. mollissima_ Bl.), only the latter proved to have much merit other than blight resistance and chestnut growing in the eastern United States in recent years has been confined almost entirely to the Chinese chestnut.

About twenty-five years ago, after the first introduction from the Orient of seed nuts of blight resistant chestnut species, the U. S. Department of Agriculture distributed a few seedling trees to various interested growers in the Southeast. Some of these trees are still growing and bearing good crops of nuts and have reached rather large size. The distribution of trees produced from nuts imported at subsequent intervals was continued by the U. S. Department of Agriculture until rather widely scattered planting of several species under varied soil, climatic, and cultural conditions was attained. As time passed it became clear that only the Chinese chestnut had promise as a commercial crop for the production of nuts. As a timber tree none of the introduced species has as yet shown outstanding merit.

[Footnote 2: Formerly Associate Pomologist, U. S. Pecan Field Station, U. S. Department of Agriculture, Albany, Georgia.]

+General Observations+

The Chinese chestnut grows well throughout the southern part of the natural range of the American chestnut and southward to the Gulf Coast, and possibly even into central Florida. Farther north it apparently grows and produces better crops along the Atlantic Coast than inland, thus indicating the need of this species for a long growing season and freedom from late spring and early fall frosts. In the plantings in Georgia, from Atlanta to the southward, no loss of crop from late spring frosts has ever been noted. In the Gulf States and northward along the Atlantic seaboard the Chinese chestnut tree is vigorous, healthy, and productive, coming into bearing at a fairly early age and thereafter producing regular crops. The trees grow to be rather large in size, developing a somewhat rounded form with a spread of branches about equal to the height. Without pruning when young many sprouts usually develop near the ground so that the mature tree has numerous trunks of about equal size, with the lower lateral branches resting on the ground.

Nearly all of the Chinese chestnut trees being grown at the present time are seedlings and exhibit a wide range of tree and nut characteristics. A few trees develop a somewhat more upright type of growth than that commonly seen, but this type is generally less productive than trees of more spreading habit, and the nuts are smaller and less desirable. Some trees showing the most upright type of growth

originated from nuts imported from the more northern provinces of China and may represent a distinct strain or form of _Castanea mollissima_. The degree of incompatibility exhibited when southern China strains are grafted on northern China strains would indicate the same conclusion. Unfortunately, several different species or strains have been included in the plantings of most cooperators with the U. S. Department of Agriculture so that seedlings resulting from cross-pollination of these types may exhibit an even wider range of characteristics and performance from the standpoint of commercial production than is commonly seen at present. A few of these hybrids may be superior to pure _C. mollissima seedlings_ in certain important respects because of hybrid vigor, but taken as a whole the best types of _C. mollissima_ seedlings are superior to the other blight resistant species for purposes of nut production.

The earliest introductions of blight resistant chestnuts from the Orient are represented by very few trees in the Southeast, but a small number of plantings of trees distributed in 1926 have been observed. These are producing good nuts and the trees are quite healthy, regardless of conditions of planting except when they have been given no attention of any kind. In one planting the trees were planted about 10 feet apart on the square with the result that they are tall and spindly with nut production only in the tops and very light on a per tree basis, which indicates the need of adequate spacing if the trees are to be vigorous and productive. Incidentally, this close spacing has not resulted in a desirable timber type of growth.

In two other plantings the trees are planted in cleared areas in cut-over timber and then given no further attention. In both locations a few trees are still living but are of no value either for timber or nut production. In still another planting on a bench about halfway up a mountain, where infrequent cultivation or mowing is practiced, the trees are growing and producing moderately well but the nuts are small. A few other scattered plantings of a few trees each are doing well around homes though receiving only moderately good care.

The distribution of trees by the U. S. Department of Agriculture in 1935 and 1937 has resulted in a few plantings that have done moderately well. In one planting the trees are growing fairly well without care but are producing few nuts. In another planting the trees are planted on rather heavy soil that is terraced; they are given applications of commercial fertilizers and infrequent cultivations and have been producing fairly good crops of nuts in recent years. Still another planting of a considerable number of trees has been entirely removed through lack of interest of the new owner. The plantings described have all been on private property.

Plantings at various experiment stations have received somewhat more attention in general than those on private property; but because of lack of keeping quality of the nuts have not for the most part been accepted as a promising crop and have been the subject of very little study.

From the foregoing observations it is evident that the Chinese chestnut cannot withstand the effects of crowding either in a solid planting or in competition with native growth. The trees have performed moderately well with a minimum of care,

but respond to good care by increased production and nut size. The rotting of the nuts soon after harvest as a result of improper methods of handling and storage has prevented an earlier acceptance of the crop as of potential economic importance in the Southeast.

+Experimental Studies at the U. S. Pecan Field Station, Albany, Georgia+

In 1926, twenty-eight seedling trees of _Castanea mollissima_ were planted in the Champion experimental block at Philema, near Albany, Georgia. These trees grew well and began producing nuts in 1932. In 1935, an additional 16 trees were planted in the same block. The trees in both plantings have shown good vegetative vigor and have been fairly productive. All the variations common to any group of Chinese chestnut seedling trees have been in evidence. One or two trees have lacked vegetative vigor but have produced heavy crops of nuts for their size. Type of bur opening has varied from free dropping of nuts to those burs from which the nuts are removed with difficulty; nut size has varied from about 35 to about 90 nuts per pound; the date of earliest and latest ripening of the nuts varies by about three weeks; nut color has ranged from light browns to dark mahogany and dark chocolate brown; and keeping quality and eating quality have ranged from good to poor. However, nut production, as shown by the data presented in Table I has been good and nut quality has been acceptable, so that with increasing knowledge of the storage requirements of the nuts the trees have paid a good profit in recent years. One of the older trees has consistently produced close to 150 pounds of nuts each year for the past few years.

Some of the trees in this planting have been topworked to selections from other plantings, including the variety Carr which showed up very poorly in comparison with most of the seedlings. Some of the trees have been culled out because of poor yield or nut size; and some have died as a result of poor drainage.

An additional planting at Philema in the Brown tract was made in 1938. The trees were planted in a portion of a five-acre block at some distance from the original plantings, with a spacing of 25 feet apart on the square in soil of rather light and sandy texture with fair subsoil drainage. The fertility was low but has been improved through the use of winter leguminous green manure crops and commercial fertilizers. Some of the trees planted consisted of trees grown from carefully selected _Castanea mollissima_ nuts imported from south China and designated by the initials MBA, MAY, MAZ, and MAX. Others carried the designating letters of "FP." The nuts from which these trees were grown were imported by the Division of Forest Pathology of the U. S. Department of Agriculture which also grew and distributed the trees. Still others were selections of _C. crenata_, the Japanese chestnut; and _C. mollissima_ selections from an experimental planting in California were also included. In 1940 the remainder of the five-acre block was planted with trees grown from seed produced by the original Philema planting.

Table I. Summary of chestnut yields at Philema, Georgia.

		HARVEST DATA	
		1926 and 1935 Planting[3]	Length

|_____| Date Harvest | | Year Harvest Period | Yield No. Trees Av. Yield | Began in Days | in Lbs. Bearing per Tree | _____|_____| | | 1932 | 14 3 4.7 | 1933 | 7 7 1.0 | 1934 | 80 16 5.0 | 1935 8-29 22 | 222 22 10.1 | 1936 8-26 33 | 379 25 15.1 | 1937 8-26 37 | 278 18 15.4 | 1938 8- 6 42 | 480 21 22.9 | 1939 8-15 42 | 995 26 38.3 | 1940 8-27 38 | 740 34 21.8 | 1941 8-14 51 | 1,467 38 38.6 | 1942 9- 3 41 | 876 32 27.4 | 1943 9- 9 26 | 1,335 38 25.1 | 1944 8-15 44 | 560 29 19.3 | 1945 8-18 34 | 1,450 27 53.7 | 1946 8-20 41 | 1,455 28 52.0 | 1947 8-26 43 | 1,975 27 73.1 | _____|_____|

_____ | | HARVEST DATA |_____ | | 1938 and 1940 Planting[4] Length |_____ Date Harvest | Year Harvest Period | Yield No. Trees Av. Yield Range in Began in Days | in Lbs. Bearing per Tree Yields _____|_____ | 1941 8-14 51 | 44 63 .7 .1-6.9 1942 9- 3 41 | 30 46 .7 .1-5.2 1943 9- 9 26 | 357 108 3.3 .1-29.7 1944 8-15 44 | 716 136 5.3 .1-37.0 1945 8-18 34 | 3,025 208 14.6 .1-50.7 1946 8-20 41 | 1,447 173 8.4 .1-48.3 1947 8-26 43 | 6,615 188 35.2 .1-108.5 _____|_____

[Footnote 3: 28 trees planted in 1926 and 16 planted in 1935, at spacing of 25 to 40 feet.]

[Footnote 4: 274 trees planted in 1938 and 60 in 1940, at spacing of 25 feet on square.]

The yield's produced in the 1938 planting have been outstanding, as indicated by the data in Table I, The trees began bearing when younger and developed heavier production than those of the 1926 planting, whether judged by age of tree or years of bearing. Many of the trees have produced nuts of outstanding size, attractiveness, eating quality, and keeping quality. There has been the usual degree of variation common to any collection of seedlings, but the best trees in this planting have been superior to any previously seen. Nut size has varied from 23 to more than 100 to the pound; the color of the nuts has varied from light tan to deep mahogany, and a few are nearly black. All have been of good eating quality. The keeping quality has varied materially, some keeping very well and others quite poorly.

Bur opening, has likewise varied so that at one extreme the nuts drop entirely free from the burs on some trees and at the other extreme the burs drop with the nuts in them and considerable work is required to remove the nuts. It is out of this group of trees that the three seedlings have been selected that the U. S. Department of Agriculture is considering worthy of variety status. These have not yet been officially released and no official description is yet available. The yield data for these three selected Seedlings are given in Table II.

Table II. Yield data by years, of three seedlings tentatively proposed for variety status, Philema, Georgia.

---------- Tree Proposed Yield in Pounds by Years Total yield No. Nuts (in Lbs.) per Lb. ---------- No. Name 1941

1942 1943 1944 1945 1946 1947 1948 from Planting

7880[5]Meiling .2 3.6 20.9 36.9 23.9 73.1 36.9 195.5 38-43
7919 Kuling 4.0 3.8 5.8 6.5 13.8 34.2 50.2 38.2 168.5 35-43
7930 Nanking .1 3.8 28.0 37.8 1.0 87.7 54.6 213.0 30-43

[Footnote 5: Meiling ("Beauty") is the first name of Mme. Chiang Kai-shek.]

The trees of the "FP" designation and, of other species were grown to fruiting, but have since been removed or topworked in entirety because of their lack of desirable characteristics and because they produced pollen for cross-pollination which would result in undesirable progeny when the _Castanea mollissima_ nuts were used for seed. Furthermore, a number of trees of the three-letter designations have been removed or topworked because they produced very small nuts, or showed poor keeping quality, or because of some other undesirable characteristic. Therefore, the nuts now being produced in this experimental orchard are of pure _C. mollissima_ inheritance of the best type, and, as such, represent some of the best and purest seed nuts available in this country today. This procedure is being continued so as to maintain the quality of the nuts for seed purposes at its present standard.

Unfortunately, many of the nuts offered in the general trade for seed purposes at the present time are coming from orchards composed of a mixture of species or types comparable to the 1938 Philema planting before culling. This is very undesirable because of the great variability in the nuts produced by trees with such an origin. When grafted or budded trees of the newer and improved varieties are available to orchardists chestnut growing for nut production may be based on the same sound practices as the other fruit industries.

In the topworking of "FP" trees at Philema with scions from other strains of _Castanea mollissima_ the degree of incompatibility has been so great, that the scion tops will have either blown out or died at the end of four or five years from grafting. At the present time this failure can only be attributed to the fact that the stocks were of mixed ancestry. On the other hand, scions of pure _C. mollissima_ placed on the same stock strains have made good unions and are entirely normal after as long as 13 years from grafting. This problem of incompatibility between stock and scion is one that yet remains to be completely solved.

The topworking of trees in the five-acre block at Philema has been generally successful where incompatibility is not a problem. Bearing-size trees topworked one spring will generally produce a few nuts in the second subsequent growing-season. Growth the first year after grafting will frequently be as much as 12 feet long and very stocky. Both cleft grafting and inlay bark grafting have been practiced, the latter method proving to be the more satisfactory from all standpoints. In this method of grafting scaffold limbs from 1 to 6 inches in diameter are cut off square across. Scions 6 to 8 inches long are prepared by making a slanting cut 2 to 3 inches long and ending about three-fourths through the scion at its basal end. A strip of bark just wide and long enough to receive the scion, with about one-half of the upper end of the bevel showing above the cut surface of the stub, is then removed from the stub.

The scion is then nailed into place with 5/8-inch nails and painted over with melted grafting wax. Two or three scions are required for most stubs. This work is done just as growth is starting in the spring and the bark is slipping well. The scions may generally be cut directly from the trees, but sometimes they may need to be cut several days earlier and stored in damp material in a refrigerator to keep them dormant.

In south Georgia the Chinese chestnut normally begins growth soon after March 1, but in some years it has started as much as a month after this date. Between south and north Georgia there is a differential in the time growth starts in the spring of one to two weeks. This differential also carries over into the date of blossoming and the date the harvest period begins. In south Georgia pollination generally occurs during the latter part of April and early part of May, and the harvest period begins about 100 days later. The peak of harvest averages 185 days after the initiation of growth in the spring. Dormancy comes only after the first frost sufficiently heavy to kill the leaves, usually about two months after nut harvest is completed. This period between harvest and leaf fall is undoubtedly an important factor in the annual bearing habit of the chestnut in the Southeast since it permits the food reserves in the tree to be replenished after the crop is mature. This is true under favorable conditions but does not hold under conditions of crowding, low soil fertility, or premature defoliation. For best growth and production the tree should be in foliage approximately nine months out of the year.

+ORCHARD MANAGEMENT+

The planting of chestnut trees in the Southeast should be done as soon as possible after the trees become dormant in the nursery. They should be planted on fertile soil which is well drained but not subject to serious drought injury. The Chinese chestnut cannot withstand a high water table, or free standing water, but appears to be somewhat resistant to drought injury when once well established. The chestnut trees have not yet reached an age at which their largest potential size has been attained, but trees of 50-foot spread have been observed. It appears likely, then, that orchards should be planted at 50 to 60-foot distances on the square, unless closer planting and subsequent thinning is resorted to in order to build up high nut production per acre at an earlier age of the orchard. Planting distances of 25 x 25 feet, 30 x 30 feet, 25 x 50 feet, and 30 x 60 feet are recommended for this reason, but only if the orchardist will plan to thin the stand at 10 to 15 years of orchard age and at later intervals as required. In no case should the branches of adjacent trees be allowed to touch as under such conditions competition between trees will reduce the yield per tree and nut size, and induce alternate-year bearing.

In planting the young tree it is usually advisable to fill the hole in which the tree is to be set with top soil, packing it firmly around the roots as the hole is being filled. Usually no fertilizer is used at the time of planting, although mixing about a handful of bone meal with the soil around the roots has given a higher percentage of living trees and has increased growth the first year. A shallow basin around the tree to facilitate watering when necessary during the first growing season, or the application

of a mulch around the tree, or both, will be helpful in obtaining a high percentage of living trees and good growth. Adding water at the time of planting is good insurance that the soil will be well settled around the roots. A wrap of newspaper tied loosely around the trunk of the young tree will aid in preventing winter injury and sun-scald.

Under conditions of little or no care the seedling chestnut tree will generally develop several trunks as a result of the forcing of multiple sprouts from near the ground line. The tree should be trained to one trunk, as such a form seems to be less susceptible to winter injury while young and makes a much more desirable orchard tree when older. Pruning of the young trees subsequent to the development of the head at a 4 to 5-foot height should be confined to the removal of crossing branches and those so near to the ground as to interfere with the necessary cultivation and harvesting work under the tree.

Most soils in the Southeast are somewhat low in fertility and must receive good care if chestnuts are to grow well. The annual application of commercial fertilizers is generally required as is the growing of a winter green manure crop, preferably a legume. One of the most satisfactory systems is to plant hairy vetch, Austrian winter peas, or blue lupine[6] in late October or early November, applying broadcast at the time of planting from 400 to 600 pounds per acre of a 0-14-10 or 0-14-7 fertilizer mixture. This green manure crop should then be disced in by April 15 of the following spring, with subsequent shallow cultivations at about six-week intervals through the growing season. The ground should be clean by the middle of August to facilitate harvesting the nuts. If such a system of culture is not feasible, as on too steep slopes or around buildings, mowing or mulching can be used to advantage, but the trees must be given annual applications of a complete fertilizer mixture, such as 4-8-6, 6-8-8, or 5-7-5. These should be made each year about a month before growth starts at a rate of 2 to 3 pounds for each year of tree age. This should be broadcast under and slightly beyond the spread of the branches.

It has not yet been found necessary to spray the trees for the control of any disease or insect. This does not indicate that control measures may not be required at some time in the future, for it is the history of horticultural crops when planted in any concentration that diseases and insects increase in number and degree of injury. As yet, the chestnut weevil has not been found at the lower elevations in the Southeast.

In a few plantings a condition causing some premature defoliation has been observed at infrequent intervals. The condition begins as a leaf scorch which may or may not develop to the point where the leaf drops. It is thought to be caused by some mineral deficiency or unbalance associated with erratic weather conditions, but the exact cause is yet unknown. A leaf spot disease has been observed but has caused no appreciable defoliation and no control measures have been thought necessary.

[Footnote 6: Blue lupine is winter-hardy only in the warmer coastal areas, not adapted north of Columbus, Georgia, Meridian, Mississippi, or Shreveport, Louisiana. Ed.]

+Harvesting and Nut Storage+

Harvesting of Chinese chestnuts has proved to have definite requirements if the nuts are to be obtained in the best possible condition. The nuts are quite susceptible to rots of several kinds and must be properly handled to keep losses at a minimum. They are also very easily and quickly injured by exposure to the sun, with the consequent, high temperatures and drying. If the nuts are to be stored for any length of time, as is necessary when they are to be used for seed purposes and as will be necessary when they are to be marketed for eating purposes during the Thanksgiving and Christmas holiday seasons, it is paramount that they be picked up from the orchard at not more than two-day intervals. Cleaning up all dropped nuts at daily intervals is most desirable.

At the end of each day the harvested nuts must be placed in cold storage at temperatures between 32 deg.F. and 45 deg.F. It has been found that a nearly air-tight container is required in order to maintain a relative humidity of 100% and prevent too much drying of the nuts. A 50-pound tin lard can with one 20d nail hole in the side near the lid has proven to be a good container for large quantities and these same cans also make good shipping containers merely by wiring on the lids. One-gallon friction top syrup cans with a single nail hole in the side make a good container for smaller quantities. In air-tight containers the nuts do not decay but germination capacity is quickly destroyed and bitter flavors develop quite rapidly. Nuts to be used for eating purposes shortly after harvest may be stored at lower relative humidities but should be placed in cold storage. A loss of about 15% in weight from the fresh weight of nuts is necessary to reach proper eating quality. Nuts dried to this extent are sweet and palatable but cannot be stored for any length of time and fail to germinate well when planted.

The experimental study of chestnut storage problems is being continued with the hope of working out still better methods. The manner of marketing chestnuts so that they will reach the consumer in a desirable condition also is still to be worked out, but it appears possible that retail cold storage and packaging in moisture-proof bags which are pervious to CO_{2} and O_{2} give promise at present. Probably the most promising aid to an increased storage life of chestnuts will come through the selection of trees for propagation and planting that produce nuts of superior resistance to storage rots. There is rather great variation among seedlings in this respect, some being-quite superior, although no completely resistant seedlings have yet been found.

+Discussion and Conclusions+

The perishable nature of the nuts of the Chinese chestnut has probably been the greatest drawback to an earlier acceptance of this crop as an adjunct to the horticulture of the Southeast. It has been only in the past few years that enough has been learned about the harvesting and storage requirements to permit the storing of these chestnuts so that they can be marketed in an orderly manner either for eating or for seed purposes. Storage losses through periods up to six months have been held to less than 10% for a mixture of nuts from all the trees at Philema. Storage tests of nuts from individual trees have shown a range in keeping quality from no loss after six months' storage to nearly 100% loss. By culling out the trees producing nuts with

a high rate of spoilage under the best storage conditions it should be possible to reduce storage losses to a minimum. Every grower of seedling trees should follow this same process of culling out or topworking trees producing nuts of poor keeping quality if the industry is to grow and prosper, since otherwise the offering of spoiled nuts for sale to the consumer will soon destroy the demand for the nuts.

There is no question but that the Chinese chestnut tree is very well adapted to the Southeast. It has proven to be healthy, vigorous, and productive. Yield records at Philema show actual yields of more than 1,000 pounds per acre and potential average annual yields of 1,500 or more pounds per acre are not out of reason. In 1947, in the Brown tract at Philema, if all the trees that bore nuts had been collected into a solid block the yield per acre would have been nearly 2,500 pounds. Crowding of the trees in the Brown tract is becoming serious at 11 years of age with a 25 x 25 foot spacing. Alternate-year bearing is becoming apparent and the stand of trees must be thinned immediately. Because of such potential yields and because rather extended storage of nuts of varied keeping quality is now economically possible the future of the chestnut industry in the Southeast is very promising.

The selection and propagation of selected seedlings is desirable as a means of advancing the industry at a more rapid rate. The propagation of selected seedlings offers a problem because of lack of compatibility between some stocks and scions. Since the chestnut is almost completely cross-pollinated it may be necessary to develop special plantings of two or three selections as a source of seed nuts for the production of stocks. Such plantings might possibly produce seedlings of quite uniform and desirable characteristics, but this prospect, is not very promising. Certainly, the evidence points to the conclusion that scion selections must be worked on stocks of the same strains if incompatibility is to be held at a minimum.

There is a further problem in the propagation of varieties on seedling rootstocks in the nursery. Only one propagator appears to be having much success in this art but others must learn it. Topworking of older trees by the inlay bark graft is generally successful and older seedling orchards can be worked over to improved selections without difficulty so long as the stocks are of compatible strains. Time will be required to work out the details of the solution for this problem but they will be worked out.

In the selection of improved seedlings for propagation the strictest attention should be paid to the important characteristics of tree vigor, precocity, productiveness, nut size, attractiveness, and keeping and eating quality, and type of bur opening. These characteristics have been previously discussed but it is well to emphasise their importance. The tree that comes into bearing at an early age seems likely to be more productive in later years. The nuts should be no smaller than 45 nuts to the pound and be attractive to the eye of the buyer. Most individuals prefer nuts with a bright and shining surface free of fuzz and with a fairly rich mahogany or chocolate color. Keeping quality is, of course, of great importance and should be carefully determined. Eating quality is generally good but distinctly superior selections may be found in the future. For the most part eating quality is dependent on the proper curing of the nuts. The type of bur opening is more important than

usually considered, as it materially affects the satisfactory harvesting of the nuts. From the commercial standpoint it appears that the most desirable bur should drop from the tree with the nuts still in it but be well split so that the nuts can be readily removed. Such a bur type prevents exposure of the enclosed nuts to the hot sun while on the tree and reduces injurious drying to a minimum yet permits rapid gathering of the nuts in the burs for later mechanical separation. Nuts that drop free from the burs are more subject to injury by drying and require more hand work in gathering. Burs that do not split readily would be more difficult to separate mechanically; and mechanical aids will be necessary for the economical daily gathering of the nuts in commercial orchards.

If is encouraging to note that many of the present new plantings in the Southeast are being made by orchardists rather than hobbyists. Many home owners are planting a few trees but the acceptance of the Chinese chestnut for commercial production by men already growing other orchard crops portends the future success of the industry. The hobbyist has been of great service and should be given full credit for his far-sighted interest in a crop that now has commercial promise, especially in the Southeast. Much experimental work is still needed by both State and Federal agencies and by individuals. This work needs be concerned now more with details of refinement rather than with basic possibilities of the crop.

* * * * *

President Davidson: Mr. Carroll D. Bush, of whom I am sure you have often heard and whom very few of you, including myself, have met, of Grapeview, Washington, will now tell us something about the Marketing of Chestnuts on the Pacific Coast. Mr. Bush.

Marketing Chestnuts on the Pacific Coast

CARROLL D. BUSH, Grapeview, Washington

Mr. Bush: Friends of the Association: There are so many here that I have known through correspondence that I have welcomed this opportunity to say something to you today. I don't think that I will add very much to anything that has been said. I hope perhaps we will have some ideas from what we have been doing on the Coast.

We were in the nursery business near Portland, and during the war we went out of it, but we are working back in trees again[7], and all this time we have been preaching the gospel of nut trees, and we find that we can't preach a gospel unless there is some reward. There is no market for chestnuts in our section of the country, and yet we had quite a few of them around Portland. We could not talk about chestnut trees when there was no market. Buyers there had been offering as low as three cents a pound or not buying them at all, and we, ourselves, had quite a few nuts to sell. So I took a trip up to Seattle and found a commission man there that would take our nuts and arranged with him, and we have sent nuts to Seattle ever since that year and got a very good price. Then a neighbor had me send some of his, and we are still sending nuts.

+Introduced on Mid-West Markets+

The next year through Carl Weschcke of St. Paul I got in touch with a reliable Minneapolis firm. They evidently had been burned and they were somewhat skeptical. They said if we would send a sample there they would look them over. So I went out and picked up a mixed sample and shipped to Minneapolis, and they said if we could send nuts as good as the sample they could use some.

We began to send them. When we shipped them we made sure we sent nuts that were considerably better than the sample, and the rewards for shipping there were also very good. Then we went on to Chicago, and we have been shipping to Chicago over since. At this time I am out here to find a little more market for some of the nuts that we have in Oregon.

At first we put the nuts in cold storage at about 32 degrees, expecting to get a better price on the Thanksgiving market. We found out that we were making a mistake and that the earliest nuts on the market brought us our best price. So now we are shipping just as early as we can ship.

We first adopted the western cranberry box as being open enough to allow a little drying off and tight enough so that it wouldn't allow too much and yet we didn't get any mold. We were very much afraid of that, because a good many of the California chestnuts had molded on the way to market. Later we turned to the splint bushel basket, and lately we have been in favor of the half-bushel basket. There seem to be buyers who don't like to stock up more than a half bushel at a time, chestnuts being of a rather high price. They dry out too fast.

We found that cold storage above 32 degrees keeps chestnuts in good condition with little dry-out. One dealer in Oregon we know of wraps his cold storage nuts in waterproof paper, keeps them that way clear on into January. A very little mold will develop on chestnuts kept in storage from 32 to 35 degrees, but not enough so we take any precaution. We have had a few batches that people have stood in sacks on damp nights, and they started to mold, especially on the open end, and we find we can kill the mold with Clorox. We have just used a little Clorox in water. We think this would prevent mold from developing on all nuts if they were put through a chlorine bath. We haven't taken the trouble to do that. I might say our walnuts, and filberts have been put through a chlorine solution, and, of course, after a chlorine solution is used you have to put the nuts through water again and wash that off.

We have on our place a nice washer. We have graded the European varieties, which we handle mostly, into three grades: standard, fancy, and extra fancy, by size. All our grading has been done by hand, except we expect to have a simple grader this year.

[Footnote 7: Mr. Bush informed the secretary by letter, early in 1949, that he did not then have any nursery stock ready for sale at his Eagle Creek, Oregon, nursery. From that location about 10 years ago he introduced, under numbers, three selections of Chinese chestnuts grown from seed imported in the early 30's. Two of these, in 1941, were named Abundance and Honan. The Abundance is now considered one of the most desirable varieties from Oklahoma to Pennsylvania, while Honan is slightly less desirable.--Ed.]

+"Sweet" Nuts Sell Faster+

We have a few "sweets." All of those on our farms are Riehl varieties, hybrids, I think. All of our European chestnuts have an astringent pellicle, heavy with tannic acid. We classify as sweets any of those that have a pellicle that is sweet enough to be eaten. We label these the sweets and mark them as they go into the market. And while, I say, we don't seem to get a better price for the sweets than for the European, they do sell faster. There are some people in the eastern cities that are grabbing these in preference to the large ones. While the large nuts sell very well, I suppose they go to the Italians and Europeans who are used to cooking them, and out on the West Coast nothing but the large nut goes; the larger the better. In the Seattle market we try to send in large nuts.

We also grade out all "cracks" by hand. They mold easily, and we have a lot of cracked nuts in our climate there, but we have been able to dispose of all of these through the Seattle market where they move off very fast and are lower priced.

+California Supplies Distant Markets+

Last winter we went to California and looked into the chestnut market there. We found them in the Sierras and found them growing in the Coast Range without irrigation, but the largest growers were in the San Joaquin Valley near Stockton. The largest grove was 30 acres at Linden owned by Caesar De Martini. He gave us our best insight into California chestnut growing. He used to grade and package his own, and he still has his cylinder grader. It has three different size holes, one inch, one and a quarter and one and a half. Anything that goes through the one-inch hole is discarded as a cull. That leaves three sizes, the size that goes through the one and a quarter, the one and a half, and the size that goes out the end, which is, of course, a class of jumbos.

All the chestnuts in California, I think, now go to buyers to do the grading and packing much as De Martini worked out. All of the California nuts have to be soaked in water just as Mr. Jones does, as they come to the packer dried out. The largest buyer that we found in California shipped about seven carloads, and he shipped them all over the world, the Philippines, Honolulu, Alaska, and other places where the chestnut hasn't been growing.

+Early Autumn Best Marketing Season+

Now, I am going to sum up what our experience has been and what we recommend as general from our experience. Your experience may be different. We clean the nuts, wash them, if necessary, grade them; large and small nuts do not sell well together. We would pack in baskets, half bushel for sweets. We are trying to make that half bushel basket the mark of the sweet nut in the markets where we sell, so that when a buyer comes in there and sees a half bushel basket he knows that's sweets. Then we ship as wet as possible, and they dry out on the way. And just as fast as we can get those nuts off the ground we pack them and ship them. Our greatest trouble now is, of course, the imported chestnut. They are beginning to come in in great quantities, and they hit the market in Chicago last year at about the 20th of October, and we tried to beat that line if we possibly can with our nuts, because just the minute the

carloads of chestnuts come in on the East Coast the market drops right down.

Without question we could use some of the preparations that we use on filberts to put a gloss on the chestnut, run them through, I think it is a paraffin mixture, put a gloss on the shell and give us a better chestnut in the market, make it look nicer and, of course, make it sell better.

+"Stick-tight" Burs Preferred for Pacific Coast+

I disagree, I think, with two of the former speakers in regard to the chestnut that falls free from the bur. I would prefer a chestnut that sticks tight to the bur. We have threshers out there that thresh them out. We can pick up those nuts in the bur with a shovel or fork, throw them into the wagon, take them in the wagon, thresh them out. You have a cleaner nut, you don't have to pick around on the ground with rubber gloves that we use, which is easy enough, but it certainly adds a great deal of work as compared to threshing them out easily after they are once picked up.

I thank you.

* * * * *

President Davidson: Thank you, Mr. Bush. We are glad to have that western angle. It is going to be very useful to us.

Next on the program is a paper on the Control of the Chestnut Weevil, the author of which is absent, but I believe Mr. Gravatt is going to read that.

Chestnut Weevils and Their Control with DDT

E. R. VAN LEEUWEN

United States Department of Agriculture, Agricultural Research Administration, Bureau of Entomology and Plant Quarantine, Division of Fruit Insect Investigations.

Failure of the American chestnut to resist the chestnut blight has resulted in the planting of a few blight-resistant species obtained from foreign lands. These foreign chestnuts would now be planted more extensively in certain districts, were it not for the fact that the nuts are injured by two species of weevils, for which heretofore there has been no practical control.

The 1947 season marks the fourth year of the experimental use of DDT for control of the chestnut weevils. During these years our knowledge of the spray and how best to use it has been advanced by conducting laboratory and field tests. Unfortunately, few chestnut orchards now exist in the Eastern States, and the scattered plantings consist mostly of a large number of Asiatic seedlings, some of which had to be top-worked to other Asiatic species and varieties. Many of these trees are grown for ornamental, shade, or timber purposes rather than for nut production. Owing to these conditions and to a series of spring frosts since 1945, it has been impossible to conduct insecticide experiments on an adequate basis of replicated plats.

Although much is to be learned regarding time of application of the sprays and the proper dosage, the use of DDT can be recommended as a standard practice, because

it has proved highly valuable in protecting chestnut trees from heavy losses due to the chestnut weevil. It is the purpose of this paper to discuss some of the experiments that have been made with DDT and the observations made on the time of egg deposition.

+Nature and Extent of Injury+

The worms attacking chestnuts are the larvae of two very similar species of weevils, one larger than the other. The adults are medium-sized beetles having extremely long, slender beaks. With these they drill through the husk of the nuts, making openings through which they insert their eggs into the nuts. From these eggs the familiar worms develop. Weevil injury varies greatly in different chestnut-growing localities. It is not unusual for 50 to 75 percent of the nuts to be wormy, and often infestation reaches 90 to 100 per cent. The small weevil does the most damage, but there are indications that this may not always be true. Because the mouth parts of the adult are situated at the end of an extremely long and slender beak, it can obtain most of its food from beneath the surface of the host plant. For this reason, stomach poisons applied to trees have not been eaten by these weevils, and hence have been of no practical value. As DDT kills by contact, it is necessary only for the body of the insect to come in contact with DDT.

+Life Histories of the Weevils+

In the vicinity of Beltsville, Md., the adults of the large chestnut weevil[8] leave the soil about August 15. The date will vary, of course, with season and locality. Both males and females soon begin to feed by piercing the burs with their long beaks. Mating begins soon after the weevils collect on the trees, and egg laying follows shortly. The eggs hatch within a few days and the worms develop within the nut. A few of the worms will complete their growth and leave before the nuts fall, but most of them emerge from the nuts after they have fallen. The worms then enter the soil, where they build cells and remain until they change to pupae the following summer. This weevil has a one-year cycle, or one generation a year.

The life history of the small chestnut weevil[9] is somewhat similar, except that in the vicinity of Beltsville the weevils leave the soil late in May or early in June, when the trees are in bloom. Several weeks later the females deposit eggs in the nuts. At Beltsville, egg laying begins late in August and continues for several weeks. After the nuts have fallen from the tree, the full-grown larvae leave them and enter the soil. Earthen cells are constructed at a depth of 4 to 12 inches, where some of the larvae remain for two winters.

The small chestnut weevil completes its life cycle in two years, and a small percentage requires three years, whereas the large chestnut weevil completes its transformation from egg to adult in one year. The large weevils pass the winter as larvae, whereas the small weevils pass one winter as larvae and the second winter as adults. With the few individuals of the small weevil which require three years for transformation, the first two winters are passed in the ground as larvae and the third in the same location as adults. This habit of the small weevil complicates control measures, as one season's spraying with DDT does not reduce the entire infestation

of weevils.

[Footnote 8: +Curculio proboscideus+ Fab.]

+Proper Time for Spray Applications+

Application of DDT sprays at the proper time is very important. An examination in 1944 of many unopened chestnut burs disclosed the fact that eggs of the small chestnut weevil were being deposited many weeks before the burs would open. It was also noted that great numbers of the larvae were leaving the nuts soon after the burs cracked open. Evidently these full-grown larvae had hatched from eggs deposited several weeks before the burs split.

In 1945, 1946, and 1947, cloth bags were tied over developing burs at various intervals during the season to prevent further egg laying in the nuts. At harvest time, the bags were removed and the nuts examined. Occasionally adults were hidden among the spines of the burs and were inadvertently enclosed in the bags; therefore, all nuts in bags containing female adults that might have continued ovipositing were discarded. The data in Table 1 show the approximate time prior to which the nuts were infested.

Because of difficulty in obtaining sufficient burs for bagging, and other orchard conditions, the results of these studies were far from conclusive. They indicated, however, that many eggs had been deposited in the nuts before the burs had reached maturity. They also suggested that the seasonal histories of the two species are closely parallel. At Glenn Dale, Md., and Fairfax, Va., the small weevils predominated, constituting about 69 to 90 per cent of the total numbers taken. At Elkton, Md., only 42 per cent of the weevils were of the small species.

[Footnote 9: Curculio auriger Casey.]

Table 1. Results of studies to determine the time of oviposition of the chestnut weevils.

Nuts Infested with

Date of Total Small Large Wormy Bagging Nuts Chestnut Chestnut Nuts Nuts Bagged Weevil Weevil Year and Orchard Number Number Number Percent 1945 July 9 52 2 5 13 Glenn Dale, Md. Aug. 1 46 4 2 13 Aug. 15 107 18 11 27 Fairfax, Va. Aug. 21 110 22 13 32 Sept. 12 123 63 11 60 1946 July 12 65 0 0 Glenn Dale, Md. July 18 40 0 0 July 26 67 0 0 Aug. 1 71 0 0 Aug. 9 29 1 0 3 Aug. 14 88 3 2 6 Aug. 23 53 18 2 38 Aug. 29 53 23 11 64 Fairfax, Va. July 26 98 0 0 0 Aug. 15 168 0 0 0 Sept. 4 164 139 16 95 1947 Aug. 15 54 5 1 11 Glenn Dale, Md. Aug. 25 38 8 0 21 Sept. 2 24 7 1 33 Sept. 9 42 18 4 52 Sept. 15 56 29 7 64 Sept. 22 90 27 11 64 Sept. 29 143 83 22 73 Fairfax, Va. Aug. 26 35 9 1 29 Sept. 10 58 25 4 50 Sept. 28 50 35 7 84 Oct. 7 217 177 22 92 Elkton, Md. Aug. 21 139 11 13 17 Sept. 4 83 22 25 57 Sept. 18 116 21 35 48 Oct. 1 108 31 44 69

+Spray Experiments in 1944+

Shortly after adults of the large chestnut weevil first appeared in the orchards in 1944, six trees isolated from other chestnuts were selected for treatment. Five trees

were sprayed with from 1 to 5 pounds of technical DDT plus 1/2 pound of sodium lauryl sulfate to 100 gallons of water, and the sixth tree was left untreated as a check. A thorough application of a coarse, drenching spray at a pressure of 400 pounds per square inch was used in an attempt to force the DDT between the many spines of the burs. The DDT used was very coarse, and difficulty was experienced in getting a proper suspension. This formula was used, however, in preference to one which contained other ingredients that might have formed a protective coating over the particles of DDT. Heavy rains prevented later spray applications.

Adult weevils obtained by jarring untreated trees were then confined in screen cages placed over the lower branches of the trees. At the end of each cage was a cloth sleeve which was tied to the limb to hold the cage in place. The treatments used and the results are given in Table 2.

Table 2. Percentage mortality of chestnut weevils placed in field cages on trees at different intervals after they had been sprayed with with DDT, 1944.

Strength of DDT (lb. per Small Chestnut Weevil Large Chestnut Weevil 100 gal.) 48 Hrs. 96 Hrs. 144 Hrs. 48 Hrs. 96 Hrs. 144 Hrs.

1 0 61 100 25 50 100 2 19 69 100 0 34 100 3 4 50 100 0 40 100 4 27 87 100 0 50 100 5 18 50 100 30 46 100 Check 0 0 0 0 7 7

Although the results obtained the first few days in the cages containing treated foliage were somewhat irregular, because of the small numbers of tests made, all weevils were killed within 6 days. The results indicate definitely that DDT is toxic to the adults of both species of weevils. No consistent differences between species were noted.

As the matured nuts dropped from the treated trees, daily collections were made, and one-third of each collection was used as a sample in determining the percentage of wormy nuts. At the time the nuts drop, the holes in the shell through which the eggs were inserted are very difficult to detect. The nuts were therefore held in wire baskets to permit most of the larvae to emerge before the final examination. All nuts not showing exit holes were cut open to find out whether they were wormy. The marked increase in clean nuts after all treatments indicates that DDT is a promising insecticide for use against the weevils. The treatment and infestation records for the sprayed trees and the check tree are given in Table 3, which also includes the results obtained in later years.

+Spray Experiments 1945 to 1947+

Spring frosts in 1945 destroyed 95 per cent of the crop of chestnuts in the Eastern States. Only six trees of different species and ages in the Government orchard at Glenn Dale, had sufficient nuts for experimental purposes. Applications of a 50 per cent DDT wettable powder in the proportions of 4 and 6 pounds plus 1/2 gallon of summer oil as a sticker to 100 gallons of water were made on August 20 and September 9.

Spring frosts again damaged the orchards in 1946, destroying about 80 per cent of the possible chestnut crop, and leaving only eight trees in the Government orchard

that were suitable for experimental purposes. The remaining trees having a small scattered crop were disregarded. A mixture consisting of equal parts by weight of DDT and kaolin 41 was used in the strength of 2 pounds of DDT to 100 gallons of water. The time and number of applications were varied.

Table 3. Results of spray tests with DDT against chestnut weevils, 1944-1947.

Larvae Emerging from Sample Reduction Nuts Small Large Wormy of DDT (per Application in Chestnut Chestnut Nuts Injured 100 gal.) Sample Weevil Weevil Nuts Year Pounds Number Number Number Percent Percent

Government Orchard, Glenn Dale, Md.

1944 1 Aug. 14 533 1896 21 44 42 2 646 402 45 25 67 3 712 421 5 18 76 4 951 814 5 22 71 5 1844 850 10 16 79 0 976 3238 100 76 1945 2 Aug. 20 & Sept. 9 660 434 38 30 57 3 305 285 58 22 69 0 297 1164 61 70 1946 2 Aug. 15 & 30, Sept. 11 621 131 12 9 90 2 Aug. 15 & 30 371 171 23 19 79 2 Aug. 30 & Sept. 11 292 87 21 26 71 2 Aug. 15 & Sept. 11 949 553 190 43 53 2 Aug. 30 1267 1407 98 43 53 2 Aug. 15 1212 3207 66 43 53 2 Sept. 11 368 1832 53 58 36 0 870 5364 134 91 1947 2 Aug. 13 & 29, Sept. 12 4084 3817 234 30 66 2 Aug. 13 & 29 2618 4255 151 52 40 2 Sept. 12 3029 9498 402 79 9 2 Aug. 13 2639 5049 198 51 41 0 974 4714 121 87

Van Reynolds Orchard, Elkton, Md.

1947 2 Aug. 21, Sept. 4 & 18 1153 264 64 14 84 2 Sept. 4 & 18 338 5 118 67 23 2 Aug. 21 & Sept. 18 149 18 59 34 61 2 Aug. 21 & Sept. 4 669 102 12 51 41 2 Sept. 18 324 63 129 77 11 2 Sept. 4 270 303 67 56 36 2 Aug. 21 500 192 127 57 34 0 338 152 118 87

Sprays containing DDT were applied in two orchards in 1947, the Government orchard at Glenn Dale, and the Van Reynolds orchard at Elkton, Md. Spring frosts injured 50 per cent of the chestnut crop at Glenn Dale and 70 per cent at Elkton, and as a result only a few trees suitable for tests were available. The remaining trees were not sprayed. Four pounds of the standard mixture of equal parts of DDT and kaolin were used to 100 gallons of water in all applications.

In Table 3 will be found information on the quantities of DDT used, the schedules followed, and the results obtained during the period 1944 through 1947.

These results indicate clearly the effectiveness of DDT in chestnut weevil control, in spite of numerous discrepancies brought about by the small number and variability of the trees available for the tests. As might be expected, programs of three applications were more effective than those of only one or two. Of the single applications, those put on during the latter half of August were much more effective than those made during the first half of September, presumably because most of the eggs had been laid by the early part of September.

These experiments gave fairly exact information on the relative abundance of the two species of weevils. At Glenn Dale the small chestnut weevil constituted 92 to 98 per cent of the population; at Elkton, 61 per cent.

The matured nuts that fell from count trees were collected daily, and one-third of each lot collected was used as a sample for determining the percentage of wormy nuts. It was possible, therefore, to obtain a rough estimate of the numbers of larvae produced on each tree. In 1946, from 1,863 nuts on a tree sprayed three times, 429 larvae emerged; and from a comparable unsprayed tree having 2,610 nuts 16,494 larvae emerged. In 1947, 1,350 larvae were produced on 9 trees with an average crop of 1,361 nuts sprayed three times, compared with 14,505 larvae from 2,922 nuts on an unsprayed tree. These figures indicate that DDT sprays bring about large decreases in the numbers of weevils and that the proper use of DDT sprays on all host trees over a period of a few years would doubtless reduce the infestation to a point where fewer applications would be necessary for effective control of the chestnut weevils.

+Tentative Recommendations+

For the benefit of those who wish to try DDT for chestnut weevil control, the following tentative recommendation is made:

Thoroughly apply +to all parts of the tree+ 2 pounds of DDT in 100 gallons of water. For example, use 4 pounds of a wettable powder that contains 50 per cent of DDT, or 8 pounds of one that contains 25 per cent of DDT. Make three applications, the first about 30 days before the first nut is due to drop, and the second and third after intervals of 12 days. Unless the entire bur, especially that portion near the stem end where most of the feeding punctures are made, +is thoroughly covered+ with a film of DDT, the weevils may feed without being affected by the insecticide. In handling DDT, one should use the same care as with such well-known poisons as lead arsenate, Paris green, calcium arsenate, and nicotine.

* * * * *

Mr. Gravatt: I might say that Mr. Van Leeuwen has used only a small section of our experimental orchard, and right near-by would be large sections not used. The weevils are not killed quickly by the DDT, they are somewhat resistant, and so we think quite a number of weevils come over and deposit eggs before they are killed by this DDT, because they don't lose any time getting to work on the nuts. He hopes to have much better results where the entire orchard is sprayed. This year we sprayed our entire orchard twice, and it is a real pleasure to go out there now and gather up nuts and not be eating weevils when we do eat them.

President Davidson: Well, Mr. Gravatt will now give us a talk on Diseases Affecting the Success of Tree Crop Plantings, and I am sure we all are on our toes to hear about that. Mr. Gravatt.

Mr. Gravatt: I only ask a few minutes to show a few slides.

(Slides shown.)

Diseases Affecting the Success of Tree Crop Plantings

G. F. GRAVATT and DONALD C. STOUT

Division of Forest Pathology, Plant Industry Station, Beltsville, Maryland.

Mass plantings of many trees of the same kind frequently result in an increase in the severity of insect pests and diseases. Leaf diseases, for instance, spread quickly through such plantings when weather conditions favor growth of the causal organisms. Plants on sites unfavorable to a specific tree species also are responsible for disease increases. Chinese chestnuts grown on a site where they are subject to early-fall and late-spring frosts will fail. Not only will crops be reduced by the killing of buds or blooms, but the twigs, or even whole trees, may be killed by freezing. The blight fungus develops rapidly on such injured trees and may mislead people into thinking that the blight fungus is the primary cause of the killing.

Still another factor that determines the damage by diseases, and thus the success or failure of nut tree plantings, is the ignoring of soil and fertilizer requirements. Trees weakened by drought, because they are on a site having a soil too shallow for good root growth, are much more subject to attack even by weakly parasitic fungi than those growing on a site with deeper soil. Innumerable dying twigs and branches with fungi growing on them are sent to the U. S. Department of Agriculture or State experiment stations with requests that the disease be identified, when the real trouble is lack of water for the roots. Weak trees are much more subject to winter injury than vigorous ones.

Trees require a good supply of plant food materials and water to produce profitable crops. Tho heaviest bearing chestnut trees we have observed were grown in an irrigated orchard in California and in a poultry yard in the East where chicken droppings actually formed a mulch under the trees. However, if you wish to kill a young chestnut tree quickly, just apply a very heavy application of chicken manure; the point is that trees must become adjusted to chicken manure by gradual applications.

Another way to damage a tree is to keep it growing late in the fall by cultivation and fertilizers so that it does not harden off properly. Many plantings, representing heavy investments, fail because of lack of organic matter in the soil. This is related to water-holding and water-supplying capacity of the soil, and lack of proper fertilizer. Dr. Harley L. Crane and his assistants, in their work with tung and pecan trees, have shown the vital need for certain elements on some soils. Trees weakened by the lack of these elements are early prey for some diseases. The element most frequently deficient is nitrogen, but sometimes boron, copper, or iron is lacking; or the elements are not in balance, because of the excess of some, or the lack of others.

By adjusting the various soil, water, and site factors necessary for a continuous, vigorous growth of trees, many so-called disease conditions are eliminated. Many fungi and viruses, however, will attack trees in the pink of condition; a few of the more important of these are treated in the following sections.

+Chestnut Blight+

The destruction by blight of the native stands of the American chestnut, and of the small eastern orchard industry based on European and American chestnuts and their hybrids is almost complete. Blight has been found in the planted European chestnut orchards of the Pacific Coast from time to time, but it has been kept under control by

eradication. Chestnut trees or nuts from the eastern States, where blight is common, should not be shipped into the Rocky Mountain or Pacific Coast States.

Finding the Asiatic chestnuts resistant to the blight, the Division of Forest Pathology sent R. Kent Beattie to Asia to make selections of chestnuts for introduction into this country. Later Peter Liu, a Chinese collector who worked with Mr. Beattie, continued to select Chinese chestnuts for introduction. These introductions, together with the earlier ones made by the Division of Plant Exploration and Introduction, were grown at Chico, Calif., Savannah, Ga., and Bell, or Glenn Dale, Md. Altogether some 300,000 chestnut trees, of pure species and hybrids, were distributed to cooperators for forest and orchard plantings. (Fig. 1.) These constituted a fine lot of material from many parts of Asia as a basis for selecting the best ones for our use. Private nurseries and State game and forestry departments are now growing these chestnuts and the Division of Forest Pathology has discontinued general distribution of trees to cooperators.

Chinese chestnuts have proved to be the most valuable for forest, orchard and ornamental use. The Japanese chestnut is being discriminated against because of the poor quality of its nuts. Orchardists having mixed plantings containing Japanese chestnuts are advised to top work the trees or remove them, if the seed is to be used for plantings. In fact, for orchard plantings, nuts should be used only from the best individual trees of the Chinese chestnut.

The Chinese chestnut should be planted on sites with good air drainage as it is very susceptible to injury from early-fall or late-spring freezes. Many persons think their trees have been killed by the blight when the primary cause of the trouble was injury to the trunk by freezing followed by growth of the blight organism over the injured parts. This fungus may grow for many years in the outer layers of the bark without doing any material damage to the tree. An important factor in resistance of the Chinese chestnuts to the blight is to keep the trees growing vigorously. Avoid late growth in the fall as this favors fall freezing damage.

[Illustration: Figure 1.--F1 hybrids between the Chinese chestnut and the American chestnut.]

+Nut Spoilage+

In the Southern States one of the most serious problems with some selections of the Chinese chestnut is the spoilage of the nuts. Marvin E. Fowler made a study of this trouble at Savannah, Ga., and found that most of the trouble in that restricted area was caused by a Gleoesporium-like fungus that infects the nuts at the tip.[10] Because spraying experiments did not give control, the more susceptible trees have been removed. In most parts of the South, however, this fungus is not the primary cause of nut spoilage and the limited work so far carried out has not revealed the cause. Part of the trouble may be due to physiological break-down. As individual trees vary greatly in susceptibility to this deterioration of the nuts, orchardists are advised to top work or eliminate the more susceptible trees. Some people have believed that exposure of the nuts to the hot sun while in the bur or on the ground may cause damage. The market for Chinese chestnuts can be ruined by shipping nuts that are

partly spoiled by the time they reach the consumer.

[Footnote 10: Gravatt, G. F., and Marvin E. Fowler. Diseases of chestnut trees and nuts. Northern Nut Growers Assoc. Rept. (1940) 31: 110-113. 1941.]

+Phytophthora Root Disease of Chestnut+

Phytophthora root disease, caused by _Phytophthora cinnamomi_, is treated briefly here, and interested nut growers can consult the detailed earlier article.[11] Briefly, this fungus is considered as introduced into this country over a hundred years ago. It killed the chestnut and chinkapin growth over large areas in the southern States. Asiatic chestnuts are highly resistant to this disease, and when grown on well-drained soils have not been damaged. Our test plantings of Chinese chestnuts growing in the same soils where susceptible trees of American and European chestnuts were killed, continue to make a vigorous growth. The European and American chestnuts and their hybrids growing in the western States are in danger from this fungus as it has now been reported in the West. This same fungus sometimes kills thousands of young nursery trees of the black walnut, but these epidemics are usually brought on by unusual weather conditions. Poor soil aeration, induced by excessive rainfall and poor drainage, makes ideal conditions for damage to the walnut and other hosts by _Phytophthora_. Even the very resistant Chinese chestnut roots are invaded by the fungus when the soil remains waterlogged for extended periods.

+Brooming Disease of Walnut+

A systemic brooming disease, observed on planted walnuts as early as 1917, has been the subject of considerable discussion during recent years, because it has now spread widely into the native black walnut growth. In 1932 Waite published that he had been observing the disease for some 15 years but that "it was unknown on the black walnut in the wild in this country or on planted trees away from the Japanese walnut." The disease has continued to increase in prevalence in recent years and is now widely distributed in native black walnut growth in Tennessee, Virginia, District of Columbia, Maryland, Delaware, and New Jersey. This extensive spread into the native growth during the last 15 or 20 years and the fact that reports indicate that all of the early cases of the disease were found near nursery-grown trees offer some evidence that the disease is an importation from another area or continent into the eastern black walnut zone. From the literature and oral reports, it seems that the disease is now present also in North Carolina, West Virginia, Pennsylvania, New York, Ohio, and Michigan. Surveys probably would uncover the disease among native wild and planted walnuts in other States.

[Footnote 11: Crandall, B. S., G. F. Gravatt, and M. M. Ryan. Root diseases of Castanea species and some coniferous and broadleaf nursery stocks, caused by Phytophthora cinnamomi. Phytopathology 35: 162-180. Illus. 1945.]

+Economic Importance and Hosts+

The black walnut is a valuable native forest tree, widely but not abundantly distributed in the eastern United States. It is extensively planted as a forest tree. The numerous plantings and natural stands around farm homes, along fences, and in

pastures are also very valuable. More and more grafted ornamentals, and orchards of black walnut are being planted. For these the per-tree investment is high.

[Illustration: Figure 2.--The brooming disease of walnut. Severe brooming on Japanese walnut.]

The ultimate effect of the brooming disease on the black walnut is not known. Dr. Waite stated, "Trees even moderately attacked soon become worthless for nut production." Some affected black walnut trees, however, continue to produce small crops of nuts. Visible symptoms have been known to disappear. In addition, some seedlings, and probably large trees also, are infected without showing symptoms. Such observations indicate the complex nature of the disease. Detailed studies are needed, but at present this Division is not in position to do more than limited, part-time work on the disease.

The butternut, a widely distributed forest tree of minor importance, is seriously injured or killed by this disease. The disease severely damages or kills the Japanese walnut, which has been planted to a limited extent but is of little importance. According to Dr. Waite's report, the Persian, or English, walnut is attacked, but very few trees of this species are planted in the eastern States. Precautions should be taken to prevent the introduction of this disease into areas where it is not now present, particularly the western states.

Symptoms expressed by infected trees are viruslike, and Hutchins and Wester[12] were able to produce the brooming symptoms on a small number of trees by means of bark patch grafts, indicating that the brooming disease probably is caused by a virus.

[Illustration: Figure 3.--Brooming disease on black walnut. Ascending type, upright, sucker growth is typical of this species.]

+Description of Symptoms+

The entire range of symptoms of the brooming disease has not been determined. Symptoms are recognizable during mid-July but they are most pronounced during September and October. Curling and cupping of leaflets, chlorosis, narrowing and basal tapering of leaflets appear to be associated with early stages of the disease. On severely affected trees there are distinct broomlike growths at branch terminals, along primary or secondary branches, or on the main stem to the ground line (Fig. 2). The broomlike growths are formed by the continuing abnormal development of normally located buds into short, succulent branches. Upright, suckerlike branches appear on primary and secondary branches and on the main stem of the affected tree. (Fig. 3).

The broomed parts usually die back during the dormant period following their appearance. The dead brooms on trees that appear to be healthy during the early months of the growing-season indicate that the trees are infected. Usually the diseased trees, even those severely affected, exhibit normal growth during the early summer months.

Evidence that walnut trees may be infected for a considerable time prior to appearance of recognizable symptoms was obtained when 37 per cent of a total of

300 severely pruned trees exhibited brooming disease symptoms. These trees had looked healthy until they were pruned. Unpruned control trees showed a 4 per cent increase in disease during the same period.

[Footnote 12: Hutchins, Lee M., and Horace V. Wester. Graft-transmissible brooming disease of walnut. Phytopathology. 37 (1): 11. (Abstract) 1947.]

+Summary+

There is strong evidence that a virus disease is active among certain species of walnut in central and eastern United States. The disease exhibits distinctive symptoms and appears to damage infected trees, sometimes severely, over several growing seasons. Present data indicate that recognizable symptoms of the disease may not appear for some time after infection, unless the host is subjected to severe shock. Thus, nursery stock may be one means of spreading the disease into new areas. It is recommended, without experimental work to back up the recommendation, that walnut nurserymen remove infected trees in the vicinity of their nursery sites.

Investigation of this disease to the present time has been limited. General observations indicate that severely broomed trees produce poor nut crops. Mortality caused by the disease appears to be quite low among black walnut trees. Butternut and Japanese walnut trees are, in general, more severely affected by the disease than the black walnut and many seem to be killed by it, although the killing process is slow. As a result of experience with other virus diseases, orchardists who have only a few infected trees among their black walnuts are advised to remove them. Whether the disease can be kept under control by repeated roguing is uncertain. If an owner has just a few trees of value as ornamentals as well as nut producers, one hesitates to advise him to remove a lightly infected tree until more information is obtained concerning the disease.

This Division will welcome information from persons having experience with the brooming disease of walnut, as it is in a position to do only a limited amount of work on the disease.

+Persimmon Wilt+

Persimmon wilt is very destructive to the native persimmon (Fig. 4). It is caused by the fungus _Cephalosporium diospyri_, which was described in 1945 by Bowen S. Crandall[13]. The fungus grows in the wood of the trees, producing discolored streaks. Most trees are rapidly killed, with yellow, wilted leaves making quite a contrast to the normal green trees.

This disease was found in spots from central Tennessee south to the Gulf, east into Florida, and up the coast into North Carolina. The American persimmon seemed to be in danger, as this quickly killing disease appeared to be spreading. The limited work on this disease was discontinued because of the war and the transfer of Mr. Crandall to Peru. However, this summer Mr. Crandall and the senior writer spent two weeks surveying some of the old infections and nearby territory, and were pleased to note that the disease had made very little progress into new territory. On several small

areas where the disease was present some six years ago practically all of the larger trees had been killed, but some new small trees were coming up. At Chattanooga National Park, where the wilt was rampant about six years ago, it is continuing to kill trees, but many new ones are coming up. No northward extension of the disease in Tennessee or North Carolina was noted in the limited time spent in inspection.

[Illustration: Figure 4.--Small persimmon trees killed by the wilt.]

What does the disease mean to the grower of grafted persimmons, both native and Oriental? The Japanese or Chinese persimmons do not grow as well on their own roots, although they are quite safe that way as these two species are very resistant to the wilt. In the East, most of the Oriental persimmons are grafted on American root stocks, and trees in one case were killed by the wilt fungus getting in on the susceptible root stock. No attempts to control the wilt have been made, and these recommendations are based on procedure with other diseases and on knowledge of the spore production of this fungus. An owner of a valuable planting of grafted trees in a region where the disease is present should watch his trees for the first indication of trouble. The planting will be safer, if there are no nearby native trees; and if native trees are growing nearby and cannot be removed, they should be given a general inspection. Prompt removal and burning of any infected trees found is advisable. The fact that usually fungus spore production does not take place until after the tree has been dead for a while makes the prospect for control better than with most diseases. Care should be taken not to bring in scions or trees from infected areas.

Most members of the Northern Nut Growers Association have only a few grafted persimmon trees, usually located outside of the infected zone and therefore in little danger. Persimmon scions and trees should not be shipped from infected to healthy regions. The disease has not been reported in nurseries, but it could occur there because it attacks small trees.

[Footnote 13: Crandall, Bowen S. A new species of +Cephalosporium+ causing persimmon wilt. Mycologia 37 (4): 495-498. 1945.]

+Thyronectria Disease of Honeylocust+

Honeylocust is widely distributed both in native stands and in plantations. Some farmers plant this species or leave native trees in their pastures for the pods, which have a high sugar content, up to 38 per cent. J. C. Moore, of the Alabama Agricultural Experiment Station, reported preliminary tests indicating a per-acre yield of livestock feed equal to that of oats.

In many areas the growth of honeylocust is seriously affected by a canker and twig fungus, _Thyronectria austro-americana_. The disease often kills many twigs and branches and sometimes results in death of the tree. In most areas it causes only slight injury. Bowen S. Crandall and Jesse D. Diller have made a few observations on the prevalence and damage by this disease, which is present from New England south into the Gulf States and west into the Great Plains States.

The fungus causing this disease is morphologically somewhat similar to the chestnut blight fungus, having two spore stages produced in reddish-brown

pinhead-size fruiting bodies on the bark. Cankers are produced on the smaller branches, but they usually are not noted until some of the affected ones wilt and die. In the exposed outer wood of a branch cut above or below the canker there are reddish-brown streaks several inches long, indicating that the fungus has grown in the vascular system.

As no control experiments are known, recommendations are based on general knowledge of sanitation. If an owner has only a few valuable planted trees, he should cut off the diseased parts a foot or more back from the lower edge of the affected bark and burn or bury them in the soil. If he has many trees scattered over extensive pasture areas, it is questionable whether any action other than elimination of the more susceptible trees is justified. We will be interested in the results obtained from control work.

* * * * *

President Davidson: Now I will turn over the chairmanship of the meeting to Mr. Chase, who will have charge of the Round Table Discussion.

Round Table Discussion on Chestnut Problems

SPENCER B. CHASE, Presiding

Panel of Experts: Max E. Hardy, Carroll D. Bush, H. F. Stoke, G. F. Gravatt, J. C. McDaniel.

Mr. Chase: Gentlemen, in the last hour and a half we have heard perhaps more about chestnuts from qualified specialists than we will ever hear in any meeting of ours, and we requested each one to withhold questions until this point. So now we will have some questions from the floor, please.

Mr. Slate: What is the present status of breeding chestnut species for timber purposes?

Mr. Gravatt: The prospects are coming along. We have one cross between a none-too-promising Chinese chestnut and an American chestnut, with a good bunch of hybrids and they are different from other hybrids. It looks like they will stand up against blight. They will have blight canker growth from 10 feet down to the ground but it doesn't go into the cambium region. It is too early to evaluate the hybrids, but they do have the upright form and rapid growth of the American chestnut.

Now when we take these first-generation hybrids, cross them back with the Chinese and get more resistance, as we have done so many times in the past, we lose that rapid and more upright growth habit of the American chestnut. But we have a lot more work to do before we are ready to say anything final on this question.

Dr. Arthur H. Graves, formerly at the Brooklyn Botanic Garden is now consulting pathologist at the New Haven, (Conn.) Experiment Station. We have been working with him and partially supporting his chestnut breeding for a good many years. He has a lot of hybrids up there. We expect to have something later, but have nothing to release yet.

A Member: Do you have any sprays to control diseases and insect pests in the tree that when they go into the soil won't destroy our ground friends?

Mr. Chase: Mr. Gravatt?

Mr. Gravatt: I don't know what insects you are after, in the first place. We have a lot of trouble with Japanese beetles. Around Washington, Dr. Crane's and my plantings there would be defoliated if they weren't sprayed for Japanese beetle control, and it is the same way with filberts.

A Member: The same sprays have a tendency to work against most of the pests, do they not? Of course, DDT will take one, the arsenate of lead takes another, Black Leaf 40 another, but if we had a spray that we can use around on--well, not limited to the chestnut--that would be neutralized in the earth. Now, we have a good deal of friendly bacteria and insects in the soil that we want to keep.

Mr. Gravatt: I would say that I am a pathologist, and insect work is out of my line.

Mr. Chase: Does anyone else have a comment on that? Dr. Cross, did you hear the question?

Dr. Cross: I didn't get his question.

Mr. Chase: Would you stand and repeat your question?

A Member: Is there a spray that we can use for combating the insect pests of our trees that when it is washed off and goes into the soil doesn't kill our soil friends. We have the friendly bacteria in the soil, as well as insect and worm life. Do we have a spray that will be neutralized as it hits the soil so we can spray the tree and not kill our lower friends?

Dr. Cross: Sorry, Mr. Chase, that's beyond me.

Mr. Gravatt: You are thinking of arsenate of lead poisoning the soil where you keep on spraying with it?

A Member: Yes.

Mr. Gravatt: I think DDT may build up a little in the soil, but it is broken down, isn't it, Dr. Crane?

Dr. Crane: Yes, DDT is broken down and it is not a fungicide and it is not a bactericide. It is an insecticide that kills insects through affecting the nervous system, according to my understanding of it. I am not an entomologist, but that's what the entomologists say. So far we haven't any evidence to my knowledge of any build-up of DDT in soils that has been detrimental. I don't know what the situation would be if DDT was used to the same extent as arsenate of lead. It was not uncommon for some growers to put on anywhere from 6 to 15 lead sprays in a season in order to control codling moth, as they used to do in certain apple orchards, particularly in the West.

I was talking to Dr. Van Leeuwen just a day or two before I had to leave for the meeting, and he is not ready yet to say anything about it, but he has already tested

some very promising insecticides as far as the control of weevil is concerned. This DDT and some of the other new insecticides are very easily decomposed, and, of course, that's one of the disadvantages of them. Under certain climatic conditions they would need to be less readily decomposed to give control over a longer period. I know that we have not had enough experience to know all about those new spray materials.

Mr. McDaniel: There has been one instance reported where DDT _in the soil_ was injurious to fruit plant growth. That was Goldsworthy's and Dunegan's work on strawberries. Where they used large amounts of technical DDT in the soil, they found that it inhibited the growth of the strawberry plant. I believe that's the only instance I've heard of, where soil application of DDT hurt growth of fruit plants. Benzene hexachloride, and some other chlorinated hydrocarbons, and parathion actually appeared to have a stimulating effect on the berry plants.[14]

Mr. Frye: Why would there be any more danger of affecting the soil in a chestnut orchard than there would in the apple and peach orchard by spraying seven, eight and ten times? That's the only question that arises with me.

Mr. Chase: Let's get back to chestnuts specifically, now, gentlemen.

Mr. Kays (Oklahoma A. & M. College): Since I don't come from a chestnut area, my impression of the nut samples supplied by Mr. Moore of Auburn, was: "I'd like them if they had salted them." I am wondering if it wouldn't have affected their rancidity if they had been treated--salting material added, prior to or in the process somewhere along the line.

Mr. J. C. Moore: I'd just like to say I have tried putting salt in the water, to boil the nuts with salt, and then I have tried shelling them and sprinkling salt, and I find that salt does not add anything to the flavor. Tasting the nuts raw, I, too, get the impression salt is what I want, but I haven't been able to add it satisfactorily. I don't say that it cannot be done.

Dr. MacDaniels: Mr. Chairman, in view of the whole situation of chestnut incompatibility of stock with scion, what would be the position that we in the Northern Nut Growers Association can take in advising people what kind of chestnuts they should plant? Should they be encouraged to try to get grafted trees? What should be our position?

Mr. Chase: Mr. Stoke, would you care to comment on that?

Mr. Stoke: You are asking me to stick out my neck, and it seems as if I have always done that. The Chinese chestnut is in the Johnny Appleseed stage, in my opinion, and we are investigating to find out the best varieties, that is, the best specimen, best performance, best quality, best in blight resistance, growth, and other qualities and when we winnow out all we have and arrive at the best, we are going to find--now, this is just my personal opinion--I will say that for myself I'd rather have one acre of the best selections we have budded or grafted--asexually propagated, than five acres of seedling trees as a financial good bet, because I say that one acre of our very best produce virtually as many nuts as five acres of seedlings. I have trees from seed I

imported through the Yokahama Nursery Company, and I think it came from Korea. The nuts run very small, and compared with those I am sure the others will pay much better, and I think it would be profitable to pay three or four or five times as much for your trees if you get good trees of good, known varieties and grafted or budded.

Don't misunderstand me. We shouldn't ask the American public to wait until those can be furnished, because they won't wait, and they shouldn't. But I say as a commercial proposition, to plant trees commercially, I would exercise caution and I would encourage my customers to exercise caution unless they are willing to follow up and do their own top working later on, and a Chinese chestnut doesn't top work as readily as a black walnut.

Mr. Chase: I don't believe that's quite the answer he wanted. The comment that I think Dr. MacDaniels is after is what position should the Northern Nut Growers Association take in regard to planting seedlings or planting grafted stock. Is that the point?

Dr. MacDaniels: Yes, it seems to be seedlings against grafted stock.

Mr. Stoke: May I answer? I don't think the Northern Nut Growers Association should take _any_ position. They should present the facts and let the buyer decide. I don't think we need to go on record, and I don't think we should. There is too much diversity of opinion.

Dr. MacDaniels: Between ourselves--and this is not an academic question--we get continual inquiries regarding the Chinese chestnuts and what should they plant and where can they get the trees, and so forth. It isn't good enough in most of these cases to write several pages explaining what the whole situation is, the _if's_, _and's_, and _but's_. But I just wonder what the opinion is of the people who know best in this regard. Who has a good orchard of 20-year-old grafted Chinese chestnuts? Where are they? I don't know: I am asking for information.

Mr. Chase: Dr. Drain, are those trees you have grafted trees or seedling trees?

Dr. Drain (University of Tennessee): They are seedling trees. They have produced a rather nice quality nut, and we have enjoyed propagating seedlings from them. That's really all we know. We haven't grafted any.

Mr. Chase: Mac, would you care to comment on this?

Mr. McDaniel: I am ashamed to say that at present we have no grafted chestnut trees on my own north Alabama farms. We have about 50 trees that are 8-year-old seedlings from imported (Chinese) nuts, growing next to a commercial peach block, and find the production quite variable on the different trees. I am aiming at top-working most of these with the named varieties, beginning this year. At present I can't answer the question of seedlings vs grafted trees. I have been advising people who are interested in trying them in Tennessee that _for their first planting_ (to test the adaptability of their locations) they can get the seedlings generally quite a bit cheaper than the grafted trees. With the experience we have had over the State and the high mortality of trees, both grafted and seedling--killing of the tops and in some cases the whole tree--the seedling might be best economically _to begin their_

experimenting with. I am _not recommending_ that anyone plant seedlings commercially, but just in a small way for trial. They are well worth a trial anywhere peaches are doing well. When we find a _suitable site_, then is the time to think about using the more expensive grafted trees.

Pres. Davidson: I just want to give a little bit of my experience along that line. Way back in 1934 I planted a few seeds that I got from Amelia Riehl. They were nuts of the Riehl hybrids. [Ed. note: Mostly American--European crosses.] She named one Dan Patch and another Gibbons. They are now about 13 years old. Each of them is bearing burs this year. They have borne burs, a few of them, in the past, but no nuts. So far in 1948, the burs that have fallen to the ground, of course, have no nuts, but whether the burs that are still on the trees have nuts I don't know. I want to know whether those trees are normal---whether a hybrid of that kind is likely to be sterile or not. That's another matter that might be discussed. Anyhow, you are taking a chance, no question about that, when you plant seedlings.

Mr. Stoke: Mr. Chairman, if you will pardon me for saying one more word, here is a suggestion I will make. Now you can check for yourself. The whole thing hinges on whether we can get _permanent_ grafts on the tree and get the characteristics in the grafted tree that the parent has--in the good selected tree. Now you take the reports sent us by Mr. Hemming; you take the reports of the station at Albany--of individual trees in those plots. You take the worst trees and you will find they are nothing but boarders. You take the best and you will find they are very profitable. You take the average and it will fall somewhere in between.

Now, why keep a lot of boarders that don't pay--free boarders--or why use run-of-mine seedlings, _if_ we can graft successfully--and some people like to dispute that--and produce nothing but the best? And you can check it on any of those tables. [Mr. Hardy's paper.] We have a few tables in our former Reports. You can check it and figure it out for yourself.

Dr. Crane: To clear up this situation I wanted to ask Mr. Hardy a question, and then I wanted to make a statement. In this report from the 1938 and 1940 planting at Albany, Georgia, in the Brown tract in 1947 there were 188 trees that bore crops, but that planting consisted of 274 trees planted in 1938 and 60 trees planted in 1940. Why weren't those 274 trees plus those 60 trees represented in the 100 with the yield records of 1947?

Mr. Hardy: Dr. Crane knows the answer, so I will let him ask the question and answer it, too.

Dr. Crane: In 1936 we planted 1,000 trees of the same Peter Liu selections on the Station farm at Beltsville, Maryland. They were of the same number and letter designations as others that were distributed to cooperators. Out of the thousand trees that we planted on the Station farm some of them came into bearing at four and five years after planting. But the nuts were small in size and were not much good. With one or two exceptions, out of that planting there were none bearing satisfactorily to suit us after ten years. In 1945 we applied the ax, because a Chinese chestnut tree, from an orchard standpoint, if it's not in bearing in ten years after planting is not

worth keeping. We haven't got time to wait. So out they came. And in addition to that we have had other trees that have done the same thing.

Now, out of this 274 plus the 60 at Albany, Georgia, we have three trees that we now figure are good enough to be raised to a variety status, plus possibly two or three more. Now, you can figure your percentage of good trees when you plant seeds.

Dr. Overholser: Mr. Chairman, may I ask a question, whether these three seedlings to which they propose to give variety status have been propagated in sufficient number that they are able to give distribution in other areas.

Mr. Hardy: Dr. Overholser, they are not available yet in quantity. That same answer is part of the answer I wanted to make to Dr. MacDaniels. The present situation in the chestnut industry is that there are very few nurserymen who know how to propagate nursery grafted trees successfully. There is going to have to be quite a bit of work done on that. If some of you here know how to do it, I would like to know, myself. There are a lot of nurserymen who would like to know, according to the reports I have, how to graft or bud a nursery chestnut tree.

As long as the situation is that way I would say to recommend seedling trees because of their low price, but--and every grower who has trees can fall in line with this--the seeds should be from properly culled-out orchards of the highest type, leaving nothing in there producing nuts or pollen but what is the highest type. I think all of you who have more than one type of chestnut in your plantings should cull them all down to the pure _Castanea mollissima_. I don't mean by cutting out the whole tree, but go ahead and top-work them. If they won't take the top, _then_ cut them out. But if you can top-work them and the grafting is good, you can increase your planting of good trees in that manner.

The improved quality of the seed will improve the quality of seedlings going to the buyer, and the chances of a higher percentage of good seedlings showing up will be greater. I think it will improve the industry through a period of years.

Dr. MacDaniels: I think I agree with his position. In fact, that's exactly what we are telling the inquiries that come in: At the present state of our knowledge, better try seedling trees.

But I didn't hear anybody get up and say they had an orchard of 20-year-old grafted chestnut trees. I have tried to get them, I have grafted successfully, I suppose, 7 or 8 different varieties on many different Chinese stocks that I have bought, or had given to me, and numbers of grafted trees. I have nothing left. They grow fine, 7 or 8 feet the first year, 3 or 4 feet the next year, then they go along for a while and then they die. In other words, there is an unsolved problem there, so that it seems to me at the present state of our knowledge we had better admit it and say, "If you are an amateur, you better get the best seedling trees that you can and wait awhile."

Mr. J. C. Moore: I just want to give some data on some of the class work at Auburn with Chinese chestnuts. We were studying Chinese seedlings, and we attempted to bud those Chinese chestnut seedlings, and on some of the larger seedlings we

top-worked. We had some 3-year-old seedlings, and we top-worked the limbs. We put in patch buds, and we put in T-buds or shield buds, and in practically every case on some of the trees the buds stuck beautifully.

In June and again in August, with another class, we had the same results, either with T-bud or shield bud or patch bud. Some of the seedlings wouldn't take the buds at all. I can't think why one seedling would take 100 per cent of the buds and another seedling growing right by it wouldn't take any buds.

Mr. Weber: The oldsters here will remember Colonel C. K. Sober, one of our former members who propagated what he later named the Sober's Paragon chestnut. It was a grafted tree and apparently it was grafted successfully on native stocks, and it grew until the blight got it.

Dr. MacDaniels: I am not talking about European or American, I am talking about Chinese chestnuts.

Mr. O'Rourke: It may affect the nursery industry. The nurserymen are looking to the Northern Nut Growers Association, Federal bureaus and State experiment stations to guide them in the propagation of desirable trees. We know now that the Chinese chestnut is becoming quite prominent, is becoming quite popular in many sections of the country, and many nurserymen are now getting requests to supply the public in their states with Chinese chestnuts. They, in turn, would like to know what they should do. If they sell Chinese chestnut trees which have been propagated vegetatively and they only grow five, eight, 10 or 15 years and then die, it's going to come back on the nurserymen. They should like to know whether they should do that or whether they should rely upon seedlings which they can develop into pure lines as best they may.

Now, that really is a serious question. I am wondering from what Mr. Hardy has told us today if it may not be an understock problem, and if it is an understock problem--if there are certain strains of understock which are compatible with certain scions, possibly we should ask for some investigations, some more research to be done in this direction.

Then possibly, on the other hand, we should also ask that certain investigations be carried out so that we will have some idea of the inheritable characters that may be "fixed" through seed selection. I really think that this seed selection should be very seriously considered, and that nurserymen in particular and the public in general would benefit greatly by such consideration.

Mr. Hardy: Mr. Chase, may I make this suggestion: I think it is something that a number of individuals could try, perhaps they should be backed up by agricultural institutions, either Federal or State. We are all interested and concerned with stocks, and I think a large part of our trouble with grafting chestnuts is a stock-scion relationship.

We have some top-worked trees 13 years old that are just as healthy, just as normal as they can be. We have some top-worked trees of various ages below that. The graft-union is good; they are just as healthy and continue to be as productive and

vigorous as the parent tree. Where there is incompatibility we run into difficulties very shortly. To a large extent I think we are involved with two problems in the trouble with incompatibility, or perhaps I should say the dying, of grafted trees. One is a stock-scion relationship, the other a mechanical problem.

I think there are these two types of incompatibilities. Now, as to the mechanical part--that can be improved through developing the art of grafting or budding, whichever works out best. The other will require quite a lot of study, perhaps the development of certain strains of the root stocks for certain scion varieties.

I have made this suggestion to two or three. I have started the work myself by putting out with friends two or three or four trees. After they get up to a size where I can top-work them, I will top-work with two varieties. Perhaps I will put Nanking and Kuling on two trees at one particular place. Two or three miles away I will put Kuling and Meiling on two others. At another place I will put Nanking and Meiling. I will get reciprocal pollination, because the chestnut is necessarily cross-pollinating.[15] I can then plant seedlings from both parents, each pollinated by the other. Then by grafting those varieties onto those seedlings stocks I can find out whether there is any reason to go into the work of developing seed orchards of two varieties whereby Meiling pollinated by Kuling may produce the best, most vigorous, most uniform seedlings on which Kuling can be propagated. And by propagating Kuling on such seedlings--the seedlings of such inheritance--we may get 100 per cent of good grafts.

The industry needs a lot of help, and I think it is a matter of time until those things are worked out, but it is going to take time and money and plenty of good effort to work out that problem. I think it probably should be worked out.

Mr. Bush: I don't like the word "incompatibility", and I hardly believe in it, and I presume most of you know that. I have Chinese on European stock, and it has been there for 20 years or more, _grafted high_. I have Chinese on Japanese grafted _under the ground_. I think a good deal of our damage is done from wind, from cold, and from sun on the graft _just above the ground_. I suspect that grafting at that point is what is the matter with many trees in the TVA plantings and others that had low survival. Of late years when I did the grafting (in the last five or six years) I cut the stock underneath the ground and stuck the graft under the ground and seemingly I got far better results. Some of those graft failures showed up. I laid that largely to mechanical damage, and again with the Japanese, particularly, I laid it on the time when the sap comes up. Call it what you will, but the timing of the growth of the two trees is different and we had trouble there. I have grafted some very widely different kinds of chestnuts on the tops of other chestnuts, and am getting them to grow. When we see the break start, we take a twig from below and break and put it above, cut through the cambium and nail it on and they will heal over and the defect disappears. So, again, it seems to be mechanical.

Mr. McDaniel: I believe from observations on a number of trees, particularly Dr. Richards' in West Tennessee, that a large part of our so-called incompatibility in this State is due to winter injury _to the stock_. So what Dr. Richards meant, evidently, was that he was rather successful in getting a "take" from last summer's propagation

but the stock then failed below the union this spring. I saw his trees, and they had the typical discoloration of bark and the dying of various bark areas--these girdling the whole tree in a number of instances. [See Richards' paper in this report.] I would agree in general with what Mr. Bush has just said, but there are certain other instances in which we think the only word for what we see is "incompatibility."

Mr. Slate: What are the prospects of planting those low-grafted trees rather deep?

Mr. Bush: I think that if the roots started to die the grafted tree would start a root above the graft. The sap is going up from the root. It will go down and the root will start above the graft and go out above the graft, thus getting the tree on its own root.

Mr. Stoke: Since we got onto grafting, do you mind if I say a word? Here is a four-branch, top-worked specimen that I chopped off and brought with me. This first tree limb was still alive and had nuts on it, the second was dying and a third dead. This fourth union was still alive, but it was badly damaged, too. That's Illinois 31 -4 on Japanese. Here is another graft of Illinois 31 -4 on Japanese in a small tree, and if that's poor union, I am no grafter!

Mr. Hardy: Mr. Stoke, may I ask you this: Is this [small graft] on the same tree as this? [Indicating larger tree first referred to.]

Mr. Stoke: No. Those four grafts, you see, all went bad. This one is in perfect condition. But I am having a hard time keeping that Illinois 31 -4 alive. I had a union on _mollissima_ three inches in diameter and as perfect as this, two years ago. Last year it began to bulge at the point of union. The top wasn't feeding back to the root, and this year it is in bad condition,--foliage very small and it put on a very full crop of burs which will never mature, and it's going to pass out. It is about four inches in diameter now.

Last year to try to beat this thing I cut out the crown of a small _mollissima_ at the below-ground level and put in several grafts of this same Illinois 31 -4, and I got a nice growth, at least four feet high. When I dug it up to transplant it--it was right in my garden--I found I had a large callus more than an inch and a half in diameter at the union but no roots. I reset it, and I haven't ventured to see whether it was all right or not. This spring I tried again.

I have four little trees, one as high as my head, the others smaller. I grafted each one on branch roots just as they lay in the ground. Didn't dig them up and they grew nicely, and along in July I went around and spaded them deeply and thought perhaps that would produce roots. About a week ago I examined one. I have a magnificent callus but no roots yet above the union. What the ultimate results will be I don't know.

With that particular hybrid I want to try one more thing. I want to grow seedlings of the European chestnut, cut them below the ground, graft Illinois 31 -4 on the root and it may make a union that will not fail, because the European is a very robust grower, and by being grafted under the ground the stock will be away from blight organisms.

[Editor's Note: Mr. C. A. Reed is naming this variety (Ill. 31-4) "Colby" in honor of the originator, Dr. Arthur S. Colby.]

Mr. Hirschi: I would like to say I put on hybrids similar to that Illinois 31 -4 and they grew the first year, and just made a bulky knot right at the point of union and died the second year.

Mr. McDaniel: What was that combination?

Mr. Hirschi: That was _mollissima_ stock.

Now, speaking about the varieties--this is in Oklahoma--I have tried practically all the older varieties and I have tried some Abundance grafts this last year. I have some Abundance grafts that are two years old that are producing. They have the most vigorous growth of anything, and in our climate we have to have vigor.

I grafted a lot of the Abundance scions on Hobson seedlings. I started out to grow an orchard from Hobson seedlings, and I found out that out of 50 splice grafts of Abundance that I put in Hobson seedlings in 1948, forty-eight grew, and they were put on rather late, in April. That's a little late for us. I have the idea--I don't know whether I am right or not--that if the Abundance proves out as our best variety, we can grow seed for stock of the Abundance and then graft the Abundance back on the seedling from Abundance. If there is so much to this incompatibility, I should overcome it by doing that very thing.

Personally I think it is a crime that thousands of trees--almost millions--are being put out by nurserymen as seedling trees, and if you will note in their price lists they have "6 to 12 inches" and "12 to 18 inches", "2 to 3 feet" and "3 to 4 feet." I venture to say that those are probably all the same age. How would you like to plant some of those 12-inch trees? Somebody is going to get hurt!

Mr. Bush: I'd like to say that you can propagate the Chinese chestnut by layering if you want to, and that will put it on its own. Put a wedge on it or girdle it and keep it damp through the summer.

Pres. Davidson: I think Dr. MacDaniels' question is still not answered. I do think that if a nurseryman sells a seedling he ought to definitely _say that it is a seedling_ and not merely that it is a "blight-resistant chestnut," or something of that sort. He should actually tell the public what he is selling.

Now, then, there seem to be reasons why in some instances a man is justified in planting seedlings when it comes to Chinese chestnuts, but when it comes to the black walnut or filbert or some of these other things, they are still selling seedlings without labelling them as such. I think we should be on record against that practice, because it takes us five or six years, or ten years sometimes, to find out that we have been gypped, and it is so easy to gyp the public when you can't find out about it any sooner than that.

Mr. O'Rourke: I quite agree with Mr. Davidson that the nurserymen should state that a seedling is a seedling when it is a seedling. And I am sure Mr. Hirschi will corroborate that the American Association of Nurserymen is exerting all the influence

they can to that end. Is that right, Mr. Hirschi?

Mr. Hirschi: Yes.

Mr. Bregger: I would like to ask, if planters for some years yet will have to rely on seedlings, is there a chance that from certain parents or certain varieties we can get a larger percentage of good seedlings than from others? How much has it been studied and is there a known result from the parent trees in the percent of what their seedlings can do?

Dr. Crane: I wish I could answer that one. It is a matter of time, to find out the seedling characteristics reproduced by a certain descendant. But we know that there is a difference in _uniformity_ of trees in the way they grow, but as far as bearing is concerned, and the type of nut produced, we haven't had enough time yet.

It's just like this: We have made selections for rootstocks in which we have selected trees that were good, strong and vigorous--the most vigorously growing trees that we have known about, and yet at the same time produced a small nut or medium-sized nut that we could use for the production of rootstocks. And we have made progress on that, and we have demonstrated that there is a very marked difference between the graftability or budability of seedlings from certain parent trees. We have demonstrated that some varieties are much easier to propagate than are others. But as for the proper combinations of stock and scion, we still haven't got enough data to recommend any. We know that there are differences, but it is going to take quite a long while, at least four or five years or more, before we know.

Now, there is just one other thing that comes up on propagation. We have found that if you bench-graft and make the graft into the transition zone between root and top just like the old method that the apple propagator used when he piece-root grafted and then plant deep, you can get a hundred per cent of the grafts to grow. In such cases the scion may root and the top will be on its own roots.

Well, there are a lot of these tricks to learn as time goes on. I don't think that we should worry too much about this graft union problem. We know that this Carr variety is a bear-cat. It is the one that gave us so much trouble. When we tried to propagate that one we had a real, nasty cat by the tail. But on the other hand, in answer to Dr. MacDaniels' question if we go out to Dr. J. Russell Smith's plantings up at Round Hill (Virginia), we can see a lot of the oldest grafted trees that I know of anywhere in the country, and the unions are just as smooth and just as slick as anyone would want to see. They are not 20 years old; I don't think there was ever a _mollissima_ chestnut grafted 20 years ago. The first grafting that I know of was about 15 years ago, maybe 18.

Mr. Stoke: In 1932.

Mr. R. C. Moore: Thomas Jefferson grafted European chestnuts.

Dr. Crane: No, I am talking about Chinese chestnuts. We didn't get in any Chinese chestnuts until 1906. We have this problem of incompatibility or graft union trouble, in apples, but do you hear anybody hollering about it? We have it in peaches, plums and cherries. One of the most important diseases they have out in the Pacific

Northwest and California on Persian walnuts, is what is called "black line disease." We mustn't get excited about graft union failure. That has been used, in my opinion, by a lot of people, to discourage the propagating of grafted chestnuts. There are thousands of people in the United States who are spending good money for seedling trees, and some of them are going to get stung. We in the Northern Nut Growers Association are going to have this thing backfire on us, just as true as I tell you. I know there are some nurserymen today that are planting unknown chestnut seeds, and they are selling the trees as Chinese chestnut. They are planting seed out of mixed orchards, too, that have _C. seguinii_ and _C. henryi_ and _C. crenata_ trees in them. The _C. crenata_ Japanese has been introduced in the United States for over 70 years and it has never made the grade.

You know, there has been many a thing that has been promoted in the United States--big for a few days and then she backfired, and then it took the industry 50 or a hundred years to recover. You can sell people gold bricks once, but you can't sell them gold bricks _all_ the time!

Mr. McCollum: Last year after Mr. Hemming's speech--you know, he is the nurseryman who sells seedlings over on the Eastern Shore--I asked him if he had been selling those long enough to have heard from customers. "Yes," he said he had, "all satisfied." Now, I don't know anything about that.

Dr. Moss: I am not an expert. They say an expert is someone who, the more he studies, knows less about practically nothing at all. That's a good deal my shape. I planted before the war Chinese seed in Kentucky and a good many of those put on burs in the nursery row. I gave them away in the community. Out of the whole bunch, some of them 20 feet tall, I know of one outstanding nut in that bunch and it's off by itself, apparently a self-pollinizer[16], and puts out a crop of good nuts.

Dr. Cross: I should like to ask Dr. Crane if it would not be possible to investigate the situation in China rather than wait to work this out. Certainly, the Chinese have sufficient knowledge of grafting and propagation to have been working on this long ago, and since these came from there, let's look into that phase of it.

Dr. Crane: I did investigate the situation in China when I was there. Unfortunately in China, although it is one of our oldest countries and longest civilizations, they don't do much grafting. They grow their trees from seed, but they have certain seed trees that they select their seed from, and within a community, within a valley, you will have a certain type of chestnut. They call them varieties. They are not varieties. That's the situation. Most all of them are different, but they have accomplished the fixing of certain characteristics.

Now, in South China the nuts are larger in size, they are stronger growing trees than they are in the North. I think that we will find that that's the situation in this country. The Chinese chestnut is one that does have a high heat requirement, just like pecan, and grown under conditions where they have high heat they are bigger in size and make more growth and probably they come into bearing sooner.

But I didn't see anything grafted in China, and I was all over the country from the most northern parts to the most southern parts where chestnuts are produced. I

could make a lot of observations myself, but I had to talk through interpreters, and sometimes you couldn't tell what the interpreter meant. But as near as I could tell, they were all seedlings. When he would tell me there was such-and-such a variety, I would ask him what it meant in English. He didn't know. When I found how they were propagated I found they planted the seed. When I found where they got the seed it was from a certain seed tree.

So we have within the valleys what they call varieties, but they are not varieties, only seedlings grown from certain seed trees.

Now, with the Japanese, on the other hand, the situation is different, because they propagated by budding and by grafting. I got a number of the Japanese publications of propagation methods and their stocks, and so forth, translated into English, and their problems are just the same as we are going through right here now. They propagate true varieties by asexual methods, but the Chinese do not to any extent at all.

Dr. Cross: Have the Russians got any?

A Member: That's the question I ask. Do we have any seed trees in this country that are better than other seed trees?

Mr. Porter: Could the gentleman tell us whether the Chinese graft _any_ chestnuts.

Dr. Crane: Yes, they do so, I was told.

Mr. Porter: Well, the industry spends a lot of money, so do other people, and so on, in a proper way to investigate that. Why don't you find out where in that country they have been doing it?

Dr. Crane: I didn't see any grafted chestnut trees over there.

A Member: You said they grafted, and then you say, "I didn't see any."

Dr. Crane: That's quite right, and I talked to their best horticultural authorities that they have. Practically all of it is produced by seed and not by budding or grafting. It is just exactly as I said with the Persian walnut. China has no varieties of Persian walnuts, although sometimes you will find some farmer that will bud or graft his trees.

Mr. Porter: They graft up on the limb?

Dr. Crane: Yes, sir. Once in a while you will find one. They have a few real horticulturists. I met one man over there that would compare very favorably with Liberty Hyde Bailey.

Mr. Stoke: Dr. MacDaniels asked for concrete evidence. He wanted to know where there was an orchard with 20-year-old grafted Chinese chestnut trees. They haven't been planted that long, but I would like to give him concrete evidence in my own experience.

In 1932 I got scions from the Department, got what ultimately became known as the Hobson, from Jasper, Georgia. I grafted a tree in my front yard which is still

bearing nicely, and in fact I have got two grafts on that tree about four feet from the ground, and it is very nice with perfect union. At the same time I grafted a Carr right at the side of my house that also has a perfect union about the same height from the ground. I grafted a scion sent me by Dr. Morris as Morris' best (which was pretty poor), and it is still living. At the present time I have perhaps five Carr trees that will average six inches or more in diameter. The oldest is the one by the side, of the house. The rest of them were grafted about 1935. One out of those five, when it got to be about six inches in diameter, in fact, about three years ago, it went bad. It is girdled and dead. It was grafted about as high as this table from the ground. The others are sound, and you'd find it very difficult to find where they were grafted.

I have Hobson, perhaps a dozen trees anywhere from six to 16 years old, and I have not had a failure on a Hobson that really was once healed over properly and got to bearing, not one. That's concrete evidence, Doctor, and that's all I wish to say.

Rev. Taylor (Alpine, Tenn.): Mr. Gravatt was about to answer a question about our seed trees, wasn't he?

Mr. Gravatt: Would you repeat that question?

Rev. Taylor: Are some seed trees better than others in the high per cent of good seedlings they produce?

Mr. Gravatt: Well, McKay has done some work and published it to show that on seedlings of certain trees you get higher percentage of bud takes than on others.

Mr. Chase: I think the question is a little confused. I think what you are after is, are there parent seed trees from which seed can be planted that would produce a good quality of seedlings.

Rev. Taylor: Yes, of good productive seedlings. No grafting to it.

Mr. Chase: I think that was answered. Apparently there are.

Rev. Taylor: Apparently there are in China, as Dr. Crane brought up.

Mr. Chase: He further brought up that those things are in the process of being tested here now, and he hopes for some information in--what was that?

Mr. Gravatt: We had Professor Beattie over in Japan, China, and Korea for two or three years, and he found in Japan that there were certain selections there, certain grafted varieties that they used for seed stock. We imported those into this country. We were getting ready to go ahead with the Japs. We also brought in a hundred varieties of Japanese chestnuts. But the Japanese varieties didn't do well here. What would produce well over in Japan didn't produce well here. But a number of those scions that we grafted in 1932 and 1933 are still living. We have had very good success with top-working chestnuts in our orchards. We have some grafts there of pure Chinese chestnuts top-worked on some worthless Japanese. Some of those have been there for 12 and 14 years, with perfect unions. But we do receive a number of reports of trees dying from blight and various other and sundry other causes and when we examine them quite frequently these have died back to where the trees had been grafted.

Rev. Taylor: I could enlarge on that question just a little bit to tie in with what Mr. O'Rourke said. If the nurserymen are going to propagate seedling trees for the trade for some time yet, where should they be advised to obtain their seed to get the best possible seedling trees?

Mr. Gravatt: In a lot of our regional distributions we sent out mixtures. In other places we would send out related seedlings, as "MY," "MZ," or "MAX," to different individuals. We have advised all nurseryman, all of our cooperators, to eliminate the Japanese; eliminate the hybrids. It gets down to pure Chinese. We have also advised again and again to take out the more worthless trees and propagate seed from the beat. But there are a lot of hybrid seeds with mixed parentage going into nursery trees.

Mr. McDaniel: How many people are going to take out trees now when they can sell the seeds for at least 50 cents or maybe even $2.00 a pound?

Mr. Gravatt: That's it. However, you take any of those Chinese trees over there at the Eastern Shore Nurseries, for example--nuts from all 19 of them have been sent over here, and they are all good eating. I have been over a lot of the seedlings of Hemming's trees. Mr. Hemming has several hundred at his own place. I have been over other orchard plantings. There is lot of variability among those seedlings. They are not as uniform as the parent tree, for some reason. Why, I don't know.

Mr. Chase: Mr. Howell, as a nurseryman, has propagated the Chinese chestnut tree. Would you care to make a few comments? Mr. Howell has Howell's Nursery in Knoxville and at Sweetwater, Tennessee, and I believe has some of Mr. Gravatt's early seedling trees and has produced a great quantity of seedlings.

Mr. Bruce Howell: A good many years ago we got from the Department five trees, and they grew and have all borne good nuts, and all chestnuts we have propagated since have been grown from seed from those five trees, and most of them are pretty good. One is a small nut, and among more recent seedlings we have got two of them that don't bear at all, or haven't so far. Now, we have got a bunch of them where they were set several years ago in nursery rows. At each end of each row the trees there bear very nice nuts, and when you get out through that row, the crowded trees don't bear at all.

I think those seedlings and those trees practically all make fairly good nuts and some of them excellent. I have got some samples. About six years ago I got a pound of imported Japanese I planted. The third year they bore and they have done very well, and all of them are about the same size chestnuts. They are as good as any _after_ they are roasted or boiled. That's about all. A good many years ago, I guess 30 years ago, I grafted Paragon chestnuts, and they did well until the blight.

Rev. Taylor: Does anybody else have this trouble? In North Central Tennessee we usually have a warm spell about the Middle of February, plowing time. We expect it every year. And then these Chinese chestnuts are the quickest trees to let the buds swell, and the bark softens up all the way to the ground on the young ones. Then we nearly always have a pretty hard freeze, afterward. So, for several years after our experimental planting was set out there they would get killed clear to the ground

next year. Is that something others have the same experience with? How do you go at correcting that?

After our trees got to be three or four or five inches in diameter they didn't kill back that way. The bark seemed to be tougher.

Mr. McDaniel: That's very common experience in Tennessee and, I might say, in north Alabama.

Rev. Taylor: Nothing you can do about it?

Mr. McDaniel: On some sites it is not nearly so bad as it is in other locations. A northern or eastern slope with good elevation seems to be best.

Mr. Frye: I have had some trouble and maybe, had a good education about frost pockets. If you get them in high elevations you escape that. I had that trouble two years ago. I got some Chinese trees from Dr. Smith, set them out. They were his best seedlings, three of them, and they started beautifully. I transplanted them. Just about that time they got nipped off. Did that three times and failed to come out the third time.

Pres. Davidson: One other remedy for that that I remember reading about, I am not quite sure in which of our Reports--maybe Mr. Becker was the author, and that is this: He said that he cultivates until August after which he plants cover crops, and he sows cover crops that grow and they hold back this vegetative growth in the late part of the year, and it is really the late vegetative growth that causes the destruction. After he adopted that plan he had very much less winter killing in his plantation. That might be one way of helping the situation.

Mr. Hardy: We have had some killing. Usually in the second year or the first year after we get killing down to the ground, if we will keep the stock pruned back to one shoot that one will make sufficient growth, become hardy enough to withstand any cold, or perhaps sun scald. Also wrapping the trunks of the trees with newspaper helps to prevent the variations in temperature, which in our section is what causes the cold injury. We don't have sufficient cold to cause absolute low-temperature injury, but we do have sudden drops just as you do in Tennessee, apparently, and wrapping with paper does help iron out those changes.

Mr. McDaniel: Wouldn't you suggest the paper wrap in the summer as well as the winter and spring?

Mr. Hardy: Yes.

Mr. Stoke: It is not only the planter of the trees that has sorrows along that line, but the nurseryman does also. I had some nursery seedlings growing on flat land, and they looked all right, passed the winter. When I went out to graft them I found that on these small stocks anywhere from the size of a lead pencil to the size of a finger, the cambium was discolored. It wasn't black nut brown. Any attempts I made to graft those failed, and yet many of those same trees grew on. They were stunted somewhat for a year or two, and they left a brown ring at that annual growth.

I would say that the best guarantee against that kind of thing is to plant your chestnut orchards--and the nurseries--to plant on land that is well air drained. _Select the same site as you would for peaches._

Mr. Chase: I will say that we should have allowed more time for discussion. However, we have used up our alloted time for this period. Supper is at six o'clock, and we are due back here at 7:30. I don't know how you folks feel about this little session, but I certainly did enjoy it.

[Footnote 14: --Goldsworthy and his associates published several items along this line in 1948 issues of +Plant Disease Reporter+. His October 15, 1948 item reported a similar result of 25% technical DDT (with 75% clay) inhibiting growth of seedling peach roots on 1-year budded Elberta trees. As low as 25 pound per acre application affected growth in quartz sand cultures, whereas with certain soils, no significant difference was noted until an 800 lb. per acre level of the DDT was reached. It was surmised that possibly some unknown constituent in the technical DDT was responsible for the suppression of new root growth, and consequent slowing down of top growth. In the case of Blakemore strawberries, and also with peaches, this effect has persisted for at least two crop years. Goldsworthy and Dunegan say, "How many other economic crop plants may be injured is unknown, but it appears certain that some caution is necessary in the promiscuous use of the chemical on ... plants, either as ... sprays or as soil additions...." In these experiments, of course, the DDT-containing material was in direct contact with all the roots. Spray residues ordinarily would be present only in the surface layer of the soil, and should have much less effect on tree roots in that case.--J. C. McDaniel.]

[Footnote 15: --Dr. McKay of the U. S. D. A. found one tree only about 2.5% fruitful to its own pollen.--Ed.]

[Footnote 16: --There is a possibility of pollination from American chestnut sprouts in his vicinity.--Ed.]

Let's adjourn.

(Whereupon, at 5:30 p. m. the meeting was adjourned, to reconvene at 7:30 o'clock p. m. of the same day.)

+Evening Session+

President Davidson: The meeting will come to order, please. We first have the pleasure of hearing from Dr. C. A. Moss of Williamsburg, Kentucky, on Greetings from a Kentucky Nut. Dr. Moss.

Greetings from a Kentucky Nut

DR. C. A. MOSS, Williamsburg, Kentucky

I am glad to see all of these beautiful ladies here this evening. We just had dinner, and I presume I should make an after-dinner speech. I have always wanted to attend a Northern Nut Growers Association convention. I am more or less of what you might call a convention addict and speak on any occasion on slight provocation. I attended a convention at Quebec earlier this year, and after that I went on to Rio de Janiero in

South America and attended another convention, but this privilege of being able to attend the Northern Nut Growers Association tops all the rest.

I am reminded of the tale of the man who rushed into the sheriff's office in Texas, and his gun was smoking, and he says, "I have killed a man." The sheriff said, "Who did you kill?" "Oh," he says, "I don't know his name. He is one of these after-dinner speakers." "You are in the wrong room," the sheriff said. "Go back in the hallway three doors to the right to the bounty room. They pay $5 a head for those."

My family fortunes, if there be any, were founded on nuts. My father when he was 16 years old was raised on Straight Creek near Pineville, Kentucky, some hundred miles away from Lexington, and they gathered up a wagonload of the old chestnuts, he and a hired man on my grandfather's place, and they took an ox team and took them to Lexington to peddle them out. It took them three weeks to make the return trip.

I come from Whitney County, Kentucky. It was named after old Colonel Whitney, the man who built the first brick house in Kentucky. It was in the fall of the year, and the mortar was freezing, and they mixed whiskey with their mortar to keep it from freezing.

When I get away from home they ask me if I am a Kentucky Colonel. That's one of the first things I hear, and I tell them that I am. And they want to know why they put that honor upon a small fellow like me, and I tell them it was on account of scientific research that I had done, that I had developed a new way of making egg-nog. I feed the chickens the whiskey mash and they lay bourbon-flavored eggs, and all you have to do is drop one in a glass of milk.

They always ask about the Kentucky Derby, and I tell them that the last I heard Mint Julep was coming in on the home stretch strong.

I am not qualified with all of these experts to get up here and talk about nuts. They say an expert is a fellow that learns more and more about less and less until he knows practically everything about nothing at all; and that's kind of my shape, sir.

Now, seriously, I have had this hobby of trying to grow nuts for a number of years. I grafted a golf club on a croquet post, and I got some wonderful golf balls. Before the war I ordered some Chinese chestnuts. I got in touch with Sakata and Company in Yokahama, and they finally came in. I didn't have any experience, and about all I had was some imagination, and I planted them out in the fall of the year like I planted any other nuts. I went out in the spring and investigated. There wasn't a darn one come up. The rats had beat me to them and eaten them all up.

I was a persistent cuss and ordered some the next year, and I put them up in fruit jars and figured I would plant them in the spring, and when the spring came they all had the dry rot.

So I ordered them the third year, and I made sacks out of fly screen wire and put those nuts outside, and in the spring they came up and I had a lot of nice sprouts about this high and put them in a seed bed with a board all the way around. My father is blind in one eye, couldn't tell a chestnut from a weed, and he pulled up the weeds

and he pulled all the chestnuts up except one.

The fourth year I had better success, and I raised that year 400-and-some-odd chestnut seedlings, and I did more or less the Johnny Appleseed stuff with those. I gave those away in the community. I am, among other things, a banker, and I figured those would be as good as calendars, and I have not been able to follow the history of them. However, there is one of them I think is exceptional. It's a self-pollinator and is bearing heavy crops, and I intend to follow that particular tree up.

A genius, he is no better than any of the rest of us. All a genius is is a fellow that's got good digestion so he can eat enough to work long hours and good eyesight so he don't get tired.

So I was reading in a magazine about the Crath English walnut. They sent the Reverend Mr. Crath over to Poland before the war, and I got four pounds of those nuts he collected, and planted them. And every spring a cold spell would come along and get them before I could cut any grafts off of them. And I planted a Nebraska pecan and got some grafts from it, and my wife said that tree never did have a chance because I kept cutting the prunes off so they couldn't grow. I got several to growing, and then they didn't fill out the nuts.

I was talking to a good doctor here from Baltimore last night. We ate dinner, at the same table here, and I told him I didn't see but one thing wrong with this Northern Nut Growers Association: It needed a lot of young people in it, because if it didn't they were going to have to hold a reunion over at the cemetery.

I have done a lot of grafting, and I am not going into the details of that. I am going to say that I am glad to be here, I give you greetings from Kentucky, and I hope that I will meet you all again.

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President Davidson: That certainly was refreshing, Dr. Moss. We enjoyed it.

Next on the program is Dr. Aubrey Richards, Whiteville, Tennessee, who is not here. Nuts for West Tennessee is the subject of that paper, and Secretary MacDaniel will read it for us.

Nut Trees for West Tennessee

AUBREY RICHARDS, M.D., Whiteville, Tennessee

At the present time I am attempting to grow 14 grafted varieties of Chinese and Japanese chestnuts, plus numerous hybrids and seedlings, eight varieties of black walnut, 5 named Persian and 18 unnamed Carpathians, 5 heartnuts, 5 hickory and hickory hybrids, 12 pecans, and 7 hazels and filberts. The total number of trees, including all varieties, is well over three hundred. A few of the trees have been under my observation for 11 years on down to some that I have just acquired.

I shall not bore you with a list of unsatisfactory varieties nor with the ones that have not had sufficient observation in this section, but shall confine my remarks to less than two dozen varieties.

Pecans I shall touch only lightly, as they are a highly specialized crop only a little farther south. Stuart and Success are favorites here. Schley and Mahan are good if scab can be controlled. Sun scald on newly planted trees is our greatest problem, which I control by a paper wrap made by cutting two inch sections from a 36 inch roll of cheap felt-base wall paper. It gradually weathers away during the second summer. I wrap from the top down in a spiral, and when I reach the bottom, I place a hand full of earth on the end of the paper. No tying is required. In this way I have reduced the mortality rate of young nut trees greatly. I am also a strong believer in cover crops and mulching, for Tennessee weather is very temperamental.

Although we get ample rainfall per annum, it is often not well distributed, especially during mid-summer. During the winter we have several days of balmy spring weather with a drop to possibly below zero occuring overnight.

Thomas black walnut grows well here, but tends to over-bear, with many poorly filled nuts on alternate years. I counted an average of 8 nuts per lineal foot of bearing wood on one tree this season.

Snyder and Stambaugh are excellent nuts, setting about all they can mature.

Elmer Myers is a beautiful thin shelled nut, but so far a little shy in bearing. I believe this can be corrected if I can find another walnut that will shed pollen late enough to catch the Myers pistils. Homeland may be the one to do it. I have set some grafts of it with the Myers to see.

Carpathian D, and a variety of unknown origin from Haywood County are the only Persian walnuts I have fruited. This tree of unknown origin grows alone, is at least 50 years old, is three feet in diameter, has a spread of 40 feet, and is about the same in height. Some years it produces a heavy crop, others, nothing. To my knowledge, it has received no care in the past 20 years.

My 18 Carpathians are all growing with varying vigor and resistance to leaf spot. None has shown winter injury.

Of all the heartnuts, Rhodes is my favorite. The nut does not appear to be as large as some, but the kernel is just as heavy, due to its compact shape which causes it to fall out when the nut is cracked. It is self-pollenizing and also a good pollenizer for all my other varieties, shedding pollen over a long period of time, although it is the latest of all in producing its pistils. It grows vigorously on black walnut stock.

Rush seems to be the best filbert for this section. Its catkins are usually hardy here.

Chestnut trees, like gray ghosts, still reach their naked arms high on many West Tennessee hillsides, and occasionally one finds a farmer splitting posts from their remains, for chestnut is an enduring wood. A few of these tenacious individuals are still sending up sprouts that may reach considerable size before they are again struck down.

I have had no serious trouble with blight in any of the named chestnut varieties, either Chinese or Japanese. I have lost some trees by its entrance into the seedling

stock, but not many. My greatest headache has been sun-scald and winter killing, or to be more exact, "early spring" killing.

One of the juvenile characteristics of oriental chestnuts is the retention of their leaves all winter. They also grow in a rather sprangling way. This is a protective mechanism, and when we prune them to an upright form, or graft, this wood having lost its juvenile characteristics, we are inviting trouble unless we protect the trunk in some other way. I prefer to use a paper wrap as described under Pecans, as it is quickly done and is inexpensive. This also gives protection to immature callus cells at bud or graft union.

Of the older Chinese chestnut varieties in my hands, Hobson has excelled, with large chestnuts (34 to the pound in 1948.) Zimmerman also produces a good nut. Colossal (Hybrid) is very productive and produces the largest nuts of any chestnut that I have seen grown in Tennessee, but the quality of the raw nut is not equal to Hobson. It refuses to grow on Chinese stock, but thrives on Japanese. It is pollen sterile. I have several newer varieties under observation and although they are growing vigorously I have not had time to form an opinion on them.

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President Davidson: The Reverend Bernard Taylor of Alpine, Tennessee, will next read a paper on The Marketing of Black Walnuts as a Community Project. Mr. Taylor.

Marketing Black Walnuts as a Community Project

THE REV. BERNARD TAYLOR, Alpine, Tennessee

The Rev. Mr. Taylor: I suppose that every community where black walnuts grow wild has a marketing of some kind, some kind of a plan of marketing, maybe just what every boy or every man who has some spare time or some of the womenfolks may do to make something out of the walnuts that are lying around.

In the community of Alpine, which is in Overton County, people used to go out on the ridge with wagons and bring home wagonloads of walnuts, and they would sell them either in the shell or they would crack them and sell them in pretty poor condition, however they could sell them. When we first began selling walnut kernels in Alpine we got 19 cents a pound for the kernels, and that was more than they were worth, I believe, because they were dirty, greasy, and they had mildew gobs in the bunches of kernels. So I don't know how the rolling stores that came around that way could make anything out of them trading them in at that price.

Then we began to study the Government bulletins on how to produce good walnut kernels, and there is a good bulletin on that; all of you are acquainted with it, probably. When we began to harvest those nuts and hull them as quickly as we could and wash them and dry them out thoroughly and then crack them before they got too dry, we organized what was called the Walnut Club. This Walnut Club mostly was composed of some of the women of the community who lived up in one little cove where the limestone outcroppings seem to favor the walnut and the air drainage or whatever it was seemed to favor the crop yields rather regularly. We don't have an every-year good walnut crop.

Well, these women got finally so that they could get 35 cents a pound for their walnut kernels, then 45 cents a pound. Then we found a good friend in Pennsylvania who would take those kernels, all we could send her, and put them up in little pound packages and sell them for whatever she could get and send us all the money. That's altogether contrary to Hoyle I guess.

You merchants, if there are some of you here, who are dealers in walnut kernels know that our people were just getting spoiled. Anytime now that a merchant says, "I will give you such-and-such a price for the walnuts and then I will sell them for such-and-such a price," he looks to them like a robber. They want to sell them for what the people pay who eat them. That isn't quite fair, maybe, but we got $1.39 a pound last year for all the kernels we could produce, and the year before it was $1.40, I believe, and it stays about that price.

That is about the story of the community project. It is a direct contact by way of a benevolent friend between people in the mountains in Tennessee and people in Pennsylvania who say that these kernels taste better than black walnut kernels in Pennsylvania taste. I don't know whether any Pennsylvanians here agree with that or not. I think they are wonderfully mild-flavored, a good many of them very light-colored kernels. Though Mr. Chase has made some beautiful exhibits of how the color changes depending on how long a time you leave them in the hull, we still have some that stay lighter than others. Some of them have rather gray-colored kernels.

There is one of those trees that Mrs. Ledbetter has, on her husband's farm. He was about to sell that tree for a log and a stump. They come along and grub the stumps out and sell the stumps and all for veneerwood. But she wouldn't let him sell it, and over the course of the last few years they sold enough kernels more than to pay for that walnut tree and it is still going to yield a good many years, probably better and better as time goes on.

I think that possibly the community angle of this is a little bit misrepresenting. It's not the entire community, but it is a little group of the community who are interested in the wild black walnut.

Last spring we were very fortunate in having some help in grafting some of the seedlings. This Mrs. Ledbetter's husband got interested in walnuts, and he planted a whole pasture with walnuts spaced every so often, and this spring we went there with the help of God and were able to graft those to Thomas black walnuts. They were just little seedlings, so we hope to go into the named black walnuts as time goes on.

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President Davidson: May I ask, Mr. Taylor, the people, of course, now comply with the Government regulations on pasteurization and so on?

Mr. Taylor: Never heard of it. You will have to tell me about that after a while, if you will, please.

President Davidson: Mr. Shadow, the County Agent of Decatur, Meigs County, Tennessee, will tell his experiences with tree crops in that county.

Experiences with Tree Crops in Meigs County, Tennessee

W. A. SHADOW, Meigs County Agent, Decatur, Tennessee

Mr. Shadow: Mr. Chairman and members of the Nut Growers Association: As President Davidson announced, I am an agricultural agent. About twelve years ago I thought it would be good to have a hobby, and since I was born and reared in the nursery world propagating fruit trees and ornamentals, and due to the fact that John Hershey came by one day and talked to me about the tree crops in the Tennessee Valley, it struck me just right, and I have made that my hobby.

You know, every man who has a job gets fed up on his job and needs to get out and play with himself, or something else, to forget his troubles. So I find in propagating nut trees, top-working them, if you will, top-working trees where I find them to named varieties, is very interesting to me.

John Hershey taught me the technique of grafting nut trees. I had grafted and budded in all kinds of ornamentals and fruits, but I needed training in nut trees. So in the spring of 1935, I guess, I grafted about a hundred Thomas black walnut on trees where I found them in the woodland. At the same time I grafted maybe a hundred Japanese persimmon of possibly a dozen varieties on the common native persimmon. I purchased three, four, maybe five Japanese persimmons and planted these trees in the spring of 1935. All these persimmons, maybe 60 or 70 of them, grew nicely. The Thomas grew very well, and the winter of 1939 or 1940, I don't recall just which, was rather severe. We had below-zero weather, and all of my persimmons were killed--I thought. The next year I found a persimmon tree up in the woods with maybe a peck of great big nice persimmons and later I found that that was a Fuyugaki persimmon. All the rest of mine were winter killed. Those that I purchased were winter killed the first year. I don't know why. I grafted the persimmon about 5 feet high. Those that were grafted at the ground I noticed winter killed the first year, and these that are grafted up about shoulder high seemed to live three or four years before they winter killed, and the one variety that survived as Mr. Kline and Mr. Chase, or someone, has told, is Fuyugaki, I believe. I have a Tamopan persimmon, a great big, nice persimmon about so big, but bitter as the dickens, and about the only thing I think it is good for is to look at. It is pretty. But the Fuyugaki is never bitter. It is very tasty even partially green, and as it ripens my lady thinks it is very good, and I think it is good, myself.

I have about two or three varieties of mulberries. I got them from Glen St. Mary Nurseries in Florida. They make awfully good pig feed and bird feed, and I don't mind eating them myself.

There are some honeylocust, Millwood and Calhoun. I purchased several seedlings of thornless honeylocust from some northwestern nursery and grafted them to Millwood and Calhoun. I also have four trees that are ten years old and they have never borne. Last year there was one tree of that hundred that bore heavily, and the rest of them are barren. It must be lack of pollenization, or something. I am not getting fruit from my honeylocust.

Someone asked me what I am going to do with all this stuff, and I said, "Well, the squirrels and I will have lots of fun anyhow, and the cows will eat the honeylocust if they ever bear."

I have two pecan trees that are bearing nicely. One is a Posey and the other is a Greenriver, bearing very nicely. They are about ten years old. I have some Schley and Delmas and Mahan, and they are not bearing. I don't know why. We are out of the realm of the southern pecan and too far south for the northern pecan, I am afraid.

My Persian walnut, heartnut and Japanese walnut think it is spring too quick, and every year they burst out and grow about so long, and then they fall down and die from freezing, and then they grow out, and this time of the year you look at them and you say, "That's a beautiful tree," But they freeze just enough to get the fruit each year.

Mr. McDaniel came by last spring a year ago and left with me a little scion of a Carpathian walnut, the Bayer selection. I wasn't present, but he left it with my lady and suggested to my lady that I would know what to do with it. I put it on a common black walnut grafted about so high, and it is ten feet high now growing nicely, but this spring I noticed that it, too, thought the spring was here before it was here. I don't know how it is going to bear. I may have to take it out on top of the hill and re-graft it on a high place where it has more air drainage.

Of the Chinese chestnut, I planted about a hundred, but I planted them in a cut-over woodland that was full of native chestnut sprouts. You know how the chestnut sprouts will do. They grow up and blight out and die down, and another sprout comes from the stump. They have been doing that for 30 years over in my part of the country. I planted these chestnuts purposely in that grove where there was lots of blight. Out of that hundred I have eight trees that are alive. The rest of them have died from blight. They are bearing very nicely, but I haven't learned how to care for those fruits so that they are good a long period of time. Someone just told me that you had someone on the program this morning who would tell us that. It is a very interesting subject for me.

And the Thomas walnut is a nice black walnut. The trees are a little bit peculiar about their bearing; sometimes they bear heavily and again they forget to bear. The Stabler doesn't bear at all for me. I just know they are Stablers because someone told me so. I have them labeled. I have Creitz black walnut. I got five from TVA four or five years ago, and they just literally bear themselves to death. They're about so high and bear every year, very nice nuts. I will have to pull the walnuts off long enough to make them grow up and make real trees. I think they are going to be all right.

Mr. Chairman, I am not an expert. I use my hobby to keep from bothering about the troubles that I have with other things, and when I get mad at a neighbor I go to playing on my trees, and it gets me well. I recommend it as a very soothing hobby.

Now, some day we will make a business out of tree crops when we in Tennessee get the bugs out of it and get them so we will have the right varieties to produce. I am not satisfied with the Thomas. Someone suggested it was a wonderful nut. I am not satisfied with it. We need a better walnut than the Thomas. But it's the best I have.

There is a native walnut I found in the valley near Watts Bar Dam. I named it Pineland. It is just a seedling. It is a most wonderful nut if it wasn't for its hard shell. It's hard as the dickens. It is a wonderful bearer, has borne every year for nine years. It happens to be in unusually good soil. But I have grafted a few up away from the river, and the grafted trees are bearing nicely. The trouble is it is hard, but it is a wonderful good kernel and it is a big nut.

Groups like this working with tree crops and nuts over a period of time will develop the right varieties, and if we can get some youngsters interested--and I am in my county getting some youngsters interested in grafting--and tell them not to expect too much but get a whole lot of satisfaction out of the fun of producing something, I think this will be the beginning. Or rather, you have been going a long time. This is a means of progress in tree crops that I am well pleased to take a part in. Mr. Chairman, I think that's about all that I have.

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President Davidson: I know we all wish we had more county agents like that, interested in trees and interested in young folks. Those two things should go together. I wish you would just sort of propagate that idea when you meet other county agents, won't you, Mr. Shadow?

Now, then, Mr. Frye of Pleasant Dale, West Virginia, will tell us something about Nut Hobbying in Eastern West Virginia.

Nut Hobbying in Eastern West Virginia

WILBERT M. FRYE, Pleasant Dale, West Virginia

Mr. Frye: After hearing such wonderful speeches as we have had, with your reading, Mr. McDaniel, I wish I could be all of us, but as it is, I am just myself. I don't know how many know where Pleasant Dale is, but anyhow, you know where Washington, D. C. is; I live just along U. S. 50 and my section is 103 miles west of Washington, D. C. That will locate where we are.

This section of the country is composed of a lot of long ridges with steep hills, narrow valleys, some of them very fertile. These valleys form bases where you will get the draft off these hills down into the valleys. You must keep all the fruit and most of the nut trees out of those places, or you have these frost spots that I have been telling some of you about.

As far back as people can remember that country has been covered with all kinds of nuts except the European (Persian or "English") walnuts, and the early people coming in there used these nuts for food, and the chestnut was their main one. Whenever a person clearing the land found a nice tree he would save it. Then he would show much pride in having a good tree, and it kept on going until there became a rivalry as to who had the best chestnut tree. Some had an orchard of them.

When the blight hit the country I had an orchard of chestnut trees. When I saw the first blight in the top of a tree I didn't like the looks. I kept noticing that. It kept on coming down the tree, and it killed the base. The total result was everybody lost their

hobby trees, and then soon they changed to something else.

Now, when the blight took the chestnut out of the country the people began to pride themselves on the walnut, who had the best walnut, who had the best shagbark in the country.

Some distance from where I am is a two-acre grove, a wonderful grove of our larger nuts. Some places it is called kingnut and some places they call it under the name of this big one in the show room, shellbark. Anyhow, there were two acres there and real moist meadows, and every once in a while the frost would kill those nuts, and the next year they would have a wonderful crop. So the climate determines whether we have an annual crop or an intermittent crop on these trees.

Then I always liked to mess around with hobbies with nature. I became interested, got to wondering who did have the best of the best. Then I began to go out and visit all of these farms and ask them for a certain number of the best, and I began to send them around to Mr. Reed and Mr. Zarger and other people to take their word on it. And, of course, I have located some that cracked very well. But every once in a while somebody tells me they have got a better one yet, and the other day I ran across a fellow a hundred miles away--he happened to hear about me, and I have a neighbor who knows him--who has a black walnut that looks like a Persian walnut. So you see, I have a trip of a hundred miles to make to see what he's got. I wrote to him just before I left. I wrote to him to send me at least 20 of those nuts, and just as soon as this fellow sends me the nuts I would come up and see him and later on would try to get some grafting wood and send down to Mr. Zarger of the TVA group.

My job is not to keep them to myself but to put out the best. So we have those different nuts, and now it is time to consolidate the best in what we have and get them in the hands of the nut growers groups and those who will put them out and really make use of them. But first we want to see these best trees all over the country. Some of them are not as good for timber as the others, but I like to incorporate the timber with the nut production.

We talked about the black walnut earlier today. The speaker was not saying much about flavor. That's one thing we want to do in all of our nut work, get as good a flavor as we can. So why not get the best and go putting it out to give it to everybody. Why keep anything within ourselves? That's the main thing we can do.

A brother was talking a while ago about this nut job, a community nut job. Now, two years ago--I will have to use my dad, who is 82 years old, as a little reference--my dad cracked 83 pounds of black walnuts from just the best of them, you might say. Sold them at a price of $1.49 a pound. So that wasn't bad, was it? I thought that was right good.

Last year we didn't have a nut in there because we had a freeze on the 31st of May of around 26 deg. to 28 deg., depending on where you were and the location. But then in the fall on the 23rd of September we had another drop just when everything was in full growth, due to a dry spell and then a rain. But in the fall on the 23rd of September we had a drop down to 20, so that was what happened to all the remaining nuts in that country. They were just frozen like black mummies.

I had what they call the Texas Thinshell black walnut. I have one tree that is about eight or nine feet high, maybe ten feet high, had 45 nuts on it, nice big ones, and they just looked like mummies, and it made me heartsick, of course. I went out there and looked at the things, and they fell off the tree. I thought, "Well, I might just as well experiment. I will dig me a little trench here along the garden, I will put these in and see what happens." To my surprise 20 of them came up after being frozen. So that might be a question: Will things sprout or germinate without reaching maturity?[17] I don't know how much maturity they had. They certainly weren't in full growth when they were frozen. That's one thing we want to see.

My main aim is just to grow things, for hobby purposes and see just what will grow. Last year we had such a hectic year from that late spring freeze and early fall freeze it discouraged me here where I am, in this frost pocket at an elevation of 1,050 feet. And I said, "Now, on the hill about 4 miles away and 300 feet higher they have a wonderful place for peaches." I have a friend who lives up there, and he has so many peach trees missing in his old orchard. I said, "How about setting out some nut trees in your peach orchard?" Ho said, "Go to it." I set out a nut tree wherever there is a peach tree out. So that gave me a chance to see what they would do. Last spring I started that too late, but I set out 45 or 50 trees, filberts, Persian walnuts, pecans, chestnuts and persimmons, and I will just see what they will do.

And today my kind friend who gave a talk on the nut trees from down in Alabama gave me seed to plant. I expect to put a row of those out and see what they will do. The land I am planting them on at one time was just a great mass of chestnuts, and this friend there on one of those sections, of about three acres, had cut 35,000 feet of this dead timber after the chestnut blight killed them.

That blight was a terrible shock to us. One thing I did note when it came on, prior to the chestnut blight in that country there were these little chipmunks, which, everybody knows, eat chestnuts. You couldn't hear yourself think for the little chipmunks chipping all over the country. You know, they carried off all the nuts. You had to be smart to beat them to them. When the chestnuts disappeared, the chipmunks disappeared, and there were eight or ten years when you were lucky if you got to hear one. In the meantime those little fellows have changed. They died, a lot of them, but now they have learned to eat something else, and now they are coming back.

That little chipmunk always amused me, because I loved to go out and play with the squirrels and things like that. Anyhow, it's just pure hobby work, and as Mr. Shadow says you can get over a mad spell and get out close to nature, because in this nut work you can't get any closer to God's work than to get out and get something better. I think that's all I have to say.

[Footnote 17: Some other members have reported similar behavior of frost-bitten and poorly filled black walnuts.--Ed.]

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President Davidson: Those of you who know Mr. Frye know that he works as well as he talks, and that's pretty good.

Is Mr. Tatum here? (No response.) In that case I am told that Dr. Rohrbacher will read a paper by Mr. Tatum of Lebanon, Kentucky on "A Look, Backward and Forward, Into Nut Growing in Kentucky." Dr Rohrbacher.

A Look "Backward and Forward" into Nut Growing in Kentucky

W. G. TATUM, Route 4, Lebanon, Kentucky

The lumberman's ax, the chestnut blight, forest fires, and the "new ground" hill farmer, together, have destroyed many thousands of our beautiful Kentucky forest acres. Much of this one time "nature lover's paradise" is now ugly, barren, and eroded, and too poor to give a living to either man or beast. Wanton destruction of God-given treasure and beauty is a sin and a shame. Thanks to the men of vision and foresight of the U.S.D.A., state agricultural colleges, and our own fraternity of nut tree lovers, this slaughter is coming to a halt at last. Our fellow citizens are being awakened to the real value of their woodlands. Much reforestation of these steep barren wastes is already under way.

We, of THE NORTHERN NUT GROWERS ASSOCIATION, INC., can look back to many mistakes we have made in the selection of varieties for our respective climates and soils. Our dates and methods of grafting, budding, and transplanting have not always been right. We have gotten hold of scionwood that we were most sure would not grow when we used it, but we did use it, hoping, and most of it did fail, as we expected.

In our Association, we have a large group of wise experimenters on varieties and methods, well placed all over the U. S. and I have every confidence that, in time, many commercially profitable varieties, and better methods will reward their research. But in the meantime, we should all keep ever on the alert for a new and better idea, or variety.

Here in Central Kentucky, of the many black walnuts I have under test, only Thomas, Victoria, and Eureka have the tendency toward young and heavy bearing. These three do show great promise in my section as young and heavy croppers. And they are all top-bracket nuts, according to tests made by expert testers. There may be newer ones better than these, and we hope there will be yet better ones turn up continually in the future.

There are at least a few Persian walnuts that show promise in my location. Of varieties I have of bearing age, only four are worthy of mention. These are Broadview, Elmore, "Crath-Dunstan No. I" and "Crath-Edmunds No. 3." All of the above have borne well on two year old grafts on large black walnut stocks. Their nuts are in my opinion excellent.

Wright and Walters heartnuts seem well adapted here, and are doing equally well for me on Japanese, butternut, and black rootstocks. These are the only two I have old enough to bear, and they are bearing their first few nuts each this season. I would like to add here, that the wild nut crop in general in my section, is very light, and these nut trees that I mention as bearing this season, are the more to be noticed for their crops in this year of bad nut crops. I am trying "buartnuts" and butternuts,

which are growing satisfactorily, but not large enough for a crop.

This is wonderful natural chestnut territory. All of the many Chinese seedlings I have, and the few grafted ones, are growing nicely, and quite a number have burs on them when only about belt high to an average man. I am anxious to get graftwood of superior individuals as they come out, for propagation here in my own planting. I believe this to be a good home for any good chestnut. No blight is showing to date in either my seedlings or grafted ones.

I live on rather deep, fertile upland, and am quite hopeful of good results from many of the Northern pecan varieties that I am trying. The oldest trees I have are only five years old, on small seedling stocks and hardly old enough to yield a crop for at least another five years. Major, Greenriver, Busseron, and Fisher are my oldest, and are making rapid growth. Stuart, of the Southern group, is bearing quite well for my friend, Lewis Edmunds, a few miles southwest of me, and he says it matures its nuts well before frost, but insects cause a goodly part of the crop to fall prematurely.

I have quite a collection of the better known grafted shagbarks on my woodland. These are mostly on wild shagbark stocks. They are all growing well, but I have had no nuts from them as yet. Grainger is the fastest grower of the lot.

To make my nut tree project complete, I have quite a long row of filberts and hazels, set hedge row fashion, which include quite a list of varieties. Those that bear quite regular and heavy crops include four "Jones Hybrids," Winkler hazel, two un-named hazels, and Barcelona filbert.

I have persimmons, too, both American and Chinese named varieties. My Chinese are young and not bearing yet, but doing well. Kansas and Josephine are my choice of the natives.

I am trying Millwood and Shessler honeylocusts for the first time this year. They are beautiful grafts, and I am looking forward to the pleasure and profit of adding them to my hill cow pasture in a year or two.

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President Davidson: Thank you, very much, Dr. Rohrbacher. We have 15 minutes before the next order comes on the program. Suppose you take a recess right now.

(A recess was taken.)

(Mr. William J. Wilson from Georgia showed moving pictures of his pecan orchard.)

President Davidson: The next order of business, we will now hear a report of the Committee on Black Walnut Standards and Judging by Dr. L. H. MacDaniels.

Round Table Discussion on Judging Schedule for Black Walnuts

DR. L. H. MacDANIELS, Chairman

Dr. MacDaniels: During the year your committee has worked on the problem of setting up a judging schedule for black walnuts, mainly through correspondence. Unfortunately, it has not been possible to get together for discussion. Had we done

so, I'm sure we could have achieved close agreement upon essentials. As it is, there are several phases of the problem upon which we would like the judgment of the association members. As far as this group here is concerned, I am quite sure that we can't profitably go into a discussion of the various details and ramifications of a judging schedule. I do think, however, that we might discuss the problem of whether our point of view in developing such a schedule should be that of the value of a variety for the commercial buyer or for home use. As far as the committee is concerned, Mr. Chase favors the home use angle. Clarence Reed and Mr. Stoke have not expressed themselves definitely one way or the other. Mr. Stoke is here, and I expect that he will say something about it.

I would like to open discussion on this question at this time, unless you want to go back to the consideration of whether it is desirable or possible, to have any such schedule, at all. May we assume that this is desirable?

Mr. Weber: Could we have a double standard, one for the commercial grower and one for the home grower?

Dr. MacDaniels: In my judgment it would be better to try one at a time.

Another schedule can be developed later.

Mr. Weber: Have you any particular preference, Dr. MacDaniels?

Dr. MacDaniels: I personally feel that the new and improved varieties will find their best use as a home proposition rather than in the commercial orchard, because apparently with a modern cracker the common wild nuts can be cracked in pieces that are satisfactory for the commercial trade, and crackability is of little moment.

Have you any comments as to which point of view the committee should take?

Dr. Crane: I would like to inquire as to the purpose for which this numerical score or method of evaluating these nuts is to be used. Is it to be used for show purposes, or is it for determining the value of a variety of nut to grow?

Dr. MacDaniels: The purpose of setting up a schedule is to provide a standard by which we can determine differences between samples in contests, and to give a basis for comparison in determining the value of a variety for growing in various climatic zones and of different varieties grown in the same place. For instance, the variety, Thomas, in one zone would be a very good nut and have a score of, say, 89. In another it might have a score of only 45, and in another a score of 55. The score would be directly related to the adaptability of this variety to a climatic zone or to a system of cultivation or to variation in any other environmental condition.

Mr. Weber: How do the other members of the committee feel about it? What is their preference? It seems to me that if you are unanimous, all we have to do is approve your report and leave out the discussion.

Dr. MacDaniels: We are not unanimous. Mr. Reed, who I regret is not here, rather doubts that any kind of schedule is either possible or desirable. Would you think that is a fair statement, Mr. Stoke?

Mr. Stoke: Yes.

Dr. MacDaniels: Mr. Chase believes that a schedule is both possible and desirable and that we should work along the general ideas advanced in the paper on judging schedules published in the last volume of the report. As I understand Mr. Stoke's position, he would go along with that in general with possibly the addition of the factors of taste and color. Is that right?

Mr. Stoke: Yes, taste and color for domestic use.

Dr. MacDaniels: I have already stated my position. I feel that unless we confine the schedule to characteristics that can be weighed or measured successfully its value and usefulness will be little.

A Member: Dr. MacDaniels, if a man has a $20,000 machine for cracking walnuts and he has a choice between the Thomas walnut and a good wild one, he will pay a little bit more for Thomas walnuts, will he not?

Dr. MacDaniels: The question raised is that if a cracking plant which cracks thousands of pounds can get more kernels out of a hundred pounds of Thomas nuts or any other grafted variety, would the operators pay something more for them? I think undoubtedly they would, but would they pay enough of a differential over the wild nuts to make it worthwhile to the grower? I don't know.

Dr. Crane: If you take pecans which are our best example, 95 per cent of all nuts produced in the United States are marketed as shelled kernels, and there is a very substantial price differential between seedlings and budded pecans, and the crackers will pay the difference based on the yield of kernels. That is their only interest. The thickness of shell, how well it cracks, or any other factor is of no importance. If the kernels are there, they will get them out.

Dr. MacDaniels: That is the crux of this whole matter. Are we interested in developing varieties for cracking in which we care little about the size of the pieces recovered or about the ease of extraction, or do we want nuts for home use that will give a high yield of large pieces? These machines, as I understand it, will crack the walnuts and get the kernels out in small pieces regardless of how they crack in a Hershey cracker.

Mr. Weber: As I understood Mr. Mullins, he favored having a lot of Thomas if he could get them.

Dr. MacDaniels: Would he pay the difference? I don't know. Dr. Crane says he would.

President Davidson: When I talked to him--we passed through there and saw the plant--he said he thinks well enough of the better nuts to come here for the purpose of learning where and how to manage a plantation of his own of Thomas and the other budded varieties for his own cracking plant. In his own cracking plant the yield for the amount of labor expended is so much better on the improved varieties that he wants to make a planting of his own. He will pay more, but just how much more, I don't know.

That brings up another matter. As I have said before, our state authorities should be urged again and again and again to buy _good_ seed nuts for distribution to the public so that we can get these better quality nuts into the woods. Some of them are agreeing to that. Some of them are doing it. But so far not very much has been done.

Dr. MacDaniels: I think that before your committee goes ahead we must get a decision on this point, for the approaches are quite different. If you are developing a schedule for home use, the size of the nuts is of importance. In general, the bigger the nut the easier it is to handle, the easier it is to shuck and crack. The percentage of kernel is relatively less important than it is in the commercial cracking. The size of the particles recovered is more important for home use. If they come out easily and in large pieces, they are much more desirable.

On the other hand, in commercial cracking the percentage of kernel is important. The commercial buyer wants to know how many pounds of kernels can be expected from a hundred pounds of nuts. He is not much interested in the size of the nuts or the size of pieces that are recovered. This is an entirely different approach to the problem. We have got to decide between the two before the committee goes further.

Dr. Crane: There is another angle to the problem. A lot of the black walnuts today are used in the bakery trade and in the ice cream trade. But I visualize a market for black walnut kernels to be eaten out of hand. There are many people in the United States that like the flavor of black walnut kernels to eat in this way. I know I am one of them, and I don't want to eat crumbs. I don't want to eat small pieces. I like to have at least quarters.

I think that if we were to gather from the status of our other native nut industry that there is going to be a premium paid for the larger pieces, then cracking quality _would_ enter into the matter. Our pecans are sold on count of whole kernels per pound or per ounce. Almonds are sold the same way. Walnuts the same way--that is, Persian or English walnuts. The number of kernels or pieces per pound is an important matter, notwithstanding the situation as it exists in the black walnuts today. So I do think that we can't take the present status of the industry as one which will prevail generally and in the future.

Mr. Weber: Would the majority report favor the side of the home consumer rather than the commercial buyer?

Dr. MacDaniels: I think it depends on what Mr. Stoke would think about the majority. We didn't get a chance to get together, because Mr. Stoke was so busy with exhibits.

Mr. Weber: We might end by moving the adoption of the majority report and let it get at that.

Mr. Stoke: I know I brought up that matter of whether we should judge by standards acceptable to the commercial buyer or to the ultimate consumer. The confectioner doesn't care about the size or color at all. When they are put up in candy or in chocolate cookies, color doesn't mean anything. It's a black walnut, and it doesn't have to depend on anything else. So I think those two points of view are

pertinent.

I never expressed any preference, and I don't know that I have any. I think it might be just as well to leave that up to this body. But the producer, or those anticipating producing must be considered. Mr. Hirschi can give us the word on marketing kernels.

Mr. Hirschi: I do not market kernels. I just crack the nuts and sell them by the pound cracked.

Dr. MacDaniels: Shell and all?

Mr. Hirschi: Shells and all. I sold about a ton and a half each winter for the last four or five winters. They are Thomas walnuts. I get 35 cents a pound with the horse shoe nail in the package.

Mr. Stoke: That man wants good color, good flavor, kernels easy to pick out, and of good size. That goes with the retail buyer. If the commercial buyer gets 30 per cent kernels from good nuts compared with 15 per cent from run-of-mine nuts, he doubtless will be willing to pay a considerable premium for the better nuts if he can get them. But unless the good nuts are in considerable quantity they go right in with the others and no more will be paid for them. That's my point of view. I don't want to express my particular opinion, because I have no particular opinion. But you might consider both, the commercial nut, and the home nut.

I think we might vote and determine what action, to take tonight as to setting up a standard, or if you want to set up a double standard.

Mr. Weber: Mr. Mullins does get a better price for a larger kernel. He separates them and treats them differently than the general run of small pieces. It's been my observation that the cracking machines do a remarkably good job with the ordinary run of seedling nuts. Kenneth Dick gets the kernels out in rather large pieces, and from what we saw up at Mullins' place he gets the same thing. He sifts out the larger pieces and gets a better price for them. So the preference is for the larger pieces. It's like buying hamburger; you prefer your hamburger ground up out of larger pieces rather than odds and ends that the butcher has around the shop and grinds it up and hands it to you.

Mr. Stoke: But isn't it true that he sells the kernels in two separate classes?

Mr. Weber: But the preference still seems to be, after we see them, for the larger pieces. They have better kernels; otherwise, they would break up in small pieces.

Mr. Korn: I believe that as long as there are very few commercial orchards, we should approach it from the angle of the people who have just a few pet nut trees around their yards, because I don't think the commercial orchards of the improved grafted black walnuts are going to be large enough to color the picture very much for a few years to come. As long as they haven't been too profitable, I think it is going to be some time before we have to worry much about commercial orchards. Therefore, we are interested in getting a superior product in kernel; it has to be large, has to be of good color and good flavor. It seems to me that would be one of the first things to

consider. Then, if orchards get more plentiful and profitable, we can take up the other angle.

Mr. Chase: Mr. Chairman, I'd like to make a few remarks on this business of commercial cracking and large pieces that I hear mentioned by my good friend, Mr. Weber. I had hoped to have the two largest shellers in the country present at these meetings, but was unable to get them here. In this area the commercial walnut cracking industry is related directly to the type of machinery necessary to recover the kernels. For example, the two or three cracking plants in Nashville handle an estimated ten million pounds of nuts each year and turn out roughly 1.2 million pounds of kernels. These kernels go directly to confectionary syrup and ice cream plants. Therefore, they are not interested in size of pieces. In fact, if they are too large, the commercial users have to chop them up. So what we are doing here, ladies and gentlemen, is confusing what we want to do in the way of judging nuts, it appears to me. There is little reason to assume that the Thomas, if they could get 10 million pounds of Thomas, would be more valuable to the commercial crackers. But that doesn't necessarily interfere with our judging system that we are trying to design to tell which nut is the best to grow.

I specifically asked these buyers of millions of pounds of nuts: "If I came in with some Thomas nuts would I get some more money for them?" Their reply was, "No, sir. We pay a flat rate per hundred pounds of nuts. We know that some of them are going to be excellent; we know some of them are going to be poor, but we intend to get from 12 to 15 per cent kernel recovery out of them."

In 1940 we brought quantities of improved varieties to the cracking plant in Knoxville and ran them through Mr. Smalley's machine. He was amazed. He didn't believe it; didn't believe his eyes. They came out in large pieces. But under present conditions they'd be chopped up. None of these kernels moving out of Nashville vicinity go to retail trade, except a few that go to confectionary stores in 25-pound boxes and are sold a pound at a time for cooking purposes, not for eating out of the hand.

People like Mr. Korn and Mr. Hirschi, who are interested in selling kernels at a much higher price than the commercial crackers, have to have large pieces, attractive kernels, properly cared for, properly colored, and of mild flavor. Is it this group we are trying to assist by this judging system or the commercial cracker?

The number of acres planted with Thomas sufficient to yield enough nuts to operate one of these machines would be tremendous. There are several examples of where the machine has been purchased to be used on Thomas but hasn't been used. It has been stored away. They prefer to crack the Thomas nuts by hand.

So my point is this: It appears to me that we are interested in the grower of several trees around the farmstead. At least, in this section we are. Everyone here gathers and cracks walnuts. Our idea of acquainting them with the Thomas variety is to make their job easier in cracking and picking them out. It seems to me that's also the problem that we have as a group elsewhere, and I believe that in order for us to make headway on this judging schedule, which I think is necessary and desirable, we must

view it from the home viewpoint at this time. That does not shut out the commercial viewpoint for later years. But now we are primarily interested in the home raising of nuts, unless I am in the wrong group. Thank you.

Mr. Weber: Mr. Chairman, I agree heartily with what Mr. Chase has to say, or otherwise we might as well quit now and raise seedling nuts to the best of our ability and sell them to the commercial crackers and let it go at that. But, if we do that, what's the use of searching out better varieties?

Dr. Cross: Mr. Chairman: I believe that if a nut acceptable to the home consumer, one which extracts easily and is attractive and palatable and is productive--if that type of nut is scored and comes to the attention of a sufficient number of growers, then I think the commercial people _will_ utilize it. So I don't believe there is anything to this argument. I believe if you go ahead on the basis of the home consumer and develop a nut that will be desirable for his purpose, and if in addition to these factors that have been discussed it is adaptable and productive, then it is going to be eventually the nut that the commercial man will utilize, because, after all, what we are growing nuts for is the kernel.

Mr. Weber: To bring it to a head, I move that we adopt that part of the report that favors the home consumer as against the commercial consumer, or we will be here all night talking about it.

Dr. Rohrbacher: I second the motion.

Dr. MacDaniels: You have heard the motion, which was seconded. Any remarks?

(Vote taken on the motion, carried unanimously.)

Dr. MacDaniels: That will be the basis on which the committee will work.

There are several other points to be considered. I would suggest the committee be asked to make further tests with the schedule as proposed in order to get additional data to determine if it is a usable schedule and can be used by different people with reasonably similar results, and if it does differentiate the things that we want to have a schedule differentiate in a test.

This last year we had hoped to do this, but there weren't enough samples of nuts available to be worth testing. I spent about $10 personally buying nuts from this source and that, and there wasn't a good sample in the lot, except one, which Sterling Smith gave me.

I think that if we have another season to work the schedule that has been proposed, we at least can demonstrate whether or not it is differentiating between varieties in a manner which is satisfactory.

I believe a motion is in order to bring this matter to a decision and end this discussion. Have you any further comment, Mr. Chase?

Mr. Chase: If it is not out of order, I move that we adopt for further trial, the scoring schedule proposed in the paper by Dr. Atwood and Dr. MacDaniels in the 1947 Report of the Northern Nut Growers Association.

President Davidson: I second the motion.

Mr. Stoke: May I make one remark? Does not that schedule ignore the factors of color and taste?

Dr. MacDaniels: It does, as not being objective characters.

Mr. Stoke: In other words, this motion approves something from the commercial slant rather than from the personal use slant.

Dr. MacDaniels: I wouldn't say that; it simply limits the judging schedule to those characteristics which can be objectively handled and are not a matter of opinion or judgment. That's the point here, I think.

Mr. Chase: Mr. Stoke and I don't quite agree--I don't think we are the only two--on flavor and color. However, in our exchange of correspondence we fully appreciate the advantage of light-colored, mild-flavored kernels. But I don't see any method by which we can place a numerical value on the color and flavor. Can we not describe the color and flavor along with the rating that describes the kernel and still have you on our side?

Mr. Stoke: Personally, I think we are splitting hairs. When we can't agree as to which color class a sample belongs, it must be somewhere near the border-line. Ordinarily the average human being will agree pretty well as to a blonde or a brunette or one that's neutral. And I think in the judging of walnuts, there can be no exact value based on the color. If you consider color and make a scientific test, your test wouldn't be the same as my test. But if it is a dark kernel, you can recognize it, and so can I, if we have any common sense.

Also in the matter of flavor, you and I can tell what we like and what we don't like. And I think there are those two limitations. We can't do this scientifically, because the human factor is here. But after all, it's humans that eat them and produce them for eating! And I rather, in the schedules last year, brought up objections to it. I didn't say I objected, and, of course, I don't now.

Mr. Chase: I'd like to just say this, and I am going to call on my good friend, J. C. McDaniel here, for agreement. A long time ago we prepared, did we not, various judging systems?

Mr. McDaniel: Yes.

Mr. Chase: We found that--you can correct me if this is wrong--by manipulating five points for flavor and five points given for color we could change the position of a variety of a list a great deal, and we also found that the points given for color were not related to inherently bad color but simply the result of poor handling, which also affects flavor. This is my reason for eliminating color and flavor from the schedule: it is _not_ to get away from the mild-flavored, pretty-colored kernels.

President Davidson: Mr. Chairman, I must say that I am inclined to agree with Mr. Stoke, for this reason: Even though color and flavor are very frequently the result of poor handling, we all know that we will say that the Stabler has the characteristic that is distinctive of quickly coloring up and quickly becoming rancid as distinguished

from the Thomas, which does not. Now, those things are inherent in the two varieties, I think, and I don't think this committee should ignore altogether the matter of color and flavor, although I do think, perhaps, not so much weight might be given to those two qualities as had been given to them in the past. But they certainly decidedly influence the marketability for kernels from the point of view of home consumption. I think there is no question about that. I should be inclined to agree with Mr. Stoke, that those two qualities should not be ignored by the committee.

Dr. MacDaniels: I think the point would be to ignore them in their simply not being objective; you can't weigh or measure them. There is a motion properly seconded before the house. Are there further remarks?

Mr. Weber: Wouldn't there be just a certain amount of trial and error connected with it, and as you go along you will either add to or take off, and then you will get a correct system of judging? You have to start out with one system and if it is wrong, change it.

Dr. MacDaniels: I think it's a matter of doing something rather than nothing, for a schedule is always subject to improvement.

Mr. Stoke: I wish to point out we have made some tests together, and your personal tests and my personal tests ran very close together.

Dr. MacDaniels: That is right.

Mr. Stoke: And one member of the committee is always very conservative and his tests never run as high on any series as the others. I make a test and he makes a test, and his are always lower. Maybe, he doesn't recover as much; perhaps he isn't as expert a cracker. (Vote taken on the motion; carried.)

President Davidson: Let us adjourn until 8:30 tomorrow morning.

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+A Picture from Our Most "Northern" Member+

John Davidson wrote in our 1943 report: "If any man deserves a bright NNGA medal, it is A. L. Young, of Brooks, Alberta." By planting his trees near enough to irrigation ditches in his "desert, cactus country," and protecting them from livestock, Mr. Young is able to get nuts on the hardier trees, but he reported that the nuts, "while of fair size, do not have fleshy kernels ... Butternuts are very sweet with fair size kernels ... Giant hickory from Ontario seems hardy but particular about the kind of soil ... Carpathian walnuts killed back quite a lot, so did most of my hybrid walnuts ... Some Manchurian walnuts ... got a setback with spring frosts ... Heartnuts got a rough deal last winter [1942-43.]" Mr. Young wrote to Dr. J. Russell Smith in 1948: "I have been using pollen of Broadview and Carpathian [Persian walnuts] on my blacks and while there are a lot of hybrid seedlings, none have fruited yet. On Peace River hazel [far Northern] I have been using Barcelona, Du Chilly and Gellatly pollen. Some of these hybrids look good, hardy, and produce good nuts ... A few varieties of oak are promising and fruiting."

At his location, Mr. Young expects winter temperature of -45 deg., and the lowest known [before 1940] was -62 deg.F. Summer temperatures go above 100 deg.F.

[Illustration: Fruiting black walnut grown at Brooks, Alberta, Canada, by member A. L. Young. The seed came from Ontario.]

+Tuesday Morning Session+

President Davidson: The only way to get started is to start. We are going to be given a look at the honeylocust situation in the South by Professor Moore of the Department of Horticulture of the Alabama Polytechnic Institute of Auburn. Mr. Moore.

The Present Outlook for Honeylocust in the South

J. C. MOORE, Department of Horticulture, Alabama Polytechnic Institute, Auburn, Alabama

Mr. Moore: Mr. Chairman, ladies and gentlemen: Before I start this discussion, just in case some of you are not familiar with honeylocust, its habit of growth, the size of the pods and the possibility of its yield, I'd like to take time out just to show a few slides, then I will go ahead with the discussion and give you some data on that honeylocust production. I believe if you would look at these slides before we start the discussion it would give you a good idea what the tree looks like, how it grows, the age at which it starts bearing and something about its general habits; it will help you a lot to understand what I have to say about it.

(Slides shown.)

This is the Millwood honeylocust. The pods will vary in size from about 12 inches to 14 inches in length, from one and a half to one and three-quarters inches in width, and the back part of the pod, something that I can't show on this particular type of picture, is very thick, and this back part of the pod, the thick part of it, is very rich in carbohydrates. We have the Calhoun and Millwood selections that have run as high, the Millwood a little over 36 per cent sugar and the Calhoun a little over 38 per cent sugar. The Millwood is a much higher yielding tree than the Calhoun. I will bring that out in a few minutes' time.

This is a borrowed slide and I don't know the history of these trees, but I judge that the tree is about three years old. We have had good yields on three-year-old trees at Auburn.

Here is a group of trees growing with a ground cover, and again I am not familiar with the ground cover, but just judging from the general appearance it looks like a picture that came from our files. If that is true then I know the story. The tree in the background is a Calhoun tree and the tree in the foreground is a Millwood growing in _Lespedeza sericea_ and I will bring out some points in a few minutes in the general discussion on the value of these two plants growing together as a combination.

I believe this is another tree that grew on my farm, and the year this picture was made this particular tree, eight years of age, bore 250 pounds of those luscious pods.

A close-up again, giving you the general size of the pod, how they are produced in masses, and you get quite a bit of weight in some of those thick-backed pods that you don't get from the thin pods that grow normally on seedling trees. The TVA has done quite a bit of work in selecting and developing the honeylocust, and I believe we give that particular organization credit for the development of both the Millwood and the Calhoun.

I thought it would be very valuable to give you just a glimpse of the habit of growth of those trees before I start with my general discussion so that you would understand something about what I am talking about.

Mr. Weber: Are these thornless?

Mr. Moore: These are thornless honeylocusts. The original parent trees of the Millwood and Calhoun had thorns. By vegetative propagation--they went out and cut scionwood on the limbs above the thorns and propagated the thornless twigs on thornless root stock--we now have a thornless honeylocust.

There has been quite a bit of disturbance in Alabama, especially in the northern part of the state, caused by native honeylocust. We have two or three characteristics that I think ought to be brought out about honeylocust. Some of out trees in the northern part of the state of Alabama have triple thorns. It is known as _G. triacanthos_ and the "tri-" part of that particular word, of course, gives us an idea of three thorns, and I have seen thorns at least 12 inches long that you could catch in your hand and use for a dagger, and it would be very dangerous. Now, some of those trees growing in the northern part of the state are very serious pests in pastures. Cows and horses and hogs are very fond of those lucious pods, and they will go around the trees and pick up every pod that falls, and occasionally a horse or cow will get close enough to the trunk of the tree and get speared with those thorns, and when the thorn pierces the skin there is a little tip on the end that breaks off and is left inside. When the usual infection that it carries get started from the part of the thorn that is left in the flesh, you get pus and, of course, later on the amputation of the leg, if it happens to be in the leg, of the horse. With the thornless type that is completely eliminated.

Then this other thing that I think ought to be brought out, the thornless or near-thornless type as a general rule has a better quality of pods than the ones with the long thorns. Now, it is true that the parent seedling trees of the Calhoun and the Millwood both had a small quantity of thorns when they were growing wild. After they were propagated vegetatively the thorns, of course, were eliminated by taking scion wood from above the thorns. But in general in our state, the thornless trees--and we do have a lot of thornless trees growing wild--have a higher sugar content in the pods than do the trees with thorns.

I just wanted to give you a general idea of what we have done with honeylocust in Alabama. In 1938 the TVA sent down some Millwood and Calhoun for test planting. We put those trees in two different types of planting. We had an integrated planting where we were trying to select at that time some good pasture plants, and, of course, we had something like a hundred different species in the one planting. The trees

were planted relatively thick, but the larger trees were planted longer distances apart, and the intermediate trees intermediate distances apart, and then we had shrubs coming in under those. It was supposed to have been a three-story type of planting, black walnut in the upper story, honeylocust as an intermediate and shrubs for the ground. We were using different types of plums for the understory; then on the ground we had _Lespedeza sericea_. But from that we did get several different plant materials that did look promising, and we put the Calhoun honeylocust and the Millwood honeylocust in with that planting for trial, and they did so well that we expanded the honeylocust into another planting. I am very sorry that this latter planting had to be taken out.

Hillculture research went under in June of 1947, and the Horticulture Department took this work over, and they thought they could not support the honeylocust pasture program in Horticulture, and the plot, of course, was pulled out and planted in peaches.

Anyway, we do have some information I'd like to give you. The Dairy Department of the Alabama Experiment Station carried out quite an extensive feeding test over a two-year period to find out the value of these pods in the dairy ration. They substituted the honeylocust pods ground. Professor Eaton of the Dairy Department assures me that none of the seeds in those pods were cracked. They ground the pods with corn in order to take up some of the excess honey that is in the back of these pods so that they'd grind well, and they ground them in a hammermill, and the burrs were running far enough apart so that he assures me that very few of the seeds, if any, were ever cracked.

That has been somewhat of a discussion, among feed producers especially, recently, as to whether or not it would be profitable to grind those seeds in order to get the protein and fats that the seed has. There isn't a very high percentage of food in the seed itself, but you do get a little more protein and a little more fat if you grind the seed itself.

We have found in storage that weevils get in these seeds, but the weevil doesn't destroy the carbohydrates, and the weevil will only pierce the seed and make a hole in it. Then the intestinal juices of a cow will go in through this hole and they can digest the seed. That is something that comes along with storage.

I'd like to give you just something briefly on what the Dairy Department of Alabama Polytechnic found out about the general value of these pods. They found that honeylocust pods could be substituted in a dairy ration for oats, pound for pound. Now, that means that if you can get a high yield of honeylocust pods and substitute it in a dairy ration for oats that you just about have half of the grain problem solved.

I'd like then to follow that up to give you the average yields. Before I give you these average yields I'd also like to bring out this fact about the Calhoun and the Millwood honeylocust. Those trees are very peculiar in their habits of bearing. One year they will bear a heavy crop. The next year they will bear scarcely anything. They are definitely alternate bearing, and I think that alternate bearing has a physiological background behind it. How We can eliminate that physiological reaction is something

else. But the years that the trees are heavily loaded with the fruit the amount of carbohydrates that it draws from the tree is so great that the tree doesn't have enough carbohydrates left to produce fruit the next year. I think it is the carbon-nitrogen ratio from the physiological standpoint, and, of course, if that is the case, then there is a possibility that you could eliminate or correct that carbon-nitrogen ratio by thinning during the blooming period. But when you see these results I think that you will agree that honeylocust has a place, even if they do bear only every other year.

In our planting we have some trees that will bear this year. Next year they won't bear, but their sister trees will bear. So we have pods every year from some of the trees. Over a period of five years, during which these trees were planted (the oldest trees that we have in 1938, and in 1942) the average production of the Millwood was 58.3 pounds per tree. In 1943 there were no pods produced on the Millwood variety. We had a cold spell in the spring that completely eradicated all of the fruit in that year. In 1944 the average yield--and that is taking the average yield of 10 trees of the oldest ones that were put in--the average yield was 146 pounds of pods per tree.

Mr. McDaniel: That's for both varieties?

Mr. Moore: That's just for Millwood. I will give you the Calhoun in a minute.

Then in 1945 the average yield was 39.5 pounds per tree. In 1946 we had an average of 180 pounds per tree. In 1947 we had an average of 12 pounds. Now, note the break there in averages from year to year: 58, none, 146, 39, 180, 12. You get from that that we have almost definitely alternate bearing in those trees.

Now, this other thing is interesting. If you take the five-year average from 1942 through 1946 inclusive, and convert that to 35 trees--this is 10 trees--but when you convert that to an average of 35 trees per acre you get the equivalent of 92 bushels of oats per acre. Now, understand, with this yield of pods we were cutting two and a half tons of hay from the _Lespedeza sericea_ each year. So we were getting our hay crop and our grain crop from the same source.

Now, to give you just briefly what the Calhoun variety did during those years, in 1942 the Calhoun trees--the same age planted under the same conditions on the same soil--averaged 26.4 pounds of pods per tree. In 1943 the Calhoun followed closely with the Millwood; on account of a freeze they didn't produce anything. In 1944 they produced 32.4 pounds of pods per tree. In 1945 they produced 63.8 pounds of pods per tree. In 1946 they produced 22 pounds of pods per tree, and in 1947 they produced 46 pounds of pods per tree.

Now, if you will take the average of those, contrast it with the average for the Millwood, you will find that the Millwood tree over a period of five years produced almost three times as many pods as the Calhoun. The Calhoun variety has a little more carbohydrates, and it always averages a little more sugar per pound than the Millwood, but the additional yield of the Millwood variety makes it very worthwhile.

I have done quite a bit of work on the blooming habits or the fruiting habits of the honeylocust over a number of years, and I find that there is quite a variation there in

the individual trees. Some trees are typically males. They never bear anything, but they have staminate catkins. Others are typically females, never bearing anything but the pistillate flowers. Then we have an integration there of perfect trees. I know of one tree in Blount County, Alabama that for nine years never missed a crop. It had perfect flowers, or rather, both pistillate and staminate flowers on the same tree. However, the flowers were borne on separate catkins, the pistillate flowers, catkins, coming out on the same node with the male and producing the pod. So you do have a large variation in the fruiting habits, and we have found those variations on Millwood selections and on Calhoun selections, even though they were vegetatively propagated.

The reason why we can take a bud off a female Millwood and put it onto a root stock and get a male tree I can't figure out, but they seem to act that way in that respect. I have had a Millwood tree that never bore anything but male flowers.[18] That is something for someone else to figure out. I can't explain it.

Just briefly I'd like to give you the observational work that we have done with honeylocust. For mules in a feeding test we fed a team of mules for 30 days nothing but honeylocust and hay, and these mules were in fine shape when they came out at the end of the feeding test. You say that's an awfully short feeding test. It is, but we had very few pods. Then for cows I have gone into it more extensively. I have a cow myself, and I have fed that cow honeylocust pods and that was all the grain she had through the winter months, and got excellent milk production. You get excellent milk flavor from these pods and an increase in milk production.

A very interesting thing happened. I went out in the community to gather pods from the wild trees for a feeding test, and there was a lady who owned a farm pretty close to our project. I went over and talked with her about getting the pods from her trees to feed to my cows for feeding tests, and it was O. K. But when I left she got to thinking the thing over, and she decided that if honeylocust pods were good for my cow they would be good for her cow! So I went back in a few days' time--the pods weren't mature when I went the first time. I went back in a few days and I didn't ask the lady if I could get the pods, I just stopped on the side of the road and we put a darky up in the tree to shake the pods off. And we saw a little darky coming across the field, just a streak. He said, "Missus says come over to the house." I went over there, and she was just a little bit embarrassed, but she said, "Mr. Moore, I have decided if honeylocust was good for the goose it was good for the gander, so I have been feeding honeylocust to my cows." And she went on with that story and said that she had been selling milk to a fraternity over in town, and the boys at the fraternity, after she had fed the cows honeylocust for a week or two, asked her what had happened to her milk, and she told them--she said honestly she was afraid she was going to lose the trade, she thought something bad was wrong with it. She told them, that so far as she knew there wasn't anything. They said, "Have you done anything to it?" "No, we haven't." They said, "Well, it's the best milk we have ever had, and we can tell the difference in the taste." And then she told them what she had done. She wouldn't tell them before.

Now, we have had story after story coming to us to corroborate that. Now, I have never seen with my cow any difference in milk flavor, either good or bad, but my wife can definitely tell, and she is very particular about her butter, because she likes to sell that. I can quit feeding honeylocust a few days, and my wife will say, "How come you quit feeding honeylocust to the cow?" It is that definite.

There are two things I want to mention: The value of a combination of a perennial ground cover with your honeylocust tree, and then I want to mention the fact that honeylocust _planted_ in a pasture will give no benefit whatsoever. You are going to have to grow your honeylocust on the outside, harvest the pods and feed them just like you would corn, or you are going to have to plant your honeylocust on a barren hillside someplace that doesn't grow anything else--and I think honeylocust will grow with a little fertilizer on about the poorest soil you have, the most eroded soil you have, with a little care--then pasture it after your trees are large enough so that the cow won't eat the limbs. There is something about the tree itself that a cow loves. They will chew the bark and chew the limbs right down to the main trunk.

We have tried planting those trees at four years of age, even, in pastures, and we just can't get them to survive. In fact, the cows and the mules in our pasture ate the trees down to the stumps in the wintertime before they ever started putting out leaves in the spring. So it has been a problem. (See Dr. Diller's pasture tree-guard paper in this report.--Ed.)

This value that you can get from growing honeylocust and _Lespedeza sericea_ on the same soil is the same as with honeylocust and alfalfa if you are in the alfalfa belt, or something like that with other perennial legumes. These are the benefits that I think you can get from a combination: In the first place, the soil is completely protected. In the second place, a concentrate and hay can be grown on the same acreage. Third, a good grazing and feeding out program can be maintained. If you plant your honeylocust on a hillside someplace and let the trees get large enough so that the cows won't eat them up, have your ground cover established, by the time that you are ready to pasture it you can put your cattle in. We had this combination, and I think it would have worked out very well if it had not been destroyed. We had our _Lespedeza sericea_ for our summer grazing crop; then we had winter annuals planted in the _Lespedeza sericea_ for our winter grazing, and the honeylocust was the fattening crop or finishing-off crop.

What we had planned to do was turn the cattle in on this last plot about January 1st, let them graze crimson clover, or bur clover, or any other winter ground cover that grows in your section until the _Lespedeza sericea_ came on in the early summer. Then they'd graze the _Lespedeza sericea_ till the honeylocust pods started falling in the fall, and they'd fatten off on the honeylocust, and you'd put them on the market just before the Christmas holidays.

Then fourth, the management cost is very low. Fifth, the weed problems in your pasture are controlled. Sixth, you get maximum production from the soil. You get your grain and your hay from the same piece of land.

Now, that's all that I plan to give on this subject. There may be some questions come up that we can discuss later.

A Member: What is the sugar content?

Mr. Moore: The sugar content of the Calhoun pods is around 38 per cent, in the Millwood about 36 per cent.

A Member: Is it different in the two varieties?

Mr. Moore: Not very much, only about 2 per cent different.

A Member: What spacing do you use in planting?

Mr. Moore: 35 by 35 feet is about the correct spacing.

Mr. Fisher: What is your labor problem? You say this is equal to oats. Can you run a combine over the field and harvest in one operation?

Mr. Moore: This one you don't harvest at all. The cow picks them up off the ground.

A Member: If you had a few hundred trees, would these pods all come on at one time, or you mentioned having somebody shake them off. Can you pick them all up at one time?

Mr. Moore: Yes, you can shake them all off at one time, rake them all up with the rake, take a pitchfork rake, carry them to the barn and throw them in storage in a dry place. You don't have to worry about weevils.

A Member: Store them like hay?

Mr. Moore: Hay or corn. I have some that have been stored for three years, and the weevil gets into the seed, but it doesn't seem to affect it. My cattle like three-year-old pods as well as the new ones--well, they like them better.

Mr. Weber: Do the pods heat up?

Mr. Moore: They won't heat up, if they aren't green.

A Member: What about the protein content?

Mr. Moore: I will give you the analysis for that, the complete analysis of ground honeylocust pods. That might be interesting. Moisture content, 12.47. Ash, 3.14. Crude protein, 8.58. Now, the crude protein has run as high as 14 per cent. I want to bring that out. This was pods collected in the wild, and this was a sample that the State Chemist ran for us on that. Fats 2.12. Fiber, 17.73. Carbohydrates total 55.96.

President Davidson: I am afraid we will have to close this if we are to get on at all. That's the most authoritative information we have ever had, I think, in this Association about honeylocust. I am sure we have been enjoying it and have been benefited by it immensely.

On the possibilities of filbert growing in Virginia, Dr. Overholser will now give you a talk.

[Footnote 18: According to botanical authorities, the honey locust is polygamo-dioecious; that is, it generally has most of its male flowers on one tree and most of the female flowers on another tree, but the trees are not 100 per cent pure in this sex division. In my personal observations of flowers on grafted trees, including Millwood and Calhoun and scores of seedlings, both "male" and "female," I never found any pollen produced in flowers of the "female" trees, but nearly all "male" trees in the Tennessee Valley will have occasional catkins with one or more perfect flowers near their terminal ends (the basal flowers being staminate on the same catkin.) The functionally perfect flowers on such "male" trees have been observed to set from one to many pods in certain years, but such pods are generally small as compared with those borne on "female" trees in the same locality, and I have never observed a heavy pod crop on any "male" tree. Grafted trees of Millwood and Calhoun selections in Tennessee were observed to set pistillate flowers, but no pods (or very few) matured on them unless there was a "male" tree in flower within insect-flight distance from them. (At Auburn, Alabama, there were wild honeylocusts, including "male" trees, within a half-mile of the Hillculture planting of grafted honeylocusts when I saw it in 1943.)

I do not argue that no pollen is ever produced by Millwood or Calhoun flowers some probably is (though its demonstration might require almost microscopic examination, in contrast with the easy finding of pods on "male" trees.) But, in the practical culture of fruiting honeylocusts, and in our present scope of knowledge of their pollination requirements, our plantings should include a handful of seedling (thornless) trees or else some grafted trees of a thornless "male" selection such as the Smith, in a ratio of about 1 Smith to 10, say, of Millwood.

It is unfortunate that the presumed male mutants of the fruiting varieties, reported above by Mr. Moore, were destroyed when the Hillculture plots at Auburn were discontinued. Perhaps similar ones will show up elsewhere, and they will be worth looking for. Meanwhile, the Smith variety (originally propagated through a mixup in scionwood collection), has been demonstrated to be a satisfactory pollinator for Millwood and Calhoun, and it, as grafted, is also a thornless tree. Perhaps any thornless male seedling honeylocust tree, if its flowering period coincides with that of the fruiting variety, might serve equally well.--Note by J. C. McDaniel.]

Possibilities of Filbert Growing in Virginia

E. L. OVERHOLSER, Head, Department of Horticulture, V. P. I., Blacksburg, Virginia

More than four-fifths of the United States filberts are grown in Oregon and nearly all the rest are produced in the State of Washington. Prior to 1933, total filbert production in these two states did not exceed 500 tons, but production has since increased steadily and in 1945 it amounted to 5,320 tons. The value of filbert production in the U. S. in 1945 was about 3 million dollars.[1]

As a wild hazel is native of Virginia and as filberts have been profitably grown, especially in Oregon and Washington the question is often raised as to whether hazelnuts or filberts could not be grown commercially in Virginia. It has been

suggested that if varieties now available are not successfull in Virginia, perhaps new varieties may be originated by crossing, including inter-specific crosses.

+American Species+

AMERICAN HAZEL. As mentioned, one species, _Corylus americana_ Walt., is native to much of Virginia. Its distribution is from the northeastern states and Canada to Saskatchewan and the Dakotas and south to Florida and the Gulf of Mexico. Its adaptation is much wider than that of the beaked hazels (_C. cornuta_ Marsh or _C. roxtrata_ Ait. and the far western _C. californica_) the two other Corylus species native to the United States and Canada. This native _americana_, species appears at least to have value from the point of view of soil conservation, as food for wild life, and for breeding purposes.

The American hazel is a large thicket-forming shrub, which sprouts very freely after cutting, and the foliage is generally dense. It is found growing on dry, well-drained sites, in both sun and shade. It, however, seldom bears fruit in the shade. The shrub is relatively hardy, withstanding mid-winter temperatures of -40 deg. to -30 deg.F. and is easily transplanted.

The nuts are available in the wild from July through September and occasionally persistent on the plant until December or even February. The nuts average about 250 per pound, with a germination of about 80 percent, producing about 60 usable plants per pound of seed.

Three of the best known varieties of _C. americana_ are the Rush from Pennsylvania, the Littlepage from Indiana, and the Winkler (most hardy) from Iowa. [See footnote following.--Ed.][19]

Incidentally, Thomas Jefferson in his list of plants native to Virginia, as published in his _Notes on the State of Virginia_, which was written in 1781, and published in 1782, in 1784-1785, and in 1787, lists among other plants the "Hazelnut (_Corylus Avellana_)", which apparently should have been called _Corylus americana_ Walters.

Breeding Filberts in the East. This brings up the question of filbert breeding in the East. Crane and Wood (1937) have fully reviewed the breeding program with filberts, and the breeding of filberts, for the East may be briefly referred to here. Tho pollen from _C. californica_ and _C. americana_ apparently does not function on the pistillate flowers of European varieties, (_Corylus avellana_ L. and _C. maxima_). Since however, _C. americana_ is useful as a pistillate parent, it is possible that _C. californica_ may be similarly used.

The workers of the United States Bureau of Plant Industry are primarily testing first-generation hybrids resulting from crosses with the pistillate parents Rush,[20] Littlepage, and Winkler of _C. americana_ and pollen from varieties of _C. avellana_ native of Europe, northern Africa, and western Asia, and of _C. maxima_, the filbert of southeastern Europe and western Asia. Other pollen parents were _C. colurna_, (Turkish hazel, native of southeastern Europe and western Asia) and _C. heterophylla_ Fisch., (various leaved hazel from eastern Asia.)

Crane and Wood (1937) suggest that varieties of high merit should be developed for home plantings over much of the region from lower New England and Great Lakes on the north, and to the Potomac and Arkansas Rivers on the south, and that much of Wisconsin, southern Minnesota, South Dakota, and Nebraska might also be included.

Dr. Crane writes, by letter dated July 27, 1948, that he has as a result of breeding work, which was started many years ago, two new varieties that have been placed in the hands of nurserymen for multiplication. These varieties are at the present time carried under the numbers of 1667 and 2336, These are hybrids between the European filbert (_Corylus avellana_ L.) and the native American hazelnut (_C. americana._) At the Plant Industry Station at Beltsville, Maryland, these two varieties have been outstanding in their yielding ability, hardiness, and quality of nuts produced. Dr. Crane does not think, however, that these varieties may very materially change the situation as regards commercial filbert growing in the East and in the South.

Because of the conditions prevailing during the last war, nurserymen have not made as much progress, in propagating these new varieties as had been originally hoped. Dr. Crane plans to release these varieties for extensive plantings just as soon as there are sufficient plants in the hands of the nurserymen to warrant their being called to the attention of the general public.

HILLCULTURE PROJECT. The Department of Horticulture of V.P.I, has what is called a Hillculture project, with Professor R. C. Moore in charge. Among the materials planted in connection with these studies are filbert varieties to determine their possible value on hill farms in the mountainous regions of Southwest Virginia as a source of additional food and supplemental income for such families. The Forestry Division of TVA has co-operated in supplying not only propagated plants of filberts, but also of walnuts and seedlings of chestnuts.

Among the filberts now being grown are six German-named varieties from the Hillculture Division of the Soil Conservation Service, Glenn Dale, Maryland, planted as rooted cuttings in 1941. The German varieties, are as follows: (1) Barr's Spanische; (2) Neue Riesennuss; (3) Fruhe von Fruendorff; (4) Schliesserin; (5) Eckige Barelloner; and (6) Vollkugel.

In addition five varieties, including two of the Jones numbered seedlings from crosses between the American hazel and the European filbert, purchased from the J. F. Jones Nursery[21] of Lancaster, Pa., were planted in 1947. These are the following: (1) Jones 185; (2) Bixby (a Jones hybrid), (3) Cosford, (4) Italian Red; (5) Large Globe and (6) Medium Long.

Seedlings of the American hazel have also been planted. Dr. Crane may be able to send the V. P. I. Department of Horticulture a few plants of his seedlings 1667 and 2336 to include among the variety plantings.

+Some Limitations of Filbert Growing in Virginia+

DISEASES. Possibly the present most serious limitation to commercial production of filberts in Virginia is the Filbert Blight or Black Knot (_Cryptosporella anomala._

(PK) Sacc.). While this fungus results in little damage to native species (_C. americana_) it does spread rapidly and with serious results to European varieties in the State. Possibly the seriousness of the disease has been lessened by the eradication of native hazel plants on roadsides, fence rows, and in the wild nearby, which serve as hosts for the disease.

It is present on the American hazel, but does little damage to the plant. The disease, however, as mentioned, is a serious menace to either European varieties or to the present hybrids resulting from _C. americana_ x _C. avellana_. The control to date is to prune off and burn affected parts. Mr. George Slate has mentioned that Mr. S. H. Graham of Ithaca, New York, has a number of hybrids between _C. americana_ and _C. avellana_ that have been subjected to severe attacks of Filbert Blight and a few of these have to date escaped, although the others have been destroyed by blight.

The bacterial blight present on the Pacific Coast apparently does not occur in the East.

INSECTS. A second limitation is the problem of the attacks of insects. Dodge and Rickett (1948) report that _Corylus_ may be affected by a leaf-damage from the feeding of leaf-hoppers (_Phepsins ishida_; _P. tinctorius_), which may involve less than half the leaf or may extend to the entire leaf. The first leaves to be infested are those next to the ground, which are affected early in July. Most of the damage ceases by the first week of August. Control is by spraying with nicotine sulphate and soap on the undersides of the leaves in late June or early July, repeating at the end of a week.

Certain nut weevils (_Balaninus spp._) attack the native hazels, but Slate (1930) reports they do not attack the European filbert (_C. avellana_). Mr. Slate reports that in Geneva where nuts are carefully picked up they do not have much of a problem with weevils.

Dr. Crane reports that the Japanese beetle severely damages the filbert. While the Japanese beetle has not yet become widely established in Virginia, it undoubtedly will eventually become a problem throughout this state. The Japanese beetle can be destroyed by using four pounds of 50% wettable DDT or two pounds of actual DDT per 100 gallons. Such sprays should be applied as the Japanese beetles begin to cause injury, and usually two applications may be sufficient.

Mr. G. F. Gravatt has reported that his filbert plantings, surrounded on three sides by woods, are badly attacked by stink bugs that sting the nuts. DDT as suggested for Japanese beetles may also be used for stink bugs.

Another serious insect pest on hazelnut is the curculio. Clean cultivation has been reported as a supplementary measure for curculio control, as they depend, upon unbroken soil in the fall for their metamorphosis. Some hybrids are reported as being relatively immune to the attacks of curculio (Weschcke, 1946). Benezene hexachloride has shown promise with other plants in curculio control and may have possibilities on the filbert.

LACK OF HARDINESS. A third limitation has been lack of hardiness in the case of European varieties. With the European varieties the staminate or the pistillate

flowers or both are likely to be killed by winter temperatures. In fact, occasional unduly low winter temperatures may kill the tree tops or even the tree trunks to the ground. The Winkler variety (_C. americana_) has been reported as more hardy in New York State than the Barcelona (_C. avellana_) or the Jones hybrids (_C. americana_ x. _C. avellana_) (Ross Pier Wright, 1944).

Under western New York conditions, Slate (1930) reported that the blooming period starts about March 20 to 25 at Geneva, and lasts about a month. In central Virginia this may well be several weeks earlier. Slate (1930) also reports that the flowers in bloom will withstand considerable frost, and that even with temperatures of 16 deg.F. during the blooming season, neither female nor male flowers, may be injured. Nevertheless, with filberts coming into bloom in late February to early March, they would be subjected to temperatures that might result in injury especially to the catkins.

Some of the more hardy varieties as reported by Slate (1930) include the following: (a) White Lambert (not of value) (_C. maxima_); (b) Red Lambert (_C. maxima_); (c) Cosford; (d) Purple Aveline (_C. avellana_); and (e) Early Globe (of little value).

Some of the varieties upon which both the staminate and pistillate flowers tend to bloom relatively late are (a) Althaldensleber, (b) Kentish Cob, (c) Red Aveline, (d) Purple Aveline, and (e) Bolwiller. Late blooming, however, does not necessarily insure escaping injury from low spring temperatures. The Cosford, Italian Red, and Medium Long are considered by Slate as good for New York. The Bixby and Buchanan are the result of crossing _C. americana_ x _C. avellana_, and appear to be of promise for home plantings in the East. Mr. H. F. Stoke is growing the Italian Red and Du Chilly (Kentish Cob) with Daviana for pollination purposes in the Roanoke area.

CROSS-POLLINATION. A fourth limitation is the fact that varieties are nearly entirely if not fully dependent upon cross-pollinization by other inter-fertile varieties that bloom at about the same time in order to insure a set of nuts. This limitation may be overcome by the proper planning of hardy varieties are inter-fertile. Colby (1944) has reported that the Winkler variety is self-fertile.

SUCKERS. A sixth limitation is the tendency of the _C. avellana_ or _C. maxima_ to sprout about the base and the labor and expense of keeping these sprouts pruned out. It is possible that this factor may be overcome by using Turkish hazel (_C. colurna_ L.) as an understock and grafting or budding thereon the varieties that sprout when on their own roots. The Turkish hazel does not sprout as badly as the two other species.

Note by Editor: An Oregon nursery, which formerly propagated European filberts on the Turkish understock, now has abandoned its use. The grafted filbert tops did not seem to survive and bear as consistently as those on their own roots, after a period of several years in orchards.

PLANTING IN VIRGINIA. In a letter dated May 17, 1948, addressed to R. C. Moore, Assistant Horticulturist, V.P.I., H. J. Pettit, Assistant Secretary of the Planters Peanut Company, Suffolk, Virginia, reported that some years ago they planted several thousand trees of filberts, which they obtained from the states of New York and

Oregon. From their experience it appears that late spring frosts destroyed the flower parts, which developed early, with the result that the yields were too low to be profitable. Hence, the filberts were removed and the land otherwise utilized. Mr. H. F. Stoke, however, in the Roanoke area has not found lack of hardiness as serious as the problems of diseases and insects of filberts.

An important nursery in Maryland has provided information to the effect that during this past 1947-48 season it sold for planting in Virginia a total of 34 filbert plants in lots of from one to ten. Its 1947-48 catalogue lists varieties of filberts for sale as follows: Barcelona, Daviana, Du Chilly, and American hazel.

Dr. H. L. Crane, Principal Horticulturist of the USDA, writes in a letter dated July 27, 1948, that he knows of no substantial plantings being made anywhere in Virginia. He has observed a few bushes or trees scattered about the homesteads, particularly in the northern or more mountainous part of the state. In most cases the performance of these filberts has not been entirely satisfactory because of leaf scorch during the summer, due apparently to high temperatures or unfavorable moisture conditions or to the winter killing of the catkins, or in some cases winter injury of the shoots. The largest plantings in Virginia that have yet come to the attention of the V.P.I. Department of Horticulture are those of Mr. Stoke in the Roanoke area.

Dr. Crane has observed the planting of a few bushes of the American hazelnut in Virginia. Their performance has been somewhat better than has been that of the European filbert, especially as to hardiness, and these American hazelnuts have borne more satisfactory crops of nuts than have the European filberts. The nuts produced by the native varieties, however, are small in size, thick shelled, and the kernels are small and lack quality. Observations by Dr. Crane, which have been made in the State of Virginia, lead him to believe that with the material that is at present available from nurserymen, there is not much hope of successful commercial filbert culture in the State of Virginia. When, however, seedlings 1667 and 2336 may become available, two varieties that are hardy and productive of fairly high quality nuts may provide material for home plantings or for local markets.

Ornamental Value. The filbert, however, also has possible value for ornamental plantings with its attractive foliage, or as a hedge, as well as for nut production, providing the home owner will control insects and diseases and maintain favorable growing conditions for our best known varieties.

Future Outlook in Virginia. With a further breeding program to combine the hardiness of the American hazel and its tolerance to Filbert Blight with some of the better qualities of the European and other species to obtain self-fertile varieties better adapted to Virginia conditions and with the better insecticides and fungicides now becoming available for insect and disease control, it may be that filbert growing in Virginia has a brighter future outlook than now appears to be true.

[Footnote 19: Tree Nuts, Acreage, Production, Farm Disposition, Value, and Utilization of Sales, 1909-45. USDA Bureau of Agr. Eco. Crop Rept. Brd.: 1-25 Oct. 1947.]

[Footnote 20: Rush, itself, is now considered a natural hybrid of American and European filberts. Many of the European varieties are derived from hybrids between +C. avellana+, +C. maxima+, and possibly other Eurasian species.--Ed.]

[Footnote 21: Now located at Erie, Ill.--Ed.]

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President Davidson: Thank you, Dr. Overholser. We have a paper from Mr. Elliott. Mr. Elliott is not here, but we are already behind our program, so I am afraid you are going to have to have that in printed form later on.

Filberts for Food and Looks in Kentucky

N. R. ELLIOTT, Extension Landscape Specialist, Department of Horticulture, University of Kentucky

Those of us interested in the landscape phase are always thinking of as many different kinds of plants as possible that may be used to create pleasing effects. Perhaps we might be criticized for overlooking several plants that would not only assist in creating pleasing effects but at the same time produce edible fruits of good quality. In my own experience I have often recommended the use of grape vines on a trellis to create a screen and at the same time produce fruit. Also in border plantings, like the shrub border, the gooseberries and currants make attractive shrubs and in addition supply fruits. In making these suggestions for plantings one needs to depart somewhat from the usual run of plants and in most instances the homeowner has never thought of using plants for effects as well as fruits.

+Filberts Good Dual Purpose Plants+

Filberts are certainly outstanding dual purpose plants, and I feel that they have not been used nearly as much as they should be. If we think of landscape from the broad point of view, we realize that screen or border plantings make up one of the most important parts of the set-up, especially in rural parts. Practically every farm home has some unattractive view near by that needs to be screened out, either partially or entirely. This view may be caused by a lot where farm animals are kept, an old, unattractive barn, or even a gullied field. Lots where animals are kept and the barn are necessary parts of the farm operations, and the gullied field may result from neglect, but regardless of the cause for the undesirable view it can and should be screened from view from the home.

In making a screen planting, two plans are possible--one, the shrub border, and the other the hedge row, and filberts are excellent to use in either planting. Where space is at a premium, the hedge offers the best form of screen. Filberts planted two and a half feet apart and pruned in such a way as to make them have a shrub appearance will make an ideal hedge and produce lots of nuts of good quality. This hedge can be counted on to be effective up to twelve feet in height.

In the shrub border filberts are allowed to produce many stems and to grow into small trees. This is done by pruning and by using groups of two or three plants in a place, planted some five or six feet apart. Different varieties may be used for different

groups, thereby producing a variation of foliage. The filberts will take their places with the well known small trees like the dogwood and the redbud, when used in this way.

Still another use for filberts in landscape work is to use them for small trees as lawn specimen plants. They have a size, shape, and foliage that makes them attractive when used in this way.

+Cross Pollination Necessary+

Our experience has been that there is need for cross pollination to get maximum yields of fruit; therefore, we suggest that different varieties be used in a planting. Barcelona, DuChilly, and the Jones Hybrids seem to us well suited for this. Of course, there are others, but our experience with varieties is limited.

When it comes to the soil for filberts, we find that a fairly rich soil that has plenty of moisture is the best. Of course, the soil must drain well because the roots of filberts seem to be very susceptible to poorly drained soil conditions. If there is a lot of sand in the soil, give the filberts more moisture and food because they are rapid growers.

So far, we have not had many complaints about filberts suffering from winter injury. This may be due to the fact that so far Kentucky is not using great quantities of these plants, or it may be due to the fact that the varieties used have been reasonably hardy. The little winter injury seen so far has been in the terminal twig growth, and removal of these twigs in the spring has not meant altering the normal shape of the plant.

I do not know whether there is any significance to it but the filberts that have been fed by using well rotted manure applied in the fall and spaded into the top four inches of soil next spring have made the best growth and produced the most fruit.

So far the filberts that we have had experience with have been free from insects and diseases. One never knows how long that condition will last.

Now, when it comes to discussing filberts as a food, all that I want to say is that at Christmas time when you buy mixed nuts you usually get a few of the filberts in the mixture. These nuts are good eating, and when the plants are grown on the home grounds everyone who has them says they are much enjoyed by all members of the family. Our experience has been that filberts yield annually and, if given reasonable care, in good amounts.

In conclusion we would like to say we feel there is not only a place for filberts in landscape work, but there is an absolute need for greater use of these plants especially in rural plantings. At present, the professional landscape artists are not inclined to recommend them as often as they could, simply because they have not been trained to think of dual purpose plants. Greater publicity as to the value of these plants would undoubtedly mean greater use of them.

* * * * *

President Davidson: We also have a paper from Mr. Reed, which is of quite a good deal of importance historically on the work of Mr. Jones. I wish you could have that. Probably you will have to read that, too.

J. F. Jones, Introducer of Many Nut Varieties

CLARENCE A. REED, Collaborator[22]

The name of J. F. Jones was once one of the best known and most highly respected in eastern nut culture. It was from Mountain Grove, Wright County, Mo., that he was first heard from in 1900, when he discovered and introduced the Rockville hican, which he named after the nearest town. It never proved of value, but that fact did not detract from the importance of being first, a habit which remained with him till his death. In 1902 he moved to Monticello, Jefferson County, Florida; five years later he moved to Jeanerette, Iberia Parish, Louisiana; and in 1912, he moved to Lancaster, Pennsylvania, where he died in January, 1928.

[Illustration]

In 1903, while at Monticello, he successfully graft-propagated the Rush Persian (English) walnut and the Weiker hickory, an intermediate form between shagbark and shellbark. Both were from Lancaster County, and he used scions sent him by J. G. Rush, of West Willow, south of Lancaster. Mr. Rush is credited with introducing the walnut bearing his name, while credit went to Mr. Jones for the Weiker hickory. Some years later, on two occasions, Mr. Jones took a visitor to the Weiker parent tree when the branches were laden with nuts so that they hung down in a manner suggestive of plums. For some reason, never explained, no other tree of the variety, so far as is known, ever bore as much as a quart of nuts, although the trees frequently flowered profusely. The variety was, however, markedly dichogamous. The parent tree, which stood in the yard of Mr. Christ LeFever of Lampeter, about two miles east of the Jones home, was blown over in a heavy gale many years ago.

Mr. Jones graft-propagated a considerable number of Hales shagbark while at Monticello, with scions that came from the original tree near Ridgewood, New Jersey. However, this variety was first propagated by Henry Hales of Ridgewood, in 1879. He also had Kirtland from Yalesville, Connecticut, but like many others since that time, both it and Hales proved to be light bearers. Other hickories may have been propagated by Mr. Jones while at Monticello but these are the only ones of which there is record. The Kirtland was first propagated in 1897.

[Footnote 22: U.S.D.A. Bureau of Plant Industry, Soils and Agricultural Engineering, Plant Industry Station, Beltsville, Md.]

+First Carload of Grafted Pecan Trees+

When he went to Florida, there were few pecan trees of bearing age in either that State or Georgia and none to speak of in the Carolinas. The "fast" trains went no more than 30 or 35 miles an hour, and a minimum of three days was required to see even an occasional planting or a single tree. Within the next few years, nurserymen everywhere propagated their own varieties and listed them in their catalogs. Mr. Jones was discriminating and propagated only varieties that then had the best

reputation, such as Schley and Stuart, and some others that have not stood the test of time. In one way, he was distinctively first; he shipped the first carload of pecan trees ever to go to one address. This was in January of 1906, when 10,000 trees were shipped to Professor H. E. Van Deman who was then establishing a 900-acre orchard near Ferriday, La. A picture of the car appeared in the American Nut Journal, published by W. N. Roper, Petersburg, Va., Vol. III, No. 50, March 1906, (Van Deman had been the first Pomologist in the Department of Agriculture, 1886 to 1892).

Mr. Ray Simpson of Vincennes, Ind., went to Mr. Jones to learn how to graft pecan trees. He offered to work without pay if Mr. Jones would teach him the art. He had graduated at Cornell in 1905, and had been inspired by John Craig, Professor of Horticulture there. Craig himself later invested somewhat heavily in pecan orchards both near Monticello and at Albany, Georgia. Mr. Simpson was taken on and proved as good a propagator as the best hand and received the same pay.

While at Monticello Mr. Jones began to feel that the region might not be the best place for pecans. Perhaps he had made a mistake. It was 300 miles to middle western Alabama, where there were the nearest native trees. A disease was appearing among many of the trees planted in the East, which was then poorly understood (rosette). Pecan wood for budding and grafting was scarce and Mr. Jones would trust no one to cut it for him. He went to the trees himself.

One man who then had an abundance of wood and who could be relied upon was B. M. Young of Morgan City, La., and Mr. Jones went to him for wood several times. Once he became confused as to the trees from which he had cut a couple of bundles, so both were thrown in the river and he went back for more. Mr. Young was greatly impressed, so much so that he remembered the incident, as we shall see.

+The Move to Louisiana+

Back in Florida Ray Simpson wished to buy and Mr. Jones wished to sell, so a deal was soon made. Mr. Jones went to Louisiana where the pecan is native and there were many large trees, probably as many as could be found in any one place in the entire South. Mr. Young knew of a group from St. Paul, Minnesota, who were about to buy and plant a thousand acres near Jeanerette and who were looking for an experienced man to take charge. Mr. Jones was recommended and was soon at work. For another five years, he worked harder than almost any other white man in the State. Great odds were against him. Being from the North, he did not associate exclusively with whites, and presently the southern white people left him severely alone. That was not all; he could not raise as good nursery trees as he had in Florida. The trees grew slowly in the cold, heavy soil of Louisiana, and the fibrous root system failed to materialize. The excellent reputation he and his trees had enjoyed in Monticello began to deteriorate. He worked harder than ever and waited for a break. When it came, he did not hesitate.

+Jones Shifts to Pennsylvania+

The St. Paul crowd fell into a squabble and divided into two factions, each wishing control. A man went south to see if Mr. Jones would sell his stock. Would he? He knew when to keep his mouth shut and he meekly made a deal. He was probably

never more glad over anything in his life. He came north, lock, stock, and barrel. But he was far from being without a place to land. Since his Monticello days, he and Mr. Rush had been good friends. Mr. Rush knew a farm of 20 acres with buildings, which could be had for $8,000. It was four miles south of Lancaster, and at a point where two main highways leading into the city came together. It sloped eastward enough so that it did not get the full force of west winds. It was two miles from Mr. Rush's home, with the town of Willow Street between.

Mr. Jones then began eight or 10 years of lean hard work. He modernized buildings, planted an orchard of nut varieties most of which were purchased from W. C. Reed of Vincennes, Ind., and W. N. Roper of Petersburg, Va. From Roper he bought both seedling and grafted trees. Some of the "seedlings" had been budded and then not cut back to force the buds. The latter were still dormant and when the trees were properly cut back, the buds pushed forth. T. P. Littlepage, of Washington, and Prof. W. N. Hutt, of Raleigh, N. C., had a good laugh at Roper, but as the trees bore no labels, they were no more valuable than seedlings and were treated as such. All three men are now deceased.

Thomas black walnut trees came from E. A. Riehl, Godfrey, Illinois. The variety had originated in eastern Pennsylvania and was first grafted in 1881 by J. W. Thomas and Son, at King of Prussia, Pa. The parent tree had been destroyed some time before by the Pennsylvania Railroad, in extending its lines. The Thomas is today the most widely planted variety, although it has rarely borne well. Mr. Jones selected and grafted the Ohio walnut, but the owner of the seed-parent tree was given credit for its introduction, although she probably knows nothing of the incident, to this day. She was a Miss Clark, McCutcheonville, Ohio, and it was felt that it would help more to give her name as originator if one were ever to locate the tree.

[See Ohio black walnut original tree photos, NNGA Rept., 1946.--Ed.]

The Stabler eastern black walnut, introduced in 1916 by Mr. T. P. Littlepage by means of a paragraph inserted in the _Country Gentleman_, was also propagated by Mr. Jones, but he early found it disappointing in its habits of bearing. He also found that about 80 percent of the nuts from the parent tree had single kernels, while with young trees 80 percent had double kernels. Most planters have long since discontinued using this variety. However, Mrs. Jane Baum, Douglassville, Pa., reports that her customers like the Stabler best. Others she has are Thomas, Ohio, and Ten Eyck.

Other varieties were tested by Mr. Jones, but he pushed none of them, rightly thinking that 4 leaders were as many as a nursery could afford to carry. He insisted that a new variety would have to prove its superiority before he would insert it in his catalog. Among other varieties was the Peanut from southern Ohio, the nut of which had single lobes; but apparently there was some mistake along the line, as nuts from grafted trees were indifferent and had 2 half kernels. He also had Creitz from Indiana, which Mr. H. F. Stoke, 1436 Watts Avenue, Roanoke, Va., thinks well of at this time. It was a prize winner in the 1926 contest of the NNGA. Neither Creitz nor Peanut was a Jones introduction.

+His Work with Hickories+

Among the hickories, there was the Stanley from Indiana in 1916, which was quite a favorite with Mr. Jones for some time. But did any one ever see a shellbark that bore well and filled the nuts? Shellbark trees are beautiful to look at, have enormous leaves, seven to nine leaflets, but they leaf out early in spring and the flowers are frequently killed back by spring frosts. Part of its flowers are killed outright with too great frequency for it to be worth growing for the nuts. These are very large, the hulls split entirely to the base, and what kernel there is, is of sugar-like sweetness. The shells are mostly thick and the kernels seldom well-filled.

The Glover shagbark hickory, from Connecticut, which was introduced by Mr. Jones in 1918, is undoubtedly one of the best shagbarks yet propagated. The nuts are of medium size and shell thickness. The flavor is very good. Most shagbarks have five leaflets; this one has seven quite as often, and the leaf is about a foot long.

There were other hybrids, or what are supposed to be hybrids. The Pleas hickory, introduced in 1916, was perhaps first successfully grafted by Mr. Jones, but credit for introduction went to the owner of the parent tree, Dr. E. Pleas, Collinsville, Oklahoma. It was a beautiful tree, shapely, with an air of considerable refinement, making it a graceful lawn tree. It bore fairly well, although not heavily. The nuts were thin-shelled and also had thin hulls that split entirely to the base. So far as most laymen are concerned, the Pleas may be but an edible, or semi-edible bitternut. On the grounds of the Plant Industry Station, at Beltsville, Md., there were once two trees of Pleas, but they were given to the Wild Life Service for planting 10 miles away, although there are many native bitternut trees just over the line fence in neighboring woods. We fancied that we could detect bitternut flavor in good shagbarks about the plantings, due to xenia influence, as in the case of chestnuts.

Burlington was another hican first propagated by Mr. Jones, in 1915. It came from eastern Iowa, and for a time was confused with Marquardt, which never was propagated, or apparently not. Burlington makes a fine appearing tree and serves well for ornamental purposes. It bears fairly well while young, but soon develops faulty nuts, few being well-filled and the majority weevil infested. It is also subject to shuck-worm and twig girdler injury.

Mr. Jones once wrote that he had given up with the hickories "in disgust." So far as is known, he never used any stock for hickories other than pecan, which grew well, made good unions and generally outgrew the scions. John Hershey, however, says this is not a good combination, but there are too many trees of Jones' propagation about the country, to accept Hershey's verdict altogether. Carl Weschcke[23], of St. Paul, uses bitternut largely or entirely; if it is a mistake, it will be expensive. Hickories are slow to grow and one gets too few nuts at best. It takes a lifetime to get even small crops, and for our part, we want no bitternuts on the place. Too often shagbarks fail to unite with bitternut and frequently they are short-lived.

In 1916 Mr. Jones propagated and introduced the Beaver hickory, from central Pennsylvania, a supposed bitternut-shagbark cross. It proved of little value and soon disappeared. The Fairbanks from northeast Iowa, a similar cross, was introduced the

same year. It was one of the prettiest of all hybrids and stood up about the longest, but it had too much bitterness in the pellicle encasing the kernel and was much subject to weevil injury.

+Efforts with Persian Walnuts+

Many varieties of Persian (English) walnut were propagated and brought into bearing. Mr. Jones included a majority of the varieties brought into the country from France by Felix Gillet, of Nevada City, Calif., as early as 1870. There were Franquette, Mayette, Meylan, Parisienne, and a cutleaf variety which appears to have had no other name. A California variety of which he thought well for a number of years was Eureka, a western introduction of 1908. He propagated a number of eastern varieties such as Lancaster (Alpine) in 1913, although credit went to Mr. Rush; Boston, from Massachusetts, also in 1913; Ontario, from Canada, in 1914; and probably others. He obtained Chinese walnuts, from P. Wang, Kinsan Arboretum, Shanghai, and sold seedlings at wholesale. These were an Asiatic form of _Juglans regia_. He limed the soil, and thought the effects were beneficial. In this he was warmly supported by T. P. Littlepage and more recently by growers in Northern Ohio; but lately liming has not been found beneficial in Italy. All in all, however, the Persian walnut was not particularly dependable, and during the last few years the nursery which he left discontinued selling Persian walnut trees. In the East, the trees of older varieties usually were little more than interesting novelties.

+He Tried the Chinese Chestnut+

The Chinese chestnut was tried for a few years; but as so often happens with this species, nursery trees died badly in winter and Mr. Jones thought it due to blight, a disease which was then sweeping his part of the country, taking its mortal toll of both American and European species. However, blight does not seriously attack young trees and it is more likely that death was caused by a combination of summer drouth and winter cold; but no matter, the trees perished and the result was the same.

+First Heartnut Grafts+

Mr. Jones tried the butternut and there is still one tree in the experimental planting east of the residence. It is Aiken, from New England, and was first propagated by him in 1918. It proved disappointing. He grafted the first heartnut ever grafted of any kind insofar as is known, the Lancaster, in 1918. The only other heartnut for which he received full credit for first propagation was Faust, obtained from a dentist, Dr. 0. D. Faust, Bamberg, S. C., in 1918. Others that he was doubtless first to propagate, but for which credit went to the owners of the parent trees, were Bates and Stranger in 1919, both from R. Bates, Jackson, Aiken County, S. C., and Ritchie, a Virginia variety found by John W. Ritchie of Flemington, N. J., in 1918.

However, heartnuts are seldom heavy bearers and the trees do not grow large or live long. In Japan the wood is sometimes used for gunstocks but only because better material is unavailable. Heartnuts have practically no market where other kinds of nuts can be had and the trees are much subject to "bunch" disease. To an enormous extent the trees have been sold to unsuspecting people of the South and East as "English" walnuts.

[Footnote 23: See Weschcke's paper, elsewhere in this report.--Ed.]

+The Filbert+

Mr. Jones had a tree or two of the Turkish filbert, a species sometimes reaching a height of 60 feet and attaining a trunk diameter of three feet or more. Bixby found the species hardy in central New Hampshire. Mr. Jones obtained his seed from three trees in Highland Park, Rochester, New York, which are believed to be the oldest in the country. In some years, the Rochester trees bear freely, while in others there is not a nut. This is a valuable ornamental species, as it is green from early spring till the last thing in fall; specimens must be selected for such use, as often the trees are unshapely. Like all filberts, they are subject to Japanese beetle attack and must be sprayed or otherwise protected in beetle infested zones. Filbert foliage may be destroyed by these insects as many as three times in a summer and the trees die down to the ground. The nuts are too small to be of value; but the wood is white, very hard, and makes good turned articles.

+His Greatest Contribution+

It was with the filbert that Mr. Jones made his greatest contribution to nut culture. In 1917 he tried crossing European varieties with pollen of the native Rush. There were no results, and he tried again in 1918 with no better luck. In 1919 he reversed the order of crossing and nearly every nut set. He had discovered that native pollen was not effective on European stigmas, but that the reciprocal cross worked. By 1924 he had a fine lot of fruiting plants. The great majority were of no value, but his No. 200 apparently was well worth while. It was named Bixby in 1937, four years after another seedling, No. 91, had been named Buchanan. The explanation of this belated selection is that the soil about the Bixby tree had so eroded that the tree was starved for a time; but with a couple of years of heavy application of stable manure, it came back, so much so that it is now considered the better of the two. Both are rather small as compared with the large filberts of the Pacific Northwest; but when fully mature, they are sweet and agreeable.

After Mr. Jones was gone, the place was managed by his daughter, Miss Mildred Jones. She kept plants of her father's filbert varieties and the best of the crosses. The latter are now called the Mildred filberts, a name applied in _Standardized Plant Names_ to the entire group of crosses between Rush American and any European filbert. Mr. Jones hoped to have these called after himself but there was an old variety of Jones "hazel" and so his own name could not be used. He once sent specimens to Dr. C. S. Sargent of Arnold Arboretum and somehow gained the impression that the name Jones was given to the cross. Later, however, Sargent's successor, Mr. Alfred Rehder wrote that Sargent had not used the name in either correspondence or on specimens placed in the herbarium.

The example of Mr. Jones in breeding filberts has since been followed by others, as the Department of Agriculture, the New York State Agricultural Experiment Station at Geneva, and. Mr. Carl Weschcke of St. Paul, Minnesota. The last has copyrighted his crosses under the designation "hazilbert," which is a good name; but with the issue of _Standardized Plant Names_ in 1942, the name "hazel" was dropped for all

members of the family. For a time, an effort was made to distinguish between the two by calling small-fruited ones "hazels" and those with large fruits "filberts," but there is not exact dividing line and so now all are called filberts.

Buchanan and Bixby are the only varieties of Mildred filberts thus far fully released by anyone and although neither variety is entirely hardy in the northernmost parts of the country, they do well as far south as eastern Tennessee. The nuts of both are too small to compete in the market with the large filberts of Oregon and Washington, but that is not the purpose for which they have been bred. It is for home planting, a use for which they are admirably adapted. Neither variety should be judged until after they have cured fully, at least a month or more. Then the flavor is excellent.

Of the various introductions made by Mr. Jones, the ones most likely to endure are the Ohio black walnut, the Glover shagbark hickory, and the Mildred filberts. The first has already lasted 32 years; the second 30 years; and the Mildred filberts are only nicely started.[24]

[Footnote 24: Except for the last two paragraphs, this paper was read and approved by Miss Mildred Jones in Pavilion, N. Y., on September 2, 1948. The following day, or September 3, she became Mrs. Wesley Langdoc, of P. O. Box 126, Erie, Illinois.]

+Mr. Reed Comments on Seedling Trees+

+Editor's Note:+ The next two paragraphs should be read in connection with the "Round Table" on chestnut problems, elsewhere in this volume.

In a broad sense, it must be remembered that every variety of seedling tree, of any species and every hybrid form that has ever been planted, or grafted on another tree, has been worth something. This is still a free country and every man has the inalienable right to plant whatever he pleases. Even the hybrids of various forms, hickory, walnut, and chestnut, are all worth something. All are trees and it is better to plant a poor kind of tree than not to plant anything, particularly if it is a nut tree. Whatever prompts a man to plant a tree is worth while.

Hybrid chestnuts bred by crossing Chinese chestnuts of unknown performance record as to habit of bearing, size or flavor of nut, shape of tree, resistance to blight, or spring freezes, and other characteristics which combine to make good nuts, with the inferior and largely inedible Japanese chestnuts, are unlikely to do the damage to the industry that is sometimes predicted. They are now so mixed up that few will be planted by themselves, and there is considerable evidence that the xenia influence of good Chinese chestnuts with which the trees are being planted will render nuts from these hybrid trees fit to market and eat.

[Illustration: MILDRED AND WESLEY LANGDOC]

President Davidson: The value of nut trees in Tennessee, then, will be discussed by Mr. F. S. Chance of the Tennessee Agricultural Experiment Station in Knoxville.

The Value of Nut Trees in Tennessee

F. S. CHANCE, Vice-Director, Tennessee Agricultural Experiment Station, Knoxville, Tennessee.

Mr. Chance: Mr. President, ladies and gentlemen: As a representative of the Agricultural Experiment Station of the University of Tennessee I want to say it is a great honor to have this distinguished group meet here in Tennessee, especially on the banks of beautiful Norris Lake, which is one of the tributaries to the dammedest river in the country. We are something like 600 miles from the Gilbertsville Dam, or Kentucky Dam near Paducah, Kentucky, and this area here is the beginning of a chain of lakes that run just about that far.

For those of you who are from a distance, you may know that in making a chain of lakes out of this great Tennessee Valley that we covered up lots of good land. We have developed lots of good power. Now, I am not just sure why I was put on this program, because, really I am not a nut tree specialist, as I see most of you people are. I will admit that I have been associating with experimentation for the last eight or ten years and have become slightly nutty, but really my big interest is timber. I am still a blockhead. So in discussing and talking with you this morning for a few minutes about the value of nut trees in Tennessee I want you to just keep in the back of your minds that the thing in the timber world that I think is the prettiest when it comes to furniture is black walnut.

So in some plantings that we made several years ago with the help of Spencer Chase at our various substations and at the parent station at Knoxville, when we began to prune those trees I wanted to go to pruning for timber and he wanted to go to pruning for nuts. He won. So as we developed these plantings we are sure that we are going to have some very excellent nut trees.

Tennessee ranging in altitude from something over a mile high down to some 300 or 350 feet at Memphis on the Mississippi gives us a very, very wide range of climate. This wide range of climate gives us the possibility of growing a very wide range of timber trees. A great part of that area is soil from a limestone formation. Nearly all parts of Tennessee are well adapted to the production of the black walnut. The tree as a nut tree has not in the past been looked at with such great interest. However, there are farms in Tennessee that have been purchased with walnut kernels. Over the period of years, why, thrifty families, especially in Eastern Tennessee sections, have gathered up the walnuts in the neighborhood round about, cracked them and sold the kernels and from year to year made certain accumulations of that kind, funds, and saved them with enough in the bank or in the sock to buy a farm. I knew one particular person who bought a nice farm in just that way.

Now, a great many of the people in the same neighborhood did not save their walnuts. These walnuts were gathered from everybody's trees without any objection on the part of anyone. But it was a means of those people getting ahead with their savings from their other farming operations, and this wintertime work that they could put in, why, that kind of thrift is the kind that gets people ahead who want to get ahead and have vision.

I might say a few words about pecans in Tennessee. We have throughout the state quite a few scattered native pecans that are used, especially in all except the more western sections of the state. As a whole they are for home use. Now, in the extreme western section of the state we have a certain amount of seedling pecans, mostly, that produce a considerable income to a limited number of people. In the 1945 census something over 4,000 farms reported some income from pecans--this was mostly in the western section of the state--the value of which was something over $32,000, which at the present time would be a considerable under-valuation.

This tree is found, I might say, throughout the state. I recall a few years ago coming off of the Cumberland Plateau down in Warren County into the cove there around Viola and seeing a beautiful grove of pecans along a stream. I hadn't been through that country before, but I had known a family that lived there, and I stopped at a house to see just what those pecans meant. And there was an old lady on the porch who owned the property, and I asked her some questions about it, and she told me how they got there and knew when they were not there. She had been raised on that place but she said, "I want to show you something." So I went with her around the side yard into the back yard, and she had a couple of pecan trees there that were loaded with pecans until the limbs were hanging over just like pear tree limbs, heavily loaded pear tree limbs. I said, "My, what a crop of pecans you have here. That's really wonderful." Those were the budded pecans, the type that is grown farther south of us. She said, "Just wait a minute, now. I don't know whether I have any pecans or not." I said, "What do you mean?" She said, "If the frost is two weeks later than usual we will have a wonderful pecan crop, if we have a late frost. If we have an early frost we don't have any pecans."

It was quite interesting to me to see that wonderful crop hanging on the tree and yet she wasn't at all assured that anything of value would come from it.

We have on some of our holdings at the University experimental Stations some wonderful Chinese chestnut trees. I can't get overly excited over them, remembering the chestnut as we had it once in Tennessee with the long, slender body, wonderful telephone poles and wonderful timber of other kinds, and to see that a tremendous economic loss has come to this country through disease that was and probably is not controllable. But from the nut standpoint we have at the present time some trees that look as though they are going to be the equal of our own native chestnut that covered Tennessee from the mountain top to the river bank. So we are very much in hope that again Tennessee will have a supply of chestnuts which will be equivalent, probably, to the harvest of chestnuts we once had. However, that's going to be many, many years off.

From the experimental standpoint I have been very much interested in the timber type of tree, hoping that our native chestnut trees, at least one out of the billions, maybe would prove to be resistant. However, watching these growths come up from time to time and attain an age sufficient to produce nuts and then have my hopes blighted by going back the next year and finding that the tree was blighted has become rather discouraging. I hope that some of you people will find just such a tree, one that will bear an excellent nut and at the same time produce excellent timber.

Now I am coming to our big asset in the way of nut trees in Tennessee, as I see it. I was rather interested here in Professor Moore's discussion of the honeylocust, that detestable tree which was such a thorn in my flesh as a child, and having heard someone championing it with such a story as he had, I have heard everything now. Everybody, though, has a champion. Even my mother loved me, regardless.

Black walnut is, as I said in the beginning, native to all sections of the state, and I think that through the collection of the better yielding or better cracking nuts by the Tennessee Valley Authority we are going to find in this crop a very potent asset to the state of Tennessee through the income from sale of nuts. We have in the state about four cracking plants. One of them is located in Morristown. Down in the basin part of the state where walnuts do particularly well, three others are in the city of Nashville. There were something like 10 million pounds of walnuts in the shell delivered in Nashville this last year, yielding about 1,200,000 pounds of kernels. Now, this is no mean return from a crop which was really just gathered up with very, very little attention given to the planting. It is just one of these free crops, so to speak.

If we were to add to that income the great income which we have been receiving through the years from the sale of timber trees, we would run the value of the black walnut into considerable proportions, with income from the sale of black walnuts in the kernel and in timber.

I see no particular reason why that crop cannot be increased ten, twenty or a hundred fold by just a stimulation of interest in the black walnut. I recall back just previous to World War I, or about that time, there was a tremendous demand, as usual, for black walnut for gun stocks. I happened to be free for a month or so at that time so I could give some attention to the purchasing and delivery of both veneer stock and walnut for gun stocks. It was quite interesting to me as I went over a couple of counties in which I made some purchases, to see that someone in the 40, 50 or 60 years back had had a vision of what the walnut tree would be worth to them on their tracts of land and how we were at that time reaping the harvest of the person who had a vision of the value of the walnut tree. A great many of those trees were trees that had been set or walnuts that had been planted years before by some far-seeing person, and it had gone on without any interruption, probably without the slightest bit of protection, until the time that it was needed and desperately needed for economic purposes.

We have some work going on also in connection with the planting of walnuts in pasture fields. The returns _from the pasture_ in the planting of walnut trees have been just practically the same, maybe a little bit better in favor of the walnuts than where we did not have walnuts in the pasture. This work is being conducted down at the Middle Tennessee Agricultural Experiment Station of the University of Tennessee at Columbia. We are using the walnut tree and also the black locust in this experiment. We don't know what the future of it is going to be, but those walnut trees have grown large enough so that they have had to be thinned to keep them from putting too much shade over the ground.

I made a statement several years ago in the presence of quite a distinguished agronomist or horticulturist that I had never seen a walnut tree growing in the open, whether it was in the blue grass region or outside of a blue grass region that did not have blue grass growing under it. He looked at me askance, and I said, "Do you believe it?" "Well, I don't know," he answered.

So we happened to be coming out of Quincy, Florida, up through southern Georgia outside of the blue grass region, and we were both sitting in the back seat of the car. Our driver drove up to a filling station, and I saw this fellow looked up at a walnut tree over in the yard not very far away, in fact, the next yard to the filling station. I somehow or other sensed what he was thinking. He pushed his door open, got out. I pushed my door open, went around the car and followed him. He walked up to that walnut tree, turned around and said, "Well, it's there." He turned around and walked back.

Now, of course, a condition may prevail in dense shade, where that does not happen in young walnut trees, but I just happened to be right. There is a symbiotic relationship between plants--I don't want to get into that subject--but this one thing I am thinking, and that is that the reason why they were able to get this good grazing from under these walnut trees is that there is a relationship there between those two plants that makes it ideal for the production of pasture grass, and blue grass over a great many of our states is our leading grass.

I might say to the gentleman from Virginia that I had a letter from up there a few days ago. I don't know why they wanted to write to me, wanting to know if the walnut tree was _a legume_. So I presume that that was the reason, that the grass grew very nicely under those trees.

I have taken too much of your valuable time. It certainly has been a pleasure and an honor to be here and talk to you these few minutes. Thank you.

* * * * *

President Davidson: Thank you, Mr. Chance. We will take a short recess.

(Recess taken.)

President Davidson: The meeting will now come to order, please. The embryo development of the black walnut will be illustrated and discussed by Dr. L. H. MacDaniels of Cornell University.

(Paper to appear in next volume.)

Dr. Crane: I was very glad Dr. MacDaniels' paper preceded mine, because it does give you a very much better picture of the development of all of our oily nuts, excepting the filbert and, of course, the almond to some extent. But we take in pecans and the hickories and for the walnuts the situation is quite general.

Now, this paper that I am going to read is one that our staff in nut investigations has been working on for the past twenty or more years, and we feel we know a lot about the growing and the development and filling of nuts. And there is a lot in this paper that I think will be of value to all nut growers regardless of the kind of nuts that

we are trying to grow.

The Development and Filling of Nuts

H. L. CRANE, Principal Horticulturist, United States Department of Agriculture, Agricultural Research Administration, Bureau of Plant Industry, Soils, and Agricultural Engineering, Division of Fruit and Vegetable Crops and Diseases.

All nut growers are confronted with the problems involved in the production of nuts of large size with well filled kernels that are "bright" or light colored. Unsatisfactory development and filling of the kernels is more often a cause of complaint by growers than any other single factor affecting nut production. This is because all of our commercial nuts now sold in the shell are priced on a basis of size and the degree to which they are filled. The size and degree of filling of the nuts varies not only from year to year, but from district to district, orchard to orchard, and even in the same orchard, because nuts of one variety may fill well and those of another poorly. This is true even though the kind and variety of nut being produced is grown in a locality usually having suitable climatic conditions for normal nut production.

+Climatic Conditions+

Prevailing climatic conditions in any locality determine how well a particular kind of nut will fill. For example, the pecan is native to the southern part of the United States and a small area in northern Mexico. In its native habitat the summers are long and the day and night temperatures are uniformly high, with little difference between maximum and minimum daily temperatures. When the pecan is grown under conditions of shorter summers, or where there is a marked difference between night and day temperatures, the nuts do not grow to proper size and the kernels fill poorly, if at all. Although pecan trees are quite hardy and may be grown successfully well north of their native limits, the normal development of the nuts and the filling of them cannot be expected there.

Good examples of the climactic effects can be cited. At Davis, California, the pecan tree grows, flowers, and sets fruit satisfactorily, but the nuts fail to grow to proper size, fill poorly, and may not mature before frost. At Davis there is an average length of growing season of 242 days; the day temperatures are high, but the night temperatures are comparatively low. Pecan trees are hardy even in Connecticut, but the trees fail to bear because of the short growing season and the great difference between day and night temperatures. The pecan is truly a hot weather crop and is not suited for culture under mountainous conditions. On the other hand it cannot be grown under subtropical conditions because of insufficient cold during the winter to meet the chilling requirement of the trees. Under such conditions, tree growth starts very late in the spring, and, although the trees may flower, few nuts may set and those that stick may be very poorly filled at harvest if they mature.

The pecan is probably more exacting in regard to its climactic requirements than are our other kinds of nuts, but the filbert or hazlenut is probably a close second in this respect. The filbert, however, represents the opposite extreme in that it does best under conditions of mild winter and moderate summer temperatures. These

differences are pointed out for the reason that many amateur nut growers want to grow certain nuts outside of their native range in places where unsuitable climatic conditions prevail, and they cannot understand why success is not possible.

+Growth and Fruiting Habit of Nut Trees+

Since the growth and fruiting habits of our different kinds of nut trees are closely related, it is desirable to point out some of these relationships. All of our different species of walnuts, the pecan and all hickory nuts, as well as hazelnuts and filberts, are borne terminally on shoots of the current season. In other words all walnut species, pecan, and all hickory species bear the pistillate flowers that develop into nuts at the terminal end of the shoots produced the same year that the nuts mature. The staminate or pollen-producing flowers of all these species arise from lateral buds on shoots that grew the previous year. In the case of hazelnut and filbert the pistillate flowers are borne in lateral buds on shoots of the previous season, as are also the staminate flowers or catkins. In this case, however, the pistillate flowers are formed and pollinated before the current year's shoot growth is made. Almonds are borne laterally on shoots produced the previous season. All chestnuts are borne laterally on shoots produced the same season as the nuts.

The chestnut bears most of the staminate flowers separately in staminate catkins whereas the pistillate flowers are in mixed catkins, but all are formed laterally on shoots of the current season. The almond, which has perfect flowers, produces these in lateral buds on shoots of the previous year. Both the hazelnut and the almond flower before any current-season growth is made, whereas all of the other kinds of nut trees mentioned produce almost all normal shoot growth before flowering occurs. These differences in growth, flowering, and fruiting habits provide a basis for the explanation of why growth of almond trees, for example, is harder to maintain than is that of walnut or pecan. Flowering and early development of the fruit before shoot growth is made tend to check such growth, so that flowering and fruiting trees will not make as much new growth as they would have made had flowering and fruiting been prevented.

In general, it can be stated that, in the case of bearing trees, the longer the shoot growth and the greater its diameter in proportion to length, the greater is the number of pistillate flowers that may be formed at its terminal. Furthermore, the set of nuts and the size that they attain are in proportion to the length and diameter of the shoots bearing them. In other words, the number of flowers formed, the nuts set, and the size that they attain are directly correlated with the vigor and growth of the trees. As trees attain age, fewer long, strong shoots and more short, weak shoots are formed. Hence the average size of the nuts produced decreases because of the reduction in average shoot growth. Furthermore, under normal conditions, the degree to which the nuts are filled is related to the vigor as it is measured by the length and diameter of the shoots bearing them. Strong, vigorous shoots usually produce the best filled and earliest maturing nuts.

+What Is a Nut and of What Does It Consist?+

Webster gives a general definition of a nut as "a fruit consisting of a kernel or seed enclosed in a hard woody or leathery shell that does not open when ripe, as in the hazel, beech, oak, chestnut." Technically speaking, it is a hard, indehiscent, one-seeded dry fruit resulting from a compound ovary. In horticultural language the fruit consists of the hard or leathery nut containing a kernel, together with the husk, hull, or bur that surrounds the nut shell. This kernel consists of the embryo plus the endosperm or its remains. In all of our important nuts, such as walnuts, pecans, hickory nuts, almonds, and filberts, the kernel is essentially the embryo with its thickened cotyledons or seed leaves, as the endosperm has been absorbed except for a thin membrane.

At the beginning of its development, growth of the embryo is slow, and in very early stages it is merely a rounded mass of cells. Later, the meristems of the epicotyl (stem or top) and root axis develop, but the whole embryo is still microscopic in size. Still later the cotyledons (seed leaves) start development from the apical meristem and their growth in length is rapid, but they are very thin and follow the contours of the seed coat. Growth in length of the cotyledons may be arrested by unfavorable nutritional conditions during the time of elongation. In such case, the lobes of the cotyledons may not attain the full length of the seed coat, or pellicle, which surrounds them. After the cotyledons have attained full length, growth in thickness begins in the area nearest the epicotyl and proceeds toward the margins. This growth in thickness results from cambium-like meristem with the formation of new cells. The formation of well developed or solid kernels that completely fill the cavity within the shell is dependent upon meristematic activity continuing almost to maturity. The weather conditions, the nutrition of the tree, or other factors that affect the synthesis and translocation of elaborated food materials from the leaves and shoots to the kernels at this time determine the degree to which the cotyledons are thickened, or in other words how well the nuts are filled.

+Periods of Development+

In the development of the nuts there are three periods or stages: (1) The period of growth in size; (2) the period of nut filling or development of the kernel; (3) the period of maturing.

What takes place during these periods of development determines the size the nuts attain, the degree to which they are filled, and finally the quality at harvest. These three developmental stages are interdependent, because the size of the nuts may affect the degree of filling, and that, in turn, the time and nature of their maturity. They are not entirely separate and distinct but overlap in that there is more or less development of the kernel, varying with the species, while the nuts are growing in size. In general, however, there is not appreciable kernel development until after the nuts have attained approximately full size, except in the chestnut.

The outstanding example of this situation is the pecan. There is practically no growth of the kernel until after the shell of the nut has started to become hard. At that time growth of the embryo, which constitutes the kernel, become rapid. The major portion of the kernel is formed during a period of approximately one month,

starting at Beltsville, Maryland about the middle of September. The final stages of filling occur just before the nuts mature, and the first nuts to fall usually have the best filled kernels. Later maturing nuts are generally poorly filled; their shells and kernels are often discolored, and the shucks fail to open properly, if at all.

The development of walnuts, hickory nuts, and filberts, so far as is known, is in all essentials the same as that described for the pecan nut except that the kernel or embryo begins to grow somewhat earlier in the season. However, the major portion of the filling, which consists in the thickening of the cotyledons, takes place late in the season, and only a month or a little more before the nuts mature.

The period of the maturing of the nuts generally closely follows the completion of the filling of the kernels. During this period in the pecan, certain other species of hickory, the Persian walnut, chestnut, and others, food reserves are transferred from shucks, hulls, or burs to the nuts. Abscission layers are formed and shucks, hulls, or burs split open on drying out, thus partially or wholly releasing the nuts. There is a very direct relationship between the degree to which the nuts are filled and their time of and normality of maturing; well filled nuts mature early and normally, whereas poorly filled nuts mature late, if at all, and shucks, hulls, or burs fail to open properly.

+Growth in Size+

The size of the nuts produced by a tree is determined by a number of factors, one or all of which may operate during the course of the season. These are: (1) Age of tree; (2) position of the nuts on the tree; (3) fertility of the soil and moisture supply, or the nutritional status of the tree; (4) size of the crop borne.

In general, old trees bear smaller nuts than do younger trees. Hence size of nut for a particular variety is only relative. The first few crops produced by a tree usually consist of nuts large in size for the variety; and then, as the tree attains age, nuts become smaller in size. Young trees make longer and thicker shoot growth than do older trees. There is, then, under normal conditions, a direct relationship between the growth made by a tree and the size that the nuts attain. The more vigorous trees not only produce larger nuts than those produced by less vigorous trees, but the hulls and shells of such nuts are thicker and constitute a higher total percentage of the total weight of the fruit.

The position of the nuts on a tree has an important effect on the size that they ultimately attain. In general, the nuts in the top are larger than those nearer the ground; and those on the strongest and most vigorous shoots of the top or lateral branches will attain a larger size under normal conditions than those located on weaker and shorter shoots or on the inside of the tree. Here again there is a direct relationship between growth of the tree and growth in size of nuts. All normal trees make longer and stronger shoot growth in the top than they do on the terminals of lateral branches, and the shortest and weaker shoots as well as the smallest nuts are generally on the lateral branches inside of the tree top.

Fertility of soil and moisture supply determine in large measure both the growth made by the tree and the size of nuts. The nuts borne on trees growing on fertile soils adequately supplied with moisture are generally much larger in size than those borne

by trees on infertile soil or soil poorly supplied with soil moisture. Deficiency of either nitrogen, or moisture, or both is particularly effective in limiting the size of nuts produced. Pecans grown under soil conditions in which both nitrogen and moisture were deficient have been known to attain only about one-fourth the size of nuts of the same varieties grown in the same orchard but under conditions of clean cultivation and supplementary nitrogen applications. A prolonged drought during the time that the nuts are increasing in size very frequently causes them to be much smaller than they would have been had the moisture supply been adequate.

The size of the crop borne by a tree determines in a very large measure the size that the nuts attain at maturity. There is generally an inverse relationship also between the number of nuts borne in a cluster on a shoot and the size they attain. In this respect nut crops are little different from apples and peaches, which, too, are sold on the basis of size. In order to produce fruits of large size having a high market value, the crops are thinned in years of a heavy set of fruit. In the case of pecans, for example, thinning the crop at the time the nuts are growing in size on heavily producing trees is a very effective method of increasing the average size of the nuts allowed to remain on the trees. The earlier the thinning is done the more effective it is; however, it will increase the size of the nuts even when done as late as when the shells have started to become hard. No practical and economical method of thinning the crop of nuts has as yet been found; nevertheless it is well to bear in mind that a large crop borne by a tree generally means reduced average size of the nuts at harvest.

+Filling or Development of the Kernels+

In general, the fruits (nuts) of a nut-bearing tree are what might be termed storage organs. In them are stored mineral elements and such elaborated food materials as carbohydrates (sugars and starch), oil, amino acids, and proteins that have been removed from the leaves and wood of the tree. These materials are stored for future use of the embryo in the nut to sustain respiration, to permit germination, and to maintain the seedling until it has produced enough leaf area to become self-sufficient.

The question may be asked, why is it so important that nuts be well filled? The answer is very simple, because the quality of the oily nuts is determined by how well the kernels are filled. All but one of our most important nuts--almonds, filberts, hickory nuts, pecans, and walnuts--are oily nuts; and well filled kernels contain from 50 to 75 percent or more of oil, depending upon the species. Chestnuts are starchy nuts and contain less than one percent of oil. The relationship between the degree of filling and the composition of the kernel in oily nuts is outstanding, in that the better filled nuts have a higher content of oil and a lower content of protein, carbohydrates, water, and undetermined constituents than do poorly filled nuts. Highest quality of the kernels is directly associated with highest oil content and highest degree of filling. Nut kernels that are poorly filled are often hollow, shrunken, shriveled, and chaffy. When eaten they may taste sweet, but are lacking in the oily flavor characteristic of the particular species of nut eaten. It is only in the best filled nuts that highest quality, flavor, and oil content are found.

The degree to which nuts are filled or how well the kernels are developed at harvest is determined by a rather large number of interrelated factors: (1) Size of crop, or ratio of number of leaves per nut; (2) average size of nuts; (3) condition of leaves; (4) amount of second growth of the trees; (5) size of preceding crop and how well the nuts produced were filled; (6) disease and insect injury to the nuts; (7) weather conditions; (8) heterosis or effect of cross-pollination on embryo size.

+Size of crop:+ Nut growers want their trees to bear large annual crops of nuts. It is very seldom that one hears a nut grower express the opinion that a certain tree is carrying too many nuts for the crop to attain proper size and fill well, yet this is very often the case. Furthermore, the production of a large crop of poorly filled nuts one year is almost certain to result in a light crop or none at all the following year. There is a very close inverse relation between the size of the crop produced and the degree to which the nuts are filled at harvest, namely, the larger the crop the less the nuts will be filled. It has been pointed out above that nuts are storage organs, and the food materials required to grow and fill them must be made in the leaves. When too many nuts are set and carried through to the filling period, in proportion to the number of leaves or the leaf area of the tree, it is not possible for the leaves to synthesize the large amount of food materials required to fill the nuts. In pecans, for example, it has been shown that six to eight leaves are required normally to fill a nut properly and 10 or more leaves per nut if the tree is to flower and set a crop the following year. Other ratios for number of leaves or leaf area exist with other kinds of nuts. It is general experience that large crops of nuts remove such large amounts of food materials and minerals from the trees that a light crop or no crop at all is produced the following year. This is especially true if the nuts are not especially well filled in the "on crop year."

+Size of nuts:+ Almost everyone prefers large nuts to small ones, and that is one reason, why the larger sizes command a higher price on the markets. Many remember how popular the McCallister hican was a number of years ago because of its extremely large size. Such varieties of the pecan as Nelson and Mahan were very popular because the nuts produced were generally much larger than those of other varieties. These varieties remained popular until experience in growing them showed that they were very often poorly filled at harvest. As a general rule, large nuts are more difficult to fill properly than small nuts. This is obvious, because much more food material must be made by the leaves and transported to fill the kernels of large nuts than is required to fill an equal number of nuts of smaller size. In seasons with conditions favorable for both tree growth and growth in size of the nuts, it is often the experience that the nuts are poorly filled at harvest. On the other hand, if the weather is dry during the period in which the nuts are growing in size, they are much more likely to be well filled at maturity. In fact, the writer has seen several instances in which, because of severe drought in the spring, pecans were undersized, yet the kernels developed and filled so well that the shells of the nuts cracked at maturity.

+Condition of leaves:+ To produce well filled nuts, nut trees must bear a large leaf area and the leaves must be in good health and vigor. If they are to produce annual crops, the trees must carry their leaves until cold weather in the late fall, undamaged

by insects or diseases. The importance of a large leaf area free from injury or abnormal condition is so great that it can hardly be overemphasized in connection with nut production. It can be definitely stated that under normal conditions the size of the crop produced and the degree to which the nuts are filled is directly related to the leaf area and the length of time it is carried by the tree.

If the leaf area is to be large, the trees must make good, strong, vigorous shoot growth, and this means that proper attention must be given to fertilization to insure that the trees have adequate amounts of nitrogen, potassium, magnesium, zinc, and boron, as well as all other essential elements. The elements mentioned have been found most likely to be deficient in the soils of eastern and southern United States. In those regions their lack may be expected most frequently to limit tree growth or the filling of the nuts because of their effects on the leaves and the consequent inability of the leaves to make food materials. Deficiency of one or more of these elements results in leaves that are not able to make food materials in anywhere near such amounts as do normal leaves well supplied with all essential elements. In severe cases, deficiency of one or more of these elements results in chlorosis of the leaves, still later in leaf scorch, and finally in premature leaf fall. Trees having leaves in such condition cannot be expected to fill the nuts borne by them.

Most nut trees grown about home or farmstead are deficient in nitrogen, as the trees must compete with grass, weeds, shrubbery, or other trees. Frequently there is not enough plant food for all. A deficiency of nitrogen limits the growth and the leaf area carried by a tree. A deficiency of potassium or magnesium very greatly limits the amount of food material made by the leaves and hence greatly decreases the filling and the oil content of the kernels. Zinc or boron deficiency has a similar effect.

Hence, to insure the production of well filled nuts, one must be certain that the trees are well fertilized and that the fertilizer elements applied are in proper balance one with the others.

Injury to the leaves resulting from attacks by diseases and insects is one of the most common and important causes of poorly filled nuts. Every species and variety of tree nut suffers from at least one disease or insect pest that damages the leaves and hence limits or curtails the amount of elaborated food materials they can make. In most cases the fungi or bacteria causing foliage diseases infect the leaves in early spring at the time they are unfolding and growing in size, although the infection may not be noticeable until later. These infected areas, even though they are small and not numerous enough to cause the leaf to fall, seriously impair the functioning of the leaf out of all proportion to the area directly affected. Should the infection be so severe as to cause premature defoliation, the damage will be great even though only a small percentage of the leaves falls. The disease of eastern Mack walnut known as leaf spot, or anthracnose, is one of these defoliating diseases that causes untold damage from poorly filled nuts in the current crop year, and results in a small crop or none at all the following year. The development and spread of these diseases is gradual, and unsuspecting growers do not realize the damage they cause.

On other hand, the injuries caused by such insects as the webworm, the walnut caterpillar, the pecan leaf case-bearer, the Japanese beetle, and others are somewhat spectacular in that the leaves may be partly or completely consumed on portions of the trees. The injury caused by the walnut aphis, the walnut lace bug, the pecan black aphis, and others, on the other hand, is less conspicuous; but the end result is far more serious than it usually is with the leaf eating insects, because the damage caused is more widespread, almost all of the leaves on a tree being affected. These sucking insects are small in size and may be overlooked until premature defoliation takes place. If nut trees are to bear satisfactory crops of well filled nuts, the diseases and insects that attack and cause injury to the leaves must be controlled. Under normal conditions the size of the crop produced, the regularity of bearing, and the quality of the nuts harvested is proportional to the leaf area of normal leaves carried by the tree from early spring until freezing-weather in the fall.

+Second growth of the trees:+ Certain of our nut trees, such as pecan and walnuts, under some conditions have two or perhaps more periods of shoot growth during the same growing season. The first, or main period of growth, starts at the time of foliation in the spring and ends soon after the shoots flower. The second period of growth, if it occurs, may begin any time after the nuts are set, and may end any time later. This second growth seriously affects the filling of the nuts, in that food materials are consumed in producing this second growth rather than in the growth and filling of the nuts. Generally this second growth is not made until late in the season, and it usually follows a period of dry weather, when conditions again become favorable for growth. Usually this is at the time the kernels should be developing, and hence the degree of filling is affected. The seriousness of the effect on the filling of the nuts is largely proportional to the amount and duration of this second growth. A third period of growth may occur later if weather conditions are suitable.

+Preceding crop:+ It has already been pointed out that nuts are storage organs and in their growth and development large amounts of food materials and minerals are removed from the tree. Under conditions of heavy crop production, the reserves of these materials left in the tree at the time of harvest are likely to be very low; and unless the trees are growing on a fertile soil and carry their leaves until frost, these reserves of minerals and elaborated food materials are not likely to be restored. Under such conditions, in the following spring the reserves are low and although there may be enough to initiate flowering and the set of nuts, they are not sufficiently high to produce well filled nuts. It is for this reason that the nuts produced in an "off crop year," even though the crop may be much lighter, may be less well filled than those produced in an "on crop year."

Such nuts as pecans, hickory nuts, and walnuts transfer large amounts of potassium from the tree itself into the shucks or hulls. The kernels of such nuts are high in nitrogen, phosphorus, and magnesium as well as in oil, which is one of the most concentrated food materials and has the highest calorie value. Nitrogen reserves in the trees are readily and rather quickly replaced if adequate amounts are applied, as this element is not fixed by the soil. This is not true of phosphorus and potassium, as they are apparently taken up by the trees much more slowly than is

nitrogen. Furthermore, certain soils have a high fixing power for these elements and hence they are slowly, if at all, available.

+Insect and disease damage to the nuts:+ Certain insects and diseases attack the nuts, causing them to be poorly filled at harvest. Although these pests may destroy, or cause a certain percentage of the crop to drop before harvest, and hence serve as a thinning measure, the affected nuts remaining on the tree may not be well filled at maturity. Examples of such insects are the pecan or hickory shuck worm, the walnut husk maggot, and the codling moth. Infestations by these insects occurring before the shells of the nuts have become hard cause the nuts to drop. However, infestations taking place after the nut shells have become hard do not cause the nuts to drop. These late-infested nuts may be poorly filled because the insect larvae mine the hulls or shucks, severing the conducting tissues that transport food materials from the fruit stem or peduncle through the shuck to the kernel. The damage caused not only results in poorly filled nuts but also interferes with the natural separation of the shucks or hulls from the shells.

Examples of diseases that attack the nuts and cause them to be poorly filled at harvest are pecan scab and walnut bacteriosis. Pecan scab may also attack other species of hickory. It is the most destructive pecan disease, causing a high percentage of the nuts on highly susceptible varieties to drop prematurely and those that stick to the tree to be poorly filled at harvest. Walnut bacteriosis or blight is the most important walnut disease in the West and unless controlled causes severe losses from premature drop or from nuts both poorly filled and having discolored kernels at harvest. It is obvious that if large crops of well filled nuts are to be produced, these insects and diseases must be controlled.

+Weather conditions:+ Many growers are inclined to blame the weather for all small crops and poor nut quality because they realize it can have such important effects. In reality its direct effects are generally much less than they are thought to be, and its indirect effects are usually much greater than is usually realized. Weather conditions have a very great effect on the development of insects and diseases and on the damage caused by them, so that most often these are of major importance.

It has already been pointed out that a prolonged drought may adversely affect the size of nuts when it occurs while they are growing in size. Similarly, the degree to which nuts are filled at harvest is affected by the moisture supply during the filling period. A moisture deficiency within the tree probably affects the translocation of food materials to the nuts to a greater extent than it affects leaf functioning, for under such conditions the leaves will withdraw so much water from the developing nuts that the shucks and hulls become wilted. Under conditions of prolonged drought the kernels do not fill properly, maturity of the nuts is delayed, and the shucks or hulls do not open normally.

Under drought conditions the temperatures of the air and of surfaces exposed to the sun are often very high, and this sometimes results in sun-scald or burning of the hulls or shucks. In severe cases the injury extends through the hull or shuck to the shell and kernels. The pellicle, or skin of the kernel, turns brown or amber color, as

does the portion of the kernel that has developed at the time of injury. Further development of the affected portion of the kernel is arrested; and on drying it becomes shriveled because of lack of filling. The greatest amount of damage from sunburn occurs on the south and southwest sides of the trees. Little can be done to prevent this type of injury other than to grow good, strong, vigorous trees that bear a heavy dense foliage that shades the nuts.

+Heterosis or hybrid vigor:+ The pistillate flowers of certain nut species, such as the almond, chestnuts, and filberts, must be cross-pollinated with pollen from another variety if satisfactorily crops of well filled nuts are to be produced. These species are self-sterile or self-unfruitful. On the other hand all walnut, pecan, and hickory species are self-fertile and cross-fertile, but may be self-unfruitful because of dichogamy, because they may shed their pollen either before or after the stigmas of the pistillate flowers are receptive to it. In all nut species cross-pollination is generally recommended so as to assure a set of nuts. With cross-pollination a better set of nuts is to be expected than with self-pollination, as well as better filling of the kernels. It has recently been found that when the pistillate flowers of a certain variety are cross-pollinated with a pollen from another definite variety the embryo or nut kernel is larger and better filled. This is a manifestation of hybrid vigor, or heterosis. Heterosis has been found in the chestnut and in the pecan. It likely will be found in other nut species. Some day the principles of selected and controlled parentage underlying hybrid vigor may be utilized in producing superior nuts, as these principals are now so widely used in producing hybrid seed corn.

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President Davidson: That paper was so extremely important that I hesitated very much to stop it, but we are already at the point where we should have adjourned. Now, unfortunately, we have some very important things, I think, yet before us, but if the speakers can give their talks from now on in the form of, shall we say, syntheses of the whole thing and give us the conclusions rather than the details, it will be appreciated by us all. Mr. Wilkinson is going to give us a very important talk on what he has done with the propagation of the Lamb curly walnut. Mr. Wilkinson.

The Grafted Curly Walnut as a Timber Tree

J. FORD WILKINSON, Rockport, Indiana

Our native trees are and have always been one of the most valuable resources of this country, and one of the greatest heritages ever to fall to a nation.

Wood has been used by our people since the landing of the Pilgrims, for almost every comfort and purpose in life, from the making of cradles to caskets.

Wood is still one of the principal materials in building homes and furniture, and is used for railroad ties, for paper, and in so many other ways that we could scarcely get along without wood.

The United States is the native home of many species of trees, of which a number are superior in some certain ways for some special purposes. The hickory has no equal for ax handles. As a building-timber where strength and durability are needed

the oak ranks among the first. Other species are equally as important for some other uses.

Not to be overlooked are nut trees. They serve the twofold purpose of producing both food for man and wild life, and valuable timber.

+Black Walnut Has Great Value+

Of the nut tree group, the black walnut is one of the most important. It ranks among the first for lumber, furniture, cabinets, and finishing material. It has no rival in use for gun stocks and airplane propellers; as walnut wood is light, strong, will not get rough, but wears smoother with use. Neither will it splinter when pierced by a bullet. Walnut wood has been largely responsible, at times, for keeping us a nation of free people.

The black walnut tree is an aristocrat of forest and field. It can justly be proud, for no other tree can fill its place. As the late author A. H. Marks said, "Who has not noticed the look of contended usefulness which a nut-bearing tree wears? It is of use to the world and knows it."

Walnuts, like other species of trees, are not all alike, either as to nut production or in the grain of the wood.

+The Lamb Black Walnut+

Several years ago an unusually highly figured, and very valuable, black walnut tree was discovered by Mr. George N. Lamb, then Secretary-Treasurer of the American Walnut Manufacturers Association of Chicago, Illinois.

When the logs from this tree came into the mill, and their value was realized, Mr. Lamb went to the place where the tree had grown. He secured some twigs from the branches of this top and sent these, as I have been informed, to Dr. Robert T. Morris and Mr. Willard G. Bixby, knowing of their interest in propagating better varieties of nut trees.

This wood had been taken from the top many days after the tree was felled, and so was dry and nearly dead. I believe Dr. Morris succeeded in getting only one graft to grow, and Mr. Bixby two. This variety was then named in honor of Mr. Lamb.

Several years later Mr. Bixby sent me a very small stick of graft wood from one of his trees, from which I made two grafts. One of them grew, giving me a start of this variety. I have annually propagated a few trees of it ever since, though with little encouragement, and even much discouragement from others, including State and U. S. Government authorities.

On one occasion I thought I practically had an order for a quantity of these Lamb walnut trees for a reforestation project. However, the prospective purchaser, before placing his final order, wrote to government authorities, then wrote me as follows:

* * * * *

" ... Sept. 30, 1940 ...

"Following some investigation in connection with the so-called curly walnut varieties, we have been advised by government authorities that these trees do not form, or grow into, a curly walnut tree at any time during the growing stage.

"We took it for granted that the wood formation would be of a curly nature, and for that reason we were interested in that particular variety.

"In view of this information which we have concerning these trees, we would not be interested in growing them as we have plenty of native black walnut here...."

* * * * *

This and other discouragements, from both government authorities and individuals, had about as much effect on me as King George's advice to the American people not to use tobacco; they smoked calmly on, and I continued to propagate Lamb curly black walnut trees.

I have been propagating nut trees since 1910, and have never yet known one of my propagated nut trees to fail to carry the characteristics of the parent tree, as to habits of growth, bark, bud formation, foliage, texture of wood, or quality of nuts. The Deming Purple walnut tree, when asexually propagated, reproduces the purple wood, so I reasoned the Lamb variety would reproduce figured wood. Nature seldom blunders.

+Value of Original Lamb Walnut Tree+

When I was in Grand Rapids, Michigan, a few years ago, doing some tree work for the late William J. Wallace, he took me a few miles to the location of the original Lamb tree. It was near a small river in a gravel loam soil near Ada, Kent County, Michigan.

The following is an extract of a letter received from Mr. Lamb as to the original tree:

"November 27, 1929

Lamb Figured Walnut-- Cut into veneers @ 6 to 18c per sq. ft. (1/28") Use: Furniture Amount of veneers 60627' [Value: $8,637.62 to $10,918.86 (Prewar!)--Ed.] Logs produced: 8' x 21"--144 Log Ft. 6' x 18"-- 73 " " 10' x 36"--640 " " 14' x 30"--591 " " 10' x 32"--490 " " Stump --500 " " _____

2438

Location of tree--Ada, Kent County, Michigan. Location--River flat 20 rods from river. Soil-Gravel loam. Type of tree--Open grown. Shape---Double stump. Height--40 ft. Figure--Throughout the tree."

Mr. Lamb further states in his letter: "Unquestionably it was one of the most thoroughly figured trees ever discovered, and if figured wood will propagate itself this stock should, certainly should, do so."

He further states, "The figure in this tree was quite apparent, even in the small branches, while the Forest Products Laboratory found evidence of a developing figure

in the twigs not over five years old."

The wood specimens I now have on exhibit here were taken from one of my 12-year-old grafted trees that I cut, and in them you will find figure visible to the naked eye, or easily noticeable by touching with a finger, in wood from branches not over 7 years old.

Comparing age at which figure shows in the wood of the two trees, this young tree seems to be developing figure at an early age, as in the parent tree.

My confidence in this outcome had never been shaken by the doubts of others. Few seemed to share this belief with me, and for this reason I have never pushed the sale of Lamb trees. Now I do not hesitate to state that curly figure will reproduce in any propagated Lamb trees, as the evidence before you here is stronger than any argument.

One purpose of the Northern Nut Growers Association is to encourage the perpetuation by propagation of the better varieties of nut trees. I consider the Lamb variety one of the best walnut trees known from a timber point of view, and until a better variety is found I shall continue to propagate Lamb black walnut trees.

+Ed. note:+ The nuts on Lamb trees, as seen at Norris, Tennessee, during this meeting, appear to be of at least average size and have better than average shell structure. They probably would be well adapted to machine-cracking. Thus the Lamb would not be a bad variety to grow for its nuts. Or we could double-work the trees, to have each tree with a good trunk of the Lamb wood growing beneath a fruiting top of any desired walnut variety. One or two of our members already have made a start on this latter scheme of propagation.

+Author's Note:+ The Lamb variety is a rapid and upright grower and should be well adapted as a stock for the purpose suggested.

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President Davidson: I don't think one can minimize the importance of what Mr. Wilkinson has done with the Lamb curly walnut. There are possibilities here that are of immense value to those who are interested in timber. Now, I am very, very sorry to put off the rest of this program until this afternoon. Possibly we can work a part of it then. Meantime, we had better adjourn.

Mr. Chase says that he has arranged for a group picture to be taken at the Community Building at one o'clock. Let's everyone be there at one o'clock. That means, of course, that you are going to cut the sandwich and coffee pretty short.

All right, let's adjourn.

(Luncheon recess was taken.)

+Tuesday Afternoon Session+

President Davidson: Come to order, please. The first speaker on the afternoon program is Mr. Shivery. I think I will get Mr. Chase to say a word.

Mr. Chase: Our next speaker is Mr. George Shivery, Extension Forester for the University of Tennessee, and I know that the interest of this Association is in the planting of improved black walnuts, and I simply can say this man arranged for the planting of more Thomas walnut trees than any other man in the world. George Shivery.

The Black Walnut Situation in Tennessee

GEORGE B. SHIVERY, Extension Forester, University of Tennessee, Knoxville, Tennessee.

Mr. Shivery: Mr. Chairman and members of the Association: I certainly appreciate that compliment made by Mr. Chase, and I want to assure all of you that we certainly are interested in the black walnut in Tennessee. In the past we have had to depend pretty much on the wild black walnut, and we will for years in the future. But we have done everything possible to get distribution on this Thomas improved black walnut which has been propagated here through the efforts of Mr. Chase, Mr. Zarger and other members in his division.

It seems to me that this black walnut kernel industry is sort of a tradition, particularly in East Tennessee. If you have lived in this state as long as I have, you have become curious about its history. Well, in the early days there were no railroads in this state, and commerce moved pretty much by means of wagon team, and the supply center seemed to be Baltimore, Maryland. Now, I can visualize very well that on outbound trips they doubtless carried black walnut kernels, and on the way in, of course, they'd bring clothing and other materials that were not produced here at home.

In the early days they produced tremendous amounts of maple sugar and maple syrup. Doubtless this was consumed at home and nowadays we don't have any evidence of that, because the climatic conditions in New York State and other northern states and New England are much better suited to the flow of the sap. The weather, I believe, is not so changeable up there. Our weather is changeable. We may have a very severe cold week, and then in ten days it will be balmy and pretty weather. We haven't made any effort to bring back the sugar maple industry. We don't consider it economic in this state, because cane sugar in the past has been cheap in price, and then we have another product that some of you may not be familiar with, sorghum molasses. That serves as dessert lots of times in many meals, hot biscuits and sorghum to finish up the meal.

Now, I might mention something about the size of the black walnut industry in this state. We estimate that there are eight million pounds of uncracked whole walnuts produced on the average in a normal crop year in Tennessee, and there is another five million pounds that is never gathered, never hulled, never enters the market, never used, and the value of this crop in a normal year would be around $750,000. That is for the nuts, the fruit, the kernels. If you speak of timber it will amount to $960,000. That is in the form of lumber and veneers, and if you figure that in the form of a log at the shipping point, we'd reduce that figure and say it would be $480,000.

I think to understand this state you have to give some consideration to physiographic regions, and if you will bear with me I'd like to sketch through these regions of the state, because they have a bearing on production of black walnut. Here in the east we have the East Tennessee Mountains, and proceeding westward we have the Great Valley of East Tennessee. It goes all the way down to Chattanooga, up through Bristol, on up through Virginia to Hagerstown, Maryland, all the way up to Harrisburg, Pennsylvania.

We have fine soil, and we also have different kinds of shale in that valley. Then we proceed westward. We come to the Cumberland Plateau, and the elevation of this plateau is around 2,000 feet. It is higher than this valley. Then we cross that and we reach this area (indicating on map). That is what we call the Highland Rim. That is made up of limestone soil of a different character, usually, than that in this East Tennessee valley. That is what we term the Eastern Highland Rim, and this around here (indicating) we term the Western Highland Rim. And this red portion would be the Central Basin in which Nashville is situated. Then you would travel through this central elevation, come up on the Western Highland Rim, and then you come up here and you cross the Tennessee River flowing north. Then you get into West Tennessee.

Now, that is coastal plain soil, and as you approach the Mississippi River here you have a covering of what the Germans call loess, fine, wind-blown material, silt loam. So that very sketchily gives you some idea of the physiographic regions in this state.

Now you want to know where these black walnuts are grown. Well, up about here (indicating the northeast) we have the towns of Greeneville and Rogersville and Morristown and Jonesboro, the counties of Washington County, Greene County, Hawkins County, say, ten counties; radiating around those ten counties you have in the past had great quantities of walnut kernels produced and sold. Now, go on down this valley past Knoxville, and McMinn County (southeast) has some years produced heavy crops of walnuts. So you have heavy production all through the valley.

There's another center, we might term it, of about six counties in this central basin. But I don't want you to get the wrong impression, because walnuts grow in almost any county in this state, but I am mentioning these greater producing areas. And this County of Williamson south of Nashville in years past has sent plenty of walnuts to market. So that's a walnut producing area. And up here in this Highland Rim we have some counties by the name of Pickett and Overton and Clay County. Well, they produce walnuts, and the people up there have in the past cracked out a lot of walnuts. And in Montgomery County they produce walnuts. So the normal trade centers where these walnuts move is really to a great degree here at this town of Morristown in East Tennessee, and Nashville in Middle Tennessee, and this Middle Tennessee center draws from Kentucky. In fact, these four or five large shelling concerns know about the walnuts pretty much all over the entire walnut producing territory.

Through the years the Agricultural Extension Service, University of Tennessee, with which I am connected, has been keenly interested in assisting in any way we can to get additional income out of walnut kernels, and in recent years the whole

uncracked walnut. And even though I am a forester I can see the possibilities of this, and we like to carry it along. In fact, I consider walnut as kind of a dual-purpose tree, fine for timber production, also for production of nuts, walnut meats or kernels. You might term it a triple-purpose tree. I don't think there is any better tree than that for a shade tree in pastures, in the field, and around the home, because for one reason it makes what we term in this state a "cold shade," and it is not a hot shade like you get under a sugar maple. The maple has a dense foliage. And as Mr. Chance indicated this morning, walnut is usually associated with blue grass. Blue grass will grow under it.

I guess some of you here remember the years of the depression, and I remember in 1932, for example, we had a heavy crop of black walnuts in the state. Then I believe the price for kernels of 15 cents a pound would have been a good price during that year, and some of them probably sold for less. So if we had the time we would follow through all the years, beginning with 1927, but just to make it as brief as possible, I will leave those out, but I would like to mention the year 1941. It sort of disrupted things in the kernel industry, because at that time the Pure Food and Drug people came in here and set up regulations, and it interfered with the merchandising of these kernels, because the producer had to satisfy certain sanitary regulations, and it really sounded worse than it was. Anyway, it confused our people, and probably that is about the year in which we had this big shift from the production of walnut kernels cracked out at home to a sale of uncracked walnuts to these shelling plants.

Then another year that I think of (we always think of these as walnut crop years) was 1945, and that year we got better prices, probably, than ever before or since, and a lot of our country people were able to sell hulled uncracked walnuts as high as $6 per hundred pounds.

We will continue to be interested in this industry, but, of course, nowadays the wage scale is higher and money is not worth as much as it was in the past, so it really seems to me that in order to get out this crop we just have to try to make the price a little more attractive.

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President Davidson: "We are now going to hear from Mr. Shessler of Ohio, on his method of grafting, and I wish to assure you that he knows what he is talking about. He has done a lot of it.

Grafting Walnuts in Ohio

SYLVESTER SHESSLER, Genoa, Ohio

In 1934 the Ohio Nut Growers Association conducted a black walnut nut contest. I read about it in the _Ohio Farmer_. As soon as the names of the winners appeared in that publication, each owner was contacted for some nuts from the prize winning trees. Answers were received from nine of the 10 winners. I did not receive nuts from the Hoover tree. The Brown nuts I planted came up in 1935 and the trees are now 22 feet high, with spread of 22 feet, and are 27 Inches in circumference. The Tritten prize nuts were planted in a fence row. These did not come up the first year. The next

year I plowed and disked the patch of ground and planted potatoes. To my surprise the Tritten seedlings came up with the potatoes. I let them grow and I now have five trees from these nuts. All of these trees produce nuts which resemble the original Tritten nut and have good cracking quality. One in particular fills out nicely, has a very thin hull, and is a little larger than the original Tritten. I have named it the Shessler. The Brown seedling trees also produce good nuts. The seedling trees from the Cowle nuts produce nuts with rough shells.

Following my nut planting project I began to collect scions from all of the original trees. Mr. Homer Jacobs, of Kent, Ohio, supplied me with scions from the Tritten tree. The next year Mr. Jacobs asked me to send him scions from the Brown tree as he intended to bench-graft some. I have planted nuts along a road 80 rods long, so that I could have many stocks to top-work. I began to graft in 1935, using the seedling trees as stock. I now have 200 seedling black walnut trees, 100 grafted black walnut trees, 25 grafted Persian walnuts, 20 chestnut trees, two "buartnuts," 15 heartnuts, six pecans, one butternut, 20 grafted hickory trees and five persimmons. Some of these trees are planted in orchard form, others are scattered along fence rows.

For grafting, I cut scions so that there is about four inches of two-year-old wood at the base and some one-year-old wood with small matured buds. These small buds will grow, as a rule. The scions are kept in damp sawdust until used. I like the stock to be a half to one inch in diameter. I wait until the trees are in full leaf before I graft. After leafing out the stock does not bleed. If I find that the stock is bleeding hard when I cut back, I wait a few days before grafting. It is a waste of time to graft when the stock is bleeding. I have grafted very early when the bark would not bleed at all. I just dug down into the cambium layer and put in the scion. I tried one Persian and three black walnuts like this and all grew. I use the slot bark method of grafting, as described in Mr. Reed's bulletin [U.S.D.A. Farmers Bull. 1501]. The stock is cut straight across and I put the lower bud just above the bark on the outside. I roughen the bark of the scion that fits just behind the bark of the stock. A small nail is pushed through the bark and scion with the handle of my knife. I generally tie with cord but sometimes when the bark is heavy I do not use cord. A two-pound paper sack with a hole on the earth side is placed over the graft and the sack is tied at the bottom. This serves as a "hot house" and protects the scion from rain. As soon as leaves appear on the scion, the sack is removed and all the new sprouts are broken off below the graft. I put only one scion on each graft. I use Beck's cold wax. It is easy to thin with water and I just flatten a stick for my brush. I never wax the bud but wax scion well on top.

I cannot give an accurate count of my grafting success but estimate that 75 percent of my grafts live. Rather than keep records I use that time to graft more trees. I am not an experimenter--I simply like to have grafted nut trees. My own trees are scattered over a two-mile area. I have grafted trees in Toledo and Grand Rapids. Every Sunday I attend church, then in the afternoon I graft trees. My aim is to try all the promising trees and select the best and weed out the poor ones. I am saving only the trees that bear nuts every year.

In 1947, I grafted the Ohio 1946 prize winning black walnuts. I achieved survival on all except Nos. 5 and 8. The scionwood of these two was in poor condition and I

did not think they would live. I also have No. 54 which looks promising to me. I am looking forward to other contests in Ohio and elsewhere so that we can uncover some more superior black walnuts.

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President Davidson: Thank you, Mr. Shessler.

Mr. Slate, will you say a word to us on grafting? That's right along the same line.

Grafting Walnuts in the Greenhouse

GEORGE L. SLATE, State Experiment Station, Geneva, N. Y.

Walnuts have been grafted in pots in the greenhouse at the Experiment Station at Geneva, N. Y. for a dozen years or more and the practice is successful and very useful. This method was adopted for two reasons. First: Under field conditions results are often uncertain, owing to the vagaries of the weather or neglect at a critical time. The inexpertness of the operator made it desirable that the work be done under as favorable conditions as possible, with the hope that a favorable environment might overcome in part the lack of skill. Second: The work can be done in March before the field work begins, whereas field grafting in May would often not get done owing to the pressure of other work at that time. This method is not original with the writer, but is similar to the method used at the East Malling Research Station in England and described by Witt in 1928 [1].

The rootstocks, two year old black walnut seedlings raised from nuts planted in the nursery, are dug in the fall, stored in the nursery cellar until late February or early March, at which time they are potted in six or eight inch pots, depending upon the size of the rootstocks. The roots are cut back so that the plant will fit in the pot. At this time the tops are cut off, leaving the stem about 8 inches high. The pots are placed in a warm house, watered as needed, and in about 10 days the buds begin to break.

The Jones modified cleft graft is used. The stub is cut off at grafting time and the cleft is made by cutting, not splitting, the stock with a large grafting knife. The scion is tied in place with nursery tape, half-inch size, with a short wick leading out of the cleft. The scion is painted with grafting wax.

+Care of the Grafted Plant+

The pot is set in a propagating frame about 18 inches deep, with bottom heat, and covered with glass, plus lath or cloth shade. An inch of peat in the bottom of the frame is desirable, to hold moisture and maintain high humidity. The temperature of the frame is kept in the eighties, but is not allowed to go above 90 deg.F. Under these conditions of warmth and high humidity, growth activity is rapid and in about two weeks the buds break, although, some may not start for a month. This spring adventitious buds developed on several scions. Many suckers arise from below the graft, and these are rubbed off two or three times a week. As soon as the shoots from the scion are two or three inches long the plants may be removed to a cooler house, where there is less danger of overheating on hot spring days. Later, they go to the

cold frame for hardening off, and when danger of frost is over after May 21st, they are set in the nursery for two years. First year growth is not over eight or ten inches, but the second year the plants grow to three or four feet or even more in a favorable season.

The percentage of grafts starting depends largely on the scion wood. Wood cut from vigorous young trees which is grafted the same day will give a 90 percent stand or better, but wood from other sources varies according, to the age and vigor of the tree from which it is cut and the percentage of success may be much less.

This method is useful for small scale operations where a greenhouse is available and it is desirable to do the grafting before outside work interferes with it. For one not skilled in nut tree grafting success is probably more certain than with nursery grafting.

+Literature Cited+

1 Witt, A. W. The vegetative propagation of walnuts. Ann. Rpt. of the East Malling Research Station 14th and 15th Yrs. 1926-1927 II Supplement pp. 60-64.

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President Davidson: There are plenty of us who don't know much about grafting, and I did want you to hear Mr. Slate's method. It is certainly worth trying and would come at a pleasant time of the year, would be easy to do, and any of us could try it out.

We now should like to hear from Mr. Clarke on Nut Investigations at Pennsylvania State College.

Nut Investigations at the Pennsylvania State College

WILLIAM S. CLARKE, JR., State College, Pennsylvania

Our present work in nut growing at the Pennsylvania State College was begun in 1946. Some work had been started many years ago, and a small number of trees were planted, mostly black walnut; but a site was selected which proved to be very cold and frosty, and most of the trees soon died. Further work had been planned at a later date, but the depression and lack of labor and land prevented us from getting under way then.

When the present project in nut growing was approved, the country was just beginning to recover from the recent war, and materials of most kinds, including nut trees, were very difficult to obtain. Therefore, in order to learn as much as possible about nut trees, we started at the beginning, with the seed. About two bushels of hulled black walnuts were collected from fence-row trees; some were planted out in the ground that autumn, and some were placed in soil in a box and kept over winter on the outdoor porch of the packing house. Some hickory and pecan nuts were bought and also stored in a similar box. The only nuts which grew were those planted out in the ground. They gave us a good germination, while not a single nut stored in the boxes grew.

At the present time we have about 200 black walnut seedlings in the nursery. When they are a year or two older, they will be grafted to several of the named varieties of black walnuts, and those that take will be planted out in a nut orchard. These seedling trees were transplanted after one year's growth. About four or five times as much of the walnut plant was underground in the root as grew above ground where we could see it.

Since the first year's work we have made a few purchases, and planted a few more nut seeds. At the present time we have planted five pairs of named varieties of filberts, four Chinese chestnuts, of which three survive, four Persian walnuts, three of which survive, and two Japanese walnuts. We also have a few seedlings of Turkish tree hazel obtained from nuts sent to us by one of our friends in the state of Washington and a few butternut seedlings grown from nuts of a tree on the college campus.

Future plans include an orchard with many of the named varieties of black walnuts and also, we hope, some of the new hardy strains and selections of the Persian walnut being introduced by the United States Department of Agriculture. Representative specimens of a wide range of nut species will be collected. Some further work on chestnuts and filberts may be attempted if they prove to be hardy here. Plans for the more distant future include studies in soil fertilization and in spraying for disease and insect control.

+Cold Injury in 1947-48+

This past winter has been very hard on nut trees, and on some other trees as well. In the first place, the cold weather of the autumn began very suddenly after six weeks of uninterrupted warm weather without any cool nights to harden the wood. In late September a few days of cool weather came, and then three nights in five with temperatures near 20 deg.. The walnut foliage and some of the youngest wood turned black. Next came a winter with extremely low temperatures, with the minima ranging from 18 to 23 degrees below zero over our orchard land. Our four Persian walnut trees were killed back to the ground; three of them have sprouted this summer from the roots. Considerable leaf bud killing occurred on Chinese chestnut. One Japanese walnut died back to the ground and has sprouted from the roots. The other tree lost most of its younger wood, but some buds near the base of last season's growth have sprouted out to make a new top. Several specimens of the golden chinkapin (_Castanopsis_) of the Pacific Coast, which had made one year's growth here, were killed outright.

Most of the terminal buds and youngest wood of our nursery trees of black walnut were killed, but the trees have grown well this year from the lateral buds. In the woods some black walnuts which had been cut down about four or five years ago, and which had made sprout growth now about fifteen feet high, were killed back from two to four feet by the winter. A twenty-year-old Stabler black walnut on our lawn lost many of its top limbs, though the lower limbs survived the winter all right. Some other types of trees were also badly damaged: some locust trees were killed to the ground, and many others were killed to very old wood. A ginkgo tree on our lawn was

killed back to the main trunk. This was one of the few times that I have ever seen injury on this species.

One of the five named varieties of filberts, Pal, escaped winter injury. DuChilly and Italian Red each have one good tree and one that was killed back to the ground, but is now sprouting from the roots. Of Medium Long, both trees have been killed way back. One tree of Cosford was killed completely, and the other tree has been badly damaged.

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President Davidson: Thank you, Mr. Clarke. Our family stopped on the way at a shelling plant where they were handling nuts by the ton, not the bushel, just the ton. I am not exaggerating. You have all heard the hill-billy program from Renfro Valley, no doubt, and we have with us today the man who is running that cracking plant and also this hill-billy chorus, Tom Mullins, who will tell us about what he is doing down at Renfro Valley.

Black Walnuts: A New Specialty at Renfro Valley

TOM MULLINS, Renfro Valley, Kentucky

Mr. Mullins: As Mr. Davidson said, I come from a little hill-billy section up in Kentucky known as Renfro Valley. Up until about a year ago the main commodity there was hill-billy music and a lot of noise on Saturday night. About last August our boss there kind of got interested in black walnuts. There were a lot of them going to waste all over the county due to the fact that most of our locals up there are kind of lazy. They don't like to get up there and stomp them out.

His original idea was to set up a hulling plant and hull the nuts and then buy the walnuts from the locals after they were dried. One thing led to another, and we talked to Mr. McCauley there, and Dad bought a big walnut plant to process black walnuts all the way through. He was new to it and so was I. He said, "Let's buy a million pounds of black walnuts." I didn't any more know what a million pounds of black walnuts was than I know how many grains of sand is in three or four buckets. It didn't take me very long, I think it was 31 days, and I bought 1,030,000 pounds. That's a whole lot of walnuts in anybody's language.

One of the local boys on our radio program came up with the bright idea that before in Renfro Valley we used to be just half nuts; now we are walnuts.

We started cracking these things along about the 15th of October, and last Saturday we cracked our last 10,000 pounds. Our machine is capable of cracking approximately 10,000 pounds in an 8-hour shift, and we carry the walnut all the way through to remove any of the field litter that it may have when it is picked up, and through cleaning air blasts and into a cracking machine that does darn near all the work. The only thing we haven't been able to figure out yet is how to get this machine to tell a bad kernel from a good one. We have to leave that to some of the girls who do the work on the picking belts.

Our future plan for this fall is to buy a million and a half pounds this year and process them. I believe one of these gentlemen a while ago mentioned something about the pure food laws. They are pretty rough on us. We have to pasteurize our walnuts. The state law of Kentucky requires 190 degrees of heat for an hour and a half. That's a lot of heat.

We package our nuts in two-ounce packages and in 35 and 50-pound cartons for the wholesale trade.

That has created quite a little industry there in our county. We have one county there, Clark County--Winchester, Kentucky, is the county seat of it--and out of that one county last year alone I bought 800,000 pounds of walnuts. That was, walnuts in the hull that the farmer had picked up and brought to us in trucks.

Our success was not too great in this method of hulling green walnuts to get our supply. We weren't adequately fixed up to dry the walnuts and take care of them in storage. We lost a few of them that way, but I think this year we have a little better sense and will let the farmer stomp them out.

We are working now on an educational program, both newspaper and radio, to persuade the farmers in our locality to let their walnut trees grow. We tell them nearly all the walnut trees will produce enough kernels or shelled walnuts to bring in as much money as they would if cut down and taken to the mill and used for saw logs. That is our main problem now, to try to keep the black walnut industry working there in our community. And our future plans call for plantings of black walnut seedlings and convincing the farmer and the 4-H Club members and all the boys in the Future Farmers of America and organizations like that to protect and cultivate their black walnut trees.

I am kind of on the fence this year. I stuck my neck away out the other day and bought a farm. After checking the farm I found I had about 600 walnut trees. Now, then, I am hollering on one hand for an increase in prices of raw material, and as a sheller I am hollering on the other hand to get the prices down. But I believe as a producer for next year I am going to try to forget about the shelling and let the prices go to the devil.

Mr. McDaniel: Would you mind telling us what you had to pay for the walnuts in the shell?

Mr. Mullins: Our average last year was $4.33. We went as high as $4.80. Some of those we bought hurriedly--

President Davidson: In the hull?

Mr. Mullins: No, that's dry shell. Our walnuts in the hull we paid a dollar and a quarter a hundred for, and if we had had good success we'd have made some money on it at that angle.

There is one question I'd like to put before you gentlemen. Maybe some of you know a little something about it. I was reading an article not long ago in Popular Mechanics Magazine about some plant on the West Coast that is developing the

Vitamin C content of the walnut hull itself. It is very high, the Vitamin C content in the walnut hull.

Another thing we did last year. After we hulled all of these walnuts we had a mess of hulls on hand, and our farmers were a little reluctant to come and get them. We tried to talk them into using them for fertilizer. They are kind of like some of the boys, they have got to be shown. They have to see somebody else do it before they tackle it.

Out of curiosity I laid my garden off and divided it in half, and on one half I put a top dressing of these dried-out, pulverized walnut hulls, and I firmly believe that the side that had the walnut hulls on it produced twice as much. And some of the boys in the neighborhood kind of noticed what kind of garden I had, and we don't have any hull problem anymore. They carried them all off.

Same way with the shells. We tried to get them to haul the shells off to use them on the fields for tobacco land and to grow blue grass, and they found out that was pretty good, so they are bothering us now about our shells.

We have another by-product. It is too small a granule kernel to go through, and we can't remove the shell from it. We have tried that out on chickens and hogs and some other farm animals, turkeys, ducks and geese. One boy that works for me there in the cracking plant had 28 hens. He had them in a pen, and he was getting six and eight eggs a day. So I talked him into taking some of these granules home and feeding them to his chickens, and in two weeks his 28 hens were producing 20 to 24 eggs a day. That kind of settled that problem, too. Some of the boys kind of got an idea they'd like to have some of that.

A lot of you folks are here from the North, and you possibly would be going back along Highway 25 going home, and I'd like to extend an invitation now to stop off tomorrow or the next day and look over our plant. It's quite interesting, quite a complicated piece of machinery. Mr. McCauley at Chicago is the gentleman who designed the machine, and he will have something to say about it.

One of the local farmers came in to see that machine one day, and it was operating, just batting the kernels out right and left. He looked up at it, gandered it all over, and I asked him what he thought it was. He said, "It's a damn lie. That thing can't do it."

So come see us.

* * * * *

President Davidson: Thank you, Mr. Mullins. Next, Marketing Black Walnut Kernels. This fits in with what Mr. Mullins has said. Mr. McCauley from Chicago will tell us about it. Mr. McCauley.

Marketing Black Walnut Kernels

F. J. McCAULEY, McCauley Company, Chicago, Illinois

Mr. Chairman, ladies and gentlemen: Tom has got me on the spot here. I came here to speak to you about the marketing of black walnuts. Machinery is a hobby of mine, and that thing there was just one of those off-shoots of an infertile brain. But

Tom is having a lot of trouble, and a lot of fun with it, so if you people would like to see that machine, that particular machine, I am glad that he invited you up there. It may give you a little different idea of what the sheller is up against in the salvaging of black walnut kernels.

You are interested in growing the black walnuts and other nuts in the shell, but they do have to be prepared for the public, and Tom's job, and other people's that are in the shelling business, is getting them out. The machines are made at Knoxville, Tennessee, and you can get a fairly decent idea about the shelling of black walnuts from the machine Smalley has. Tom's is a much larger size.

Now we will get down to this thing I came here to talk to you about, the marketing of black walnuts. My speech is divided into three parts; the first is about nuts, the second is about nuts and the third is about nuts, and I am nuts. Yes, that's more true than you think. My nickname throughout the United States is "Nuts" McCauley, and I am proud of it. It is a good nickname to have for a man that's in the nut business. And I most certainly am in the nut business, machinery on one hand and the selling of various types of nut kernels on the other.

You people probably don't know it, but you have the best advertised nut in the United States that you are working with, black walnuts. There are very few people in the United States that don't know what a black walnut kernel is, or a black walnut. In fact, I would say that 75 per cent of them at some time or other have gathered black walnuts, have hulled them. You know those pretty stained hands you have, and I can remember back in those days when I was a kid when I used to get those hands of mine just so brown and black from the hulling of black walnuts that my mother would almost want to turn me over her knee and spank me. But when wintertime came I always had a bunch of black walnuts that we could sit down and crack and put in those cookies or in that fudge.

I have talked to a good many of you people here, and I have a prepared speech, but I am going to ramble a little bit and I am going to ask you to ask me questions, because I found out that I don't know so many things, or the speech that I was going to make to you might not be as interesting as your asking me questions. I do want to say a few things, and I will go through quickly.

The first is the marketing of black walnuts in the shell. We find in the marketing of any product that there is a tremendous amount of waste due to poor sacking, due to a little dishonesty on the part of the people who are selling merchandise. You know, if there is a brick in a bag, the brick weighs a pound, that costs the man who buys the black walnuts money. In other words, out of that pound of brick he intended to get a small quantity of meats to sell, so his cost immediately goes up. You'd be surprised at how many bricks and how much iron there is in black walnuts and pecans! It's universal throughout the United States. There is a lot of chiseling that goes on. Your bags should be good. Black walnuts must be held for some time before they are processed, and one black walnut bag used one year can't be used another. If you can get by with one year's use of a bag to hold a hundred pounds, or whatever is put in it, of black walnuts, you are very fortunate. Usually they break out before the year is

over, and that causes waste. So start out with a decent bag.

I made a little note here to talk to you about California black walnuts. The standard throughout the United States to people who actually buy black walnut kernels is what we call in the brokerage field Eastern black walnuts. That means Kentucky and Tennessee. Those are Eastern blacks, they are the blacks with the flavor, the blacks that stand up. From my home state they have Missouri blacks, but the quality isn't there. The flavor doesn't hold up. But you people down here grow the finest blacks in the world. California, yes, California grows and shells a lot of black walnuts, but they don't have a black walnut flavor. The flavor is gone. Where it went, I don't know. But there isn't any black walnut flavor in California blacks. [A different species, _Juglans hindsi_--Ed.]

So some unscrupulous people buy California blacks and mix them with Eastern black walnuts. Then they can't call them Eastern blacks. They are just black walnut kernels. But black walnut kernels that are 100 per cent Eastern black walnut kernels should be the standard of black walnuts through the country.

Now, Tom has told you something about the process of shelling. I am just going on to say that the average sheller gets about 10 to 11.7 pounds of black walnut kernels to the hundred pounds. So you can realize there again what a problem he has.

Well, the marketing of black walnuts is the selling of black walnuts in the shell or shelled. We have very little demand in the Chicago markets for in-the-shell black walnuts. I probably sell, oh, maybe 5,000 pounds a year on South Water Market, and they go out to the various stores, and they, in turn, sell them to the homes that like to crack black walnuts instead of buying the kernels.

The American public buy with their eyes. Consequently, the packaging of black walnut kernels or the packaging of any merchandise is very important. I made a statement this morning that has always been interesting to me. You know, Chicago is the biggest candy center in the world, and we do a lot of experimenting with candy. Now, your industry is tied very closely to candy, because a lot of the black walnuts, hickory nuts, and the like, go into the making of candy. But to prove my point, a number of times friends of mine who are interested in the sale of merchandise have taken quality candy and packed it in a common box, and they have taken an inferior quality of candy and packed it in a fancy box and set it on the floor and put the same price on both products. The American public, remember, buys with their eyes. So they buy something that is well dressed and they buy that inferior product, twice or three times as fast as they would that quality product in the common box.

I am bringing this out to illustrate a point. _Well packaged merchandise, sightly merchandise, always pays._ Quality to you people who actually crack black walnuts in your homes is something that will pay dividends. Separate your big kernels. Offer them to the public and they will pay for them.

I was talking to Dr. Jones of Pennsylvania about the sale of black walnut halves. He says that he gets a good many of them. Well, there are throughout these United States of ours a good many very fancy stores that will buy merchandise of this type. But the quantity that anyone gets is very small, so the suggestion that I made to Dr.

Jones is that he take his quarters and mix them with his halves. That's not cheating or anything like it. It is making a product that is superior. And you know they say if a man makes a better mousetrap the world will come to his door. And that is generally true. Sometimes it takes a long time to bring it to the American public or to your buyers, to make them realize that you have a superior product, but that's the thing that it takes.

Now, there are a number of ways they sell blacks in this country. They sell them in two-ounce cellophane bags, they sell them in six-ounce cellophane, they sell them in eight-ounce cellophane, but the greater quantity of the blacks are sold in bulk, as Tom told you, in 35- and 50-pound cases, and they go to the candy manufacturer, they go to the ice cream manufacturer, and chiefly throughout the southern part of the United States for ice cream, believe it or not. The Southern States buy more black walnut ice cream than any other division of the United States. In the Central West, too, black walnuts are quite popular for use in ice cream.

Now, if there is anyone that has any questions, I'd like for you to ask them, and I will try to answer them, I won't promise that I can, about the marketing of black walnuts.

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A Member: What's the retail sale on those cellophane bags?

Mr. McCauley: What would be the retail sale price?

A Member: Yes.

Mr. McCauley: Well, the cellophaning of walnuts is quite an expensive proposition. We will say right now the kernels are worth 70 cents a pound. The cellophaner has to add a dime a pound to that price, so he figures his cost at 80 cents and the cost of cellophane, and he sells that merchandise so that he makes a 15 per cent profit. Let me see if I can tell you, a two-ounce bag--

Mr. Mullins: It sells for from 18 cents to 25 cents.

Mr. McCauley: Yes, 18 cents in the chain stores. An 8-ounce package at A & P in Chicago will sell for 59 or 69 cents. I have forgotten now just what it is. I can't keep these prices in my mind, although I will tell you this now. If any of you ever come to Chicago, I have an experimental plant in Chicago. If you could remember McCauley, it's "McCauley Company," or "McCauley Machinery Company," and in that plant I also have a new machine for bagging nuts, cellophane bagging. It makes the bag, fills it and seals it in one operation, and we have operated that machine at the rate of 100 bags per minute, 2-ounce or 6-ounce, it doesn't make any difference. The only trouble is the people couldn't handle the bags that fast, so we had to cut it down to 58 a minute. It's quite an operation, and at this time it is an experimental operation. But I would be more than pleased to have any of you drop in on me in Chicago. If I am not there someone in my organization will be glad to show you, if you tell them what you came for.

I have a "California" walnut, or Persian, as you call it. I was much surprised to see all these samples of walnuts down here. I have a walnut shelling plant in Chicago, I do at this time. Maybe when you get there it will be a pecan shelling plant, or maybe it will be a _Macadamia_ nut plant. How many of you people have ever heard of _Macadamia_ nuts? (Several hands raised.) More than I thought for. Well, we are working on a plant to shell _Macadamias_ now. Of course, that is a tropical nut, grown chiefly in Hawaii and Australia. The Australian nut is not nearly as good as the Hawaiian nut. But to those of you who are not familiar with the nut, I have given it to any number of people and asked for their reaction, and some said it tasted like a filbert, others said it tasted like cocoanut, and the third one named was Brazil nut. So it's a very pleasant nut to eat, but very, very expensive.

Dr. Moss: I live in Williamsburg in Whitley County not far from you, and we have no market there for black walnuts at all and got quite a lot of them there. I wonder if it would be practical to have a collection center.

Mr. Mullins: It certainly would. In the southeastern part of Texas we have one.

Dr. MacDaniels: A question, Mr. McCauley. You said that you are able to recover about 11 per cent in the cracking plant on the average, I think you said 10 to 11.7 for ordinary run quality. Now, if you had walnuts that would run 25 to 28 per cent kernel, how much would your processing plants recover out of that, I am just-wondering?

Mr. McCauley: Well, I would like to say two per cent less than the hand-cracked weight. In other words, if you had a total, hand-cracking total kernel content of 25 per cent, I would like to say 23, but I think that is just a little bit strong. In Tom's early processing of black walnut kernels at Renfro Valley his first average was 16 per cent on wild nuts. I don't know where he got those nuts. They must have been Thomas variety. But as he told me today, he is down to 10.7.

Mr. Mullins: Those nuts I talked about, Mac, that ran up that high percentage were from over in Clark County around Winchester. And I have quite a few of them that I pick-up that are even larger in size than some of these Thomas nuts that are lying in here, out of that particular locality. They are very big.

Mr. McCauley: You will find that that is true. Your percentage varies over the country. I like to think that the wild seedling black walnut has a possibility of about 18 pounds in a hundred. I may be wrong.

Dr. MacDaniels: Is that loss in the cracking procedure; I mean, that the things don't crack out?

Mr. McCauley:. The loss is in the cracking, but on an 18 pounds possibility we would probably get between 14 and 15 per cent with this new method of cracking and processing.

Dr. MacDaniels: Now, if you had a nut that would run hand-cracked 24 per cent, you lose 2 in your cracking procedure, and you recover 22. Would you pay twice as much for nuts of that quality as you would for common grade?

Mr. McCauley: Yes, I think that I would. If I had assurance that I was going to get 22 per cent kernels I would be very pleased to pay double. It would pay me, if I were shelling, to pay twice as much for that variety for the simple reason that I only have one cost of picking. Now, the average cost of picking black walnuts kernels is about 11-1/2 cents a pound. At least, that's the best I have ever been able to do with them. And if you sold me a walnut that would give me twice as many kernels with one cracking and one picking, I'd make money and I could pay you twice as much money for that nut.

Dr. MacDaniels: What volume would a cracker have to have to make it interesting? What quantity would have to be produced and offered to a cracker to make it interesting? That is, say I have 50 bushels of Thomas. That isn't any good to you, because your cracking plant--

Mr. McCauley: Why wouldn't it? If I had 50 bushels, that is 2500 pounds, right? All right, Tom could run his plant for two hours and a half, we will say, on 2500 pounds, and in that two hours and a half he would be getting as much kernels as he would otherwise in five hours. That's good business.

Mr. Chase: I'd just like to comment on that 18 per cent kernel you mentioned as the average you'd like to think of. Mr. Zarger has run a study on the sample trees in the Tennessee Valley to measure the kernel content in some 130 trees for about seven years running, and it pans out to about 18 per cent. I thought you'd just like to know.

Mr. McCauley: I didn't want to make a definite statement and then have somebody throw something back in my face. That's why I said I'd like to think.

Mr. Fisher: Since this question has come up and a cracker is here, and that is the question, whether the commercial cracker would be willing to pay a premium price for premium nuts, I wonder if Tom will answer the question, would he pay twice as much?

Mr. Mullins: Certainly.

Mr. Chase: Mr. Acker is another sheller who operates in Morristown, Tennessee, and Broadway, Virginia, who just dropped in on us. I'd like to ask him that same question.

Mr. Acker: What is the question again?

Mr. Chase: Would a considerable quantity of walnuts such as Thomas be worth more to you?

Mr. Acker: We'd be inclined to buy them according to the value we get out of them.

Dr. MacDaniels: What would you do, run sort of a pilot test on them?

Mr. Acker: If I can buy something for a dollar and make money on it, I am willing to try.

Mr. Chase: I made a little unscheduled talk last night in which I said from my information from shellers in Nashville, particularly, that they at this time would not

be able to pay any premium price for higher quality nuts simply because they do not have time to examine each bushel, each hundred pounds that comes in and see whether they would pay a special price for better walnuts. Is that the general situation?

Mr. McCauley: Yes, I think generally that is. They take everything at the price of seedling variety. However, you gentlemen who are growing cultivated nuts shouldn't be too disturbed, because of the fact that there is going to be a time in the not-too-distant future where you can dispose of those nuts according to the kernel content within the nut.

President Davidson: I think that's the answer. Eventually it will come.

Mr. McCauley: It's good business. Stop and consider. We go right back to that point where we are going to get twice the amount of merchandise out of a hull which must be broken, which must be picked, which must be cleaned, which is one operation. In a two and a half hour period, which is what it would take, we will say, to run 2500 pounds, you would get the net content on a Thomas variety that you would ordinarily receive in five hours of actual operation. You are saving two and a half hours labor, you are saving two and a half hours machine time, and you are getting just twice as much.

Mr. McDaniel: You'd have twice as many girls on the sorting belt, wouldn't you, to examine that volume?

Mr. McCauley: No, not necessarily. When it gets to that point it isn't necessary. Sometimes the machine gets too far ahead of them, but the machinery is fast getting to a point where it is going to be more or less mechanical. It's an inspection proposition.

Mr. Taylor: May I ask you this question? In other branches of farming you have what you call seed certification, as with certified potatoes, and people who certify those potatoes. Wouldn't it be possible for the same Government agency to certify growers of walnuts so that when you bought from certain members of this association they would be certified so you would know what you were getting? Would that be possible?

Mr. McDaniel: Certification has to do with planting stock.

Mr. Taylor: I mean a different type of certification.

Mr. McDaniel: What you have in mind probably is U. S. Grades on fruit. For instance, if it is stamped "U.S. 1" it should be considerably better than orchard run, and I don't know why it shouldn't be possible for nuts in the shell. It is used in California.

Mr. McCauley: It is in peanuts. All peanuts are Government graded, and that's in the shell. But this black walnut situation is going to take a little longer than that. But I am sure that there are people in the shelling business who would buy Thomas variety or the other varieties if you just go ahead and tell them that's what you have. People are always looking for something better, and I am sure that your cultivated

varieties are going to be better, but you are going to have to keep talking them up all the time and getting them to the people who will buy them.

President Davidson: Right. We'd all like to go on with this, but we must really go on with the program, too. We will next hear something about pasteurization. The Production of Bacteria-Free Walnut Kernels will be discussed by Mr. Pease of West Virginia University. Mr. Pease.

Production of Bacteria-Free Walnut Kernels

ROGER W. PEASE, Assistant Hillculturist, West Virginia University, Morgantown, West Virginia

Mr. Pease: Before I go into any detail about the construction of the pasteurizer, I am going to review the bacillus that causes the trouble very briefly. Most of you will know more about it than I do, but some of you may know less.

When the farmer takes the hulls from the black walnuts he generally spreads his hulled crop to dry almost anywhere. Rats will go over them, and these rats or mice infect the hulled walnuts with an organism called _Bacillus coli_ that is on the outside of the shell. They go from there to the cracking plant, go through the cracker which thoroughly mixes up the infected nuts with the clean ones. They go from there to the separator, which does a better job at spreading the bacteria. Then they go on the market. If they are shipped from one state to another they are subject to inspection by Federal authorities. If they find this organism in the kernels, they may at their discretion heave the whole shipment into the river. They don't always do it. They haven't worked out yet a definite scheme to follow. In other words, they will not tell us, "If your kernels have a certain number of these _B. coli_ in them we will let them by." As it reads, there should be not one organism there, and I can assure you that's almost impossible to get if a rat has crawled over those things.

Now, to get rid of poison ivy the best way is not to get it, and it's just the same with this organism. The place to get rid of it would be for the farmer to store the nuts to dry where the rats and mice cannot get to them and for the cracking plants to do the same. Unfortunately, this isn't done and sometimes isn't practicable. The next place to hit them would be before they are tumbled, that is, before the black powder on the outside of the shells is shaken off in a tumbler, or immediately after that to disinfect the shell without hurting the kernel.

That is where we should have started at West Virginia, but we didn't. We began at the other end after the thing was through and began studying pasteurization. The Government had recommended, I believe, temperatures of up to 300 deg.F. for pasteurization. We found out right away--that is, I didn't, Dr. Colmer and Harvey Erickson, who are now--one of them--in Baton Rouge and the other one in Seattle, and they would know about it. They found out that after temperatures of over 300 deg. the nuts tasted toasted and they would not keep nearly so well or so long as an unpasteurized nut.

After inspecting what pasteurizers they could get access to they concluded some work was necessary, so they spent 12 months and found that at a temperature of 160

and humidity of 80 per cent a more efficient job of pasteurization was done, and at the same time the kernel was not hurt at all. The taste was identical with an unpasteurized nut, and it would keep just as long. At that point one of them, as I say, went to Louisiana and the other went to Washington, and the research fell on my shoulders, that didn't know much about it.

We started to construct the machine. Meanwhile, Mr. Erickson told me he had developed a new strain of bacteria which was much more hardy and 160 degrees at 80% humidity would not kill the thing. So we constructed our machine to run a temperature of 180 at 70% humidity for 30 minutes, and that will kill them.

Now, in 15 minutes I can't give you anywhere near all the details of construction of that machine. I can give you a few of the principles. On the outside, of course, is a well insulated box. The nuts are fed through the top with a revolving drum with fins on it. They comes down to a belt that travels this way for six feet, drops to another, travels back, a series of five belts. It takes them just half an hour to go through. The layer of nuts is perhaps three-eighths of an inch thick. The temperature is kept up with electric coils. It is regulated with a thermostat.

We had some difficulty with the humidity. Try it and see. As we raised our temperature it was hard to keep our humidity up. Finally we went back to the simplest thing, which usually works. We just took a pan of water, with a solenoid valve and float such as you have in the modern hot air furnaces and put a magnetic switch on it. As the water boiled it helped raise the temperature, and it gave off vapors. The automatic switch and the wet and dry bulb from the thermometer and thermostat will shut the water off and shut the heat off automatically when you get the required temperature and the required humidity. In that machine our nuts start at the top, take 30 minutes to travel through. From the time they start at the top until the time they get to the bottom they have a standard temperature of 180 deg. plus the 70% humidity.

Then the second problem, if you want to make one, is to get that temperature standard in all places. I know one man who made one of these machines and put four fans in at different places, and when he closed it up and got it to working, the center of his machine was still cold, because your hot air acts differently from free air. We put at the bottom a shelf with a tube in it and a big fan in the middle. The air is drawn down from the top here, driven through there, hits some baffles and comes across each belt. In that way it works.

Now, if you want, any of you, to get the details of the pasteurizer you could write to Mr. Erickson, College of Forestry, University of Washington, Seattle 5, and he who designed it would be very glad, I believe, to help with your problems, or you could write up to our Agricultural Engineering department, and they would do the same.

I will tell you this, that after we drew up the plans, I took the plans to several manufacturers, and the cheapest bid I got was $5,000 to make it. We made it ourselves for a little less than $1,200 not counting labor. Not that they would have made that much profit, but I tell you that to show you it's a rather inexpensive machine. On the other hand, you can save considerable money by getting it made up

yourselves.

I am going to stop with the thing there. If there are a few questions that you wish to ask, I will try to answer.

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A Member: What is the name of the bacteria you are killing?

Mr. Pease: _Bacillus coli_, that's the chief bacteria, and the others that cause the damage are similar to that, and they are always spread by the rats or the mice.

A Member: Do the kernels properly pasteurized show any brownness of kernel?

Mr. Pease: No, they are identical with an unpasteurized kernel at that temperature.

Mr. Korn: I buy kernels at the plant in Nashville, and some of them have been toasted.

Mr. Pease: They have 350 degrees.

Mr. Kays: You mentioned you should have started on the other end a while ago. Could you treat those nuts before they are cracked and do the same thing for less money?

Mr. Pease: I believe you could.

Mr. Kays: The other question I have is how about using ultra-violet light?

Mr. Pease: I have written to a good many authorities, and some of them say yes and some say no.

Mr. Kays: In pecans that is one of the practices.

Mr. Pease: I believe you could use it in our present machine.

Mr. Stoke: Isn't this heat to remove contamination? After the nuts are cracked is there any examination of the nuts?

Mr. Pease: No, there is not.

Mr. Stoke: Could there be any possible value in sterilizing the nut before it is cracked?

Mr. Pease: Yes. You see, the bacteria is on the shell, on the outside. Then when you crack it, it gets on the nut.

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President Davidson: Thank you very much, Mr. Pease. I do hate to cut these discussions short. You get as much out of them or more, sometimes, than we do otherwise. There is just one thing I'd like to say before we take a five-minute recess. Mr. Acker is here. He is another man that you might talk to in addition to talking to Mr. Mullins during the recess.

(Recess taken.)

President Davidson: The meeting will come to order. The first thing on the program is a talk by Dr. Cross, Head of the Department of Horticulture, Oklahoma A. & M., Stillwater, Oklahoma, on Pecan Selection in Oklahoma. Dr. Cross.

Pecan Selection in Oklahoma

DR. FRANK B. CROSS, Head, Department of Horticulture, Oklahoma A & M College, Stillwater, Oklahoma

Dr. Cross: Mr. Chairman, ladies and gentlemen: The present status of the pecan industry in Oklahoma is the result of close cooperation between the growers and the experiment station combined with a resource which we have in that state consisting of thousands of native pecan trees which may be quickly and economically changed into producing trees instead of just wild forest trees.

I am going to utilize my time this afternoon to discuss, first, briefly the present situation as we find it with reference to pecans in Oklahoma, because there is the important phase of nut growing which we follow in that state. We do grow some walnuts and we have a great many men interested in walnuts, but far and away our major interest is in pecans.

We might divide the work and interest in the state into two phases. First, but of least importance, is that connected with the planting and production of varieties. We have a great many men in the state who wish to plant land to pecans, and, of course, in cases like that the varieties which are available are always selected for planting, and nursery trees, of course, are utilized. The latest phase of that type of development is the planting of apple trees for filler trees with the expectation that the apple trees will be removed after 15 or 20 years, thus leaving the pecan trees at a large size to fully occupy the ground, and in the meantime the apple trees, of course, have produced a profitable crop.

Our growers, however, and the industry in the state are far more concerned with the utilization of the native trees. To talk about these native trees is almost--well, we might borrow a Texas expression--these trees grow both in Oklahoma and Texas--and the Texans say whenever a Texan tries to tell the truth everybody knows he is lying. That's the way everybody knows about some of these native trees. When we think of a huge, tall tree 20 or so feet in circumference over a hundred years of age and realize that the white man has occupied that particular territory for only a little over 50 years, we wonder about the history of that tree for the first 50 years of its life when wild Indians were roaming the territory and buffalo were grazing under these trees which were getting started.

These trees occur along the streams, very seldom out away from the streams for any considerable distance, as one of the native forest trees and in sufficient number so that when all other trees are removed the stand of pecan trees remaining is in many cases more than adequate to make a complete stand of pecans for commercial production. So that after having removed the oaks and elms and cottonwoods and willows and the other native trees, we have the opportunity of making a considerable selection of desirable native or seedling trees by observing the type of nut which each tree produces.

We are not, in making this selection, concerned so much with the size of the nut produced as we are with the kernel percentage which will be yielded by the nut upon cracking and extracting the kernels and by the ease of separation. Within comparatively recent years many cracking and shelling plants have been established throughout the state, and the history of the industry I think will record that the establishment of these cracking plants in the territory where the pecans are produced will be a great stimulus to the production of that kind of nuts.

I don't know whether I have made the picture clear or not. Throughout the eastern part of the state, that part which you in your old geographies knew under the name of Indian Territory, and particularly concentrated in the middle of the state there are native trees which if properly handled, that is, cultivated and sprayed and thinned so that each tree stands out individually by itself, will produce in paying quantities.

On the experiment station we have a half mile of such territory lying between cultivated fields on both sides of a creek which had eroded a considerable basin. The area was unsatisfactory for cultivation, and so it was fenced out. Back some years ago the area was cleared of grape vines and other trees, and we have since that time pastured sheep in this tract of land. It Is narrow, not over three or four hundred feet wide at any place and, of course, varying in width from one end to the other, and the creek meanders along. There really is more than a half mile of total length.

The potential production of that half mile is now, in terms of dollars and cents, about $2,500 to $3,000, and before wheat and cattle attained their present prices that was no mean income for a quarter section of land. Naturally, with that opportunity prevalent over a great part of the state, we in Oklahoma are interested in the production of native or seedling pecans to be sold to the cracker. We feel that the future of the pecan industry is undoubtedly headed toward the utilization of pecans as kernels and not nuts in the shell. Such being the case, we are not interested particularly in large size. We are interested in kernel yield and in the potential production of each individual tree.

There are a great many problems connected with the industry, and we have more or less taken those into consideration and classified them under insects and diseases and marketing and harvesting and varieties. I will not have time to touch upon very many of these. Our harvesting situation is completely chaotic. Within the last two ot three years shaking machines have been developed, and we are indebted to the West Coast growers for these inventions, which are very helpful. Previous to that a, long bamboo pole was used to knock the pecans from the trees, and then they were picked up off the ground. There are two machines now waiting for the present crop to be harvested which are supposed to pick up the nuts by vacuum picking.

If the industry can be mechanized in that manner, getting away from harvesting pecans as we have been harvesting them, it is just like cradling wheat as compared to the present-day 12-foot, self-propelled combine that cuts the wheat so rapidly. If this mechanization can be put into effect, then the native seedling territory in Texas and Oklahoma will be able to produce pecans at a price which the market will accept.

I don't know whether you know it or not, but the pecan market situation has apparently reached a condition of saturation. It was very difficult to sell pecans last fall, not because there is over-production, no, but because there is under-consumption.

There are two things which will remedy the situation. The pecan is unquestionably the finest nut that is produced in the United States. If the people of the North can be acquainted with the pecan, there is no question in my mind but that it will be possible to vastly increase consumption. The Oklahoma growers and buyers hope to put before the legislature a proposition to assess a tax of a quarter of a cent or something like that per pound, which will be used in an advertising campaign to advertise pecans outside of the state, so maybe you folks in New York and elsewhere, if the campaign is successful, will hear more about Oklahoma pecans in the future.

Well, these seedling trees--I must get on with my story--are cultivated and sprayed. We are sometimes accused of producing wild nuts at no cost. This is not the situation distinctly. It costs just as much to produce these native seedling nuts as it does to produce the varieties, the advantage being that we start with a large tree which is capable of producing from 50 to 200 or 300 or even 400 pounds of nuts within four or five years after the operation is started instead of waiting 20 or 25 years to get good commercial production.

As I said, a selection is made of the trees at the beginning. The selection is continued with each succeeding year as the trees grow larger and additional trees are thinned out so that they stand eventually a hundred or 150 feet apart, giving to each tree adequate room.

Throughout the state we have a great deal of interest in propagation by topworking of varieties of pecans. The experiment station made the serious error for 15 or 20 years in the early development of the interest in the work in centering on the idea of changing these natives over to varieties. We now are swinging back to a proper evaluation of the native nuts, and nobody is satisfied with the present varieties, our interest of developing and the exploration and discovery of new varieties being such that the Northeast Oklahoma Pecan Growers Association arranged two years ago to finance a contest for the discovery of seedling nuts which could be utilized in that territory and be more profitable than any variety that we now have.

We don't like the Stuart because of its low quality. We don't like the Stuart because it doesn't come into production until it reaches a considerable age. We just simply will not have the Mahan, because it doesn't fill. We do not like the Success because it has a tendency to over-bear every other year and does not fill. We cannot use the Squirrel's Delight which for ten years or so we had at the top of our list, because a special strain of scab fungus came in and completely wiped them out, and so on throughout the list of varieties that we have.

Well, these growers decided to take the matter into their hands and in cooperation with the experiment station have been, during the past two years, attempting to find some nuts which would be more desirable, and I thought those of you who are in the

walnut exploration work would be interested in learning how this is worked out.

I don't suppose you can see this. It Is an entry blank for the grower. Annual prizes of $50, $25, $15 and $10 are awarded. Ten awards are made each year, and the ten winning growers this year will have their particular nut automatically entered in a grand prize contest hoping that some of those nuts will be worth naming, and if any should be worth naming, after further study, naming and introducing, the grower will be awarded a prize of $1,000.

Four of the Tulsa, Oklahoma, crackers are financing this work by putting in $75 each annually. The college is cooperating in this respect, and when I say the college, I also mean the Extension Division. The Extension Director is pushing the matter and tells the county agents to pay attention to these entry blanks when they come, and get as many growers in each county to send in samples as possible.

The contest closes on November 25th. Those samples are sent to the college, and in three or four days--and those of you in colleges will recognize the Thanksgiving holiday--in three or four days' time those nuts are cracked and evaluated and placed. Last year, the second year of the contest, there were over 200 entries, and it was no small job to finish in time to get them on display at the annual meeting and show of the Oklahoma Pecan Growers Association in early December.

We are not content with the evaluation of the nut. It is just one phase of successful production to have a nut which is satisfactory for cracking and consumption; unless those trees are free from disease and productive and otherwise satisfactory we could never think of introducing a variety. And so the staff at the college, as soon, as the show is over, goes out and locates each of these trees individually and puts a tag on it. We visit each of those trees a sufficient number of times during the year to properly evaluate the tree.

The things that we are looking for, of course, are productiveness, freedom from disease and other characteristics of that type. If, after five years of observation, the tree characteristics are satisfactory, then the nut will be certified as worthy of propagation.

We are getting some place with this program, as evidenced by the data on last year's cracking contest. Normal seedling pecans yield about 33 per cent kernel to the packing plant. In last year's contest, as I say, there were over 200 entries, and I was just looking to see what the low was. I really haven't paid enough attention. The lowest entry apparently was about 33 per cent, and the highest entry was 59 per cent kernel. Over 30 of these seedling nuts yielded better than 50 per cent kernel, and that is better than most popular varieties.

These nuts are relatively small. The cracker doesn't care how small they are, he wants a nut that handles well in the cracker, a nut that is the shape of a football. A miniature football is an ideal cracking type of nut. The cracking docks come together from the ends. We cannot use a round nut. About two-thirds of these good nuts which yielded over 50 per cent kernel were so round that the machinery in cracking would not place the docks on the ends, but they were apt to hit anyplace. So they had to be discounted.

It is quite a job to evaluate these nuts. We have been arbitrary about it. We haven't developed any scoring system, because there are so many variable factors that it seems to be almost impossible to do so. In our general plan of operation in the state we expect this native grove improvement program that I described to continue, and as the trees get larger the growers will topwork sprouts which develop from the trees which have been removed so that the thing goes on and on with a constant improvement in the quality of the nut.

We also have many, many acres of nuts being propagated by topworking to varieties rather than by letting the seedling continue to produce. That is the reason why we are so much interested in getting a better type of pecan.

One man who makes it a commercial practice puts on thousands of scions every year. We in Oklahoma can't understand why you all seem to have so much trouble propagating nut trees. It is just as easy to propagate pecans and walnuts--not quite as easy--as apples, but then it isn't too difficult. I think it is the attitude and frame of mind in which you go about it.

Thank you very much, I appreciate the time.

* * * * *

President Davidson: Thank you. Now, then, Mr. Magill of the University of Kentucky, will give us "A Planned Program for Improving the Pecan Industry in Southwestern Kentucky."

A Pecan Improvement Program for Southwestern Kentucky

W. W. MAGILL, Extension Horticulturist, University of Kentucky

The production and marketing of seedling pecans in extreme southwestern Kentucky has been of major importance for many years.

This industry naturally extends into northwestern Tennessee and parts of Missouri directly across the state line in the Mississippi River bottom. It might be said that this industry was developed by nature, because in the Mississippi River bottoms we find seedling pecan trees which undoubtedly are more than 100 years old. Some native seedling pecan trees in this area are five feet or more in diameter; some have a spread of branches covering a radius of 60 feet, and are more than 100 feet in height.

This industry took on considerable momentum about seven years ago when a group of local business men at Hickman, the county seat of Fulton County, developed a cracking plant known as the Roper Pecan Company. They now have thirty modern cracking machines, with sorting belts, grading machines, and other complete equipment, so that they are in a position to receive and process a large tonnage of native seedling pecans, merchandise the kernels and other by-products and, therefore, are able to purchase a large quantity of seedling nuts and operate their plant for eight months each year. Not having sufficient local nuts ("Kentucky Kernels") to take care of their business, they also buy not only Kentucky nuts but also from Tennessee, Missouri, Illinois, and Arkansas.

+Young Native Trees Top-Grafted+

Realizing that this industry is here to stay and that many farmers of that district have many young seedling pecans growing on their farms, the Kentucky Agricultural Extension Service, cooperating with some of the leading farmers of the district, has launched a program of top-working these native seedling pecans with some of the known, improved varieties that have a reputation for producing good yields of high quality nuts. The beginning of this top-grafting program was in late April of 1948, when some 200 trees were top-worked mostly on the farm of the late Roscoe Stone, a farmer in this area who had considerable land holdings. It is highly pleasing to our Extension workers to observe what we think is an outstanding success in this first year of development, for we find that over 90% of the five to ten year old trees that were top-grafted have developed a new growth from the spring grafting, to the extent of from two to eight feet of new growth. John Watts, County Farm Agent of Fulton County, who pioneered this pecan improvement development, tells me that we already have requests for top-working over 500 other trees in this area for the spring of 1949.

+Northern Varieties Preferred+

The best information available was sought from such pecan authorities as Ford Wilkinson of Rockport, Indiana, Dr. A. S. Colby, chief in nut culture, Horticulture Department of the University of Illinois, Bob Endicott of Villa Ridge, Illinois, and others. They are of the opinion that this southwestern Kentucky area approaches the northern limit of successful production of known southern varieties of pecans, and that our success in our pecan grafting program can best be assured by top-working to the hardy northern varieties of pecans such as the Major, Greenriver, Niblack, Giles, Goforth, and others.

Thanks to our pioneers of this generation who located some outstanding seedling pecans in the Ohio Valley, such trees as the original Major, Greenriver, and others have proved their worth as hardy northern pecans and they have been used for propagating purposes rather than being destroyed by farm hands who burned piles of debris left by high water around many of these early trees.

+Some Superior Local Nuts+

We are of the opinion that other seedling varieties are now growing in this Reelfoot Lake area, maybe in Kentucky, maybe in Tennessee, or across the line in Missouri, that are equally as good pecans, and, we hope, better, than the already named seedlings which have been mentioned above.

During early August it was my pleasure to spend a day in the Hickman bottoms with County Agent Watts and Mr. Ernest Fields, manager of the local nut cracking factory, together with Mr. C. B. Toombs, of Hickman, at which time we inspected a number of recognized successful native pecan groves. Mr. Toombs knows that whole area and is familiar with the pecan trees of outstanding quality and yield history, just as you and I knew where every tree stood in the old home apple orchard or that of grandfather, where as boys we made frequent trips to get a pocketful of those outstanding local variety apples.

Mr. Toombs pointed out to me a tree on his own farm that he said bears a crop every year of from 300 to 400 pounds of nuts. In his own language he described the tree in detail but the thing which impressed me was the fact that he had developed standing orders for private sales to individuals from the crops of this one tree each year because they are of outstanding value. He showed us another tree on a neighbor's farm, one which produced 700 pounds of nuts one year; another tree on which the nuts were ready to harvest a month ahead of the nuts from other pecan trees in that region. (Mr. Wilkinson, it strikes me that propagation from this early maturing tree might well find a place several miles north of the Mason-Dixon Line where normal fall frosts often injure the crops.)

We are of the opinion that with organized effort we can locate these outstanding individual trees, get enough scion wood from them and put it in the hands of a good pecan grower, and in a few years develop sufficient grafting wood so that we can top-work thousands of these young native trees in the district, thereby increasing not only the number of pounds produced per tree, but have a volume of production of the very best quality of nuts. They tell me that one of the trees I observed has a cracking percentage of above 60 as compared to many of the native seedlings which have a cracking percentage of only 20-30 of nut kernels.

+First Annual Nut Show in 1948+

In an effort to locate these outstanding seedling trees in an organized way, our Kentucky Extension Service, cooperating with the Fulton County Farm Bureau, local civic organizations, the local nut cracking plant, and the Northern Nut Growers Association, through its secretary, Mr. J. C. McDaniel, has made plans for a nut show to be held at the county court house in Hickman, Kentucky, in early December of 1948. The feature of the show with be the cash prizes offered for the best seedling pecans. We request that the owners give us a history of the trees, the age, regularity in bearing, etc., with the nut show management reserving the right to cut a few sticks of grafting wood from the winning trees. Prizes will also be offered for hickory and walnut seedling trees. An educational program is also planned in connection with the day's show, and it will include a visit to the farm of the late Roscoe Stone, where a top-working program was started last spring, as well as a visit to the local nut cracking firm. This nut show is set up to become an annual affair, and we feel that the sky is the limit for the good that can come out of such an organized program as it affects the pecan industry in that area.

There are thousands of acres of excellent pecan land in this southwestern Kentucky area, that can be profitably developed into pecan groves. The land is deep, very fertile, and is already well supplied with moisture. We cannot question its being a natural home for pecan production, for nature proved this point to the public two generations ago.

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PRESIDENT DAVIDSON: Pecan Culture in South Carolina by Mr. A. M. Musser, Head of the Department of Horticulture at Clemson Agricultural College is next. Mr. Senn will read the paper because Mr. Musser is not able to be here.

Pecan Production in South Carolina

T. L. SENN, Assistant Professor of Horticulture, Clemson, South Carolina

In the southern colonies on the Atlantic coast, the pecan was first described by Thomas Walter in his publication "_Flora Caroliniana_" in 1787. He was an Englishman who had a plantation in St. John's Parish on the Santee River, South Carolina, where he made an extensive collection of southern plants. After describing the tree, evidently a nursery specimen, he ended with the words, "The fruit I have never seen." It is known now that the native range of pecan did not extend to the present state of South Carolina. One of the first large pecan plantings in the state dates back to 1890; This was a seedling planting of 1000 trees made by John S. Horlbock at Charleston. Some of these trees are still producing. The planting never proved profitable and has changed ownership several times.

There are several small plantings of black walnuts, Chinese chestnuts, and Persian walnuts in various parts of the state. Persian walnuts do well in the Piedmont soil region and in 1947 the trees there had a good crop.

+Commercial Pecan Plantings+

The pecan, is one of the most popular tree nuts and is the only one grown on a commercial scale in South Carolina. Pecans are grown in every county, although there is a comparatively small number of trees in most of the Piedmont and Mountain counties, and several counties in the lower Coastal Plain. Orangeburg County, with the largest number, had 27,528. Pickens County, with the fewest trees, had 801. The total for the state was reported as 227,027 trees.

Pecans are an important money crop of the state. During the last five years the production of pecans has averaged three million pounds, which brought farmers a yearly average of $500,000. The average yield per tree of bearing age in 1947 was only about 7 pounds, or 100 pounds per acre. Eighteen cents was the average price received for improved varieties, and twelve cents for seedlings, during the ten-year period 1935-1944. With these prices and yields per bearing tree, it is easily seen that there is plenty of room for improvement, for the production of pecans in South Carolina by the average grower has not been very profitable during the past nine or ten years. South Carolina has ranked fifth or sixth in the production of pecans of improved varieties during the past several years. While production from year to year has been up and down, the general trend is up.

There are two general classes of pecan trees grown in South Carolina: seedlings and named or improved varieties. The average crop figures over the ten-year period 1933 to 1942, show that six times as many nuts of improved varieties were produced as of seedlings. South Carolina produces about 6% of the pecan nuts of improved varieties in the United States and less than 1% of the seedlings. The seedling trees are for the most part given very little attention, receiving neither fertilizers nor sprays. They produce nuts of miscellaneous size, shape, and quality, and are usually smaller than the improved varieties. The cost of production of seedling pecans is small for they are usually grown in back yards, in chicken ranges, and in pastures.

There are a number of pecan varieties that are adapted to and grown in South Carolina. The most popular varieties are Schley, Stuart, Success and Moneymaker. A number of other varieties, including Teche, Frotscher, Mahan, Pabst, Delmas, Van Deman, and Moore are grown in some sections.

Schley is very susceptible to scab and should not be planted if a spray program is not carried out. Moneymaker, Stuart, and Success are not so very susceptible to scab and are satisfactory where a complete spray program is not used. Some years ago several growers in one county ordered Stuart trees and these trees, now bearing, turned out to be Teche, so there is some uncertainty as to the variety names in some sections.

The planting distance varies considerably, depending somewhat upon fertility of soil and length of growing season. Most of the plantings are too close, having as many as 20 or more trees per acre. Because of the longer growing season in the lower half of the state, trees grown there will be larger at a given age than those grown in the Piedmont section.

+Cultivation Methods+

Intercrops or cover crops are usually grown to increase the income of the farm. Cultivation programs vary according to the intercrop grown. Pecan trees are grown on various types of soil, which also vary greatly in their fertility. Different fertilizers are recommended for these varying conditions. Fertilizer is usually applied late in February or early in March, several weeks prior to the swelling of the buds. The exact time of application varies according to the area in the state in which the trees are grown. Many of the soils of the state are probably too acid for best growth of pecans and the necessary winter cover crops that should be grown in the plantings. In some soils that have been limed, or where the soil pH is 7.0 or approximately so, the application of zinc, to the soil has not eliminated rosette. Few such conditions exist in South Carolina, but where these conditions do prevail, zinc treatment is being tried in the form of sprays, using commercial spray materials.

Unfavorable weather at blooming time often prevents pollination. Instances of cross-incompatability occurring between the varieties grown in this state are practically unknown. Late spring frosts sometimes kill the male or female flowers or both.

The pecan in South Carolina is subject to attack by numerous insects and diseases, just as it is in other places. Scab is the worst offender. Several species of borers are found attacking the trunks, the twig girdler severing the tips of twigs, the shuck worm and case-bearer affecting the husk, and the pecan weevil affecting the nuts. Many of the trees growing in South Carolina are not planted in sufficiently large groves to justify the expenditure necessary for spray equipment. Contract spraying has been done to some extent and has possibilities in South Carolina. Where the number of trees is small this will be the only way in which growers can afford to obtain the use of high pressure equipment.

+Marketing Conditions+

South Carolina Circular 301 gives the following account of the pecan marketing situation in South Carolina. "Most of the pecans in this state are sold in small lots. The assembling at a number of locations of these small lots into lots large enough to make handling economical has been a great problem. It is believed that three auction markets properly located in the state would be the most satisfactory marketing arrangement. If each of these markets would have one sales day per week so that buyers could attend sales at each place, the cost of marketing could be greatly reduced." There are nine companies in five counties that handle pecans.

This is a rather brief discussion relating facts about the pecan industry in South Carolina, and most of the figures given are average figures. Those plantings receiving good cultural practices give more satisfactory returns. The pecans enterprise can be made a profitable one if the grower will carry out a complete program to overcome the problems of fertilization and control of diseases and insects and not just leave the trees to fight the battle alone.

+References+

Rawl, E. H. and Nettles, W. C.--Pecan Production, S. C. Circ. 183, 1940.

Musser, A. M., et al--Pecan Production and Marketing in S. C., S. C. Circ. 301, 1947.

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President Davidson: Thank you, Mr. Senn. I think this program is just tops. I don't know how you feel, but I think it is a wonderful program.

The Storage of Shelled Pecans will be told to us by Professor Harris of the Department of Horticulture of Alabama Polytechnic Institute of Auburn, Alabama. Professor Harris.

Preservation of Shelled Pecans by Drying and Hermetically Sealing

HUBERT HARRIS, Associate Horticulturist, Alabama Agricultural Experiment, Station.

Without some special treatment, storage life of pecans at ordinary temperatures is rather short. Nuts held in common storage from fall harvest show noticeable flavor losses by spring. Staleness and rancidity are likely to follow as summer weather approaches.

These facts emphasize the need of a better method that might be used both commercially and in the home for preserving the pecan over a long period at common storage temperatures. A satisfactory method might be used not only for preserving nuts for year-round use, but for carrying them over from heavy crop years to light crop years.

This report presents the results of experiments conducted at the Alabama Agricultural Experiment Station on methods of preserving pecans.

+Storage Troubles+

Some of the common pecan storage troubles are staleness, rancidity, color changes, molding, and insect injury. Molding occurs only when the product is stored in a moist atmosphere at ordinary temperatures. Insect infestation usually results when the shelled kernels are held in unsealed containers under ordinary storage conditions. Rancidity develops during the summer months when the product is stored by common, methods at ordinary temperatures.

Rancidity is probably the most common of pecan storage troubles; it has been considered the most difficult to control. Rancidity is usually preceded by general loss of flavor followed by staleness of kernels. Color gradually becomes darker as the typical rancid condition develops. These changes are associated with and caused by certain chemical reactions that progress at a slow rate. The oils and fats are slowly oxidized to aldehydes, ketones, and other chemical compounds (10), which cause the undesirable aromas and flavors characteristic of the rancid condition.

+Review of Previous Work+

The period during which pecans will remain free of rancidity and other forms of spoilage varies considerably with storage conditions and other factors. Common storage periods ranging from 3 to 12 months have been reported. Wright (10) placed the common storage life at 3 to 5 months. Blackmon (1) suggested 6 months as the maximum period. Medlock (11) was able to keep them in edible condition for periods up to 12 months. The wide ranges of time as observed by the different investigators are due, no doubt, to storage conditions, variety, quality of the nuts, and seasonal variations.

Cold storage was effective in preserving pecans for periods up to 2-1/2 years. Wright (10) reported effective periods of 13 to 30 months, and Medlock (11) was able to keep them for more than 2 years. Shelled pecans did not keep as long either in common or cold storage as the unshelled nuts.

+Methods and Procedures+

Results of preliminary tests made at the Alabama Station in 1937 indicated that pecans kernels might be kept at common temperatures by drying them in an oven and storing in sealed containers, which prevents absorption of moisture from the air. Since storage tests reported by other workers did not include studies of the relationship of moisture content of kernels to their storage life, it was felt that this phase of the problem should be investigated.

Preliminary tests were made to determine suitable temperatures and periods for the drying process. Temperatures of 200 to 225 deg.F proved to be most satisfactory. These temperatures dried the kernels quite rapidly without appreciable scorching or discoloration. The drying period was varied to give desired moisture contents for the various treatments.

Four methods of sealing the jars were used in the experiments. They were the "cold seal," "steam seal," "hot seal," and "vacuum seal."

The cold seal was accomplished by placing cool kernels in cool jars and sealing without special treatment.

By the steam seal, air was exhausted from the jars with a small blast of steam and the jars sealed immediately. Condensation of the steam resulted in a partial vacuum in the jars and a slight increase in moisture content of the kernels.

By the hot seal, hot kernels were transferred directly from the oven pans to clean, dry, hot jars, and sealed immediately. Contraction of the air as the jars cooled resulted in a partial vacuum.

The vacuum seals were made in clamp-top fruit jars by means of a home pressure cooker. Filled jars, with covers partially clamped, were placed in the cooker. The cooker cover was lubricated at the sealing surface and screwed down tightly. The pressure gauge in the top of the cooker was replaced with a vacuum gauge. The needle valve was removed. An aspirator was attached to the water faucet and connected to the needle valve opening by means of a vacuum hose. After the desired vacuum had been pulled on the cooker, the vacuum hose was removed from the needle valve fitting thus permitting air to rush back into the cooker. The sudden change in pressure automatically sealed the jars.

+Presentation of Data+

Preliminary studies were made to determine the effect of temperature and time of drying on moisture content, color, and toasting of kernels. Results of these studies are given in Table I.

TABLE I.--Effect of Temperature and Time of Drying on Moisture Content, Color, and Toasting of Pecan Kernels.

Oven temperature	Time in oven	Moisture in kernels	Change in color[25]	Degree of toasting
225 deg.F	0	3.7	0.0	0
	20	2.6	0.5	0
	30	1.4	1	1
	37	1.2	2	2
	44	1.0	3	4
	51	0.5	6	8
	58	0.3	8	10
	65	0.2	8	10
	72	0.1	9	10
203 deg.F	30	2.9		
	41	2.4		
	50	2.2		
	60	1.9		

[Footnote 25: The numbers 0 to 10 indicate varying amounts of change in color or degree of toasting: 0 represents normal color and/or no toasting. 10 represents considerable intensification of color and/or development of typical flavor of toasted kernels.]

The first series of the processing and storage tests was started in December, 1939. The treatments together with results are given in Table II. The different samples were dried in an electric oven at 225 deg.F to moisture contents ranging from 0.1 to 3.4 per cent. They were sealed in glass jars, both with and without vacuum, and stored in a dark room at ordinary temperatures. Those dried to 2.9 per cent moisture or less were still good after 2 years in storage, whereas those with higher moisture content were rancid after one year in storage. Samples dried to approximately 2 per cent moisture were still good September 1, 1948, which was almost 9 years after processing and storing. The color was preserved somewhat better by vacuum sealing. However, the quality of air-sealed samples was practically as good as those that were

vacuum sealed. These tests did not show how long kernels might have been kept by drying and storing in unsealed containers.

Table II.--The Effect of Different Amounts of Drying and Different Methods of Sealing on the Storage Qualities of Pecan Kernels (Tests made at Auburn, beginning December, 1939.)

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Per cent No. min. moisture Methods of Sample in oven in dry sealing [28] No. 225 deg.F kernels

Cold-seal (a) 1 0 8.4 Hot seal (b) 2 20 2.9 3 30 1.6 4 44 1.0 5 51 0.7 Steam-seal (c) 6 0 3.4 7 50 0.2 8 60 0.16 9 65 0.10 Vacuum-seal (d) 10 0 3.4 11 20 2.7 12 30 1.0

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When canned After 12 mo. After 24 mo. _____ _____ _____ Methods of Sample sealing [28] No. Color Flavor Color Flavor Color Flavor [26] [27] [26] [27] [26] [27]

Cold-seal (a) 1 1 Excellent 2 Medium 3 Medium Hot seal (b) 2 1 Excellent 2 good 3 Very good 3 2 Very good, 2 Very Good, 3 Good, slightly dry slightly dry slightly dry 4 2 Excellent, 3 Very good, 3 Very good, slightly slightly slightly toasted toasted toasted 5 2 Excellent, 2 Very good 2 Very good toasted toasted toasted Steam-seal (c) 6 1 Excellent 2 Fair 2 Fair 7 3 Excellent, 3 Very good, 3 Very good, toasted toasted toasted 8 3 Excellent, 3 Good, 3 Good, toasted toasted toasted 9 4 Excellent 4 Good, 5 Good, toasted toasted toasted Vacuum-seal (d) 10 1 Excellent 1 Very good 1 Good, slightly flat 11 1 Excellent 1 Very good 1 Good, slightly flat 12 2 Very good, 2 Very good, 2 Medium, slightly dry slightly dry slightly flat

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[Footnote 26: Color ratings: Nos. 1 to 5 represent different amounts of discoloration. 1 = Normal bright yellow color of fresh kernels. 5 = Normal brown color of aged kernels.]

[Footnote 27: Flavor ratings: fair means scarcely edible.]

[Footnote 28: Methods of sealing: (a) sealed without heating; (b) hot kernels immediately transferred from oven pans to dry, hot jars and sealed; (c) air exhausted from jars with steam and sealed immediately; (d) sealed under vacuum by method described under "Procedures."]

Table III.--Effect of Moisture Content, Container, and Sealing on Storage Quality of Schley Pecan Kernels--1940.

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Moisture content Flavor _____ _____

When[29] After 6 After 8 After 12 After 18 stored months months months months

Covered 6.00 7.00 Not edible Not edible Not edible unsealed 4.43 6.85 Not edible Not edible Not edible ice cream 3.50 6.75 Not edible Not edible Not edible cartons 1.71 6.80 Not edible Not edible Not edible

_____

Covered 6.00 10.45[30] Not edible Not edible Not edible unsealed 4.43 6.70 Rancid Not edible Not edible glass 3.50 5.00 Fair Not edible Not edible jars 1.71 4.50 Good Fair Not edible _____

6.00 6.15 Rancid Not edible Not edible Sealed 4.43 4.70 Fair Not edible Not edible glass 3.50 3.30 Good Good Rancid jars 1.71 1.85 Very good Very good Very good

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[Footnote 29: The cured pecan kernels had a moisture content of 4.43 at the time the tests were made. Samples with moisture contents below 4.43 per cent were oven dried at 200 deg.F for periods necessary to reach the respective moisture levels. Samples with moisture contents above 4.43 were treated in steam to obtain the desired amount of moisture.]

[Footnote 30: Excessive increase in moisture content resulted in heavy molding of product.]

A second series of processing and storage tests was started in December, 1940. These studies included tests of effect of moisture content, type of container, and sealing on storage qualities of Schley pecan kernels. Table III shows a portion of these tests together with the results obtained. It is pointed out that unsealed samples regained moisture and became rancid within 8 months in storage.

+Summary+

Results from the foregoing experiments show that pecan kernels can be kept for nine years by drying them to about 2 per cent moisture and storing them in sealed containers. The best results were obtained by drying the kernels in an oven for about 50 minutes at 200 deg.F. The exact length of the drying period may vary somewhat with the moisture content of the undried kernels and the quantity of kernels dried at one time. The temperature of the oven could probably be reduced without affecting the drying time by using a fan for circulating the air in the oven.

This method will preserve the fresh qualities of pecans for a much longer time and equally as well as such common methods as freezing and canning preserve fresh qualities of other foods. It is felt that the process offers a practical and effective method that might well be used in the home as well as in commercial plants for preserving shelled pecans for year-round use and/or for carrying over surpluses from a heavy crop year to supplement the light crops that usually follow.

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* * * * *

President Davidson: I wonder, Professor Harris, whether those methods could be applied to other nuts in addition to pecans. Would your methods with the pecan be adaptable to other nuts or kernels?

Prof. Harris: I have not tried other nuts, but I have thought quite a bit about the black walnut, and I would like to run some experiments. It seems to me that it would be adaptable to other nuts which have constituents similar to the pecan such as black walnut, and perhaps peanuts. We intend to work some on the peanuts down there. Now, of course, in the case of the chestnut we more or less checked it out on that, because when you dry the chestnut down to low moisture content you have a hard product that is not palatable and is quite undesirable.

President Davidson: Now comes the follow-up studies on the 1946 Ohio black walnut prize winners. Mr. L. Walter Sherman has prepared something on that matter but Mr. Sterling Smith, I believe, is going to read that to us.

Follow-Up Studies on the 1946 Ohio Black Walnut Prize Winners

L. WALTER SHERMAN, Mahoning County Experiment Farm, Canfield, Ohio

In 1946 a black walnut contest was conducted in Ohio that brought out roughly 800 samples of nuts from all sections of the state. Judging on the characters of the nuts only, there were ten selected as prize winners.

Notice that I say on nut characters only. In 1950 a grand prize is to be given to the tree selected from these ten trees that has been the outstanding performer for the five year period. We want to know more about these trees. Do they produce regularly? Do the nuts fill well each year? Are the trees young or old? On what kind of soil are they located? Just what are the factors that cause them to produce such superior nuts?

In order to try to answer some of this type of questions each of the ten trees was visited in 1947 and a careful survey of each was made. This was done in August, at which time the crop prospects for 1947 could be noted. Mimeographed blanks such as the following were used to record the desired data.

Tree Name

Latitude _____

Name of Owner _____ Address _____

County _____ State _____ State Route _____

Telephone _____

TREE Isolated [] ; moderately crowded [] ; dense woods []

LOCATION Types of trees in vicinity _____ Air drainage _____

Level [] ; Slope [] ; Direction of slope _____

TREE SIZE Circumference 4-1/2 feet from ground _____

AND SHAPE Probable age ____ ; limb spread [] ; tall [] ; short [] ;

open-branched [] ; symmetrical [] ; irregular [] ;

SOIL Sod [] ; plowland [] ; bottom [] ; upland [] ; hillside [] ;

CONDITIONS clay [] ; alluvial [] ; loam [] ; sandy [] ; pH [] ;

Distance to subsoil _____ ; kind of subsoil _____

Humus [] ; lack of humus [] .

DRAINAGE Nearness to spring [] ; tile drain [] ; well [] ;

lake [] ; stream [] .

FERTILITY Fertilized [] ; manure [] ; commercial fertilizer [] ;

lime [] ; not fertilized [] .

PERFORMANCE CHART

Resistance to disease and insects:

Blight_____; Witches' Broom_____; Caterpillars_____;

1947 1948 1949 1950

1947 1948 1949 1950

Bearing: G F S F

Good; Fair; Scattering; Failure.

1947 1948 1949 1950

Season: Date of leafing out

Male: Date of blossoming

Female: Date of blossoming

Date of ripening

Date of killing frost

Last in spring; first in fall Rate of growth

Moisture; Rainy, dry, average

Clusters: Size 1947 1948 1949 1950 Range in number of nuts per cluster

Production: Size of crop in proportion to size of tree

1947 1948 1949 1950

Percentage of unfilled nuts: 1947 1948 1949 1950 R*H ----------crop pounds

* R = 1/2 limb spread. * H = height; lowest branch to top.

In addition to these data, photographs, both in black and white, and in color, were taken of the trees and often of the surroundings, and a map made so that the trees can be located in the future by any one wishing to do so.

For examination by any one wishing to do so, there are on the secretary's desk copies of the case histories, as written up, of the first and second prize winners, the Duke and the Burson.

A careful study of these ten trees has not revealed any single factor that can be pointed to as essential to the production of a superior walnut variety. They were found on good and on poor soils, on good and poor sites, in soils of a wide range of pH values from very acid to alkaline in reaction. Most of the trees were located in the southern part of the state at 39 deg. to 40 deg. North Latitude, but it is hard to imagine that the latitude has any specific effect on the superior qualities of the nuts.

In all cases where the trees were now standing in impoverished soils, low in humus, fertility, and in pH value, it was quite evident that the soil was probably in far better condition when the trees got their start fifty to a hundred or more years ago.

+Winter Killing 1947-1948+

In 1947 scions of six of these prize winning trees were successfully grafted into established ten year old black walnut seedling trees at the Mahoning County Experiment Farm at Canfield, Ohio, location 41 deg. north latitude. The scions grew nicely in 1947 but all were winter killed during the winter of 1947-1948 with the exception of one scion of Kuhn and one of Davidson. Two scions of Duke, two of Kuhn, one of Athens, one of Orth, seven of Jackson perished during the first winter after grafting. This severe killing of 1947-48 apparently indicates that winter injury to these varieties may be expected some years when they are planted under conditions similar to those at the Mahoning County Experiment Farm. The one scion of the Davidson variety came through in fine shape, so this would be the exception.

The winter of 1947-48 was unusual in the severity of the winter injury to the black walnut trees at the Mahoning County Experiment Farm. Two ten year old Stabler trees and a ten year old Jansen tree killed back to the ground level, and one year old growth of Cowle, Havice, Jansen, Murphy, Mohican, Ohio, Stambaugh, Twin Lakes, and Lisbon was badly damaged although not always completely killed.

+Winter Killing of Bench Grafts+

Bench grafts that were still in the hot bed and were not transplanted to nursery rows until spring of 1948 fared much better than the grafts growing in the established trees. As they had no winter protection but the side walls of the hot bed it is a little hard to see why they fared so much better.

One bench graft of the Duke, two of Burson, four of Kuhn, two of Davidson, three of Orth, two of Williamson, two of Penn, and six of Jackson all came through in good shape.

Indications certainly point to the conclusion that the prize winning varieties of the Ohio 1946 contest are adapted to the southern part of the state rather than to the northern part. The Davidson is a possible exception to this.

Mr. Smith: I asked Mr. Silvis why Mr. Sherman wasn't here, and he said he wasn't able to come because he was doing the same type of work this year, and it is very evidently the reason why he wasn't at the last meeting because he was preparing this work. Instead of coming and enjoying the convention, he stays home and does work that helps the Association, so I think the Association is very much indebted to him.

President Davidson: I think that is true.

That makes it possible for us to close in good time. I think this program is tops. I think it is by far the best program I remember.

Mr. McDaniel: Let's give Mr. Chase, the Program Chairman, a big hand.

(Applause.)

President Davidson: We will now adjourn.

(Whereupon, at 4:30 o'clock, p. m. the meeting was adjourned to reconvene for business session after the banquet.)

* * * * *

+Tuesday Evening Business Session+

President Davidson: There is a little business that remains to be done. In order to let Mr. Slate get away, we'd like to have a report of the Committee on Place of Meeting.

Mr. Slate: The committee consisting of Royal Oakes, myself and two others, conferred with each other. We have considered the matter of a meeting place for next year, and we think, and those we have talked with think, that perhaps Beltsville would be the best place. It does not seem feasible to have a meeting in the Middle West. The New York City region will probably be better for us a year later. The other good places we have visited rather recently. So we are recommending that the place of meeting be Beltsville.

Do you wish to consider the time of meeting now, or will we vote on the place?

President Davidson: Let's act on that now. First, may I have a motion?

A Member: I will move we hold our next convention at Beltsville, Maryland.

(The motion was seconded, vote taken and motion carried unanimously.)

President Davidson: Now, time of meeting.

Mr. Slate: It has been customary for us to have our meeting early in September, about Labor Day. Next Labor Day is the 5th of September. Now, we are not making any recommendations as to time, but if we follow our past custom we will probably meet about the 6th, 7th and 8th. Some of you might like to come later to avoid the Labor Day traffic, but that interferes with some of those who have teaching duties, registration, and so forth, at that time of the year. Personally, I do not think that the Labor Day traffic is insurmountable. It is rather unpleasant in certain areas, but we can make it all right, and we have made it. Perhaps I should recommend the dates the 6th, 7th and 8th, which are Tuesday, Wednesday and Thursday.

President Davidson: You have heard the report. What shall you do with it?

Dr. MacDaniels: I move its adoption.

(The motion was seconded, vote taken and motion carried unanimously.)

President Davidson: Next, I should say, we would like to have a report of the nominating committee.

Mr. Weber: The Nominating Committee reports for the consideration of the members the following list of candidates:

For president, H. F. Stoke from Virginia. Vice-president, L. H. MacDaniels from New York. For secretary, J. C. McDaniel from Tennessee, and treasurer, Sterling A.

Smith from Ohio.

President Davidson: You have heard the report of this committee. I should say that in this case nominations from the floor would be in order.

A Member: Mr. President, I move that nominations be closed.

(The motion was seconded, vote taken, and motion carried unanimously.)

President Davidson: That was on the motion that nominations be closed. What is your pleasure, shall we vote by ballot or shall we vote by--

A Member: Mr. President, I move that the secretary be instructed to cast a unanimous ballot for those nominated by the Nominating Committee.

Mr. Fisher: Second.

(A vote was taken on the motion, and it was carried unanimously.)

President Davidson: This gavel I should have called to your attention before. It is called to your attention at the end of our program, and so you know its story. The other one that has been used in our past meetings was sent to me by Mr. Reed. It is "An Historical Gavel, Northern Nut Growers Association." I understand from Mr. Reed that this was a piece of wood sent to Mr. Littlepage and turned by him and made into a gavel, and this little metallic name plate sunk in by Mr. Littlepage, who is one of the very early members of our association. So we have two historic gavels. Rather interesting, I think.

One other matter. The question has come up in view of the fact that the next annual report will be larger than normal and also in view of the fact that the membership dues have been raised to $3.00, whether it should not be wise and fitting to charge $3.00 for the coming 1948 report instead of the old price of $2.

Mr. McDaniel: $2.00 is the current price for the last two issues of the report to non-members. If that is allowed to continue the man who purchases a report without becoming a member will get it for one-third less than the members do.

President Davidson: What is your wish?

Mr. Weber: Mr. President, I say that they should not be given any more preference than the members, so let them pay $3.00 like the rest of us. I make it in the form of a motion.

A Member: Second.

President Davidson: Moved and seconded that the charge for the forth-coming report of this Association be made $3.00 to non-members. Of course, that report goes to all members, as you know. Are there any remarks on this motion?

Mr. Slate: Mr. President, what about the matter of supplying reports to libraries? In the past we supplied libraries at $1.00 a copy. I don't know whether Mr. McDaniel has had any special requests.

Mr. McDaniel: I haven't had any orders from libraries during the past year.

President Davidson: Shall we make a difference for libraries? What is your feeling?

Dr. MacDaniels: Mr. Chairman, if we have had a differential before I think that might be continued. I will propose a motion that libraries be allowed to purchase the proceedings for $2.00.

President Davidson: Do you make that in the form of an amendment?

Dr. MacDaniels: Yes.

Mr. Weber: I accept that amendment.

A Member: It meets the second's approval.

President Davidson: The motion is then that a charge for the forth-coming report shall be $3.00 to non-members, except that the charge shall be $2.00 to libraries and similar organizations, if that is satisfactory.

(A vote was taken on the motion, and it was carried unanimously.)

Mr. McDaniel: I have a report to read, as follows:

+Report of the Auditing Committee+

The Auditing Committee has examined the records of our treasurer, D. C. Snyder, for the fiscal year just closed and has found them correct as reported and commends him for his excellent service to our Association. [Signed] R. P. Allaman, Royal Oakes, _Auditing Committee_.

President Davidson: You have heard the report. What will you do with it?

A Member: I move that the report be accepted.

A Member: Second.

(A vote on the motion was taken, and it was carried unanimously.)

Dr. Crane: Mr. President, members of the Association: Most of the thunder that the Resolutions Committee had has been taken care of either tonight or at various meetings that we have had. These resolutions have been rather spontaneous at these meetings. However, the Resolutions Committee, for a matter of record, does make the following report:

+Report of the Resolutions Committee+

The Northern Nut Growers Association in its annual meeting assembled at Norris, Tennessee, September 13th to 15th, 1948, adopts the following resolutions:

That, our sincere thanks be extended to Mr. George F. Gant, General Manager of Tennessee Valley Authority, the members of his staff, especially to Mr. Willis G. Baker, Director of the Division of Forest Relations, Mr. Spencer Chase, Mr. Thomas G. Zarger, and others, for the courtesies extended and for making-the necessary arrangements for holding the meetings and caring for the needs of those in attendance.

That we extend thanks and appreciation to Mrs. Willis G. Baker and the other ladies of her committee who provided and served the refreshments on Sunday evening and assisted in arrangements for the banquet.

That we extend thanks to Mr. Spencer Chase and the other members of the committee for the very interesting and instructive program.

May we extend our thanks to those who presented papers and otherwise took part in the program.

We greatly appreciate the very fine work being done by our Secretary, J. C. McDaniel. _Resolutions Committee_, Stoke, Silvis, Sterling Smith, and Crane.

President Davidson: You have heard this report, and I think it is well that we have had it in the form so that it could be a part of our record. What will you do with it?

Dr. MacDaniels: Move the acceptance of the report.

Mr. McDaniel: Second.

(Vote taken on motion, carried unanimously.)

President Davidson: I have here a telegram that I should like to read to you, and this is the way it is worded: "Your generously worded telegram is greatly appreciated. I am grateful beyond all words. My greetings to everyone present tonight. C. A. Reed." We are glad to have the word from Mr. Reed.

Our business meeting is now adjourned.

(Whereupon, the program and business sessions of the Thirty-ninth Annual Meeting of the Northern Nut Growers Association were closed.)

[Illustration]

(On September 15, the members were conducted on a tour of the nursery, arboretum, and experimental nut plantings of the Tennessee Valley Authority near Norris.)

Odds and Ends

DR. W. C. DEMING, West Hartford, Connecticut

I would like to suggest, especially to the younger members of the association, three horticultural projects that I believe promise to be of importance, and on which nobody that I know of is doing any work. Only one of these projects has to do with a nut.

1. Utterly neglected and wasted, the fruit of the horsechestnut or buckeye, "said, to have been formerly used as food or medicine for horses," still might become an abundant food for animals, and perhaps for man, if a way could be found to deprive it of its disagreeable bitter taste and reputed, probably exaggerated, poisonous quality.[31]

There is one late flowering horsechestnut, _Aesculus parviflora_, a dwarf species from the Southeast, and commonly seen in Connecticut as an ornamental on lawns,

which bears a nut entirely free from bitterness, and is sometimes known as the edible horsechestnut. The possibilities in crossing this with the bitter horsechestnut tree species are evident and fascinating. [Several hybrid horsechestnuts are cultivated, but none of these apparently involves any _A. parviflora_ parentage.--Ed.]

2. In temperate zones there are, so far as I have learned, no _perennial_ legumes the seeds of which are used as food. All our immensely valuable edible leguminous seed crops are annually planted. The only exception I think of is the honeylocust, the pods of which, under favorable conditions, are sometimes used as fodder for horses and cattle. But there are thousands of leguminous plants and trees, many of them hardy. I mention the herbaceous _Baptisia australis_, several hardy perennial peas, such as _Lathyrus sylvestria_, _L. maritimus_ etc., _Caragrana_ the pea tree, and species of _Robinia_, _Cercis_; _Cymocladus_ and _Wistaria_. A collection of these, with as many more as one might wish, would be a fascinating group in which to spend hours with brush and forceps.

3. All over America thousands of "tired business men," and school boys who ought to be tending to their baseball, have to spend weekends and holidays pushing lawn-mowers. If an acceptable ground cover could be found that would have to be mowed only half as often, or one quarter as often, or maybe only once a year, or even (glory be) not at all, what a saving of time it would be for good healthy sport and non-depressing exercise.

There are many promising plants. _Pachysandra_ and _Vinca_, don't quite fill the bill but have their good points, such as growing in the shade. There is a little round-leafed plant common in Florida and, apparently, found in the north. There are many plants that could be grown experimentally in patches a yard square. Why have we so tamely limited ourselves to grasses and clover? What a chance for a man to immortalize himself by discovering variants for grasses and clover for lawns and thus become a benefactor to millions of lawn-mower slaves!

[Footnote 31: (_See letter from the American Medical Association on next page.--Ed._)]

COUNCIL ON PHARMACY AND CHEMISTRY

of the

AMERICAN MEDICAL ASSOCIATION Office of the Secretary, 535 N. Dearborn Street, Chicago 10, Illinois

January 5, 1949

Doctor W. C. Deming 31 South Highland Street West Hartford, Connecticut

Dear Doctor Deming:

This is in reply to your inquiry of December 28, 1948, regarding the toxicity of horse chestnuts.

All six of the species of Aesculus which are native to the United States have been reported as poisonous, but specific references in the literature are infrequent. The species Aesculus hippocastanum has been studied and has been found to contain saponin, tannin, and the glycoside, esculin. Esculin is used in patent remedies in the form of ointments and pastes to protect the skin from sunburn. The saponin seems to be the toxic component.

Fruit of the horse chestnut is rich in starch and oil and is a _valuable food for livestock_. The bitter taste of the nut is removed by alcohol extraction which removes the saponin, thus rendering the nut harmless. Certain domestic animals, however, seem to be able to eat the untreated nut without suffering ill effects. [_Italics are by Dr. Deming.--Ed._]

Most of the saponins are markedly irritant to the mucous membranes. They have an acrid taste and provoke a flow of saliva, nausea, vomiting and diarrhea. If injected directly into the circulation they produce hemolysis, diuresis and direct actions on the central nervous system which may be rapidly fatal. Absorption after oral administration is so poor that saponins produce only local effects. The toxicity of various saponins is ten to a thousand times higher by vein than by mouth and is generally proportional to the hemolytic action. Some saponins have a different toxicity for different species.

In experiments with rats Hindemith found that the saponin from Aesculus hippocastanum is not toxic in daily oral doses of 87.5 mg. per kg. Nonhemolytic doses injected intravenously in cats have no effect on respiration or blood pressure; hemolytic doses produce a sudden drop in pressure owing to liberation of potassium from the _erythrocytes_. The saponin increases the activity of the isolated frog heart, then stops it in systole. In frog nerve muscle preparations of this saponin reversibly interrupt stimulus transmission; recovery occurs upon washing.

For a general review of the literature you are referred to Bull. Sc. Pharmacol. 47:290 (November-December) 1940, which is available at the New York Academy of Medicine Library, 2 East 103rd Street, New York City.

Sincerely yours,

[Signed] BERNARD E. CONLEY, R. Ph. Administrative Assistant.

BEC:nr

The Birth of a New Walnut Cracker

B. H. THOMPSON

The home of the Thompson walnut cracker is the home of the maker, on the farm, five and a half miles northwest of Harrisonburg, in the Shenandoah Valley of Virginia. I live in the upland area, 1,500 to 1,700 feet up in the hills.

A man once said he killed two birds with one stone. I went him one better in one instance. I went to the back end of the farm and picked up all the walnuts and placed them on a pile, not too far from the house. Then the squirrels came to help

themselves. I got all the squirrels I wanted to eat and those that got away retreated so fast they dropped their walnut. Then I cracked what walnuts were left for cakes and candy, which we all enjoy so much.

It was while cracking these nuts with a hammer that the thought came to me: Why should there not be a faster and better way to crack nuts? Later I happened to see a walnut cracker made by a blacksmith which did a very good job of cracking, but was entirely too slow for me.

Being mechanically inclined, I have always entertained a desire to invent something worth while. I set out to perfect a cracker that would be fool-proof, easy to work, fast, simple, and strong enough to last a lifetime. This I accomplished in the Model 6. Before reaching this point, I had designed and tested five different models, made five different ways, to see which would be best. They all worked, some good, some I did not like so well. It was discouraging at times but something seemed to tell me I had the right principle.

This No. 6 walnut cracker is a success, now in its 11th season and going stronger all the time. You will find it in 37 states, from Florida to Washington State, from New Hampshire to California, from Minnesota to Texas.

Most of the crackers are sent by mail, and some of the customers mention the fact that they are members of the N.N.G.A. Others do not have trees on their premises, but collect walnuts by the roadside. One I know of has 2,000 walnut trees on his 1,200 acre farm.

Marketing of Black Walnuts in Arkansas

T. A. WINKLEMAN, Rogers, Arkansas

The Benton County Produce Company has been in the walnut business for 38 years. For the first few years we dealt only in hulled nuts, shipping carloads of them to Omaha, Chicago, several points in Nebraska, and the West Coast. About twenty years ago, as I recall, there was a large cracking plant at Kansas City and we shipped several carloads there.

Eventually we began to receive small orders for kernels. We filled them and the number of orders increased. This led us finally to the decision that we should get out of the hulled nut business and sell only kernels, and with few exceptions, that's what we have been doing for the past 25 years. During this time the production of kernels throughout the walnut region has gone up tremendously. As you know, many plants using mechanical cracking machines have become established. We have stuck to hand-operated crackers; but even so, we were able one year to turn out 13,000 pounds of kernels. At present we ship kernels to practically every state in the Union.

Millions of pounds of walnuts are available from Arkansas, Missouri and Tennessee. Here the walnut tree seems to make its best growth. It has been our experience that the better nuts come from upland trees. Those produced in the bottomlands along the larger streams lack the rich flavor typical of those coming from higher elevations. This means we get our best nuts from the Ozarks in northern Arkansas and southern Missouri. So far, few walnut trees are grown commercially.

Practically all of the nuts come from wild trees. But recently there seems to be a trend toward planting grafted walnut trees and grafting native seedlings to improved varieties. The nurseries in this area now have walnut seedlings for sale and some landowners are setting out considerable acreages. It seems like a good investment. The trees grow fast, bear nuts at an early age, and eventually yield additional income in the form of logs. We believe walnut offers better prospects for commercial production than pecan, owing mainly to the value of the walnut wood for cabinet uses.

Not much has been done here with improved varieties. There are some Thomas trees in the region and they yield very well. You get about 20 pounds of kernels from 100 pounds of hulled Thomas nuts as against an average of 12 pounds from our wild native nuts. We anticipate that within three or four years the Thomas will attain commercial importance here. In my opinion, however, _Thomas kernels do not have the flavor that the wild nuts have_; the percentage of oil seems to be less. I have also been told that wood from the Thomas trees has little value in the furniture trade. Why this should be true, or whether it is true, I don't know.

Shells are a problem with us as they are with most concerns in the walnut cracking business. We sent some samples to Iowa State College for testing and got a pretty favorable report. If available in sufficient quantity, the shells apparently can be used for gas production, oils and for other purposes.

Walnut in this region has few enemies; but one, the walnut _Datana_ caterpillar, does considerable damage. We need federal or state aid in controlling this dangerous pest.

Further Notes on Nut Tree Guards for Pasture Plantings

OLIVER D. DILLER, project Supervisor, Hillculture Research, Soil Conservation Service, Wooster, Ohio

In an article entitled, "Nut Trees for Ohio Pastures," which appeared in the 37th annual report of the Northern Nut Growers Association[32], the writer called attention to the advantages of nut trees planted in fence rows and in the interior of permanent pastures and the need for a more satisfactory cattle guard to protect the trees during their period of establishment.

[Illustration: Nut Easy Tree to Guard Install]

The writer has for several years studied various types of cattle guards and in 1946 suggested the possible use of an electric guard along permanent fence lines. This set-up worked fairly well during the first growing season, but it was found that a considerable amount of maintenance is necessary and therefore electric guards may not be practicable over a period of years.

During the summer of 1947 a prominent wire fence manufacturing company was contacted concerning the availability of a welded wire fabric which might be used as a substantial yet economical tree guard. The company made available for test purposes two 150-foot rolls 72 inches high. One roll was galvanized, 11 gauge wire, with 2 x 4 inch staves, while the other was ungalvanized 10 gauge, with 4 x 4 inch

spacing between the staves. These rolls were cut into lengths of 13.7 feet, resulting in a circular guard 4.36 feet in diameter (shown in picture). The guards were installed along a permanent fence on the pasture research farm of the Ohio Agricultural Experiment Station at Wooster during the summer of 1947. Observations made during July, 1948, indicate that these guards have not been damaged in any way by the dairy cattle on this farm during the current season, and that the trees are being adequately protected. No guards have as yet been installed in the interior of the pasture, but it would seem that only one standard fence post would be required to support such a guard.

The wire company was not able to give the exact cost of this material to the farmer, but suggested a quotation of $3.90 per hundred square feet for the galvanized wire, 2 x 4 inch spacing, 11 gauge, as compared to $3.00 per hundred square feet for the ungalvanized wire, 4 x 4 inch spacing, 10 gauge.

Assuming that the ungalvanized wire would serve the purpose for a period of ten years, the cost would be approximately $2.50 per guard if it were attached to a line fence; If placed in the interior of a field, the cost of a standard fence post would have to be added. While this cost may appear to be rather high, it is believed that it will compare favorably with another type guard which will provide equal service. The chief advantages of this guard seem to be its apparent sturdiness and ease of installation.

[Footnote 32: Diller, O. D. "Nut Trees for Ohio Pastures," Northern Nut Growers Association, Inc., 37th Annual Report. 1946, pp. 62-64.]

A Pecan Orchard in Gloucester County, Virginia

MRS. SELINA L. HOPKINS, River's Edge Flower Farm, Nuttall, Virginia

Mr. Reed has asked me to tell you of our experience with pecans in Gloucester County, very near Chesapeake Bay, on North River, a tidewater estuary of Mobjack Bay. Our house is about 20 feet from the shore, so we call it "River's Edge," which describes it very well. The pecan trees are on the lawn, in the barnyard, and in an adjoining field.

The orchard was planted by my late husband about 1915. The trees came from at least two nurseries as there are two distinct sets of varieties. There are eight varieties from the North and eight from the South. Of the northern sorts there are Busseron, Butterick, Indiana, Kentucky, Major, Niblack, Posey, and Warrick. These came from the nursery of R. L. McCoy, Lake, Spencer County, Indiana.

The southern varieties are Delmas, Frotscher, Georgia (Georgia Giant), Hale, Schley, Stuart, Teche, and Van Deman. Hale trees have been the slowest to come into bearing, and there are several which appear to be Hale which are not yet in fruit.

+Nut Crops Scanty+

The trees near the house, both on the lawn and in the barnyard are set irregularly but those in the orchard are in rows, 65 feet each way. They are beautiful in appearance, being from 40 to 55 feet tall, and are very healthy.

However, they do not bear well. We had a pretty good crop in 1943, about 500 pounds, which we sold for 30 and 35 cents per pound. Since then we have had very few nuts, as the flowers have evidently been killed each year by frost.

Most of the nuts we have had have come from trees near the river, where the air is tempered by salt air coming in at high tide. At this writing, early August, there seems to be more nuts than at any time since 1943. There was no frost that I could detect after the trees flowered, but there are few nuts on the trees farthest from the river.

The fruit trees back in the county, on what we call "the highlands," have no fruit this year. Apparently our northern varieties of pecan do not stand the cold any better than the southern sorts. In the last few years, there have been more nuts of the southern varieties. I suppose the flowers of the northern varieties came out at a time when they were more easily frozen.

We have several trees that are evidently seedlings, as they grew up from the ground after the tops died, They usually bear well, producing sweet nuts, well-flavored but small.

We have six Persian walnuts that have had only about ten nuts in all these years. One tree has a black walnut coming up from the root on which it was grafted. It is of the same size as the Persian top. Two years ago, this tree had about 30 nuts on the Persian side and 50 on the black. It is not easily accessible and I have not been to it this year.

+Behavior of Pecans+

The Posey trees are in an east-west row about one-third the distance from the north end of the orchard. Most of the Major and Busseron trees are farther south, some as much as 200 yards. A few trees of both varieties are directly south, within 100 yards, while others are the same distance away off and some farther southwest. It is stated in a recent bulletin of the Virginia Agricultural Extension Service that Posey is needed to pollinate Busseron and Major.

Since reading the bulletin, I have been thinking of our crops in the past. I remember that trees of these two varieties farthest from the Posey, do not bear as well. Until now, I have attributed this to the fact that the soil was less fertile and the trees are smaller and less vigorous. Also the trees are farther from the frost-tempering river. I am not sure yet that this is not the reason.

We are not alone in our experience of an uncertain crop, as other pecan growers in the county tell the same tale. There are a number of large old trees in this general section of Virginia, as well as a good many seedlings. In addition, there are native, bitter, large-growing water hickory (_Carya aquatica_), which is not uncommon in lowlands. (These hybridize freely with true pecans, producing beautiful trees but astringent nuts. Ed.)

One of the largest orchards was set out a few years before ours, by the late Dr. Wm. C. Stubbs, on a farm that had been in his family for many generations. It is on York River, about 15 miles from our place. It was he who encouraged my husband to set out our orchard. Dr. Stubbs was for many years Director of the Louisiana

Experiment Station near New Orleans. He spent his summers at his old home. His trees were probably the best started and cared for during his life, as he knew how to do it. I drove to see the farm recently, and talked with the present owner, who bought it in 1942. The next year, when I also had my good crop, he nearly paid for the place with proceeds from the nuts.

However, like ourselves, he has had practically no nuts since, and is so much discouraged that he plans to take out some of the trees. The varieties there are mostly Moneymaker, Schley and Success. The same varieties are also in a small orchard of another neighbor, who reports that Success does best. The trees owned at one time by Dr. Stubbs seem not to be cultivated at all, but are grazed and mowed, and the orchard is now rather a tangle of briers and weeds.

+We Grow Bulbs with Pecans+

As this is primarily a daffodil farm, and the trees have the best land, it is also used for bulb growing. The daffodils are a much surer crop with us than pecans. We sell both flowers and bulbs. The season for daffodils is in March and April which is well ahead of the pecans. The pecans do not leaf out early enough to shade the daffodils, and I can't see that they injure them in any way except in very dry years. Bulbs near the trees do just as well as those in the open field and sometimes bloom earlier.

All cultivation and fertilization that the trees get is what is accorded the bulbs. As soon as the season is ended for bulbs, we begin cultivating. We go over the bulbs about three times before the tops die back to the ground, in late May. In late July, we mow the weeds, which are high by that time. We frequently mow again later in the fall. We take up the bulbs every two or three years in June, cure them in trays in airy buildings, grade them, sell some, and replant what we need to keep up our supply. When a plot is dug, we plant it with soybeans, turn them under in late summer and replant with a winter cover crop, rye or clover usually. That crop is turned under the following late April when the rye is usually waist high. We replant again with beans which are turned under in July.

If we think the soil needs more humus, we repeat the process another year. During this rotation we apply 0-14-7 at least twice, usually with the first two plantings. The land is limed only at long intervals, as daffodils like a soil rather on the acid side. Of course, during this cultivation and planting, we plow rather close to the trees, within about four feet, and sometimes cut the roots. You may well think that this accounts for their not bearing well, but in this neighborhood there is the same story with trees that are not plowed around. I have wondered at times if they are not too near salt water, and maybe the roots go down to water, yet the trees nearest the river bear best. We have a Teche tree only about 20 feet from high tide line, and it is our surest bearer, having never missed a crop.

Our only varieties that scab to any extent are the one Georgia and the two trees of Delmas, but the man on Dr. Stubbs' place says that both varieties scab although I forgot to ask which variety was worst. (Delmas is one of worst scabbing varieties in the South.--Ed.)

Indiana Nut Shows Have Educational Value

W. B. WARD, Extension Horticulturist, Purdue University, Lafayette, Ind.

A few days ago I had a letter from Professor George L. Slate, of the Geneva station, requesting that I send you some information concerning our recent nut shows and a couple of pictures to be used in the current report.

I am enclosing two pictures--one showing a display of hicans, shellbark, shagbark, bitternut, mockernut hickories and in the background a few of the miscellaneous sorts, while the other picture shows mostly the black walnuts. (Latter picture printed on next page.--Ed.)

If you will note in the pictures, we have used a saw and cut the nuts in two for display purposes. This has been one means of classifying the seedlings to find out whether or not they were worthy of further propagation, although this alone was not the final classification. It was rather surprising to the number of visitors we have had at our shows to see the difference in the interior of the nut and believe me it has done a lot toward the education of the people when it comes to locating some of the better seedlings.

[Illustration]

The purpose of the nut program in the state of Indiana was for the betterment of native nuts and we were in hopes that we could find some very promising seedlings that would be worthy of further work. With the cooperation of the Indiana Horticultural Society and the Student Horticultural Show here at the University, we have had this nut show for the past six or eight years. Sometimes it has been in competition, other times it has been mostly for display. The show served the purpose which we had in mind for thus we have located some very nice walnuts, hickory nuts and a few good seedling pecans. After the show had been visited by some six to eight thousand annually for the past several years, we have further made displays at the annual meeting of the Indiana Horticultural Society in Indianapolis, at the A.P.S. meeting at St. Louis last winter and at the Indiana State Fair in 1948, with a display going to some of the other institutions--particularly to Oklahoma and Texas--for display at their state shows. A new collection was gathered by the students and the writer this year which, in part, will be displayed at the Indiana Horticultural Society meeting on January 19, 20 and 21 and another collection is being shown at Oklahoma A. & M. at this time. The nuts will be returned and placed in cold storage to be exhibited at the State Fair next fall and we have sufficient quantities on hand for individual displays as well as for collections.

Each plate contains from 35 to 40 nuts of seedlings or named varieties and at our recent show we had 66 plates of hickory nuts and allied species. We had 41 plates of walnuts including some very fine Persian walnuts, 16 samples of filbert seedlings, 20 plates of miscellaneous and all told 141 different plates at our show which was held on November 5, 6 and 7, 1948.

Some of our best contributors have been such as Ferd Bolten, Linton, Indiana, who sent five good Persian walnuts and one excellent black walnut. Edward Smith, of Rochester, Indiana, and Henry Buit, of Lafayette, also have found some wonderful walnut seedlings. Donald Sly, Rockport, Indiana, has produced the best seedling

filberts, about eight in number, and contributed a wonderful display of the McCallister hican. Mr. J. F. Wilkinson, Proprietor of the Indiana Nut Nursery, has contributed largely to the collection of seedling and named varieties of hardy northern pecan while W. A. Owen, Poseyville, and Clem Seib, Owensville, have been consistent winners in the large shellbark hickories. O. W. Thompson, Owensville, and William Seng, of Jasper, contributed some large size thin-shelled shagbark hickories to our show. James Stall, of Brownstown, is a consistent winner in butternuts.

Each year more interest is being shown in the planting of native nuts and some of our Persian walnuts are rather outstanding. Nolan Fateley, Franklin, Indiana, has a very fine seedling Persian walnut of large size which we are hoping to propagate. (A large Carpathian tree.--Ed.)

The Importance of Stock and Scion Relationship in Hickory and Walnut

CARL WESCHCKE, St. Paul, Minnesota

Twenty-five years of practical study and living with the hickories ought to suffice to make a success in growing these trees for their delicious product. However, it is only in the twenty-eighth year of such work that I have made an important discovery about the particular hickory with which I have had the most success; I refer to the variety known as the Weschcke shagbark hickory.

I began to graft such varieties as Beaver and Fairbanks (bitternut--shagbark hybrid) hickory on Wisconsin native bitternut hickory (_Carya cordiformis_) in 1920, and some grafts are doing very well at this time, 1948, but they are practically barren of fruit. Since then I have accumulated more varieties to test from many different sources, to continue the work down to the present day. During that time I noticed, but did not appreciate, the significance of the relationship of growth between scion and root system. True, I have been very cognizant of the so-called compatibility between stock and scion in the hickory family, and have written about this matter for publication several times, but I was then more concerned with the stock and scion living together in a harmonious state of existence and health without realizing that there was something else necessary to this relationship in order to promote heavy bearing.

+Experiments in Grafting Black Walnuts+

Parallel to these early experiments, I was grafting in the same family as the hickories, known as the walnut, or _Juglandaceae_ family, using wild native butternut (_Juglans cinerea_) as a stock for grafting to such varieties as the Thomas, Ohio, Stabler and Ten Eyck black walnut (_J. nigra_). Some of these trees, so grafted, exist today, being more than 25 years old, and they have never borne more than a hatful of walnuts to a tree, even when they became large trees. Most of them are entirely barren year after year. I often remarked to persons who were interested in this phase of my work, that the black walnut was non-productive on the butternut root system, but it was very evident that there was not completecompatibility because the walnut scion greatly outgrew the butternut stock causing a marked difference in their trunk diameters just below and above the union. This great difference, the butternut being so much smaller, was no doubt the cause of a shortage of food supply elaborated

through the bark circumference which limited the top to a mere growth of leaves, not leaving sufficient additional supply for the growth of fruit.

My observation among the hickories, with which I did far more experimental work than with the walnuts, was beclouded by the fact that many successful, apparently compatible varieties, grew and throve on the native bitternut stock without bearing fruit, except for just a few nuts occasionally; and yet there was no apparent difference between the scion diameter and the trunk diameter, nothing like the overgrowth of the black walnut when grafted on butternut. So it took many years and a different growth phenomenon to open my eyes as to what was the trouble in getting hickories to bear on foreign root systems.

The final solution of the problem was determined by my observation this year of grafted hickories of several sizes and ages were Weschcke shagbark (_C. ovata_)[33] scions and other hickory scions, such as Siers, Bridgewater, Deveaux, Beaver, and Fairbanks have been grafted on the same tree to act as pollinators for the Weschcke, which is devoid of pollen.[33] This year particularly, the difference in rate of growth between two varieties grafted on the same stock was very apparent; in every case all other varieties greatly exceeded the growth of the Weschcke hickory, but in many cases, only the Weschcke hickory had any nuts growing on the graft, and if there were any nuts on another graft, there were but a few. In practically all cases, the diameters of the scions of varieties of hickory other than the Weschcke were at least twice the diameters of the Weschcke grafts, and the growth of all varieties so grafted was healthy and vigorous and thoroughly compatible with the native bitternut hickory root system.

Several years ago I had to trim some of these other varieties back in order to allow the Weschcke graft to get more growth because it was so backward in development that it looked as though it might be crowded out of existence. It never occurred to me in those years that it was the difference in rate of growth between the two varieties which was really responsible for the difference in the diameter of the scion growth, and not some accident of propagation. Now it is very apparent, from the many examples that I have about me, that the Weschcke hickory is about one-half as fast a grower as such varieties as Bridgewater, Deveaux, Laney, Siers, and many others. This, then, accounts for the heavy bearing of the Weschcke when it starts to bear on the bitternut roots, and it also explains the lack of bearing in such varieties as Beaver, Fairbanks, Laney, Siers, Pleas, Deveaux, Rockville, Green Bay, Hope pecan, Stanley shellbark, Platman, Kirtland, Glover, Barnes, and many others which are hardy and get along well with the native bitternut root system, some of them having lived more than fifteen years grafted in such combination. The Bridgewater is the only variety which bears a fair crop of nuts as compared to the prolific Weschcke, and is the pollinator for the Weschcke when used in orchard planting.

[Footnote 33: See author's added remarks following.--Ed.]

+Are Pecan Stocks Desirable for Hickory Scions?+

It would appear, therefore, that it is necessary for stocks to be at least as vigorous as the variety to which they are grafted, and to insure this it would seem to me that

the northern pecan seeds, such as grow around Des Moines, Iowa, would be the proper seedling stock for almost any variety of hickory, as they outgrow bitternuts and shagbarks by quite a margin. I have only one Weschcke grafted on a pecan of this sort, and it makes much greater growth each year than does this variety grafted on the native bitternut stocks. However, it has not started to bear yet and the reason is that it is still very young, and is over-topped by plum brush and apple trees.

Since it requires about ten years here for a native bitternut to acquire the proper size of one-half inch to three-quarter inch diameter, which is about the size necessary for grafting, you have some idea of how slowly this native species grows. The forest trees, of which there seem to be thousands on my property, very seldom exceed a diameter of six inches, yet they appear to be very old trees. Occasionally we find one that reaches the diameter of a foot or more, and generally it is one that is located where it has plenty of space to grow, as in open pasture. The tree is rather easy to graft to many varieties of hickories. No doubt if it were grown in large numbers, in the proper soil, the time for producing seedling stock ready for nursery propagation could be cut down. But it appears more likely that some northern pecan seed can be found which will produce a hardy understock to furnish a seedling of sufficient vigor and size for propagating purposes in five years or less.

+Records of Bearing+

Our first successful grafting of Weschcke hickory on bitternut hickory (_Carya cordiformis_) was in 1927, but these grafts did not bear for about ten years. We know now that this was because there was a lack of pollen of the shagbark species to pollinate its blossoms. Now these trees are bearing profusely.

The second batch of grafts from the original Weschcke hickory, which grew near Fayette, Iowa, was made in 1934. One mature nut from grafts made that spring was gathered from the ground in the same year, about October 1, 1934, but it had been partly consumed by a squirrel. From that year to the present, these grafted trees produced each year and never failed to mature some edible nuts up to and including this year, when there is a very large crop (6-1/2 bushels). This, then, is the 15th consecutive crop of nuts of which I have a record. During two years we had such early fall frosts that the nuts were a little shriveled and not fully mature, but still edible. In other years there were some light crops, but there never has been a crop failure in all this time. The variation in bearing is also due in part to several late frosts which in the spring in some years killed back all the foliage and newly expanding buds. Yet new dormant buds opened, some of which had flowers, and so carried on the unbroken bearing record.

Last winter (1947-48) produced the most severe damage to exotic species of fruit and nut trees as well as ornamentals, including evergreens, ever recorded in this area; yet the grafted Weschcke hickory trees were so loaded down with nuts that I had to support the load by tying up branches to keep them off the ground. This tough winter caused almost every variety of apple tree to be barren, such as Wealthy, Northwestern Greening, Whitney Crab, Haralson and Malinda. Only two varieties, Lowland Raspberry and Hibernal, bore fair crops. Last winter killed outright (to the

ground) most of my Thomas black walnuts, some of which were more than 25 years old, and damaged severely such other varieties as Ohio, Vandersloot, and Ten Eyck. The winter was responsible also for the killing of several seedling Chinese chestnuts which had survived ten years of our winters and yet others of these Chinese chestnuts are growing again from sprouts near the ground surface. The mulberries suffered greatly also, but in general the hickories of many varieties came through this winter, with very little damage, and most of them are bearing a few nuts. Even the wild hazels suffered differing amounts of damage and have only partial crops of nuts because of the effects of the winter.

In conclusion, keep in mind that these experiments and tests have been conducted in severe climatic conditions in the 45th parallel at River Falls, Wisconsin, 35 miles east of St. Paul, Minnesota, and that out of more than fifty varieties of hickories and pecans and their hybrids tested, only these two, Bridgewater and Weschcke hickory, (both thin-shelled easy-cracking varieties), have succeeded to a point which can be classed as commercial; the writer can now recommend these two varieties for propagation by nursery firms capable of undertaking the propagation of hickory nut trees, the sale of which to the public is a foregone conclusion.

By request of the secretary, Mr. Weschcke sent the following additional information on the Weschcke hickory:

About ten years ago I noticed that there was no pollen coming from this tree and yet from the very beginning, even when there was no other pollen available except the wild hickory pollen from the _Carya cordiformis_, the Weschcke hickory produced nuts. Thinking that it was due to parthenogenesis I bagged clusters of pistillate blossoms, and although setting nuts they all dropped off which is typical of non-pollenization. I then bagged groups of pistillate blossoms which I pollinized with different available pollens of the _Carya ovata_ and these set nuts which started to grow, upon which I removed the bags. From this experiment I found that the Bridgewater did a very good job of pollenization and it became the tree that I considered as a compatible mate. Other trees that pollinate well are Kirkland, Deveaux and Glover; Beaver is not a good pollenizer and I have not experimented with Fairbanks to know whether it is satisfactory. The catkins grow vigorously on the Weschcke up to the time that the pollen sacs seem ready to open, then the catkin drops off. No pollen has ever matured that I know of. When dried from this state, they yield no pollen.

I told Dr. J. W. McKay about this nearly seven years ago, and he asked for fresh samples of the catkins at different periods which I mailed to him in receptacles that he furnished. He wrote me a very nice treatise on this subject for inclusion in my book which I expected to be published at that time. The book was never published, however, since Orange Judd turned it down during the war for lack of paper as the excuse. I did not try any further to get it published, and since that time many new things should be added to the hazel hybrid chapter. Dr. McKay said that he is familiar with this action on the part of nut trees. I have felt that it was phenomenal since I have had no other such experience among all the nut trees with which I have experimented. However, this loss of pollen saves vitality apparently for the

production of several times the pistillate bloom that I have seen on any other hickory with which I have worked and this apparently accounts for the prolificacy of the Weschcke when grafted on the native Wisconsin hickory. (Male-sterility occurs with chestnut and apple.--Ed.)

At first I considered the Weschcke somewhat of a hybrid nut; later I changed my mind about it and considered it a pure shagbark. I have reversed my opinion again and consider the possibility of its being slightly hybrid with bitternut blood. The parent tree at Fayette, Iowa stood close to big bitternuts. The shell, being the thinnest of all hickories (known to me) leads me to suspect the hybridity with the bitternut. It is quite smooth and the ridges are less prominent than in almost any other hickory except such known hybrids as the Beaver. Its shape is oval to long and it is flat so that whenever you throw a handful down to a smooth surface they all assume the same position, and because of this they would no doubt lend themselves to commercial cracking as they would feed through the mechanism of a cracking machine exactly in the same order.

I have not always had such a high opinion of this nut. Dr. Deming has letters from me which have a disparaging note, and although Dr. Deming considered it a valuable nut, he has letters from me in which I indicated that I was sorry that it was not productive and that it had such a small nut. Both these conditions changed with time and within twenty years this nut sometimes becomes one of the largest hickories of the cultivated varieties and its proficacy then probably depended on correct pollination which I was not aware of in the beginning.

I hope you will pardon me for dwelling so on this hickory, but after working with hickories for nearly thirty years it certainly seems remarkable to me that we have such a productive variety that is hardy this far north and west, that is perfectly at home on the native hickory roots, and that matures its nuts from September 15 to October 1, is self-hulling, that has escaped the attack of all sorts of weevils that infest our native nuts. (I have never found one wormy Weschcke hickory nut although sometimes you find empty nuts.) This variety also escapes the spring frosts so that there have been fourteen consecutive years of bearing without interruption. The foliage is vigorous, has no diseases so far; the young branches are sometimes cut off by oak tree pruners or girdlers. This happens to many kinds of trees, including all the oaks, butternut, black walnut, all the hickories and even the chestnuts. When you take into consideration the fact, that no other hickory has such a fine record it makes me very enthusiastic over this variety in spite of the fact that it bears my name. Were you to classify this hickory from casual observation, you would think it is a pure shagbark, and it is only the extreme thinness of shell and the outside appearance pf the nut shell which indicates some slight hybridity.

Progress with Nuts at Wolfeboro, New Hampshire

MATTHEW LAHTI

Inasmuch as I do not expect to be able to attend the thirty-ninth annual meeting, I thought I would report to you on the progress of my nut trees since my letter of a

year ago.

Last winter was a severe one in Wolfeboro, New Hampshire (43 deg. 36' north latitude.) We had more than the usual amount of snow, and although the temperature did not get down much lower than 25 deg. below zero at my place, it remained cold for many days at a time without relief. This, and the fact that last fall was one of the driest seasons on record, plus the fact that this spring it rained almost continuously for more than a month, resulted in considerable damage to my nut trees.

My Broadview Persian walnut graft suffered severe damage, with branches up to two inches in diameter being killed. Whether this was from frost or lack of moisture in the fall I do not know, but two Crath Persian walnuts, one of which is situated within fifty feet of the Broadview, suffered no apparent winter injury at all. Neither Broadview nor Crath bore any nuts this year, whereas last year the Broadview produced eighteen nuts.

My Gellatly heartnut also suffered severe winter injury similar to the Broadview Persian walnut, and after it leafed out it looked as if a fire had gone through it because of the dead wood. However, it is bearing thirteen nuts this year.

Strange to say, the black walnut trees did not suffer any winter injury (the Thomas, set out in the spring of 1939, having been injured in each previous winter), except that the Tasterite is barren of nuts this year against a pretty good crop last year. However, the Thomas is bearing a fair crop, but the nuts are smaller than usual.

While my hickory trees appeared to suffer no winter injury, out of possibly two dozen that I have planted since 1939 I expect to have only three left. The number had dwindled to nine last year, and six of those I am afraid will be dead by the end of next year. These six had done well for six or seven years. The cause appears to be poor circulation through the graft union. This is unfortunate as I believe hickory trees will live and bear fruit in our climate.

I had the usual experience with filberts and hazelnuts, namely that the catkins were, for the most part, Winter killed. There, are no nuts on Rush, Barcelona, Medium Long, or Red Lambert, and the Winkler bushes [self-fertile--Ed.] which bore heavily last year (although the nuts did not fully ripen), are bearing only a few nuts this year.

Native butternut trees last year bore a heavy crop of nuts. This year, the crop is very light.

As an experiment, I planted three Chinese chestnuts this spring: Abundance, Nanking, and a "straight line" seedling. Although I haven't much hope that they will survive many winters, I thought I would try them out.

Several people have inquired about nut growing in New Hampshire, and I have sent them a resume of my experiences.

Breeding Chestnuts in the New York City Area

ALFRED SZEGO, Jackson Heights, New York City

[Part of a letter to the Secretary, October 27, 1948.]

I am one of those members who have not been able to attend annual meetings. In the two or three years that I have been a member, I have derived great enjoyment from reading the annual reports and receiving information through your news bulletin.

Therefore, when I received your improved bulletin, "The Nutshell," I felt that I and others like me should write and tell you how wonderful it is.

There is much that I just want to "get off my chest." My past criticism was that the organization was a bit lethargic. But nut trees are _slow_ in showing results, despite the nurserymen's attractive visions of quick, big harvests of nuts and even timber!!! This slow patience of the black walnut has determined the tempo of much of the membership.

+Chestnut Breeding Efforts+

My main work is to attempt to breed two types of chestnuts: (1) One that is very productive with a low head and will bear nuts like the old American chestnut. (2) Another that will make a good timber stick. It is my theory that present chestnut breeders are crossing inferior material, using any specimens that happen to be in flower at the right time as long as they represent the species to be crossed.

Suppose they intend to cross _C. crenata_ x _C. dentata_. An average Japanese chestnut is usually pollinated with flowers from a poor struggling sprout on the edge of the woods that has only one thing to recommend it. That is an early bearing characteristic which is inherent, but which, according to experiments and observations I have tediously carried out, _is not totally due to ringing by the blight_.

The experiment takes place and a few hybrid nuts are produced. They are termed (_C. crenata_ x _C. dentata_). It is expected that the characteristics of the offspring will be somewhere between those of the two parents in blight resistance and nut size and quality. But what of the grandparents, the many ancestors of the American chestnut sprout that have not even the slight resistance of the sprout? Can they not express their characteristics and hand them, down to their grandchildren? And some individuals of _C. crenata_ are not reputed to be so highly blight resistant.

Of course the scientists engaged in this work are men of the highest calibre and no doubt are aware of this, but it is extremely difficult to obtain, propagate, and care for named varieties of the finest individuals of each chestnut species.

Apple, cherry, and other fruit breeders would not dream of crossing common scrub cull fruit trees and expect any degree of success.

My first task when I began, three years ago, on my coppice growth 35 to 40 year old hardwood forest, was to clear a little land and to begin planting different world species of _Castanea_.

You would be astonished to find that it was impossible for me to obtain seed or trees, at the time, of _C. crenata_, _C. seguinii_, _C. pumila_, _C. henryi_ and _C. alnifolia_. I obtained some 24 seeds of _C. mollisima_ from Dr. A. H. Graves, for which

I was grateful. At the time he didn't have a good crop, I think. Institutions and government agencies would not or did not like to release their newly developed hybrids for fear that I was a nurseryman or perhaps would sell them for "blight resistant" chestnuts, although they were not yet proven.

+Experiment at Pine Plains+

By diligent search I managed to get a few trees and hybrids of _C. crenata_ and a variety (seedling of) called "Colossal." These thrived and survived about 30 deg. below zero under deep snow at Pine Plains, New York. I also set out 2 bushes (_C. pumila_) obtained from Harlan P. Kelsey, East Boxford, Massachusetts. Dr. Graves' seed gave fair germination, and I now have seven nice young _mollisimas_ from 8" to 30" high. Of two three year old trees I obtained from a local nursery, one died (my fault) from not reducing the top, and the other died back to the ground from winterkill, but came back again as sprouts. I easily obtained seed of _C. sativa_, but the severe winter mowed the seedlings down and there are only two survivors. One is smaller this year than last but the other is about 14" high and making slow, straight growth. The _chinkapins_ are perfectly hardy and this year one of them made _3 feet of growth_.

I estimate that I have some 3,000 to 4,000 American chestnut sprouts that range anything from 1 to 18 feet in height. But more promising--I have a cluster of fine young seedlings that I have been caring for. All the woods were cleared away from them to give them plenty of light. They are watered by the old hand bucket method in dry spells. I report on them occasionally to Mr. G. F. Gravatt and Mr. Russell B. Clapper of the U.S.D.A. They are a faint ray of hope.

Four of them are about from 18 to 20 feet tall. One is about 9 feet high. One blighted and died two years ago and was removed. Another blighted at the base and I cut the canker out, but I fear it's going. One branch is dead and was removed. The others developed strong blight resistance. Small cankers formed on the lower branches but did not make headway. I cut some of these out and the trees healed nicely. As the trees become older, their resistance diminishes and the proof lies ahead. One tree that I labeled No. 1 has about two dozen, well healed _Endothia_ scars already. The trees have not bloomed for me yet but I may have some results soon. I intend to cross this clonal group with the following:

1. With _C. seguinii_ for greater blight resistance and productivity.

2. With _C. mollisima_ (var. Abundance) for blight resistance, fine nuts of medium size, and a good timber stick with good vigor.

3. With large Japanese like Austin, and their hybrids like "Colossal," for a medium size nut of fair quality and highly prolific for the general market for a cooking or roasting chestnut.

Though many people dislike the Japanese chestnuts, they are at least productive and hardy (at my place). Their chief attribute is their possibility as food for stock and wildlife. Some of the same people who dislike them (among nurserymen) recommend planting oaks which certainly do not compare with _C. crenata_. When a very "sweet"

acorn is found it is proclaimed to be "as good as Japanese chestnut."

The Chinese chestnut has its faults here. It is not very thrifty in growth here and as a rule doesn't bear until late. It is not very productive and the nuts spoil easily. I have since planted much seed from the south and it often doesn't even get here in a viable condition.

+Assistance from Beltsville+

My work has lately been facilitated by Mr. Gravatt and Mr. Clapper. I visited them at Beltsville and Mr. Clapper personally toured the orchard with me at Glenn Dale, showing me the kind of helpful courtesy that one never forgets and that is a tribute to these men.

Some promising material was given to me which will greatly facilitate my work. Mr. Gravatt suggested the use of "Ammate" as an experiment to poison trees that interfere with any American chestnut growth I wish to save. The experiment is intended to eliminate the resulting sprouts that accompany girdling. Incidentally, part of the experiment is to attempt to give light and cultivation and fertilizer to 100 native chestnut sprouts in a four acre area.

I have some information on American chestnut sprouts that may be of interest to the membership. In an endeavor to locate the best American material, I have been combing the woods and thickets on Long Island, in New Jersey, Connecticut, and parts of Dutchess County, New York (the latter not extensively). Many thousands of sprouts were examined to discover the following:

Their present status.

1. Sprouts occur almost always in woodlands.

2. They reach their greatest height and are most luxuriant at the edge of woodlands or in clearings therein.

3. They rarely exceed 15 feet in height and reach a diameter of about three inches.

4. One in many hundreds, and only where there is light in abundance, will bear flowers.

5. One in many thousands bears female or pistillate flowers which sometimes produce "blind" or empty nuts. [Unpollinated--ED.]

6. Rarely, indeed very rarely, are two flowering trees close enough to produce viable seed.

7. There are a few seedlings that are single stem upright trees (no old stump in evidence) that reach up to 20 and rarely 25 or 30 feet in height with a diameter of 6" or so.

(Mr. R. B. Clapper thinks it is probably due to the absence of an old, infected stump that this greater height is reached.)

8. Ringing by the blight does not necessarily force the flowers and nuts. The woodlands abound with chestnut sprouts in all stages of girdling without pollen or

fruit.

When I have my trees in bearing, I will be glad to furnish pollen and nuts from them to anyone that pursues the important work of trying to improve what I consider the most promising nut tree we yet know.

Winter Injury to Nut Trees at Ithaca, New York, in the Fall and Winter of 1947-48

L. H. MacDANIELS and DAMON BOYNTON, Ithaca, N. Y.

The winter of 1947-48 caused more damage to nut trees at Ithaca, New York, than any since 1933-34. It was a combination of a series of early freezes followed by sub-zero temperature in mid-winter. Apparently the most injury was done by the fall freezes. These occurred on September 25, 26, and 27. On each successive night the temperature dropped lower than the preceding, and on September 27 was around 20 deg.F. There was considerable variation in temperature related to exposure, air drainage conditions, and other factors.

On West Hill in Ithaca the minimum temperature recorded on September 27 was 23 deg.F. Injury to leaves and nuts was severe. Within a few days the leaves had shrivelled and dried on the trees. It was apparent that this early freeze came before the abscission layers were formed in the leaf bases or growth matured. Ordinarily, a hard freeze late in the season will cause the trees to drop the leaves the next day. The nuts on the trees were frozen solid and mostly turned black within a few days and began to shrivel. Development was stopped, with the result that the nuts on all varieties were very poorly filled. The cavities appeared on first cracking to be full of kernel, but on drying these shrunk so that they were practically valueless. Some of the nuts were planted in a nursery row in the fall and germinated fairly early, showing viable embryos in spite of arrested development.

During the winter the temperature fell to -25 deg.F, a temperature which ordinarily would not damage black walnuts seriously. It is impossible to separate the effect of the low winter temperatures from that of the early freeze in September. In this location the net result of the early freeze and the severe winter was to kill vigorously growing grafts on the walnut trees. Also the cambium in the main crotches of a Stambaugh tree with a trunk about 14 inches in diameter was killed. This tree was destroyed in a windstorm in August, 1948, but it is not clear that the breakage was related to the winter killing in 1947-48. None of the trees now has a good crop, which may be or may not be related to the frost in the fall. It is entirely possible that failure to form blossom buds is caused either by killing of bud primordia or more likely by depletion of carbohydrate reserves due to the loss of leaves in early fall.

One seedling of Carpathian walnut was not damaged seriously except for some slight terminal twig killing. Another tree, however, had most of the smaller branches killed. Hickories and chestnuts were apparently not seriously damaged but some seedlings of the Japanese walnut were killed to the ground.

+Walnut and Hickory Plantings+

At the orchard of the Department of Pomology of Cornell University there is a large collection of walnut and hickory varieties and other nut trees. It is not known exactly

what the temperatures were in this location but an exposed location half a mile distant had a minimum September temperature recorded of 24 deg.F. and minimum winter temperature of -20 deg.F. The planting in question is on two levels and on a hillside. The damage on the hillside and the upper level was relatively less than on the lowlands where apparently the air drainage was poor. Probably the temperature in the lowlands may have reached 20 deg.F. in September and -25 deg.F. in the winter. At any rate, the damage to the trees was much more severe than in the West Hill location where the temperature reached 23 deg.F. in September.

Injury to the black walnut on the higher land and on the hillside was mostly the killing back of the twigs and smaller branches. On some trees, the petioles of last year's leaves were still attached to the dead twigs late the following summer, showing that the freeze occurred before the abscission layers had formed. The dozen or more varieties of black walnut on the higher land showed little difference between them except that the Elmer Myers showed somewhat greater injury. On the low ground, many varieties including Murphy, Edmunds, Benton, Ohio, Todd, and Stambaugh were killed to the ground or back to the main branches of the trunk. Of three Thomas trees, about 20 years old, one was killed outright, one severely injured, and the other injured only in the twigs. Apparently the difference in these three trees was related to the size of the crop on the trees, although no definite data are available on this point. Walnuts showing little or no injury were: Mintle and Tasterite. Neither of these had had a crop in 1947.

Many of the varieties of hickory were injured as was the native bitternut, _Carya cordiformis_. This injury consisted mostly of the killing back of the lower limbs and twigs with some varieties being killed outright. Killing of the lower limbs as compared with the tops of the trees is probably related to lower temperatures near the ground due to temperature inversion and possibly to the fact that the lower branches were somewhat weaker in their growth. This sort of injury is common with fruit trees.

On the higher ground the Chinese chestnut trees planted some 20 years ago showed considerable injury. About 50% of them were killed and others were damaged in the lower branches. Chestnut trees in this planting had all survived the cold winter of 1933-34, with winter temperatures below -30 deg.F., so that it is probable that the early freeze of September 27 was responsible for their death.

Japanese walnut seedlings again showed great difference in hardiness, the more tender seedlings killing to the ground and others showing little damage.

Northern pecans on higher ground showed severe damage, the killing extending to the trunk and larger limbs. The variety Burlington, which is a hybrid, pecan x shagbark, showed little injury.

In a planting of several hundred seedling black walnut trees in another location the temperature on September 27 was probably around 18-20 deg.F. About 20% of the trees were killed to the ground. These trees were growing under a sod mulch, were not overly vigorous, and for the most part had not come into fruiting.

In the 1947-48 winter about half of the sweet cherries in the Pomology orchard were killed and peaches were severely injured. No injury was apparent on apple

trees.

Weather conditions such as occurred in 1947-48, though unusual, are to be expected occasionally in the latitude of Ithaca, and in fact throughout the northern states. Apparently the fall freeze was widespread as it was almost impossible to obtain any black walnuts that were of any value. Some of the specimens received from other sources obviously had been frozen. The possibility of such damage might well be a deterrent on planting black walnuts in any considerable acreage as a commercial venture in the north. The experience of the past year certainly emphasizes the fact that as yet our knowledge of varieties is incomplete and also that the Northern Nut Growers Association has much work to do in either locating or developing varieties of greater hardiness or with growth characteristics which provide early maturity and thus immunity from early frost damage.

What Came Through the Hard Winter in Ontario

GEORGE HEBDEN CORSAN, Islington, Ontario

For winter killing of trees I refer you to the winter of 1947-48. I had a huge elm and a very tall white ash killed. A lot of black walnuts and heartnuts and some Persian (English) walnuts were killed back the length of last year's growth. Some Persian walnuts were killed to the ground while others were not even nipped off of a bud. Very strange to say, my best Persian walnut---whose shell is very thin, whose meats are very sweet and fat, the tree itself a fast grower, prolific and self-pollenizing--not only did not show a sign of trouble but actually had a crop of most excellent nuts. _These trees only_ will I distribute in future, as well as my two types of "Rumanian Giants." The Rumanian Giants did show a little winter killing of two or three inches of the tips and showed up poorly on the crop size.

I find that all my Russian walnuts [_J. regia_, probably "Carpathian"--Ed.] run true to seed--no bitter nuts as from north China. They evidently planted the sweet nuts only, thus eliminating the bitter types; they knew and practiced no budding or grafting in [that part of] Russia. Astounding to say, filberts came through last winter in excellent shape, but the terrible, cold, late spring, froze all male blossoms but those of the "Jones Hybrid" types, which I have from seeds I sowed. These latter yielded a good crop of nuts as did Brixnut seedlings.

Not a butternut on a tree nor a beechnut! Some black walnuts were loaded while others were quite empty.

And so I predicted--last September--a mild, open winter with some cold days. [His prediction was good for his locality.--Ed.]

My "Senator Pepper" hybrid (butternut x heartnut cross) had a crop but my "David Fairchild" had some empty and some full. My "Mitchell hybrid" had a good crop and, believe me, this nut is far away ahead of the Mitchell heartnut and up against the world for cracking out clean. It will equal an almond, and as for taste, it is so far ahead of a Brazil nut that the Brazil nut would rank D 3 beside it.

I still believe in seed planting, even for speed of eventual growth. Last October I climbed up a black walnut tree I planted in mid-World War I. From the top of it I

looked away down to the tops of electric power poles!

Filberts Grow in Vermont

JOSEPH N. COLLINS, R.F.D. No. 3, Putney, Vermont

Fifteen years ago I set out a few hundred nut trees and bushes. The Chinese chestnuts are not doing very well, as they needed more attention than I could give them. Honeylocusts, in this climate, require more time. At present I can report only on seedling filberts. The seeds for these plants were collected from the four corners of the world. Some of the seedlings perished, lots of them were discarded as unworthy. At present I am setting out two acres of the ones that stood up well under the test.

The filbert (_Corylus avellana_) is a bush 15 to 20 feet tall and the bushes should be planted 20 to 25 feet apart. It doesn't mind partial shade, requires no spraying and very little pruning. Like the red raspberry, it is easily propagated by suckers. Most of my bushes started producing when they were four years old and now in their fourteenth year, drop about 15 pounds of large fine nuts each September. They stand up well under the rigorous Vermont climate, at an elevation of 1,000 feet. Knowing as much about their growing habits as I do, I believe that a steady winter with plenty of snow on the ground and a late spring that isn't fickle, is well suited for filbert growing in the Northeast. The need for wind protection and good air and water drainage cannot be over-emphasized.

There are a few reasons why I should advise against growing filberts in tree fashion--with a single trunk, as they are mostly grown on the West Coast. The catkins of the filbert develop during the summer, lie dormant through the winter, and shed their pollen very early in the spring. Should the temperature fall as low as -35 deg.F, the catkins winterkill. To overcome this shortcoming, I bend down and peg to the ground, in the late fall, a few slim shoots with dormant catkins, so that the snow, or some other mulching material supplied when there is insufficient snow, will cover and protect the catkins from winterkilling.

By the end of March, after a stretch of fair weather, two tiny red tongues appear at the tips of some of the leaf buds. These are the pollen catching parts of the pistillate flowers. If the winter was kind, the filbert bushes will be a riot of golden catkins, shedding their pollen. If the catkins remain dormant when the pistillate flowers bloom, they have been winterkilled, and the bent down reserves have to be called up. These being protected during the winter, on being bent back to their original position, will come into bloom in a few days, pollenizing the waiting pistillate flowers. Bees eagerly seek this, one of the earliest pollens. The now fertilized flowers, which always stayed inside the buds, go back to sleep for about two months; they are safe from the "North Easter," from late freezes, or from snow. When filberts are grown naturally, that is with many shoots from the ground, it is easy to harvest them by shaking the slender shoots. I hand hoed my bushes for the first three years, and gave them a permanent mulch over the whole area, adding some material each year.

I am inclined to believe that part of my success with filberts is due to mulching. In the middle of summer, I apply a 4" cover of low grade hay, and in the fall I again

cover the ground with fallen leaves. Due to the ideal conditions thus created (optimum temperature and moisture) for soil bacteria and earthworms, this material is entirely digested. The mulching material almost disappears by the middle of the next summer, indicating vigorous biological activity. By this time a new layer of mulch is spread, completing the cycle. Late in the fall a load of manure is heaped in the middle of the plantation as an earthworm refuge. This heap is scattered early in the spring. Light applications of wood ashes and super-phosphate are given yearly, late in the fall.

In conclusion, I wish to state that selected varieties of filbert nuts can be grown in the Northeast. Hybrids between the American and European filbert are good growers and producers, although I find that the flavor of the nut isn't as good as that of the pure _avellana_. I would advise the planting of a dozen bushes by each of a great number of persons further to prove the possibilities of growing this specific nut in the New England area, also to promote the idea of growing both feed and food on trees and bushes.

Report of Necrology Committee

C. E. SCHUSTER

Carl E. Schuster, horticulturist with the U. S. Department of Agriculture stationed on the Oregon State College campus and generally recognized as the nation's foremost authority on filbert production, died February 6, 1948, in Corvallis as a result of a heart attack. At the time of his death, he was 58.

Associated with the Northwest's growing nut industry for more than 30 years, he was recognized for his outstanding contributions to filbert and walnut production. One of his first and most outstanding developments was related to the pollination requirements of filbert trees. After research proved the common commercial filbert variety, the Barcelona, was self-sterile, he recommended to filbert growers that they plant DuChilly, Daviana and White Aveline filbert trees with their Barcelona to insure complete pollination. Full crops resulted.

+Organized Summer Tours+

For approximately 20 years prior to 1941, he served as secretary-treasurer of the Western Nut Growers Association. In this capacity he assisted in starting the summer tours which have been carried on continuously since. He helped guide the Association through its early years to a position of importance among the commodity groups of the state. In 1941, he was forced to relinquish his office as a result of the enactment of a federal regulation. At this time, he was given an honorary life membership in the Association.

In recent years, he devoted major attention to orchard management with emphasis on fertilization and general nutrition needs of nut trees. In this work he co-operated with Dr. R. E. Stephenson at Oregon State College. Their outstanding development was in the field Of boron deficiency in walnuts.

Walnut production of many orchards, they discovered, could be increased two and three fold by the addition of borax fertilizer. The presence of "snake heads" or sprouts in summer walnut growth and "die-back" or winter kill noticeable in some walnut trees during the winter months are now generally recognized as signs of boron deficiency.

+Wrote Many Nut Articles+

Other work in walnuts proved that fertilizer applications can and do pay. Prior to this work with resulting fertilizer recommendations, many walnut growers had not made heavy enough applications on certain soil types and felt that fertilizers were not worthwhile in walnut production.

Mr. Schuster was the author of many articles pertaining to nut culture.

In the 10 years he was on the staff of the college horticulture department before entering federal service, he made an outstanding record in teaching and research. With other scientists he worked in developing a successful pollination program for cherries. This work was carried on after it was determined that the three leading cherry varieties, Royal Ann, Bing, and Lambert, were all self-sterile and intersterile.

A native of Ohio, he came to Oregon in 1912 to attend Oregon State College after having completed two years at Ohio Wesleyan. He received a B.S. degree in agriculture in 1914 and two years later, 1916, received his master's degree.

He joined the college staff three years later and remained until 1929, when he took the federal position he held until his death. He was a veteran of World War I, having served as an infantry second lieutenant. He was a member of Alpha Zeta Sigma Xi, and Gamma Sigma Delta honor societies and was a life-long member of the Evangelical church, which has since merged with the United Brethren church.

He is survived by Mrs. Schuster and four children, Charles, Robert and Margaret--all Oregon State College students, and Flora, a high school student. A brother, Dr. Earl J, Schuster, lives at Tillamook.--Reprinted from _Better Fruit_ magazine.

MRS. LAURA SELDEN ELLWANGER

Mrs. Laura Selden Ellwanger, member of one of Rochester's pioneer families, died at her home, 510 East Avenue, Rochester, New York on September 1, 1948, after a short illness.

She was the widow of William D. Ellwanger, whose father, George Ellwanger, was a co-founder of the Ellwanger & Barry Nursery Company.

Her brother, George B. Selden, was inventor of the gasoline automobile, and her father, Henry R. Selden, was a New York State Court of Appeals judge and one-time lieutenant governor of the state.

Mrs. Ellwanger was the last survivor of 12 children in the Selden family. Her maternal grandfather, Dr. Abel Baldwin, settled in Clarkson in 1811, just a year

before Rochester was founded. She was born in a house on the land now occupied by the Highland Hospital. One of her sisters, Louise, was the wife of Maj. Gen. Elwell C. Otis, former governor of the Philippine Islands.

Mrs. Ellwanger spent many summers at her home, Brookwood, in Ontario, Wayne County.

She was honorary president of the Rochester Female Charitable Society, one of the city's oldest organizations, and a member of the Rochester Historical Society, The Rochester Garden Club, Genesee Valley Club, and the Rochester Rose Society.

She is survived by a daughter, Mrs. Kenneth H. Field, of Rochester; two granddaughters, Mrs. John F. Weis, Jr., of New York City, and Mrs. Edwin II, Atwood, Jr., Rochester, and four great-grandchildren.

(The above, is from a Rochester newspaper clipping.--Ed.)

M. M. KAUFMAN

(The secretary received the following letter from Mrs. M. M. Kaufman, dated March 17, 1949.)

"I regret to advise you that my husband, M. M. Kaufman (Clarion, Pennsylvania), a member of the Association for many years, died March 3, 1948.

"My husband was a strong, conservationist and always appreciated the work of the Nut Growers. In continuing his interests, I should like to join the Association, and I am enclosing my check for $8.00 to cover dues of $3.00.... and $5.00 as a contribution in my husband's name for furthering the work of the group."

NORMAN B. WARD

Norman B. Ward, a new member, with offices at 866 Hanna Bldg., Cleveland 15, Ohio, was reported deceased in September, 1948. No obituary notice has been received for him.

Attendance

R. P. Allaman, Harrisburg Pennsylvania Mrs. R. P. Allaman, Harrisburg Pennsylvania Stephen Bernath, Poughkeepsie, New York Mrs. Stephen Bernath, Poughkeepsie, New York Charles B. Berst, Erie, Pennsylvania Frank B. Blow, Norris, Tennessee Gertrude R. Blow, Norris, Tennessee Mrs. L. C. Brann, Knoxville, Tennessee John T. Bregger, Clemson, South Carolina Carroll D. Bush, Eagle Creek, Oregon J. Edwin Caruthers, Alpine, Tennessee Wm. S. Clarke, Jr., Dept, of Horticulture, State College, Pennsylvania B. C. Cobb, Norris, Tennessee Miss Mary R. Cochran, Cincinnati, Ohio C. E. Connally, Roanoke, Virginia Mrs. C. E. Connally, Roanoke, Virginia Thomas S. Cox, 103 Hotel Avenue, Knoxville. 18, Tennessee H. L. Crane, 6822 Pineway, Hyattsville, Maryland Frank B. Cross, Oklahoma A & M College, Stillwater, Oklahoma Mrs. Frank B. Cross, Stillwater, Oklahoma W. H. Cummings, Fountain City, Tennessee Mrs. W. H. Cummings, Fountain City,

Tennessee Helen E, Davidson, 234 E. Second St., Xenia, Ohio John Davidson, Xenia, Ohio Margaret Davidson, Xenia, Ohio Elora Donnelly, Hoboken, New Jersey John H. Donnelly, Hoboken, New Jersey Brooks D. Drain, Knoxville, Tennessee Martin D. Ehlmann, St. Charles, Missouri Mrs. Martin D. Ehlmann, St. Charles, Missouri R. W. Fisher, West Plains, Missouri A. E. France, Charleston, West Virginia Wilbert M. Frye, Pleasant Dale, West Virginia F. C. Galle, Dept, of Horticulture, Univ. of Tenn., Knoxville, Tennessee H. R. Gibbs, 803 William St., Front Royal, Virginia Mrs. Bessie J. Gibbs, 803 William St., Front Royal, Virginia Jack Godwin, Signal Mountain, Tennessee G. H. Gordon, Union, South Carolina Dr. Edward A. Grad & Family, Cincinnati, Ohio G. F. Gravatt, Plant Industry Station, Beltsville, Maryland. Dr. Clyde Gray, Horton, Kansas G. A. Gray, Cincinnati, Ohio John L. Gray, Raleigh, North Carolina Henry Gressel, Mohawk, New York Mrs. Henry Gressel, Mohawk, New York Earl C. Haines, Shanks, West Virginia Max Hardy, Sr., Albany, Georgia Mrs. Max Hardy, Sr., Albany, Georgia Max Hardy, Jr., Albany, Georgia Hubert Harris, Auburn, Alabama John F. Hatmaker, Norris, Tennessee Agnes V. Hendricks, Knoxville, Tennessee A. G. Hirschi, 414 N. Robinson St., Oklahoma City, Oklahoma C. F. Hostetter, Lancaster, Pennsylvania Mrs. C. F. Hostetter, Lancaster, Pennsylvania Bruce Howell, Sweetwater, Tennessee C. B. Howell, Jr., Knoxville, Tennessee Lilian Jenkins, Norris, Tennessee Denman A. Jones, Walnut Grove Farms, Parkesburg, Pennsylvania G. S. Jones, Rt. I, Box 140, Phenix City, Alabama Mrs. Tinman W. Jones, Walnut Grove Farm, Parkesburg, Pa. Raymond Kays, Oklahoma A & M College, Stillwater, Oklahoma J. B. Kingrohm, Knoxville, Tennessee G. J. Korn, 140 N. Rose St., Kalamazoo, Michigan Ira M. Kyhl, Sabula, Iowa E. W. Lemke, Detroit 14, Michigan R. C. Lorenz, Fremont, Ohio Mrs. R. C. Lorenz, Fremont, Ohio W. W. Magill, Lexington, Kentucky D. E. Manges, Norris, Tennessee J. C. Moore, Auburn, Alabama R. G. Moore, Dept, of Hort., V. P. I., Blacksburg, Va. Dr. C. A. Moss, Williamsburg, Kentucky John T, Mullins, Renfro Valley, Kentucky H. O. Murphy, Chattanooga, Tennessee Dr. L. H. MacDaniels, Ithaca, New York Mrs. L. H. MacDaniels, Ithaca, New York Frances C. MacDaniels, Ithaca, New York F. J. McCauley, 233 West Erie St., Chicago 10, Illinois Elizabeth L. McCollum, White Hall, Maryland Blaine McCollum, White Hall, Maryland J. C. McDaniel, 403 State Office Bldg., Nashville 3, Tenn. Mrs. Herbert Negus, Mt, Rainier, Maryland James R. Oakes, Bluffs, Illinois Royal Oakes, Bluffs, Illinois Mrs. Vincent L. Odum, San Diego, California Robert E. Ogle, Tenn. Experiment Sta., Knoxville, Tennessee F. L. O'Rourke, East Lansing, Michigan E. L. Overholser, Dept. of Hort., V. P. I., Blacksburg, Virginia Roger W. Pease, Morgantown, West Virginia Gordon Porter, Windsor, Ontario, Canada Sara M. Potts, Knoxville, Tennessee Carl Prell, South Bend, Indiana Edna M. Pritchett, 803 William St., Front Royal, Virginia Ralph H. Quick, Lesage, West Virginia G. B. Rhodes, Route 2, Covington, Tennessee Mrs. G. B. Rhodes, Route 2, Covington, Tennessee Ralph, Richterkessing, R. R. 1, St. Charles, Missouri Mrs. Ralph Richterkessing, R. R. 1, St. Charles, Missouri David Richterkessing, R. R. 1, St. Charles, Missouri John Rick, Reading, Pennsylvania W. Jobe Robinson, Route 7, Jackson, Tennessee Mrs. W. Jobe Robinson, Route 7, Jackson, Tennessee Dr. Wm. L. Rohrbacher, 811 East College St., Iowa City, Iowa Mrs. Wm. L. Rohrbacher, 811 East College St., Iowa City, Iowa Ralph Schreiber, Sr., 245 Cherry St., New Albany, Indiana Ralph Schreiber, Jr., 245 Cherry St., New Albany, Indiana T. L. Senn, Clemson, South Carolina W. A. Shadow,

Decatur, Tennessee Maurice E. Shamer, M. D. & Son, Baltimore, Maryland Sylvester Shessler, Genoa, Ohio Mrs. E. D. Shipley, Knoxville, Tennessee G. B. Shivery, Knoxville, Tennessee Raymond E. Silvis, Massillon, Ohio Frances Simpson, Norris, Tennessee George L. Slate, Geneva, New York Barbara Sly, Rockport, Indiana Donald R. Sly, Indiana Nut Nursery, Rockport, Indiana Louesa M. Sly, Rockport, Indiana Raymond E. Sly, Rockport, Indiana Sterling Smith, 630 W. South St., Vermilion, Ohio H. F. Stoke, Roanoke, Virginia Mrs. H. F. Stoke, Roanoke, Virginia Bernard M. Taylor, Alpine, Tennessee Clifford R. Von Gundy, Cincinnati, Ohio Ford Wallick, Peru, Indiana Arthur Weaver, 3339 South St., Toledo, Ohio Harry R. Weber, Morgan Road, Rt. 1, Cleves, Ohio Mrs. Martha R. Weber, Morgan Road, Rt. 1, Cleves, Ohio J. F. Wilkinson, Indiana Nut Nursery, Rockport, Indiana Mrs. R. Allen Williams, Chicago, Illinois William J. Wilson, Fort Valley, Georgia Mrs. William J. Wilson, Fort Valley, Georgia T. G. Zarger, Norris, Tennessee Mrs. T. G. Zarger, Norris, Tennessee Mrs. G. A. Zimmerman, Harrisburg, Pennsylvania

Membership Listings

Miss Helen Lewis, of the secretary's office, has corrected the following alphabetical list of members by states and countries, up to May 1, 1949, and further additions up to press time will be added below "Wisconsin", if space permits. We are listing also the members' occupations, so far as they have been furnished, and ask that other members who want them listed include this information when they pay their dues for the coming year. Please check on your own listing now, and notify the secretary if any correction in the name or address (including zone number) should be made.

Northern Nut Growers Association

Membership List as of May 1, 1949

*Life Member

**Honorary Member

ALABAMA

Campbell, R. D., Route 1, Stevenson. +Farmer, mine operator.+ Dean, Charles C., Route 3, Box 220, Anniston Orr, Lovic, Route 1, Danville. +Farmer, chestnut and peach grower, merchant.+

ARKANSAS

Clawitter, A. T., Route 3, Box 210, Little Rock Hale, A. C., Route 2, Box 322, Camden Van Arsdale, D. N., Route 4, Berryville Williams, Jerry F., Viola Winn, J. B., West Fork

CALIFORNIA

Armstrong Nurseries, 408 N. Euclid Avenue, Ontario. +General nurserymen, plant breeders.+ Gaston, Eugene T., Route 2, Box 771, Turlock. +Nut nurseryman, Turlock

Nursery+ Haig, Dr. Thomas R., 3344 H. Street, Sacramento. +Surgeon+ Kemple, W. H., 216 West Ralston Street, Ontario. +Nurseryman, plant, breeder and research horticulturist.+ Nicholson, Thomas B., 1017 N. Ophir Street, Stockton Parsons, Charles E., Felix Gillett Nursery, P. O. Box 1026, Nevada City. +Nurseryman.+ Pozzi, P. H., 2875 South Dutton Avenue, Santa Rosa Serr, E. F., Agri. Experiment Station, Davis. +Associate Pomologist.+ Walter, E. D., 899 Alameda, Berkeley Welby, Harry S., 500 Buchanan Street, Taft. +Private & Corporation Horticulture.+ Williams, Edward L., Sheepranch

CANADA

Brown, Alger, Route 1, Harley, Ontario. +Farmer.+ Cahoon, Dr. E. B., 333 O'Connor Drive, Toronto 6, Ontario Casanave, John A., 209 Patterson Rd., Lulu Island, Vancouver, B. C. Cornell, R. S., R. R. No. 1, Byron, Ontario Corsan, George H., Echo Valley, Islington, Ontario. +Nurseryman, nut breeder.+ Crisp, Dr. Allan G., Suite 204, 160 Eloor Street West, Toronto, Ontario Elwood, H., 78 Trans Canada Highway West, Chilliwack, B. C. +Nurseryman.+ English, H. A., Box 153, Duncan, B. C. +Farmer, fruit and nut grower.+ Filman, O., Aldershot, Ontario. +Fruit and vegetable grower.+ Gellatly, J. U., Box 19, Westbank, B. C. +Plant breeder. Fruit grower & nurseryman.+ Giegerich, H. C., Con-Mine, Trail, B. C. Goodwin, Geoffrey L., Route 3, St. Catherines, Ontario. +Fruit grower.+ Harrhy, Ivor H., Route 1, Burgessville, Ontario Housser, Levi, Rt. No. 1, Beamsville, Ontario, +Fruit farmer.+ Maillene, George, R. R. 1, Saanichton, B. C. Manten, Jacob, Route 1, White Rock, B. C. *Neilson, Mrs. Ellen, 5 Macdonald Avenue, Guelph, Ontario Papple, Elton E., Route 3, Cainsville, Ontario Porter, Gordon, R. R. No. 1, Harrow, Ontario. +Chemist+ Snazelle, Robert, Forest Nursery, Dept. of Industry & Resources, 140 Cumberland St., Charlotteville, P. E. I. Trayling, E. J., 609 Richards Street, Vancouver, B. C. +Jeweller.+ Wagner, A. S., Delhi, Ontario Wharton, H. W., Route 2, Guelph, Ontario. +Farmer.+ Willis, A. R., Route 1, Royal Oak, Vancouver Island, B. C. Young, A. H., Portage La Prairie, Manitoba Young, A. L., Brooks, Alta.

CONNECTICUT

*Deming, Dr. W. C., 31 S. Highland, West Hartford 7. +(Dean of the Association)+ Giesecke, Paul. R.F.D. 3, Pinewood Road, Stamford. +Physicist.+ Graham, Mrs. Cooper, Darien Graves, Dr. Arthur H., 255 South Main Street, Wallingford +Consulting Pathologist, Conn. Agr. Expt. Station, New Haven, Connecticut. *Huntington, A. M., Stanerigg Farms, Bethel McSweet, Arthur Clapboard Hill Road Guilford. +Industrial Engineer.+ *Newmaker, Adolph, Route 1, Rockville Pratt, George D., Jr., Bridgewater White, George F., Route 2, Andover

DELAWARE

Brugman, Elmer W., 1904 Washington Street, Wilmington. +Chemical Engineer.+ Wilkins, Lewis, Route 1, Newark. +Fruit grower.+

DENMARK

Granjean, Julie, Hillerod. (See New York.) Knuth, Count F. M., Knuthenborg, Bandholm

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ECUADOR SOUTH AMERICA

Colwell, P. A., Institute of Inter-American Affairs, c/o American Embassy, Quito.

FLORIDA

Avant, C. A., 960 N. W. 10th Avenue, Miami. +Pecan grower.+ Estill, Gertrude, 153 Navarre Drive, Miami Springs. +General Manager.+

GEORGIA

Eidson, G. Clyde, 1700 Westwood Ave., S.W., Atlanta Hammar, Dr. Harold E., U. S. Pecan Field Station, Box 84, Albany. +Chemist, U.S.D.A.+ Hardy, Max, P. O. Box 128, Leeland Farms, Leesburg. +Nurseryman, farmer.+ Hobsen, James, Jasper Hunter, Dr. H. Reid. 561 Lake Shore Drive, N. E., Atlanta. +Teacher and farmer.+ Neal, Homer A., Neal's Nursery, Rt. 1. Carnesville. +Farmer, nurseryman.+ Noland, S. C., P. O. Box 1747, Atlanta 1. +Owner of Skyland Farms.+ Wilson, William J., North Anderson Avenue, Fort Valley. +Fruit and nut orchardist.+

IDAHO

Baisch, Fred, 627 E. Main Street, Emmett Dryden, Lynn, Peck. +Farmer.+ Kudlac, Joe T., Box 147, Buhl. +Orchardist.+ McGoran, J. E., Box 42, Spirit Lake. +Nurseryman.+ Steele, A. A., John Steel Orchards, Parma. +Manager of Orchard.+ Swayne, Samuel F., Orofino

ILLINOIS

Albrecht, H. W., Delavan Allen, Theodore R., Delavan Anthony, A. B., Route 3, Sterling. +Apiarist.+ Baber, Adlin, Kansas Best, R. B., Eldred. +Farmer.+ Bradley, James W., 1307 N. McKinley Ave., Champaign Bronson, Earle A., 800 Simpson Street, Evanston Churchill, Woodford M., 4323 Oakenwold, Chicago Coe, John E., 2024 Sunnyside Avenue, Chicago 25 Colby, Dr. Arthur S., University of Illinois, Urbana

Dietrich, Ernest, Route 2, Dundas. +Farmer.+ Dintelman, L. F., State Street Road, Belleville Erkman, John O., 103 N. Lincoln Street, Urbana Fordtran, E. H., 8700 Fullerton Avenue, Chicago 47 Frey, Frank H., 2315 West 108th Place, Chicago 48. +Assistant to V. P., C B I & P R. R.+ Frey, Mrs. Frank H., 2315 West 108th Place, Chicago 48. +Housewife.+ Gerardi, Louis, Route 1, Caseyville. +Nut and fruit nurseryman.+ Grefe, Ben, R. R. 4, Box 22, Nashville. +Farmer.+ Haeseler, L. M., 1959 W. Madison St., Chicago Heborlein, Edward W., Route 1, Box 72 A, Roscoe Helmle, Herman C., 526 S. Grand Avenue, W., Springfield. +Division Engineer, Asphalt Institute.+ Hockenyos, C. L., 213 E. Jefferson Street, Springfield. +Business man.+ Johnson, Hjalmar, W., 5811 Dorchester Avenue, Chicago 37 Jungk, Adolph, 817 Washington Avenue, Alton Kreider, Ralph, Jr., Hammond Langdoc, Mrs. Wesley W., P. O. Box 136, Erie. +(J. F. Jones Nursery)+ Oakes, Royal, Bluffs (Scott County) Pray, A. Lee, 502 North Main Street, LeRoy. +Attorney.+ Seaton, Earl D., 2313 6th, Peru. +Machinist.+ Sonemann, W. F., Experimental Gardens, Vandalla. +Lawyer and farm operator.+ Whitford, A. M., Farina. +Horticulturist.+

INDIANA

Arata, J. W., R. R. 2, Box 28, Osceola. +Mechanical Engineer.+ Bauer, Paul J., 123 South 29th Street, Lafayette Behr, J. E., Laconia Boyer, Clyde C., Nabb Buckner, Dr. Doster, 421 W. Wayne Street, Ft. Wayne 2. +Physician and Surgeon.+ Clark, C. M., c/o C. M. Clark & Sons Nurseries, R. R. 2, Middletown +Fruit & nursery stock.+ Eagles, A. E., Eagles Orchards, Wolcottville. +Apple grower.+ Eisterhold, Dr. John A., 220 Southeast Drive, Evansville 8. +Medical Doctor.+ Fateley, Nolan W., c/o Campbell Oil Co., 2003 Madison Avenue, Indianapolis 2 Garber, H. C., Indiana State Farm, Greencastle Gentry, Herbert M., Route 2, Noblesville Glaser, Peter, Route 18, Box 463, Evansville Hite, Charles Dean, Route 2, Bluffton Hunter, J. Robert, 215 So. Broadway, Peru Prell, Carl F., 1414 E. Colfax Avenue, South Bend 17 Richards, E. E., 2712 South Twyckenham Drive, South Bend. +Studebaker Corporation.+ Russell, A. M., Jr., 2721 Marine Street, South Bend Schreiber, Ralph, 245 Cherry Street, New Albany Skinner, Dr. Charles H., Route 1, Thorntown. +Teacher and farmer.+ Sly, Miss Barbara, Route 3, Rockport Sly, Donald R., Route 3, Rockport. +Nurseryman, nut tree propagator.+ Wallick, Ford, Route 4, Peru Ward, W. B., Horticulture Bldg., Purdue University, Lafayette. +Ext. Horticulturist, Vegetables.+ Wichman, Robert P., R.R. 3, Washington Wilkinson, J. F., Indiana Nut Nursery, Rockport. +Nurseryman, farmer.+

IOWA

Anderson, Donald, Welton Junction Berhow, Seward, Berhow Nurseries, Huxley Boice, R. H., Route 1, Nashua. +Farmer.+ Clayton, Donovan, Route 1, Coin Cole, Edward P., 419 Chestnut Street, Atlantic Ferguson, Albert B., Center Point. +Nurseryman.+ Ferris, Wayne, Hampton. +President of Earl Ferris Nursery.+ Harrison, L. E. c/o Harrison Lake Shore Orchards, Nashua. +Orchards.+ Huen, E. F., Eldora Inter-State Nurseries, Hamburg. +General nurserymen.+ Iowa Fruit Growers' Assn., State House, Des Moines 19. +Cooperative buying organization+ Kaser, J. D.,

Winterset. +Farmer.+ Kivell, Ivan E., Route 1, Greene. +Farmer.+ Knowles, W. B., Box 126, Manly Kyhl, Ira M., Box 236, Sabula. +Nut nurseryman, farmer, salesman.+ Lounsberry, C. C., 209 Howard Avenue, Ames Martazahn, Frank A., Route 8, Davenport McLaran, Harold F., Mt. Pleasant. +Lawyer.+ Rodenberg, Henry, Guttenberg. +Farmer.+ Rohrbacher, Dr. William, 311 East College Street, Iowa City. +Practice of Medicine.+ Schlagenbusch Brothers, Route 2, Fort Madison. +Farmers.+ Snyder, D. C., Center Point. +Nurseryman, nuts and general.+ Tolstead, W. L., Central College, Pella Wade, Miss Ida May, Route 3, LaPorte City. +Bookkeeper.+ Welch, H. S., Mt. Arbor Nurseries, Shenandoah White, Herbert, Box 264, Woodbine. +Rural Mail Carrier.+ Williams, Wendell V., Danville

KANSAS

Baker, F. C., Troy Borst, Frank E., 1704 Shawnee Street, Leavenworth Breidenthal, Willard J., Riverview State Bank, 7th & Central, Kansas City. +Bank President.+ Funk, M. D., 612 W. Paramore Street, Topeka. +Pharmacist.+ Gray, Dr. Clyde, 1045 Central Avenue, Horton. +Osteopathic Physician.+ Harris, Ernest, Box 20, Wellsville Leavenworth Nurseries, Carl Holman, Proprietor, Route 3, Leavenworth. +Nut nurserymen.+ Mondero, John, Lansing Thielenhaus, W. F., Route 1, Buffalo Underwood, Jay, Uniontown

KENTUCKY

Alves, Robert H., Nebi Bottling Company, Henderson Magill, W. W., University of Kentucky, Lexington. +Field Agent in Horticulture+ Moss, Dr. C. A., Williamsburg. +Physician and Bank President.+ Mullins, Tom, Renfro Valley. +Radio entertainer, commercial walnut cracker.+ Rouse, Sterling, Route 1, Box 70, Florence Tatum, W. G., Route 4, Lebanon. +Commercial orchardist.+ Whittinghill, Lonnie M., Box 10, Love. +Growing nut trees, evergreens, fruit trees.+

MARYLAND

Crane, Dr. H. L., Bureau of Plant Industry Station, Beltsville. +Principal Horticulturist, U.S.D.A.+ Eastern Shore Nurseries, Inc., Dover Road, Easton. +Chinese chestnuts & ornamentals.+ Gravatt, Dr. G. F., Plant Industry Station, Beltsville. +Research Forest Pathologist.+ Hogdson, William C, Route 1, White Hall Hoopes, Wilmer P., Forest Hill. +Retired farmer.+ Kemp, Homer S., Bountiful Ridge Nurseries, Princess Anne. +General nursery.+ Lowerre, James D., Dist. Training School, Laurel McCollum, Blaine, White Hall McKay, Dr. J. W., Plant Industry Station, Beltsville. +Government Scientist.+ Negus, Mrs. Herbert, 4514 32nd Street, Mt. Rainier Porter, John J., 1199 The Terrace, Hagerstown. +Farm Owner.+ Shamer, Dr. Maurice E., 3300 W. North Ave., Baltimore 16

MASSACHUSETTS

Babbit, Howard S., 221 Dawes Avenue, Pittsfield. +Service Station owner and part time farmer.+ Bradbury, Capt. H. G., Hospital Point, Beverly Brown, Daniel L., Esq., 60 State Street, Boston Bump, Albert H., 160 Standish Rd., Watertown Davenport, S. Lathrop, North Grafton. +Farmer, Fruit Grower.+ Farrell; Charles, 46 Pratt Street, Tanaton Pitts, Walter H., 39 Baker Street, Foxboro. +General Foreman, Instrument Company.+ Feitse, Ernest, Osterville Kendall, Henry P., Moose Hill Farm, Sharon La Beau, Henry A., North Hoosie Road, Williamstown. +Steam engineer.+ Rice, Horace J., Box 146, Wilbraham. +Attorney-at-Law.+ Russell, Mrs. Newton H., 12 Burnett Avenue, South Hadley Short, I. W., 299 Washington Street, Taunton Steward O. W., 15 Milton Avenue, Hyde Park 36, +Fire Protectors Engineer and Manager.+ Swartz, H. P., 206 Chincopee Street, Chicopee Wellman, Sargent H., Esq., Windridge, Torsfield. +Lawyer.+ Weston Nurseries, Int., Brown & Winter Streets, Weston. +Nurserymen.+ Weymonth, Paul W., 183 Plymouth Street, Halbrook Wood, Miss Louise B., Poeassett, Cape Cod.

MEXICO

Compean, Senor Federico, Gerente, Granjas "Cordelia", Escobado No., 76, San Luis Potosi, Mexico. +General Manager of "Cordelia" Farms.+

MICHIGAN

Achenbach, W. N., Petoskey Ainsworth, Donald W., 5851 Mt. Elliott, Detroit 11 Andersen, Charles, Andersen Evergreen Nurseries, Scottsville Barlow, Alfred L., 13079 Flanders Avenue Detroit 5 Becker, Gilbert, Climax Hoylan, P.B., Cloverdale Bradley, L. J., Route 1 Springport. +Farmer.+ Bumler Malcolm R., 2600 Dickerson, Detroit 15. +Insurance Trustee.+ Burgart, Harry, Michigan Nut Nursery, Box 33, Union City. +Nurseryman.+ Burgess, E. H., Burgess seed & Plant Company, Galesburg Burr, Redmond M., 820 S. 5th Avenue, Ann Arbor. +General Chairman, The Order of Railroad Telegraphers, Pere Marquette District, C&O Ry. Co+ Cook, Ernest A., M. D., c/o County Health Dept., Centerville Corsan, H. H., Route 1, Hillsdale. +Nurseryman.+ Emerson, Ralph, 161 Cortland Avenue. Highland Park 3 Estill, Miss Gertrude. (See under Florida. Summer Address: Rt. 4, Box 762. Battle Creek,) Grater, A. F., 820 Liberty Avenue, Buchanan Hackett, John C., 3921 Butterworth Rd., S. W., R. R. 5, Grand Rapids 6 Hagleshow, W. J., Box 314, Galesburg. +Grain farmer. Odd contract jobs.+ Hay, Francis H., Ivanhoe Place, Lawrence Healey, Scott, 200 Sherwood Street, Otsegu **Kellogg, W. K., Battle Creek King, Harold J., Sodus. +Farmer and fruit grower.+ Korn, G. J., 140 N. Rose Street, Kalamazoo 12. +Shop worker.+ Lee, Michael, P. O. Box. 16, Milford Lemke, Edwin W., 2432 Townsend Avenue, Detroit 14. +Engineer, and nut orchardist.+ Miller, Louis, 417 N. Broadway, Cassopolis. +District Forester.+ O'Rourke, Dr. F. L., Horticultural Dept., Michigan State College, East Lansing +Professor of Horticulture.+ Pickles, Arthur W., 760 Elmwood Avenue, Jackson Prushek, E., Route 3, Niles Scherer, Milton E., M.C.M.T., Qts, 20, Sault Saint Mario Stahelin, C. A., Stahelin Nursery, Bridgman. +Nurseryman.+ Stocking, Frederick N., Harrisville Tate, D. L., 959 Westchester Street, Birmingham Taylor, Merrill W., Trust Dept., First

Natl. Bank & Trust Co., Kalamazoo Whallon, Archer P., Route 1, Stockbridge Wiard, Everett, 510 South Huron St. Ypsilanti Zekit, Arnold, 1958 Catalpa Court, Ferndale 20

MINNESOTA

Ruer, Eldred, Route 3, Canby Hodgson, B. E., Dept, of Agriculture, S. E. Experiment Station, Wasaca Mayo Forestry & Horticultural Institute, Benjamin F. Dunn, Supt., Box 498; Rochester Skrukrud, Baldwin, Sacred Heart Tulare, Willis E., 800 3rd Avenue, S.E., Rochester Weschcke, Carl, 96 S. Wabasha Street, St. Paul. +Proprietor, Hazel Hills Nursery Co.+

MISSISSIPPI

Meyer, James R., Delta Branch Experiment Station, Stoneville. +Cytogeneticist (Cotton.)+

MISSOURI

Bauch, G. D., Box 66, Farmington Blake, R. F., c/o International Shoe Co., 1509 Washington Ave, St. Louis 3. Fisher, J. B., Rt. 1, Pacific Fisher, Richard W., Box 112, West Plains Glesson, Adolph, River Aux Vases Hay, Leander, Gilliam Howe, John, Route 1, Box 4, Pacific Huber, Frank J., Weingarten. +Farmer.+ Hudson, Perry H., Smithton James, George, Brunswick Johns, Jeannette F., Route 1, Festus Logan, George F., Oregon Nicholson, John W., Ash Grove. +Farmer.+ Nicholson, Kadire A., Ash Grove Ochs, C. Thurston, Box 291, Salem. +Foreman in garment factory.+ Richterlessing, Ralph, Route 1, St. Charles. +Farmer.+ Stark Brothers Nursery & Orchard Co., Louisiana. +Fruit and general nurserymen.+ Tainter, Nat A., 714 N. Fifth Street, St. Charles. +Factory worker and Nursery owner.+ Van Erp, George D., 7 East 85th St., Kansas City Weil, A. E., c/o Dow Chemical Company, 3615 Olive St., St. Louis 8. +Representative on agricultural chemicals for Dow.+

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NEW JERSEY

Anderegg, F. O., Raritan Blake, Dr. Harold, Box 93, Saddle River Bottoni, R. J., 41 Robertson Road, West Orange. +President of Harbot Die Casting Corp.+ Brewer, J. L., 10 Allen Place, Fair Lawn Buckwalter, Mrs. Alan R., Flemington Buckwalter, Geoffrey R., 20 Cedar Street, South Bound Brook. Chemist. Cumberland Nursery, Route 1, Millville. +Nurserymen.+ Donnelly, John H., Mountain Ice Company, 51 Newark St., Hoboken Dougherty, William M., Broadacres-on-Bedens, Box 425, Princeton Ellis, Mrs. Edward P., Strawberry Hill, Route 1, Box 137, Keyport Franek, Michael, 323 Rutherford Avenue, Franklin Hyper Humus Company, Newton *Jacques, Lee W., 74 Waverly Place, Jersey City Kelly, Mortimer B., Route 2, James St., Morristown McCullouch, J. D., 73 George Street, Freehold McDowell, Fred, 905 Ocean Avenue, Belmar Ritchie, Walter M., Route 2, Box 122R, Rahway. +Landscape nurseryman.+ Rocker, Louis P., The Rocker Farm, Box 196, Andover. +Farmer.+ Sheffield O. A., 288 Hamilton Place, Hackensack Sorg, Henry, Chicago Avenue, Egg Harbor City Sutton, Ross J., Jr., Route 2, Lebanon Van Doren, Durand H., 310 Redmond Road, South Orange. +Lawyer.+ Yorks, A. S., Lamatonk Nurseries, Neshanic Station

NEW YORK

Barber, George H., Route 1, Stockton. +Farmer.+ Barton, Irving Titus, Montour Falls. +Engineer.+ Bassett, Charles K., 2917 Main Street, Buffalo Beck, Paul E., Beck's Guernsey Dairy, Transit Road, East Amherst Benton, William A., Wassaic. +Farmer & Secretary, Mutual Insurance Co.+ Bernath's Nursery, Route 1, Poughkeepsie. +Nut Nursery.+ Bernath, Mrs. Stephen, Route 1, Poughkeepsie Bixby, Henry D., East Drive, Halesite, L. I., +Executive V. P., American Kennel Club, New York City.+ Brook, Victor, 171 Rockingham Street, Rochester 7. +Sales Engineer.+ Brooks, William G., Monroe. +Nut Tree Nurseryman.+ Bundick, Clarkson U., 35 Anderson Avenue, Scarsdale Button, Arthur J., Lock Box 348, Olean Carter, George, 428 Avenue A., Rochester 5. +Textile weaver and tree grower.+ Cassino, Augustus, Valatie, Columbia County Cowan, Harold, 643 Southern Building, The Bronx, New York 55 Elsbree, George, R.F.D., Stanfordville Feil, Harry, 1270 Hilton-Spencerport Road, Hilton. +Building Contractor.+ Ferguson, Donald V., L. I. Agr. Tech. Institute, Farmingdale Flanigen, Charles F., 16 Greenfield Street, Buffalo 14 Freer, H. J., 20 Midvale Road, Fairport Fribance, A. E., 139 Elmdorf Avenue, Rochester 11. Fruch, Alfred, 34 Perry Street, New York 14. Artist. Graham, S. H., Bostwick Road, Ithaca. +Nurseryman.+ Granjean, Julio, c/o K. E. Granjean, 9406 68th Ave., Forest Hills Gressel, Henry, Route 2 Mohawk Haas, Dr. Sidney V., 47 West 86th Street, New York 24. +Physician.+ Hasbrouck, Walter, Jr., 19 Grove Street, New Paltz Iddings, William A., 1931 Park Place, Brooklyn Irish, G. Whitney, Valatie Knorr, Mrs. Arthur, 16 Central Park, West, Apt. 1406, New York Kraai, Dr. John, Fairport. +Physician.+ Larkin, Harry H., 189 Van Rennsselaer Street, Buffalo, 10 *Lewis, Clarence, 1000 Park Avenue, New York Little, George, Ripley. +Farmer.+ *MacDaniels, Dr. L. H., Cornell University, Ithaca. +Head, Dept. of Floriculture & Orn. Hort.+ Miller, J. E., J. E. Miller Nurseries, Canandaigua. +Nurseryman, fruit grower.+ Mitchell, Rudolph,

125 Riverside Drive, New York 24 *Montgomery, Robert H., I E. 44th Street, New York Mossman, Dr. James K., Black Oaks, Ramapo Muenscher, Prof. W. C., 1001 Highland Road, Ithaca. +Prof. of Botany, Cornell University, also grows black walnuts.+ Nelson, Howard F., 350 Elmwood Avenue, Buffalo 18 Newell, P. F., Lake Road, Route 1, Westfield Overton, Willis W., 3 Lathrop Street, Carthage Owen, Charles H., Sennett. +Superintendent of Schools.+ Page, Charles E., Route 2, Oneida Rightmyer, Harold, Route 4, Ithaca Salzer, George, 169 Garford Road, Rochester 9. +Chestnut nurseryman.+ Schlegel, Charles P. 990 South Avenue, Rochester 7 Schlick, Frank, Munnsville Schmidt, Carl W., 180 Linwood Avenue, Buffalo Shannon, J. W., Box 90, Ithaca Sheffield, Lewis J., c/o Mrs. Edna C. Jones, Townline Road, Orangeburg Slate, Prof. George L., Experiment Station, Geneva Smith, Gilbert L., State School, Wassaic. +Nut Nurseryman.+ Smith, Jay L., Chester. +Nut Tree Nurseryman.+ Steiger, Harwood, Red Hook Szego, Alfred, 77-15 A 87th Avenue, Jackson Heights, New York Timmerman, Karl G., 123 Chapel Street, Fayetteville Waite, Dr. R. H., Willowaite Moor, Perrysburg. +Physician.+ Wichlae, Thaddeus, 3236 Genesee Street, Cheektowaga (Buffalo) 21 *Wissman, Mrs. F. De R.--no address.

NORTH CAROLINA

Brooks, J. R., Box 116, Enka Burch, O. L., Route 2, Roxboro Dunstan, Dr. R. T., Greensboro College, Greensboro Finch, Jack R., Bailey. +Farmer.+ Parks, C. H., Route 2, Asheville. +Mechanic.+ Wagner, J. M., Turner Manufacturing Company, Statesville

NORTH DAKOTA

Bradley, Homer L., Long Lake Refuge, Moffit. +Refuge Manager.+

OHIO

Glen Helen Department, Antioch College, Yellow Springs Barden, C. A., 215 Morgan Street, Oberlin. +Real Estate.+ Bitler, W. A., R.F.D. 1, Shawnee Road, Lima. +General Contractor.+ Brewster, Lewis, Swanton Bungart, A. A., Avon Cinade, Mrs. Katherine, 13514 Coath Avenue, Cleveland 20 Clark, R. L., 1184 Melbourne Road, East Cleveland 12 Cook, H. C, Route 1, Box 12, Leetonin Cornett, Charles L., R.R. Perishable Inspection Agency, 27 W. Front St., Cincinnati. +Inspector.+ Craig, George E., Dundas (Vinton County) Cranz, Eugene F., Mount Tom Farm. Ira Cunningham, Harvey E., 420 Front Street, Marietta Daley, James R., 400 W. South Street, Vermilion. +Electrician.+ Davidson, John, 234 East Second Street, Xenia. +Writer.+ Davidson, Mrs. John, 234 East Second Street, Xenia Davidson, William J., Old Springfield Pike, Xenia Diller, Dr. Oliver, D., Dept. of Forestry, Experiment Station, Wooster Dowell, Dr. L. L., 529 North Ave., N.E., Massillon Dubois, Miss Frances H., 6938 Miami Road, Cincinnati 27. +Landscape gardener and newspaper columnist.+ Emch, F. E., Genoa Evans, Maurice G., 335 S. Main Street, Akron 8 Fickes, Mrs. Ada C., Route 1, Wooster Foraker, Major C. Merle, 2545 Romig Road, Akron Foss, H. D., 875 Hamlin Street, Akron 2 Frederick, George F., 3925 W, 17th,

Cleveland 9 Garden Center of Greater Cleveland, 11190 East Blvd., Cleveland Gauly, Dr. Edward, 1110 Euclid Avenue, Cleveland 15. +Ocullst.+ Gerber, E. P., Kidron Gerstenmaier, John A., 18 Pond S. W., Massillon Goss, C. E., 922 Dover Avenue, Akron 20 Gard, Dr. Edward A., 1506 Chase Street, Cincinnati 23 Greib, Louis W., 1150 N. Limestone St., Springfield Hawk & Son Nursery, Rt. 2, Beach City. +Chestnut trees.+ Heena, Carl R., Route 2, New Richmond Hill, Dr. Albert A., 4187 Pearl Road, Cleveland Irish, Charles F., 418 E. 105th Street, Cleveland 8 Jacobs, Homer L., Davey Tree Expert Company, Kent Kappel, Owen, Bolivar Kintzel, Frank W., 2506 Briarcliffe Avenue, Cincinnati 13. Kobelt, Don, Route 5, Medina. +Insurance Adjuster.+ Kratzer, George, Route 1, Dalton Krok, Walter P., 925 W. 29th Street, Lorain. +Research and Development Engineer.+ Laditka, Nicholas G., 5322 Stickney Avenue, Cleveland 9 Lashley, Charles V., 216 S. Main St., Wellington Lehmann, Carl, 1601 Union Trust Building, Cincinnati, 2. +Attorney at Law.+ Lorenz, R. C., 121 North Arch Street, Route 5, Fremont Machovina, Paul E., 1228 Northwest Blvd., Columbus 12 Madson, Arthur E., 13608 Fifth Avenue, E. Cleveland 12 McBride, William B., 2398 Brandon Road, Columbus 10 McKinster, Ray, 1682 South 4th Street, Columbus 7 Metzger, A. J., 724 Euclid Avenue, Toledo 5 Miller, Ralph J., 251 Westview Avenue, Worthington Nicholson, Jonathan J., 175 W. Columbus Avenue, Mount Sterling Nicolay, Charles, 2259 Hess Avenue, Cincinnati 11. +Accountant.+ Oches, Norman M., R.D. 1, Brunswick. +Mechanical Engineer.+ Osborne, Frank C., 4040 W. 160th Street, Cleveland 11 Pomerene, Walter H., Route 3, Coshocton. +Agricultural Engineer, Hydrological Research Station+ Ranke, William, Route 1, Amelia Rieck, C., 522 S. Main Street, Findlay Rummel, E. T., 13618 Laverne Avenue, Cleveland 11. +Sales Engineer.+ Schaufelberger, Hugo S., Route 2, Sandusky Seas, D. Edward, 721 South Main Street, Orrville Scitz, M. B., 975 Nome Ave., Akron. +Auto dealer.+ Shelton, Dr. E. M., 1468 W. Clifton Blvd., Lakewood 7 Sherman, L. Walter, Mahoning County Experiment Farm, Canfield Shessler, Sylvester M., Genoa Silvis, Raymond E., 1725 Lindberg Avenue, N.E. Massillon. +Realty.+ Slutz, Russell C., Box 504, 123 High St., Navarre Smith, Kenneth, 642 Collins Park Avenue, Toledo Smith, Sterling A., 630 W. South Street, Vermilion +Telegrapher, N.Y.C.R.R. (Treasurer of the Association.)+ Spring Hill Nurseries Company, Tipp City. +General nurseryman.+ Steinbeck, A. P., Box 824, Route 7, North Canton Stocker, C. P., Lorain Products Corp., 1122 F. Street, Lorain Thomas, Fred, Route 1, Bedford Road, Masury Thomas, W. F., 406 South Main Street, Findlay Toops, Herbert A., 1430 Cambridge Blvd., Columbus 12. +College Professor.+ Urban, George, 4518 Ardendale Road, South Euclid 21. +Mayor.+ Van Voorhis, J. F., 215 Hudson Avenue, Apt. B-1, Newark Von Gundy, Clifford E., R.F.D. 13, Cincinnati 30 Walker, Carl F., 2851 E. Overlook Road, Cleveland 18 Weaver, Arthur W., 3339 South Street, Toledo 4 *Weber, Harry R., Esq., 123 E. 6th Street, Cincinnati. +Attorney, Farm owner.+ Weber, Mrs, Martha R., Route 1, Morgan Road, Cleves Williams, Harry M., 221 Grandon Road, Dayton 9. +Engineer.+ Willett, Dr. G. P., Elmore Wischhusen, J. F., 15031 Shore Acres Drive, N.E., Cleveland 10 Yates, Edward W., 3108 Parkview Avenue, Cincinnati 13 Yoder, Emmet, Smithville

OKLAHOMA

Butler, Roy, Route 2, Hydro. +Farmer, cattleman.+ Cross, Prof. Frank B., Dept. of Horticulture, Oklahoma A & College, Stillwater. +Teaching and Experiment Station work.+ Gray, Geoffrey A., 1628 Elm Ave., Bartlesville Hirschi's Nursery (A.G.) 414 North Robinson, Oklahoma City. +Dry cleaning business, nurseryman.+ Hartman, Peter E., Hartsdale Nursery Company, P. O. Box 882, Tulsa 1. +Nurseryman.+ Hughes, C. V., Route 3, Box 614, 5600 N.W. 16th Street, Oklahoma City Meek, E. B., Route 2, Wynnewood Pulliam, Gordon, 1005 Osage Ave., Bartlesville Ruhlen, Dr. Charles A., 114 North Steele. Cushing. +Dentist.+ Swan, Oscar E. Jr., 1226 E. 30th Street, Tulsa 6. +Attorney, Mid-Continent Petroleum Co.+

OREGON

Carlton Nursery Company, Forest Grove. +Nurserymen and Nut Orchardists.+ Miller, John E., Route 1, Box 912-A, Oswego Osborne, W. L. H., Mont Alto, Idylyld Route 275, Roseburg Pearcy, Harry L., Rt. 2, Box 190, Salem. +H. L. Pearcy Nursery Co. (Nut trees.)+ Sheppard, Charles M., Tucker Road, Hood River

PENNSYLVANIA

Allaman, R. P., Route 86, Harrisburg Bangs, Ralph E., Route 2, Spartansburg. +Farmer.+ Banks, H. C. Route 1, Hellertown Beard, H. K., Route 1, Sheridan. +Insurance Agent.+ Berst, Charles B., 11 W. 8th Street, Erie. +Inspector, Lord Mfg Co., Erie, Pa.+ Bowen, John C. Route 1, Macungie Breneiser. Amos P., 427 North 5th Street, Reading Brown, Morrison, 342 East Cooper Street, Slippery Rock. +Teacher.+ Clarke, William S., Jr., P. O. Box 167, State College Creasy, Luther P., Catawissa Damask, Henry, 1632 Doyle Street, Wilkinsburg. +Telephone man.+ Eckhart, Pierce, 5731 Haddington Street, Philadelphia 31 Etter, Fayette, P. O. Box 57, Lemesters. +General Line Foreman for an Electric Company.+ Gardner, Ralph D., 4428 Plymouth Street, Harrisburg. +Assistant State Fire Marshall.+ Good, Orren S., 316 N. Fairview Street, Lock Haven. +Retired.+ Gorton, F. B., Route 1, East Lake Road, Harbor Creek. +Electrical Contractor.+ Hammond, Harold, 903 South Poplar Street, Allentown Heckler, George Snyder, Hatfield Hostetter, L. K., Route 3, Lancaster. +Farmer, black walnut grower.+ Hughes, Douglas, 1230 East 21st Street, Erie Johnson, Rooert F., 1625 Greentree Road, Pittsburgh 5 Jones, Mildred M. (See Mrs. Langdoc--under Illinois) Jones, Dr. Truman W., Walnut Grove Farm, Parksburg Kaufman, Mrs. M. M., Clarion. Kirk, DeNard B., Forest Grove. +Engineer.+ Knowse, Charles W., Colonial Park, Harrisburg. +Coal Dealer.+ Laboski, George T., Route 1, Harbor Creek. +Fruit Grower and Nurseryman.+ Lambert, E. A., Box 76, McKean Leach, Will, 406-410 Scranton Life Bldg., Scranton 3. +Lawyer.+ Mattoon, H. Gleason, Box 304, Narberth. +Consultant in Arboriculture.+ Mecartney, J. Lupton, Room 1 Horticultural Building, State College. +Teacher.+ Mercer, Robert A., Rt. 1, Porkiomenville Miller, Elwood B., c/o The Hazleton Bleaching & Dyeing Works, Hazleton Miller, Robert O., 3rd and Ridge Street, Emmaus Moyer, Philip S., U.S.F. & G. Building, Harrisburg Nicderriter, Leonard, 1726 State Street, Erie. +Merchant.+ Nonnemacher, H. M., Box 204, Alburtis. +Line Foreman, Bell Tel. Co. of Pa.+ Oesterling, Howard M., R.D. 1, Marysville Ranson, Flaval, 728 Monroe Avenue,

Scranton 10. +Farmer.+ Reidler, Paul G., Ashland. +Manufacturer of textiles.+ Rial, John, 528 Harrison Ave, Greensburg *Rick, John, 438 Pennsylvania Square, Reading Rupp, Edward E., Jr., 57 W. Pomfret Street, Carlisle. +Draftsman--Tree Surgeon.+ Schaible, Percy, Upper Black Eddy. Laborer. Smith, Dr. J. Russell, 550 Elm Avenue, Swarthmore. +Geographer-Nurseryman-Author.+ Sofianos, Louis S., Greenfields, Reading. Gardener. Starr, Miss Charlottee Churchill. R.R. 1 Bucks County, Quakertown. +Artist & housewife.+ Stewart, E. L., Pino Hill Farms Nursery, Route 2, Homer City Theiss, Dr. Lewis E., Bucknell University, Lewisburg. +Writer-Retired College Professor+ Twist, Frank S., Box 127, Northumberland Washick, Dr. Frank A., S.W. Welsh & Veree Roads, Philadelphia 11. +Surgeon.+ Weaver, William S., Weaver Orchards, Macungie Weinrich, Whitney, P. O. Box 225, Wallingford *Wister, John C., Scott Foundation, Swarthmore College, Swarthmore. +Horticulturist.+ Wright, Ross Pier, 235 W. 6th Street, Erie Zarger, Thomas G., Route 3, Chambersburg Zimmerman, Mrs. G. A., R.D., Linglestown

RHODE ISLAND

*Allen, Philip, 178 Dorance Street, Providence Rhode Island State College, Library Dept., Green Hall, Kingston

SOUTH CAROLINA

Bregger, John T., Clemson +Research Supervisor (Soil Conservation), Orchard Erosion Investigations.+ Gordon, G. Henry, 13 1/2. Main Street, Union. +Retired Mariner.+ Henderson, E. P., Bath. +Manager, Kaolin Mines & Ornamental Nurseryman.+ Hundley, P. C. & Son, Woodruff. +Orchard supplies.+ Poole, M. C., Cross Anchor. +Beach grower.+ Senn. T. L., Horticultural Dept. Clemson College. Clemson. +College Teacher.+

SOUTH DAKOTA

Richter, Herman, Madison

TENNESSEE

Acker Black Walnut Corporation, Morristown. +Walnut processors.+ (See also under Virginia.) Alpine Forest Reserve, Alpine. +Presbyterian Church project.+ Boyd, Harold B., M. D., 905 Kensington, Memphis 7. +Physician.+ Boyd, Robert W., Boyd Nursery Company, McMinnville. +General nurseryman.+ Chase, Spencer, T.V.A., Norris. +Horticulturist.+ Cox, T. S., 108 Hotel Avenue, Knoxville 18. Dunlap, Dr. William B., 912 E. Main Street, Union City. +Optometrist.+ Garrett, Dr. Sam Young, Dixon Springs. +Surgeon.+ Holdeman, J. E., 855 N. McNeill, Memphis 7 Howell Nurseries, Sweetwater. +Ornamental and chestnut nurserymen.+ Kingsolver, J. B. Route 2, Concord Lowe, Dr; Jere W., c/o Mrs. Murphy Webb; Westover Drive, Nashville McAlexander, Kenneth J., Cedar Grove. +College student.+ McDaniel, J. C, Tenn. Dept: of Agriculture, 403 State Office Bldg., Nashville 3. Horticulturist, farmer.

McDaniel, Mrs. J. C, 1421 Kirtland Avenue, Nashville 6 McDaniel, J. C, Jr., 1421 Kirkland Avenue, Nashville 6 McQueen, S. S., Box 1262, Mountain Home Murphy, H. O. 12 Sweetbriar Avenue, Chattanooga. +Fruit grower.+ Parsley, G. B., Route 1, Smithville. +Nurseryman.+ Rhodes, G. B., Route 1, Covington. +Farmer.+ Richards, Dr. Aubrey, Whiteville. +Physician.+ Roark, W, F., Malesus. +Farmer, chestnut grower.+ Robinson, W. Jobe, Route 7, Jackson. +Farmer.+ Sammons, Julius, Jr., Whiteville Shadow, Willis A., Decatur. +County Agricultural Agent.+ Shipley, Mrs. E. D., 3 Century Court, Knoxville Smathers, Rev. Eugene, Calvary Church, Big Lick. +Pastor.+ Southern Nursery & Landscape Co., Winchester Sutherland, W. B., 520 Clearview Street, Knoxville 17 Zarger, Thomas G. (Temporarily in Pennsylvania)

TEXAS

Arford, Charles A., Box 1230, Dalhart Arp Nursery Company, (Clark Kidd) 5th and Wall St., P.O. Box 867, Tyler. +Wholesale Nursery.+ Bailey, L. B., Box 1436, Phillips. +Chemist.+ Brison, Prof. F. R., Dept. of Horticulture, A. & M. College, College Station Florida, Kaufman, Box 151, Rotan Price, W. S. Jr., Navarro County, Kerens Romberg, L. D., U. S. Pecan Field Station, Box 539, Brownwood Winkler, Andrew, Route 1, Moody. +Farmer and pecan grower.+

UTAH

Petterson, Harlan D., 2164 Jefferson Avenue, Ogden. +Highway Engineer.+

VERMONT

Aldrich, A. W., Route 3, Sprinfield. +Farmer.+ Collins, Joseph N., Route 3, Putney Ellis, Zenas H., Fair Haven. +Perpetual member, "In Memoriam."+ Foster, Forest K., West Topsham. +Fruit grower.+ Ladd, Paul, Putney. +School Teacher.+

VIRGINIA

Acker Black Walnut Corporation, Broadway. +Walnut processors. (See also under Tennessee.)+ Burton, George L., 728 College Street, Bedford Case, Lynn B., Route 1, Fredericksburg Dickerson, T. C., 316-56th Street, Newport News. +Statistician, farmer.+ Dudley, Charles, Glen Wilton Gibbs, H. R., 808 William Street, Front Royal. +Carpenter, wood worker.+ Gunther, Eric F., Route 1, Box 31, Onancock. +Retired business man.+ Lee, Dr. Henry, 806 Medical Arts Building, Roanoke 11 Pinner, R. McR., P.O. Box. 155, Suffolk Stoke, H. F., 1436 Watts Avenue N.W., Roanoke Stoke, Mrs. H. F., 1436 Watts Avenue, N.W., Roanoke Stoke, Dr. John H., 21 Highland Avenue, S.E., Roanoke 18. +Chiropractor.+ Thompson, B. H., Harrisonburg. +Manufacturer of nut crackers.+

WEST VIRGINIA

Cannaday, Dr. John E., Charleston General Hospital, Charleston 25. +Surgeon.+ *Frye, Wilbert M., Pleasant Dale Gold Chestnut Nursery, c/o Mr. Arthur A. Gold, Cowen. +Chestnut nurseryman.+ Haines, Earl C., Shanks Long, J. L., Box 491, Princeton Mish, Arnold F., Inwood Reed, Arthur M., Moundsville. +Proprietor, Glenmount Nurseries.+ Shepler, Harvey, Oxford

WASHINGTON

Altman, Mrs. H. E., 2338 King Street, Bellingham 9 Barth, J. H., Box 1827, Route 3, Spokane 15. +Watchmaker and farmer.+ Bartleson, C. J., Box 25, Chattaroy. +Office worker.+ Biddle, Miss Gertrude W., 928 Gordon Avenue, Spokane 12 Brown, H. R., Greenacres Bush, Carroll D., Grapeview. +Chestnut grower and shipper.+ Denman, George L., 1319 East Nina Avenue, Spokane 10. +Dairyman.+ Eliot, Craig P., P. O. Box 158, Shelton. +Electrical Engineer, part time farmer.+ Hyatt, L. W., 2826 West La Crosse, Spokane 12 Kling, William L., Route 2, Box 230, Clarkston Knight, J. C., W. 723 Sinto Avenue, Spokane 12. +Retired.+ Latterell, Misa Ethel, Greenacres Linkletter, F. D., Route 2, Box 722, Mercer Island Naderman, G. W., Route 1, Box 381, Olympia Shane Brothers, Vashon Shepard, Will, Chelan Falls Tuttle, Lynn, Nursery, The Heights, Clarkston

WISCONSIN

Cox, Irvin W., P.O. Box 2632, West Allis Koelsch, Norman, Jackson Ladwig, C. F., 2221 St. Lawrence, Beloit Mortensen, M. C., 2117 Stanson Avenue, Racine Talbot, Harold W., Rt. 7, Hex 198, Milwaukee 13

NEW MEMBERS ADDED SINCE MAY 1, 1949

Carlisle, Francis, 819 Second Street, S.E., New Philadelphia, Ohio Gehring, Rev. Titus, P. O. Box 668, Grants, New Mexico Keathly, Jack, Marland, Oklahoma Koeferl, Alots J., 2835 North 20th St., Milwaukee 6, Wisconsin O'Brien, Howard C., 25 Irvington Street, Boston 16, Massachusetts Spears, Ernest G., 4326 Forest Avenue, Norwood 12, Ohio Warnecke, Martin H., 714 S. First Avenue, Maywood, Illinois

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+Subscribers and Standing Library Orders+

Brooklyn Botanic Garden Library, 1000 Washington Avenue, Brooklyn 25, N. Y. Clemson College Library, Clemson, South Carolina. Cornell University, College of Agriculture Library, Ithaca, New York. Detroit Public Library, 5201 Woodward Avenue, Detroit 2, Michigan. Jones, G. S., Route 1, Phenix City, Alabama. Library, University of Miami, Coral Gables 34, Florida. Library, University of New Hampshire, Durham N. H. Oregon State College Library, Corvallis, Oregon. Peachy, Enos D., P. O. Box 22, Belleville, Pennsylvania. Rhode Island State College, Library Dept., Green Hall, Kingston, Rhode Island (membership). Rutgers University; Agricultural Library, Nichol Ave., New Brunswick, N. J. St. Louis Public Library, Olive, 13th and 14th

Streets, St. Louis, Missouri.

EXHIBITS AT THE NORRIS MEETING, 1948

Noah Abernathy, Marble, North Carolina. Chinese chestnuts.

Benton & Smith Nut Tree Nursery, Wassaic, New York. Shagbark hickories, hybrid hickory, Persian walnut.

Dr. R. T. Dunstan, Greensboro College, Grensboro, North Carolina. Persian walnuts.

A. G. Hirschi, Oklahoma City, Oklahoma. Pecans, black walnuts, heartnut, Carpathian Persian walnut, Chinese chestnuts, Oriental persimmons.

Jaynes Hobson, Jasper, Georgia. Chinese chestnuts.

Howell Nurseries, Sweetwater, Tennessee. Chinese chestnuts, Japanese chestnut.

Dr. G. S. Jones, Phenix City, Alabama. Chinese chestnuts.

G. J. Korn, Kalamazoo, Michigan. Shagbark hickories, shellbark hickory, black walnuts, butternut, collection of photographs.

R. C. Lorenz, Fremont Ohio. Pecan, Persian walnut.

Dr. C. A. Moss, Williamsburg, Kentucky. Black walnuts, Persian walnut, pecans, shellbark hickory, Chinese chestnut, filbert.

New York State Agricultural Experiment Station, Geneva, New York. Turkish tree hazel; Chinese tree hazel, native hazel, European hazels, hybrid hazels, black x Persian hybrid walnut.

G. B. Rhodes, Covington, Tennessee. Pecans, heartnut, Persian walnut.

Dr. Aubrey Richards, Whiteville, Tennessee. Chinese and hybrid chestnuts, heartnuts, black walnuts.

Sylvester Shessler, Genoa, Ohio. Persian walnuts, black walnuts, heartnut.

H, F. Stoke, Roanoke, Virginia. Persian walnuts, black walnuts, butternuts, heartnuts, shellbark hickory, shagbark hickory, filberts, Chinese, Japanese and hybrid chestnuts, hybrid hazels, graft unions, photographs.

TVA Forestry Relations Department (Norris Nursery), Norris, Tennessee. Large collections of black and seedling Persian walnuts; Chinese chestnuts, heartnuts, filberts, American hazel, pecans, shellbark hickory, Oriental persimmons.

U.S.D.A. Pecan Station, Albany, Georgia. Named varieties of Chinese chestnut.

J. F. Wilkinson, Rockport, Indiana. Pecans.

Dr. W. C. Willett, Elmore, Ohio. Heartnuts.

William J. Wilson, Fort Valley, Georgia. Black walnut, hican.

(List compiled by H. F. Stoke)

ANNOUNCEMENTS

40th Annual Meeting at Beltsville, Maryland September 6, 7 and 8, 1949

Dr. H. L. Crane, Plant Industry Station, Beltsville, Maryland, is chairman of the local arrangements and program committees for the 40th Annual meeting this year, to be held at Beltsville, a suburb of Washington, D.C. Other committee members are listed in the front of this volume; They will welcome your suggestions on things to be included in the program and the tour near Washington. _Members will receive the advance program._

+Older Reports of the Northern Nut Growers Association, Inc., Are Available+

The Association, which was organized in 1910, has published a report of its annual meeting each year except two, beginning with 1911.

Sets of reports lacking only volumes for 1923, 1925, 1926, 1935, 1940, 1941, and 1944 may still be purchased, These sets, consist of 29 reports through 1948 and contain over 3800 pages of material pertaining to nut culture in many stated and Canada. The price of the set of available reports is $12.00. (A very few complete sets through Vol. 39, including an index to the first 30 volumes, are available to agricultural and other libraries only at $17.00). Single numbers are $1.00 each, except the current number and the preceding one: 1948 at. $3.00 and 1947 at $2.00 each. Orders should be sent to the secretary accompanied by remittances made payable to the Northern Nut Growers Association, Inc.

Libraries and other institutions desiring to receive the reports regularly without the bother of ordering them every year may have their names placed on a special mailing list to receive each report regularly when published. A bill for $2.00 will accompany the 1949 report, when sent to such institutions.

+Other Publications on Nut Growing+

1. Bush, Carrol D. _Nut Grower's Handbook._ Orange Judd Publishing Company, New York, 1941. $2.50.

2. Smith, J. Russell. _Tree Crops, A Permanent Agriculture._ Revised edition on schedule for 1949 publication. Inquire of author at Swarthmore, Pa.

3. Smith, J. Russell, _How to Graft Nut Trees._ May be purchased from Walnut Lane Press, Swarthmore, Pa. Illustrated with diagrams. 9 pp. 25c.

4. Smith, J. Russell. _The Planting, Fertilization and Care of Nut Trees and Persimmon Trees._ Available from Sunny Ridge Nursery, Swarthmore, Pa., price 25c.

5. Reed, C. A. _Nut Tree Propagation._ U. S. Department of Agriculture Farmers' Bul. 1501. For sale only. 5c (coin) from Supt. of Documents, U. S. Government Pointing Office, Washington 25, D. C.

6. Mattoon, W. R. & Reed, C. A. _Planting Black Walnuts._ U. S. Department of Agriculture Leaflet 84. Free from Department of Agriculture; Washington, D. C.

7. Moznette, G. F. et al. _Insects and Diseases of the Pecan and their Control._ U. S. Department of Agriculture-Farmers' Bul. 1829. May be had from U. S. Department of Agriculture, Washington, D. C.

8. Sitton, B. G. & Akin, E. O. _Grafting Wax Melter,_ U. S. Department Leaflet 202. Free from U. S. Department of Agriculture, Washington, D. C.

9. Sitton, B. G. _Pecan Grafting Methods and Waxes._ U. S. Department of Agriculture Circ. 545. May be had from U. S. Department of Agriculture, Washington, D. C.

10. Sitton, B. G. _Vegetative Propagation of the Black Walnut._ Mich. Sta. Tech. Bul. No. 119. Sept., 1931. Available from Michigan State College, E. Lansing.

11. MacDaniels, L. H. _Nut Growing._ Cornell Univ. Ext. Bul. 701. From College of Agriculture, Ithaca, New York.

12. Haseman, L. _The Walnut Caterpillar._ Missouri Exp. Sta. Bul. 418.

13. Talbert, T. J. _Nut Tree Culture in Missouri._ Mo. Exp. Sta. Bul. 454. May be had from Agr. Exp. Station, Columbia, Mo.

14. Schuster, C. E. _Filberts._ Oregon State College Ext. Bul. 628. May be had from Oregon State College, Corvallis, Oregon.

15. Schwartze, C. D. _Filbert Culture._ Washington State Col. Ext. Bul. 263. May be had from Extension Service, Washington State College, Pullman, Wash.

16. Sherman, L. W. and Ellenwood, G. W. _Topworking and Bench-grafting Walnut Trees._ Special Circ. 69. May be had from Agr. Exper. Sta., Wooster, Ohio.

17. Slate, G. L. _Filberts._ N. Y. State Agr. Exp. Sta. Cir. 192. Free from Experiment Station, Geneva, N. Y.

18. _DDT to Control Pecan Weevil._ Multigraphed, 1948. Available free from Division of Horticulture, Tenn. Dept. of Agriculture, Nashville 3, Tenn.

19. Blake, M. A. and Edgerton, L. J. _Experience with Blight Resistant Chestnuts in New Jersey._ Bul. 717 N. J. Agr. Exp. Sta., New Brunswick, N. J.

20. Yerkes, Guy E. _Propagation of Trees and Shrubs._ U.S.D.A. Farmers' Bul. No. 1567, available from Supt. of Documents, U. S. Gov't Printing Office, Washington 25, D. C., price 10c (coin).

21. Cox, John A., et al. _Top Working Pecan Trees._ Ext. Circ. 209. Available free from Louisiana State University, University, Louisiana.

22. Hilton, R. J. _Frameworking Fruit Trees._ Farmers' Bulletin 136 of the Dominion Department of Agriculture. Available from Dominion Department of Agriculture, Ottawa, Ontario, Canada. (Framework grafting technique is adapted to top-working large seedling nut trees).

23. Snyder, John C. _Pollination of Tree Fruits and Nuts._ Ext. Bul. 342. Washington State College, Pullman, Washington.

24. Smith, Gilbert L. _Practical Nut Growing._ 60 pp. illus. $1.50 from author, Wassaic, N. Y.

Note: In addition to the above publications, the horticultural departments of many state and provincial agricultural experiment stations and agricultural colleges have free circulars or bulletins listing the recommended varieties of fruit and nut trees for their areas. The prospective tree planter is advised to place more reliance on the local recommendations (where available) than on those from distant states where the soils, the climate, and the adapted varieties may be quite different.

The NNGA list of some nurseries which sell hardy, named varieties of nut trees is revised each winter. The secretary, will send copies of the next revision free on request.--J. C. McDaniel, _Sec'y._, Nashville 3, Tenn.